

MFIA User Manual

500 kHz / 5 MHz Impedance
Analyzer



Zurich
Instruments

MFIA User Manual

Zurich Instruments AG

Revision 24.01

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CE Declaration of Conformity



The manufacturer

Zurich Instruments
Technoparkstrasse 1
8005 Zurich
Switzerland

declares that the product

MFIA, Impedance Analyzer , 500 kHz / 5 MHz, 60 MSa/s

is in conformity with the provisions of the relevant Directives and Regulations of the Council of the European Union:

Directive / Regulation	Conformity proven by compliance with the standards
2014/30/EU (Electromagnetic compatibility [EMC])	EN 61326-1:2013, EN 55011:2016, EN 55011:2016/A1:2017, EN 55011:2016/A11:2020 (Group 1, Class A and B equipment)
2014/35/EU (Low voltage equipment [LVD])	EN 61010-1:2010, EN 61010-1:2010/A1:2019, EN 61010-1:2010/A1:2019/AC:2019-04
2011/65/EU, as amended by 2015/863 and 2017/2102 (Restriction of the use of certain hazardous substances [RoHS])	EN IEC 63000:2018
(EC) 1907/2006 (Registration, Evaluation, Authorisation, and Restrictions of Chemicals [REACH])	-

Zurich, October 20th, 2022

Flavio Heer, CTO

UKCA Declaration of Conformity



The manufacturer

Zurich Instruments
Technoparkstrasse 1
8005 Zurich
Switzerland

declares that the product

MFIA, Impedance Analyzer , 500 kHz / 5 MHz, 60 MSa/s

is in conformity with the provisions of the relevant UK Statutory Instruments:

Statutory Instruments	Conformity proven by compliance with the standards
S.I. 2016/1091 (Electromagnetic Compatibility Regulations)	EN 61326-1:2013, EN 55011:2016, EN 55011:2016/A1:2017, EN 55011:2016/A11:2020 (Group 1, Class A and B equipment)
S.I. 2016/1101 (Electrical Equipment (Safety) Regulations)	EN 61010-1:2010, EN 61010-1:2010/A1:2019, EN 61010-1:2010/A1:2019/AC:2019-04
S.I. 2012/3032 (Restriction of the Use of Certain Hazardous Substances Regulations)	EN IEC 63000:2018

Zurich, October 20th, 2022

A handwritten signature in black ink that reads 'Flavio Heer'.

Flavio Heer, CTO

1. Change Log

1.1. Release 24.01

Release date: 31-Jan-2024

- Sweeper: The sweeper grid now includes the exact start and stop sweeping points.
- Threshold Unit (TU): Indicators to display the saturation of TU input at lower and upper bounds.
- AM/FM MOD Option: Phase-adjust capability to nullify the measured phase of demodulated signals.

1.2. Release 23.10

Release date: 31-Oct-2023

- Sweeper: Dual-plot mode as the default configuration of Sweeper tab.
- Sweeper: Setting the start and stop points of the sweep parameter from the x-axis cursors in the Sweeper tab.
- Scope: Recording of scope streaming samples in HDF5 format from the Config tab.

1.3. Release 23.06

Release date: 30-Jun-2023

- Equivalent circuit model for dielectric material characterization.
- Open-circuit mode for user compensation.
- Impedance phase compensation for any cable length.

1.4. Release 23.02

Release date: 28-Feb-2023

- Aux Outputs: Changed the default scale factor from 1 to 10.
- LabOne API: Added support of Python 3.11.
- Sweeper Module: Improved phase unwrap feature.
- LabOne: Dropped support for Windows 7 and Windows 8.1 as they have reached their end of life.

1.5. Release 22.08

Release date: 31-Aug-2022

- Impedance Analyzer: Added input monitor and signal-add control of signal output to IA tab.
- Lock-in Amplifier: Added Tip Protect feature to adjust the Aux Output signal.
- Scope Module: Added Aux Output trace to Scope input to display Aux Output value after Limit and Tip Protect units.
- PID/PLL Option: Added Threshold Unit output to the list of signal inputs for PID controllers.

1.6. Release 22.02

Release date: 28-Feb-2022

- Impedance Analyzer: Option to enable or disable all confidence indicators at once.
- LabOne Software: 'Flat Top' window function for FFT mode of Scope, DAQ, and Spectrum.
- LabOne API: Support of Python 3.10.

1.7. Release 21.08

Release date: 30-Aug-2021

- Impedance Analyzer: Current Zone Ranging is added, which allows you to manually set the switching frequency limits for all eight current input ranges.
- Impedance Analyzer: Impedance Ranging is deprecated to avoid confusion.
- Impedance Analyzer: User compensation sequence is improved. The Open step is moved into the Advanced tab.

2. Getting Started

This first chapter guides you through the initial set-up of your MFIA Instrument in order to make your first measurements. This chapter comprises:

- A Quick Start Guide
- The package contents list
- Essential handling and safety instructions
- Instructions for connecting to the MFIA Instrument
- Instructions for running LabOne on a separate PC
- Instructions for getting started with the LabOne APIs
- Troubleshooting guidelines

This chapter is delivered as a hard copy with the instrument upon delivery. It is also the first chapter of the MFIA User Manual.

2.1. Quick Start Guide

This page addresses all the people who have been impatiently awaiting their new gem to arrive and want to see it up and running quickly. If the MFIA Instrument is to be integrated into a LAN with DHCP server, it is ready to go without further need of software installation. Please proceed along the following steps:

1. Inspect the package content. Besides the Instrument there should be a country-specific power cable, a USB cable, an Ethernet cable and a hard copy of the user manual [Getting Started](#).
2. Check the Handling and Safety Instructions in [Handling and Safety Instructions](#).
3. Connect the Instrument to the power line. Turn it on and connect it to a switch in the LAN using the Ethernet cable, or directly to the host computer with the USB cable. After flashing green for a short time, the front panel LED will show a steady blue color. If the LED remains flashing green, power-cycle the MFIA instrument again and wait for the LED to turn blue. If the LED does not turn blue, please contact Zurich Instruments for assistance.
4. **Ethernet:** open a web browser (Chrome, Edge, Firefox, Safari, or Opera) on a computer in the LAN and type the following text in the address bar:
`http://<instrument-serial>/`
 where `<instrument-serial>` is the serial number of the instrument that can be found on the instrument back panel. Using the `http://` prefix and trailing slash prevents the triggering of search engines. Examples for the browser address bar (left) and for the instrument serial sticker (right) are shown below. In this particular example the serial number is **MF-DEV3026**.



Alternative ways to address the instrument are: `http://<instrument-serial>.<domain>/` where `<domain>` is the network domain in which the instrument is running, `http://<instrument-serial>.local/` and, in case the IP address is known, e.g. `http://192.168.11.2/`.

USB (Windows): when connected via USB, the MFIA acts as a flash drive similar to a USB stick. Find the suitable USB driver for your operating system on that flash drive, and install it e.g. by double-clicking **MF-Device-Finder64.msi**. [Universal Serial Bus \(USB\) Connection](#) contains more detailed instructions. Under Windows 10, the installation of the driver will create a Start Menu entry¹ in **Start Menu → Zurich Instruments → MF Device Finder**. Click this link to open the MF Device Finder app which lists all available instruments. Double-click on the entry for your instrument to open the default web browser with the LabOne User Interface start-up screen.

USB (Linux/Mac): Linux computers usually don't require a USB driver installation. On Macintosh computers, install the USB driver according to the installations in [Universal Serial Bus \(USB\) Connection](#). Open a web browser and enter `http://<instrument-serial>/` into

2.2. Inspect the Package Contents

- the address bar, where the instrument serial of the format **MF-DEV3026** is found on the back panel of your instrument.
5. The LabOne User Interface start-up screen will appear. Click the **Open** button on the lower right of the page. The default configuration will be loaded and the first measurements can be taken. If the user interface does not start up successfully, please refer to [Running LabOne on a Separate PC](#).

If any problems occur whilst setting up the instrument and software please see the [Troubleshooting](#) at the end of this chapter. For turning off the instrument after use, it is recommended to use the soft power off functionality by clicking on the  button at the bottom left of the user interface screen before turning the power switch on the back panel of the instrument.

Once the Instrument is up and running we recommend going through some of the [Tutorials](#). Moreover, [Functional Overview](#) provides a general introduction to the various tools and settings tabs with tables in each section providing a detailed description of every UI element as well. For specific application know-how the [blog section](#) of the Zurich Instruments website will serve as a valuable resource that is constantly updated and expanded.

1. Under Windows 7 and 8, the Start Menu entry is found in **Start Menu → All programs / All apps → Zurich Instruments → MF Device Finder** ←

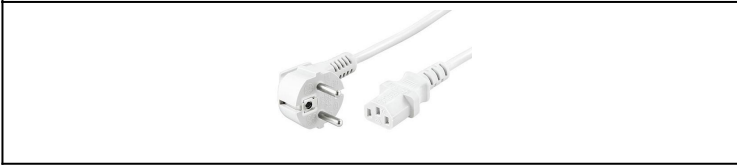
2.2. Inspect the Package Contents

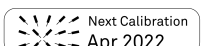
If the shipping container appears to be damaged, keep the container until you have inspected the contents of the shipment and have performed basic functional tests.

Please verify:

- You have received 1 Zurich Instruments MFIA Instrument
- You have received 1 Zurich Instruments MFITF Impedance Test Fixture
- You have received 1 power cord with a power plug suited to your country
- You have received 1 USB cable and/or 1 LAN cable (category 5/6 required)
- You have received 1 Getting Started booklet
- The "Next Calibration" sticker on the rear panel of the Instrument indicates approximately 2 years ahead in time. Zurich Instruments recommends calibration intervals of 2 years
- The MAC address and serial number of the instrument are displayed on a sticker on the back panel

Table 2.1: Package contents for the MFIA Instrument

	
	MFITF Impedance Test Fixture including DUT carriers
	the power cord (e.g. EU norm)

	USB cable
	Power inlet with power switch and fuse holder
	LAN / Ethernet cable (category 5/6 required)
	S/N and MAC address sticker on the back panel of the instrument
	the "Next Calibration" sticker on the back panel of the instrument

The MFIA Instrument is equipped with a multi-mains switched power supply, and therefore can be connected to most power systems in the world. The fuse holder is integrated with the power inlet, and can be extracted by grabbing the holder with two small screwdrivers at the top and at the bottom at the same time. A spare fuse is contained in the fuse holder. The fuse description is mentioned in the specifications chapter.

Carefully inspect your Instrument. If there is mechanical damage or the instrument does not pass the basic tests, then you should immediately notify the Zurich Instruments support team at support@zhinst.com.

2.3. Handling and Safety Instructions

The MFIA Instrument is a sensitive piece of electronic equipment, and under no circumstances should its casing be opened, as there are high-voltage parts inside which may be harmful to human beings. There are no serviceable parts inside the instrument. Do not install substitute parts or perform any unauthorized modification to the product. Opening the instrument immediately voids the warranty provided by Zurich Instruments.

Do not use this product in any manner not specified by the manufacturer. The protective features of this product may be affected if it is used in a way not specified in the operating instructions.

The following general safety instructions must be observed during all phases of operation, service, and handling of the instrument. The disregard of these precautions and all specific warnings elsewhere in this manual may negatively affect the operation of the equipment and its lifetime.

Zurich Instruments assumes no liability for the user's failure to observe and comply with the instructions in this user manual.

Table 2.2: Safety Instructions

Ground the instrument	The instrument chassis must be correctly connected to earth ground by means of the supplied power cord. The ground pin of the power cord set plug must be firmly connected to the electrical ground (safety ground) terminal at the mains power outlet. Interruption of the protective earth conductor or disconnection of the protective earth terminal will cause a potential shock hazard that could result in personal injury and potential damage to the instrument.
Electromagnetic environment	This equipment has been certified to conform with industrial electromagnetic environment as defined in EN 61326-1. Emissions, that exceed the levels required by the document referenced above, can occur when connected to a test object.
Measurement category	This equipment is of measurement category I (CAT I). Do not use it for CAT II, III, or IV. Do not connect the measurement terminals to mains sockets.

Maximum ratings	The specified electrical ratings for the connectors of the instrument should not be exceeded at any time during operation. Please refer to the Specifications for a comprehensive list of ratings.
Do not service or adjust anything yourself	There are no serviceable parts inside the instrument.
Software updates	Frequent software updates provide the user with many important improvements as well as new features. Only the last released software version is supported by Zurich Instruments.
Warnings	Instructions contained in any warning issued by the instrument, either by the software, the graphical user interface, the notes on the instrument or mentioned in this manual, must be followed.
Notes	Instructions contained in the notes of this user manual are of essential importance for correctly interpreting the acquired measurement data.
Floating shield connectors	The Signal Input connector shields are allowed to float relative to earth ground. These shields are connected to each other through internal circuitry. Attempting to drive them to a different potential can result in high currents that may damage your instrument. The Signal Input connector shields are only allowed to float relative to earth ground within a certain voltage range, please refer to the Specifications . It is recommended to discharge the test device before connecting or to enable the floating Signal Input shields only after the signal source has been connected to the instrument.
High voltage transients due to inductive loads	When measuring devices with high inductance, take adequate measures to protect the Signal Input connectors against the high voltages of inductive load switching transients. These voltages can exceed the maximum voltage ratings of the Signal Inputs and lead to damage.
Location and ventilation	This instrument or system is intended for indoor use in an installation category II and pollution degree 2 environment as per IEC 61010-1. Do not operate or store the instrument outside the ambient conditions specified in the Specifications section. Do not block the ventilator opening on the back or the air intake on the chassis side and allow a reasonable space for the air to flow.
Cleaning	To prevent electrical shock, disconnect the instrument from AC mains power and disconnect all test leads before cleaning. Clean the outside of the instrument using a soft, lint-free cloth slightly dampened with water. Do not use detergent or solvents. Do not attempt to clean internally.
AC power connection and mains line fuse	For continued protection against fire, replace the line fuse only with a fuse of the specified type and rating. Use only the power cord specified for this product and certified for the country of use. Always position the device so that its power switch and the power cord are easily accessible during operation.
Main power disconnect	Unplug product from wall outlet and remove power cord before servicing. Only qualified, service-trained personnel should remove the cover from the instrument.
Operation and storage	Do not operate or store the instrument outside the ambient conditions specified in the Specifications section.
Handling	Handle with care. Do not drop the instrument. Do not store liquids on the device, as there is a chance of spillage resulting in damage.
Safety critical systems	Do not use this equipment in systems whose failure could result in loss of life, significant property damage or damage to the environment.

If you notice any of the situations listed below, immediately stop the operation of the instrument, disconnect the power cord, and contact the support team at Zurich Instruments, either through the website form or through [email](#).

Table 2.3: Unusual Conditions

Fan is not working properly or not at all	Switch off the instrument immediately to prevent overheating of sensitive electronic components.
Power cord or power plug on instrument is damaged	Switch off the instrument immediately to prevent overheating, electric shock, or fire. Please exchange the power cord only with one for this product and certified for the country of use.

Instrument emits abnormal noise, smell, or sparks	Switch off the instrument immediately to prevent further damage.
Instrument is damaged	Switch off the instrument immediately and ensure it is not used again until it has been repaired.

Table 2.4: Symbols

	Earth ground
	Chassis ground
	Caution. Refer to accompanying documentation
	DC (direct current)

2.4. Connecting to the MFIA Instrument

The Zurich Instruments MFIA is designed to work out of the box with minimal effort on the part of the user. The Zurich Instruments LabOne software runs on an embedded PC in the MFIA instrument and is pre-installed before delivery. One of the programs running on the embedded PC is the LabOne Web Server, which can be connected to via a web browser once suitable physical and logical connections to the instrument have been established.

Note

The following web browsers are supported (latest versions)



Chrome



Firefox



Opera



Edge



Safari

There are two ways to physically connect to the instrument:

- Ethernet (no software installation required). Integrate the instrument into an existing local area network (LAN) by connecting the instrument to a switch in the LAN using an Ethernet cable. The instrument can then be accessed from a web browser running on any device in the same LAN. The Ethernet connection can also be point-to-point. This requires some adjustment of the network card settings of the host computer. Depending on the network configuration and the installed network card, one or the other connection scheme is better suited.
- Universal Serial Bus (USB). The USB connection is a point-to-point connection between the instrument and the host computer to which the USB cable is connected. This requires the installation of an RNDIS driver on the host computer. For more information, look at [Universal Serial Bus \(USB\) Connection](#).

2.4. Connecting to the MFIA Instrument

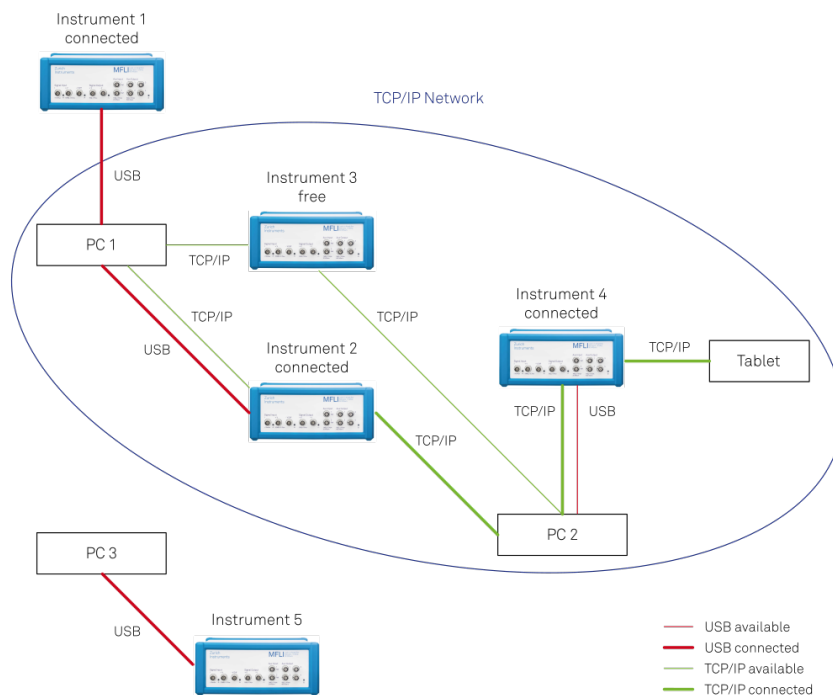


Figure 2.1: Connectivity

Figure 2.1 shows some examples of possible configurations of computer to instrument connectivity.

2.4.1. Ethernet TCP/IP Connection

The simplest connection method is to integrate the instrument into an existing LAN supporting the Domain Name System (DNS) and dynamic DNS update. In such a network the MFIA instrument can be addressed using its serial number instead of its IP address. Attach an Ethernet cable from the instrument to a LAN network switch. The LabOne User Interface can be started by typing the following text in the address bar of a web browser running on a computer in the LAN:

`http://<instrument-serial>.<domain>/, or`

`http://<instrument-serial>.local/, or`

`http://<instrument-serial>/`

where **<instrument-serial>** is the serial number of the instrument and **<domain>** is the network domain in which the instrument is running.

This approach uses DNS lookup to resolve the IP address of the instrument which is configured with this network name by default. It is possible that this approach fails due to special network policies or for other reasons. In that case, there are other options to configure the TCP/IP connection to the instrument. These are described in [Advanced TCP/IP Configuration](#).

With connection via LAN, multiple web browser and API client sessions have simultaneous access and control over the instrument. Therefore changes made to the settings of the instrument by say a Python program via the API will be seen by a web browser session connected to that instrument. The instrument data can be streamed to multiple client sessions simultaneously.

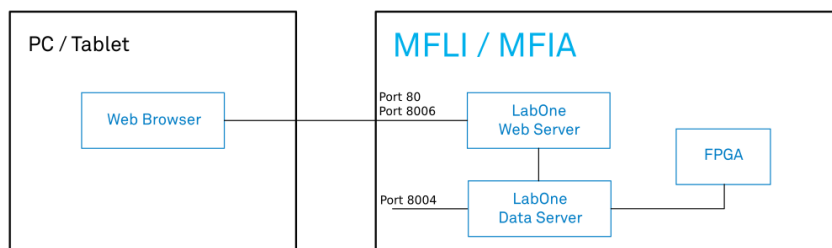


Figure 2.2: Simplest connection to MFIA instrument

2.4.2. Universal Serial Bus (USB) Connection

To control the instrument via USB, connect the instrument with the supplied USB cable to the PC you are using. Depending on the operating system you are using, it may be necessary to install a USB RNDIS device driver on the PC. The RNDIS provides a virtual Ethernet link to the instrument enabling it to be addressed using the normal IP address and host name mechanisms. The RNDIS driver installation procedure for various operating systems is described in the following.

USB RNDIS Device Driver Windows

Zurich Instruments provides a Microsoft MSI installer to simplify the Windows RNDIS driver installation. The MFIA Instrument is designed so that when the USB cable is plugged in, a flash disk partition appears with the label **MF-DRIVER**, containing the required Windows MSI installer for the USB RNDIS device driver, plus the instrument user manual in PDF form and a copy of this Getting Started chapter.

Proceed with the device driver installation as follows:

1. Log on to the PC as an administrator. Installing the USB RNDIS device driver requires administrator rights.
2. Ensure that the USB cable is connected from the PC to the MFIA Instrument.
3. Power cycle the MFIA Instrument and wait for approximately 20 seconds for the instrument initialization to complete. During this time an AutoPlay window will pop up showing the newly detected MFIA drive.
4. In the AutoPlay window, select Open folder to view files.
5. Two installers will be visible in Windows Explorer: Double click on the .msi installer appropriate for your operating system. This will be either **MF-Device-Finder64.msi** for 64 bit operating systems, or **MF-Device-Finder32.msi** for 32 bit operating systems.

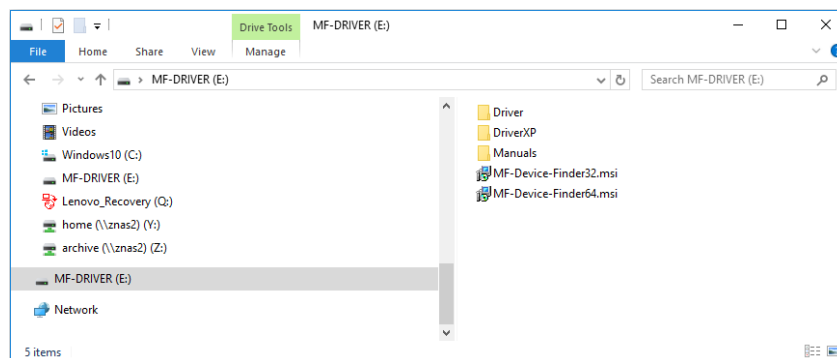


Figure 2.3: Read-only drive of the MFIA shown in the File Explorer

6. In the welcome screen click the Next button.

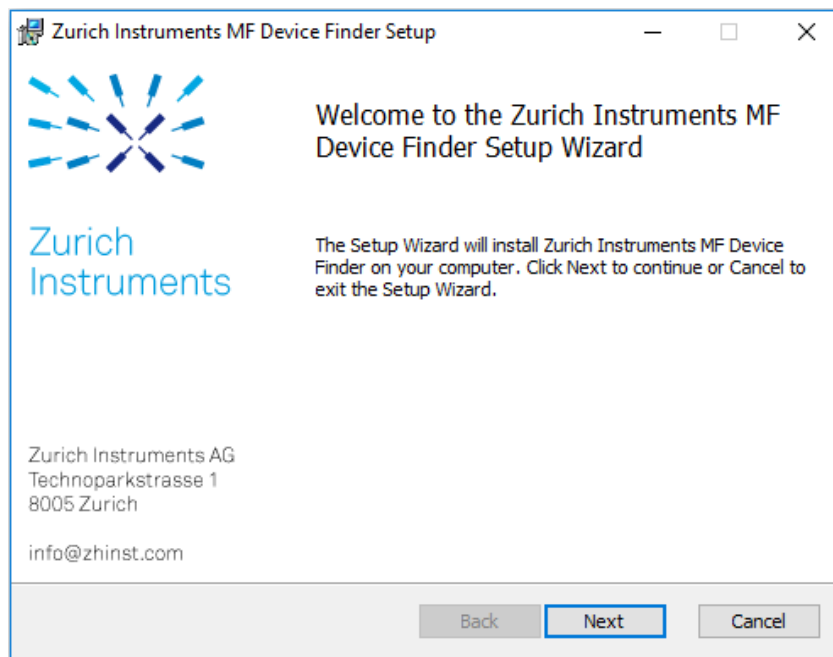


Figure 2.4: Installation welcome screen

7. After reading through the Zurich Instruments license agreement, check the "I accept the terms in the License Agreement" check box and click the Next button.

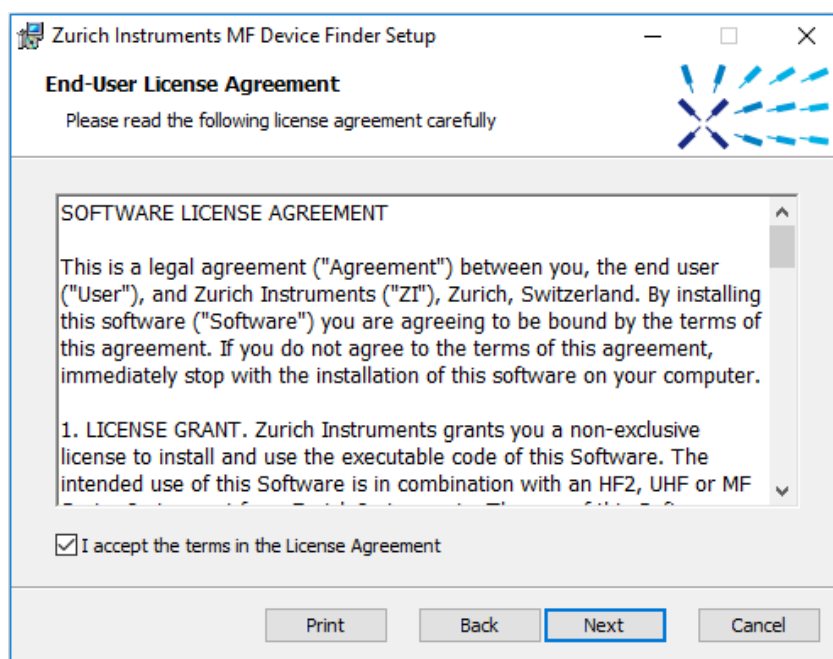


Figure 2.5: Installation license agreement

8. In the Custom Setup screen click the Next button.

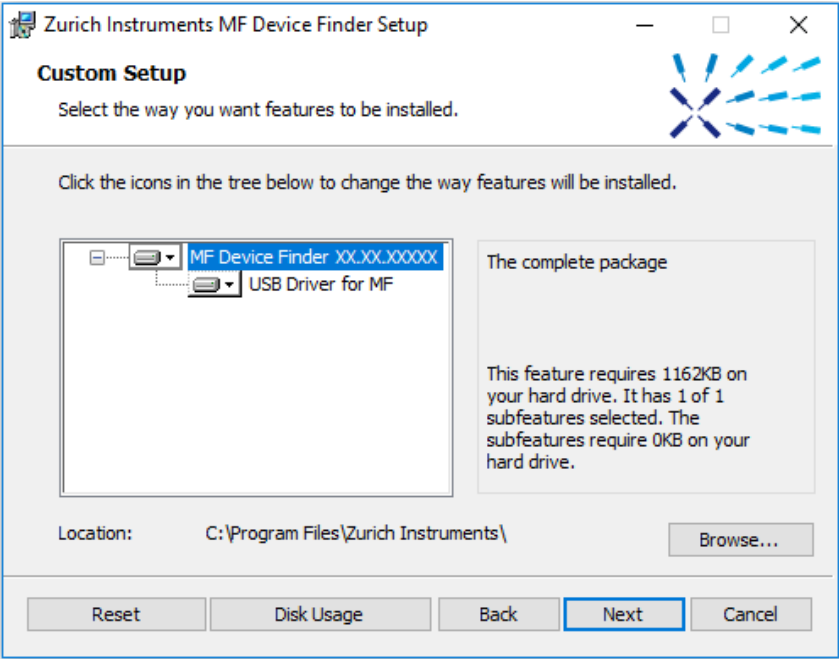


Figure 2.6: Custom setup screen

- 9. Click the Install button to start the installation.

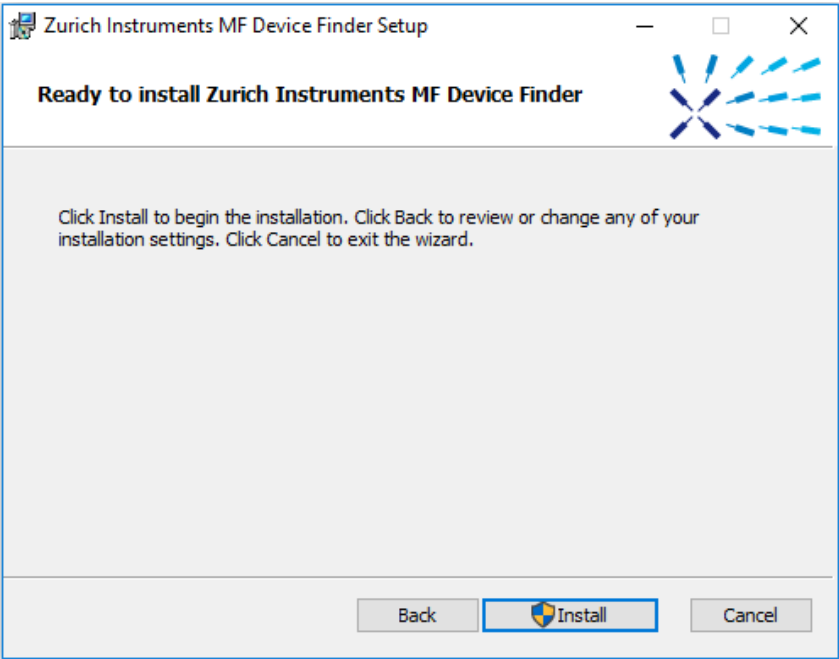


Figure 2.7: Installation confirmation

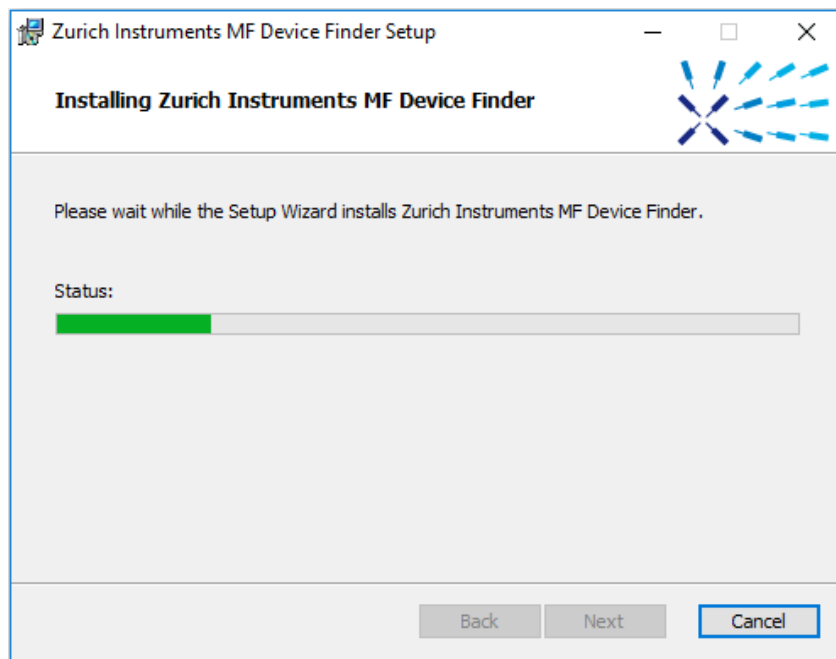


Figure 2.8: Installation progress

If the User Account Control pop-up window appears, click Yes to allow the installation program to make changes to the computer.

10. On Windows Server 2008 and Windows 7 it is required to confirm the installation of up to 2 drivers from the trusted publisher Zurich Instruments. Click on Install. You can also click the Always trust software from "Zurich Instruments AG" check box to prevent the message from appearing in the future.

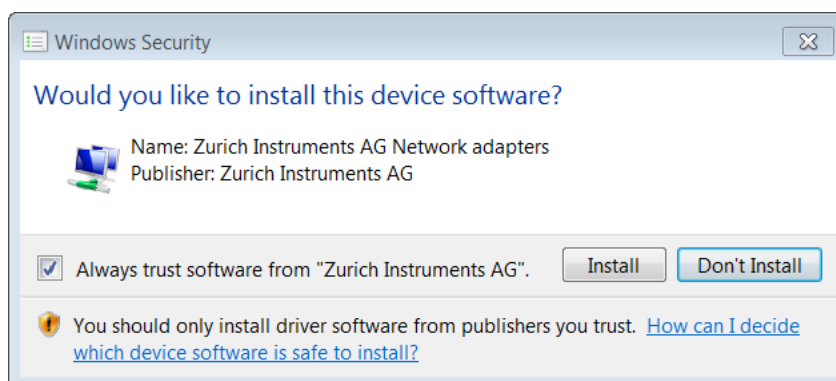


Figure 2.9: Installation driver acceptance

11. Click the Finish button to complete the installation.

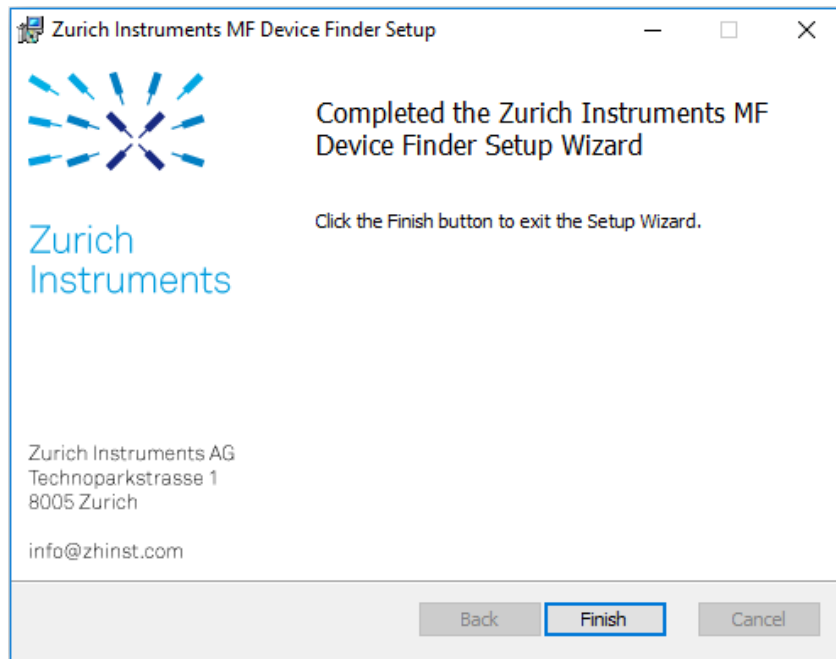


Figure 2.10: Installation complete

The installation of the driver will create a Windows Start Menu entry¹ in **Start Menu → Zurich Instruments → MF Device Finder**. This link will open the ZI MF Device Finder shown in [MF Device Finder](#). In the list of devices, identify your instrument by its serial number found on the back panel. Double-click on the entry for your instrument to open the default browser with the LabOne Device Connection dialog shown in [Figure 2.13](#) Click on Open on the bottom right of the screen to open a LabOne browser session with the default settings.

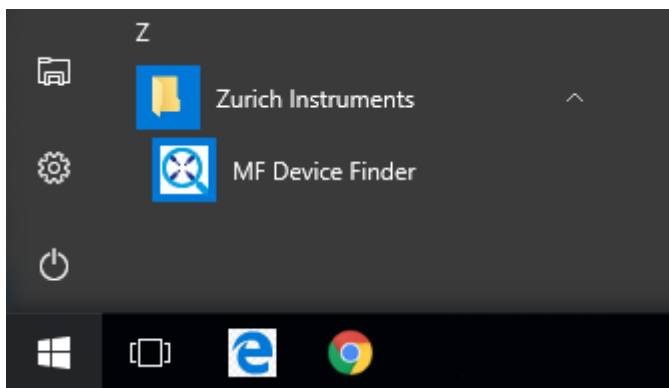


Figure 2.11: Windows Start Menu entry to start the Zurich Instruments MF Device Finder

Note

The procedure described in this section works both for 1GbE and USB connections. In case you use 1GbE, you can also connect to the Instrument without software installation using just a web browser. To that end, follow the instructions in [Ethernet TCP/IP Connection](#).

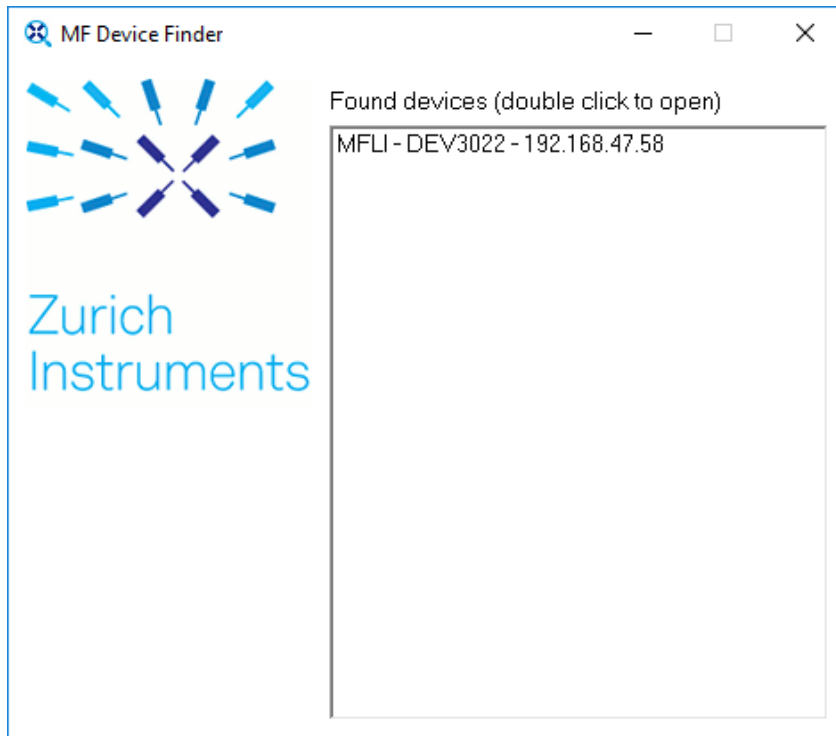


Figure 2.12: MF Device Finder

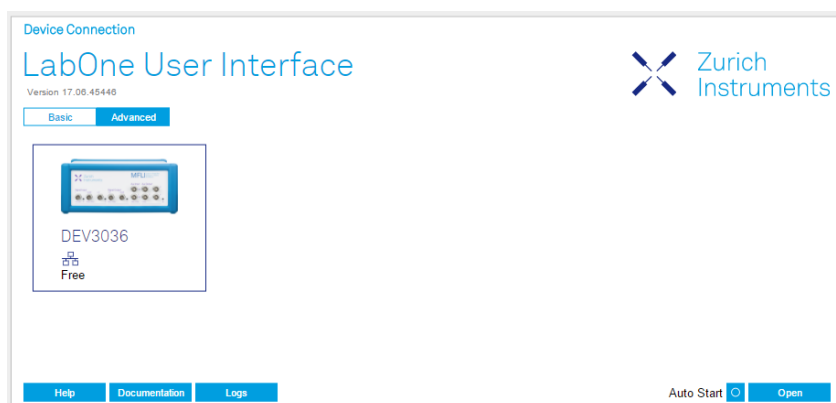


Figure 2.13: Device Connection dialog after starting the LabOne User Interface

USB RNDIS Device Driver Linux

With an up-to-date version of an Ubuntu Linux distribution there is no special installation necessary and the USB connection works out of the box. Other Linux distributions should also work but may require further installation steps. Detailed instructions cannot be given here due to the large number of Linux distributions available.

USB RNDIS Device Driver macOS

There is no current RNDIS driver available for macOS and as a result, the USB port of the MFIA cannot be used to communicate with macOS systems. Instead, one can use a USB-C to gigabit Ethernet adapter and an Ethernet cable to communicate with the MFIA via its 1GbE port on the rear panel of the instrument.

2.4.3. Advanced TCP/IP Configuration

Various schemes are possible for logically connecting to the MFIA instrument via TCP/IP.

- DHCP (the simplest scheme for the user and the default)
- Static IP (the fallback if DHCP fails)

■ Point-to-point (P2P)

DHCP is the simplest and preferred connection method. Other connection methods can become necessary when using network configurations that are in contradiction with the local policies.

DHCP

The most straightforward Ethernet connection method is to rely on a LAN configuration to recognize the MFIA Instrument. This is the MFIA Instrument's default configuration. By connecting the instrument in a LAN, a dynamic IP address is assigned to it like to any other PC by the DHCP server. In case of restricted networks, the network administrator may be required to register the instrument on the network by means of the MAC address. The MAC address is indicated on the back panel of the instrument. If the network configuration does not allow or does not support DHCP it is necessary to use a static IP setup as described below.

Static Instrument IP

Although it is highly recommended to use dynamic IP assignment method in the host network of the instrument, there may be cases where the user wants to assign a static IP to the instrument. For instance, when the host network only contains Ethernet switches and hubs but no Ethernet routers are included, there is no DHCP server to dynamically assign an IP to the instrument. It is still advised to add an Ethernet router to the network and benefit from dynamic IP assignment; however, if a router is not available, the instrument can be configured to work with a static IP.

Note that the static IP assigned to the instrument must be within the same range of the IP assigned to the host computer. Whether the host computer's IP is assigned statically or by a fallback mechanism, one can find this IP by running the command `ipconfig` or `ipconfig/all` in the operating system's terminal. As an example, [IP and subnet mask of host computer](#) shows the outcome of running `ipconfig` in the terminal.

```
Ethernet adapter Ethernet 4:

    Connection-specific DNS Suffix . : 
    Link-local IPv6 Address . . . . . : fe80::f3ad:19ae:ffd9:f8ef%17
    Autoconfiguration IPv4 Address. . : 169.254.16.57
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . :
```

Figure 2.14: IP and subnet mask of host computer

It shows the network adapter of the host computer can be reached via the IP **169.254.16.57** and it uses a subnet mask of **255.255.0.0**. To make sure that the instrument is visible to this computer, one needs to assign a static IP of the form **169.254.x.x** and the same subnet mask to the instrument. To do so, the user should follow the instructions below.

1. Attach the instrument using an Ethernet cable to the network where the user's computer is hosted.
2. Attach the instrument via a USB cable to the host computer and switch it on.
3. Open the LabOne user interface (UI) and connect to the instrument via USB.
4. Open the "Device" tab of the LabOne UI and locate the "Communication" section as shown in [Figure 2.15](#).
5. Write down the desired static IP, e.g. **169.254.16.20**, into the numeric field "IPv4 Address".
6. Add the same subnet mask as the host computer, e.g. **255.255.0.0** to the numeric field "IPv4 Mask".
7. You can leave the field "Gateway" as **0.0.0.0** or change to be similar to the IP address but ending with **1**, e.g. **169.254.16.1**.
8. Enable the radio button for "Static IP".
9. Press the button "Program" to save the new settings to the instruments.
10. Power cycle the instrument and remove the USB cable. The instrument should be visible to LabOne via Ethernet connection.

Communication

Current Configuration

Interface	PCIe
MAC Address	80:2F:DE:00:09:01
IPv4 Address	169.254.16.20

Network Configuration 1GbE

Jumbo Frames ☐

Static IP ☒

IPv4 Address	169.254.16.20
IPv4 Mask	255.255.0.0
Gateway	0.0.0.0

Program

Figure 2.15: Configuration of static IP in LabOne UI

To make sure the IP assignment is done properly, one can use the command **ping** to check if the instrument can be reached through the network using its IP address. Figure 2.16 shows the outcome of **ping** when the instrument is visible via the IP **169.254.16.20**.

```
C:\> ping 169.254.16.20

Pinging 169.254.16.20 with 32 bytes of data:
Reply from 169.254.16.20: bytes=32 time<1ms TTL=64
Reply from 169.254.16.20: bytes=32 time<1ms TTL=64
Reply from 169.254.16.20: bytes=32 time<1ms TTL=64
Reply from 169.254.16.20: bytes=32 time<1ms TTL=64

Ping statistics for 169.254.16.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 2.16: Instrument visible through ping

If set properly according to the instructions above, the instrument will use the same static IP configurations after each power cycle.

Requirements

- The chosen static IP address must be unique within the LAN.
- Needs network administrator support on networks with dynamic IP configuration (the static IP address will need to be reserved).

Fallback Device IP

When configured to a dynamic address, but no DHCP server is present in the network, e.g., device connected directly to a PC, the instrument falls back on an IP address in the local link IP range that is **169.254.x.x**. If the host computer has also an IP address within the same range, the instrument becomes visible to the LabOne data server running on the host computer. This way, there is no need to go through the process described above to assign a static IP to the instrument.

Point-to-Point (P2P)

Setting up a point-to-point network consisting only of the host computer and the MFIA avoids problems related to special network policies. Since it is nonetheless necessary to stay connected to the internet, it is recommended to install two network cards in your computer, one of them for network connectivity, the second one for connecting to the MFIA Instrument. Notebooks can generally profit from wireless LAN for internet connection.

1. Use one of the network cards and set it to static IP in TCP/IPv4 using the IP address **192.168.1.n**, where $n=[2..9]$ and the mask **255.255.255.0**, see [Figure 2.17](#) (go to **Control Panel → Internet Options → Network and Internet → Network and Sharing Center → Local Area Connection → Properties**).

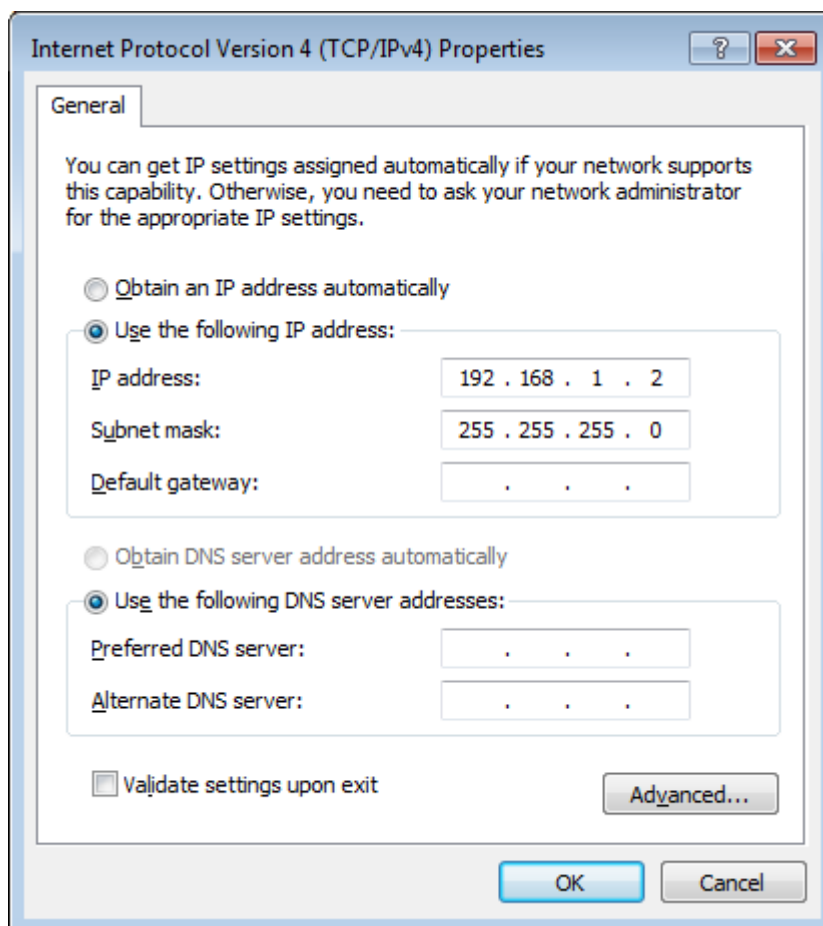


Figure 2.17: Static IP configuration of PC

2. Set the MFIA IP address to the static value **192.168.1.10** as described in the previous section. To connect to the MFIA Instrument and start the LabOne User Interface, type the following in the address bar of a web browser:
http://192.168.1.10/

Requirements

- Two network cards needed for additional connection to internet
- Network adapter connected to the instrument must be in static IP4 configuration

Note

A power cycle of the MFIA Instrument is required if it was previously connected to a network that provided an IP address to the instrument and then the user decides to run in static IP configuration.

Note

Only IP v4 is currently supported. There is no support for IP v6.

Note

If the instrument is detected by LabOne but the connection can not be established, the reason can be the firewall blocking the connection. It is then recommended to change the P2P connection from Public to Private.

Warning

Changing the IP settings of your network adapters manually can interfere with its later use, as it can no longer be used for network connectivity until it is set again for dynamic IP.

-
1. Under Windows 7 and 8, the Start Menu entry is found in **Start Menu → All programs / All apps → Zurich Instruments → MF Device Finder** ↩

2.5. Running LabOne on a Separate PC

By default the LabOne Web Server and the Data Server run on the embedded computer in the instrument. It can be advantageous to run both servers on a separate PC, especially when using high data transfer rates or recording high-resolution FFT spectra. The higher processing power of laptop and desktop computers improves UI responsiveness, allows for a higher number of frequency points in the Spectrum Analyzer tool, and reduces the probability of data loss.

To run the Data Server on a separate PC, the LabOne software must be installed on that PC. See [Software Installation](#) for LabOne installation instructions. [Software Architecture](#) gives an overview of the LabOne software architecture that is particularly useful when using multi-instrument control or the programming interfaces.

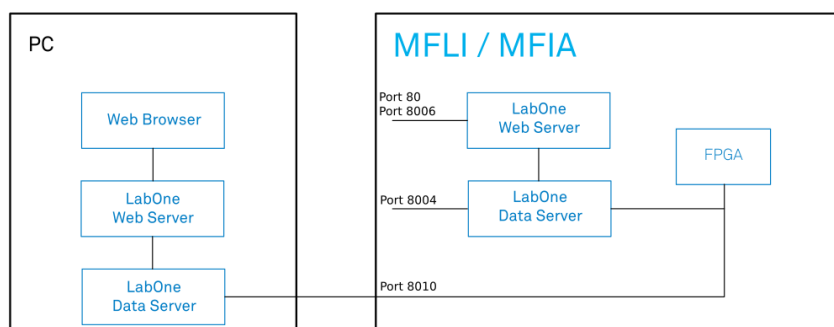


Figure 2.18: LabOne Data Server and Web Server running on separate PC

2.5.1. Software Installation

The MFIA Instrument comes with the LabOne software pre-installed and is ready to work out of the box. The LabOne software can also be installed on Windows and Linux PCs. This can be advantageous for performance reasons, and is necessary to use the MATLAB, LabVIEW or Python APIs with the instrument. To install the LabOne software on a computer, administrator rights may be required. In order to simply run the software later, a regular user account is sufficient. Instructions for downloading the correct version of the software packages from the Zurich Instruments website are described below in the platform-dependent sections. It is recommended to regularly update to the latest software version provided by Zurich Instruments. Thanks to the Automatic Update check feature, the update can be initiated with a single click from within the user interface, as shown in [Software Update](#).

Installing LabOne on Windows

This section describes how to additionally install the LabOne Software on Windows. This is only necessary if you wish to access the User Interface with a Web Server running on a PC instead of on the MFIA, itself, to use one of the LabOne APIs, or for multi-device synchronization. The installation packages for the Zurich Instruments LabOne software are available as Windows installer .msi packages. The software is available on the [Zurich Instruments Download Center](#). Please ensure that

you have administrator rights for the PC on which the software is to be installed. See [LabOne compatibility](#) for a comprehensive list of supported Windows systems.

Windows LabOne Installation

1. The MFIA Instrument should not be connected to your computer during the LabOne software installation process.
2. Start the LabOne installer program with a name of the form **LabOne64-XX.XX.XXXXX.msi** by a double click and follow the instructions. Windows Administrator rights are required for installation. The installation proceeds as follows:
 - On the welcome screen click the **Next** button.

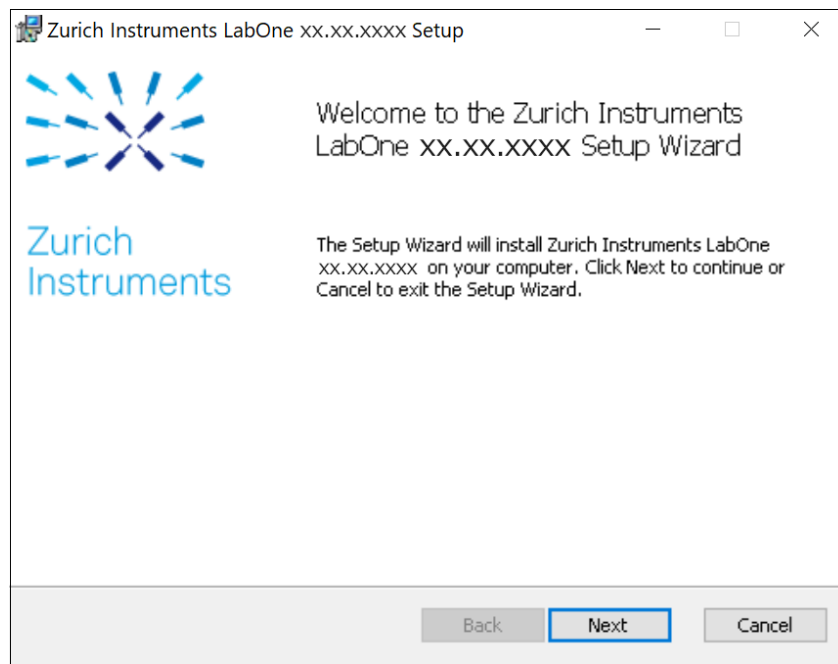


Figure 2.19: Installation welcome screen

- After reading through the Zurich Instruments license agreement, check the "I accept the terms in the License Agreement" check box and click the **Next** button.
- Review the features you want to have installed. For the MFIA Instrument the "MF Series Device", "LabOne User Interface" and "LabOne APIs" features are required. Please install the features for other device classes as well, if required. To proceed click the **Next** button.

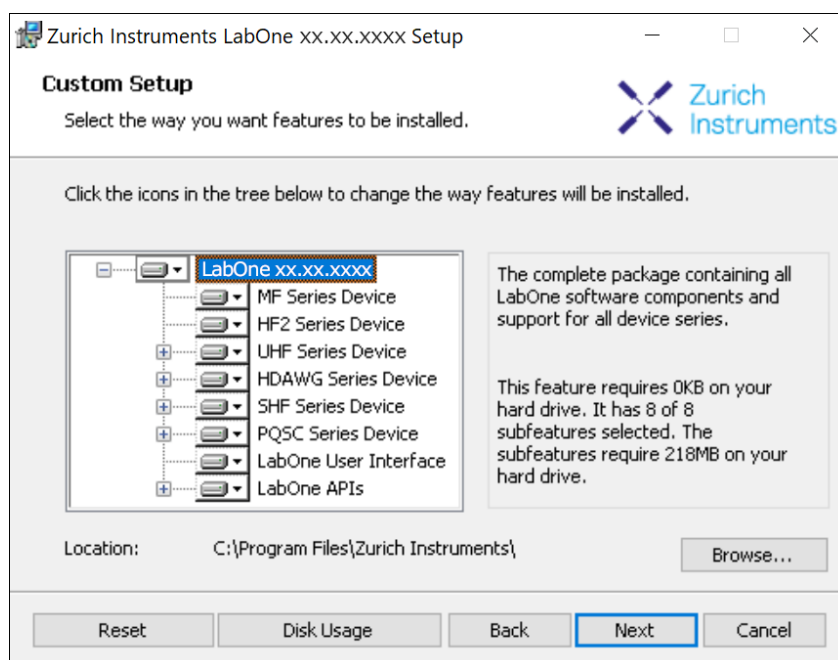
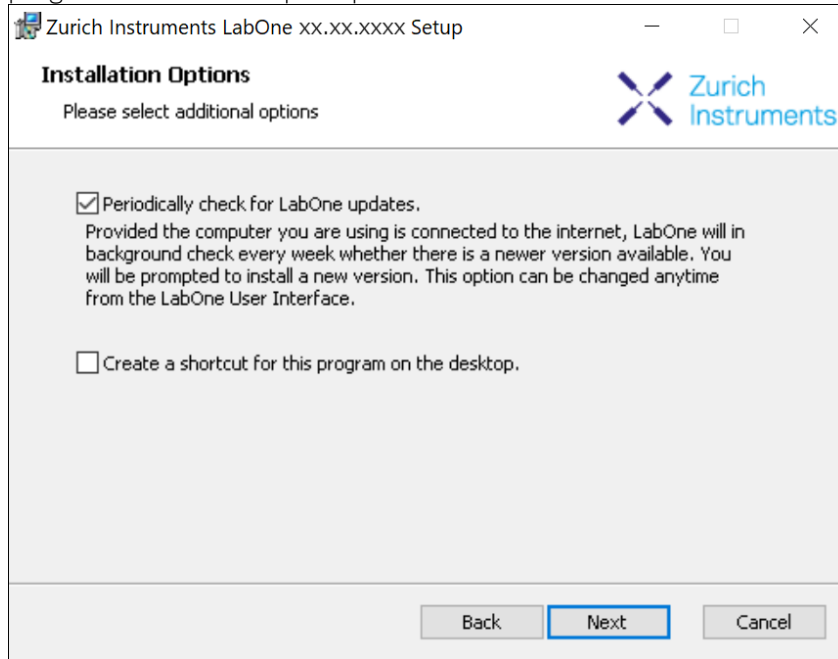


Figure 2.20: Custom setup screen

- Select whether the software should periodically check for updates. Note, the software will still not update automatically. This setting can later be changed in the user interface. If you would like to install shortcuts on your desktop area, select "Create a shortcut for this program on the desktop". To proceed click the **Next** button.



- Click the **Install** button to start the installation process.
- Windows may ask up to two times to reboot the computer if you are upgrading. Make sure you have no unsaved work on your computer.

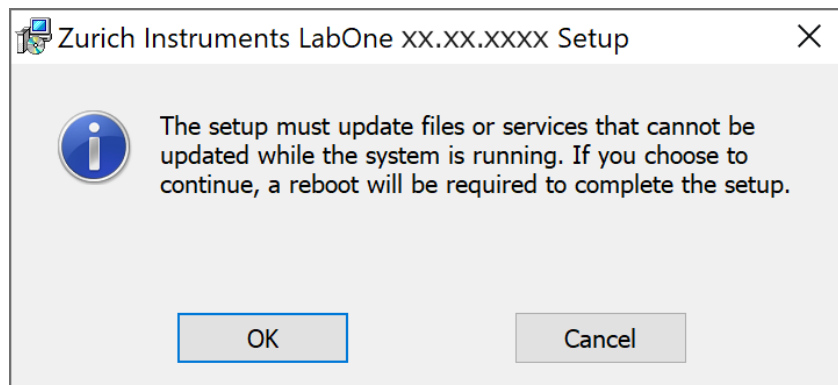


Figure 2.21: Installation reboot request

- During the first installation of LabOne, it is required to confirm the installation of some drivers from the trusted publisher Zurich Instruments. Click on **Install**.

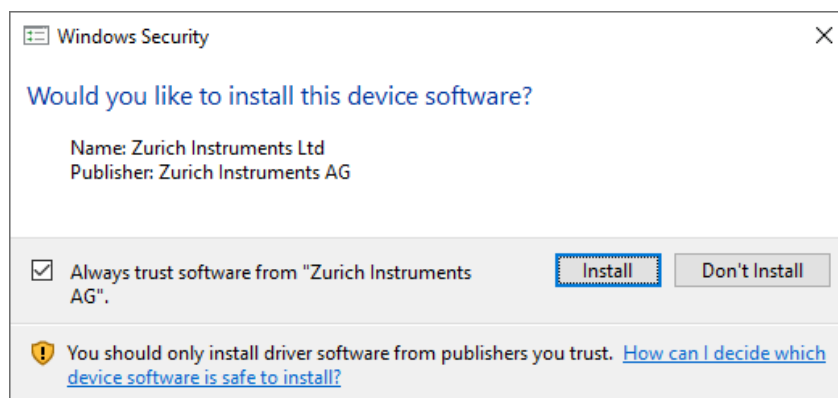


Figure 2.22: Installation driver acceptance

- Click **OK** on the following notification dialog.

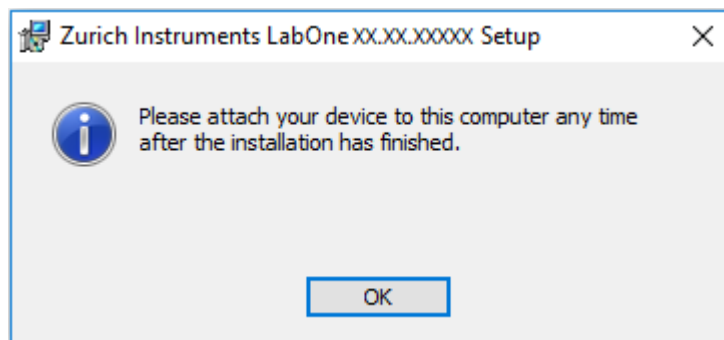


Figure 2.23: Installation completion screen

3. Click **Finish** to close the Zurich Instruments LabOne installer.
4. You can now start the LabOne User Interface as described in [LabOne Software Start-up](#) and choose an instrument to connect to via the Device Connection dialog shown in [Device Connection dialog](#).

Warning

Do not install drivers from another source other than Zurich Instruments.

Start LabOne Manually on the Command Line

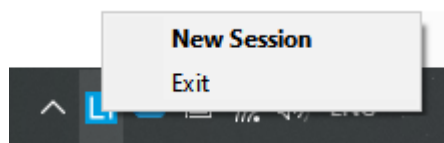
After installing the LabOne software, the Web Server and Data Server can be started manually using the command-line. The more common way to start LabOne under Windows is described in [LabOne Software Start-up](#). The advantage of using the command line is being able to observe and change the behavior of the Web and Data Servers. To start the Servers manually, open a command-line terminal (Command Prompt, PowerShell (Windows) or Bash (Linux)). For Windows, the current working directory needs to be the installation directory of the Web Server and Data Server. They are installed in the Program Files folder (usually: C:\Program Files) under \Zurich Instruments\LabOne in the WebServer and DataServer folders, respectively. The Web Server and Data Server (ziDataServer) are started by running the respective executable in each folder. Please be aware that only one instance of the Web Server can run at a time per computer. The behavior of the Servers can be changed by providing command line arguments. For a detailed list of all arguments see the command line help text:

```
$ ziWebServer --help
```

For the Data Server:

```
$ ziDataServer --help
```

One useful application of running the Webserver manually from a terminal window is to change the data directory from its default path in the user home directory. The data directory is a folder in which the LabOne Webserver saves all the measured data in the format specified by the user. Before running the Webserver from the terminal, the user needs to ensure there is no other instance of Webserver running in the background. This can be checked using the Tray Icon as shown below.



The corresponding command line argument to specify the data path is `--data-path` and the command to start the LabOne Webserver with a non-default directory path, e.g., `C:\data` is

```
C:\Program Files\Zurich Instruments\LabOne\WebServer> ziWebServer --data-path "C:\data"
```

Windows LabOne Uninstallation

To uninstall the LabOne software package from a Windows computer, one can open the "Apps & features" page from the Windows start menu and search for LabOne. By selecting the LabOne item in the list of apps, the user has the option to "Uninstall" or "Modify" the software package as shown in [Figure 2.24](#).

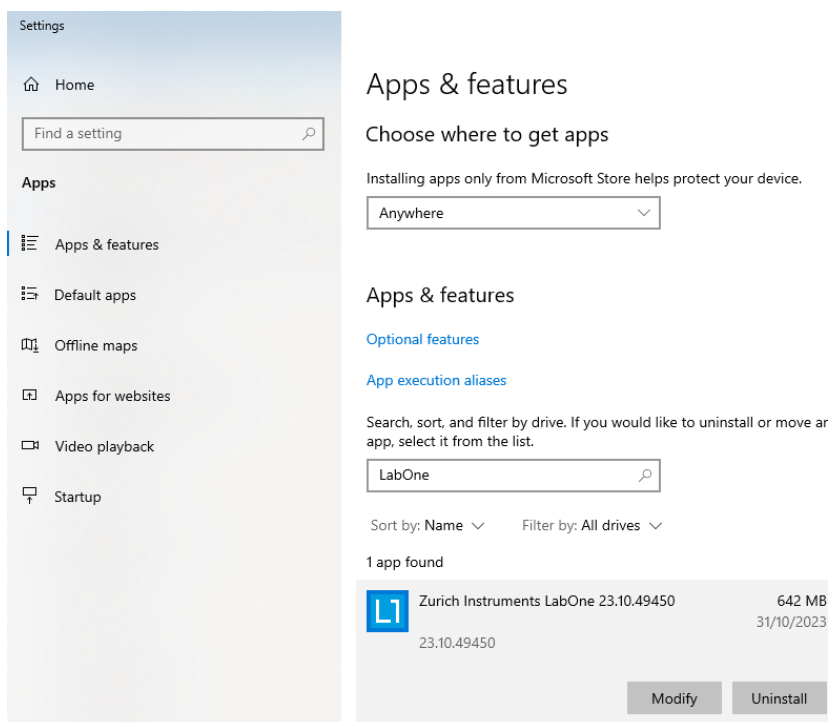


Figure 2.24: Uninstallation of LabOne on Windows computers

Warning

Although it is possible to install a new version of LabOne on a currently-installed version, it is highly recommended to first uninstall the older version of LabOne from the computer and then, install the new version. Otherwise, if the installation process fails, the current installation is damaged and cannot be uninstalled directly. The user will need to first repair the installation and then, uninstall it.

In case a current installation of LabOne is corrupted, one can simply repair it by selecting the option "Modify" in [Figure 2.24](#). This will open the LabOne installation wizard with the option "Repair" as shown in [Figure 2.25](#).

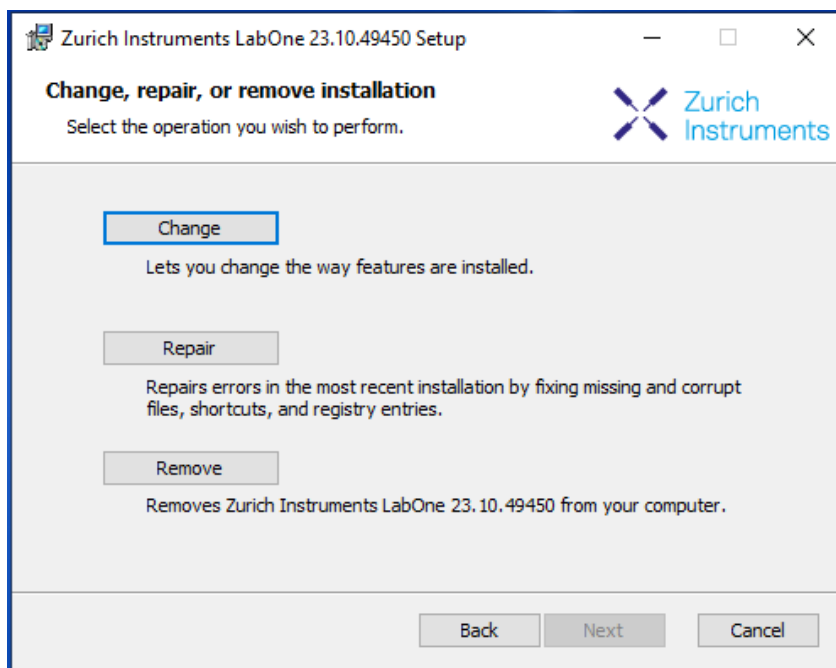


Figure 2.25: Repair of LabOne on Windows computers

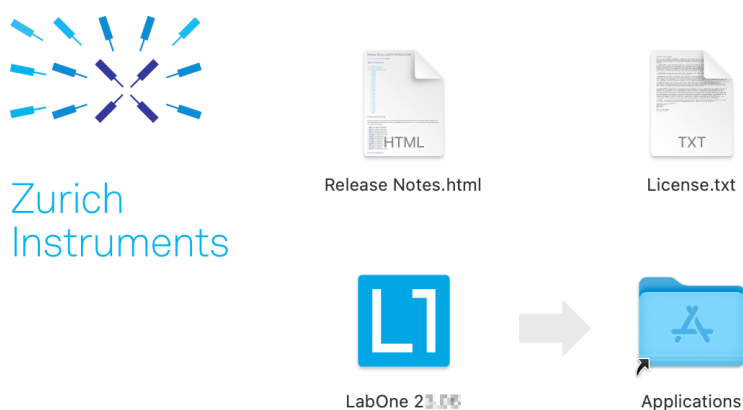
After finishing the repair process, the normal uninstallation process described above can be triggered to uninstall LabOne.

Installing LabOne on macOS

This section describes how to additionally install the LabOne Software on macOS. This is only necessary if you wish to run the Web Server on a computer rather than on the MFIA itself.

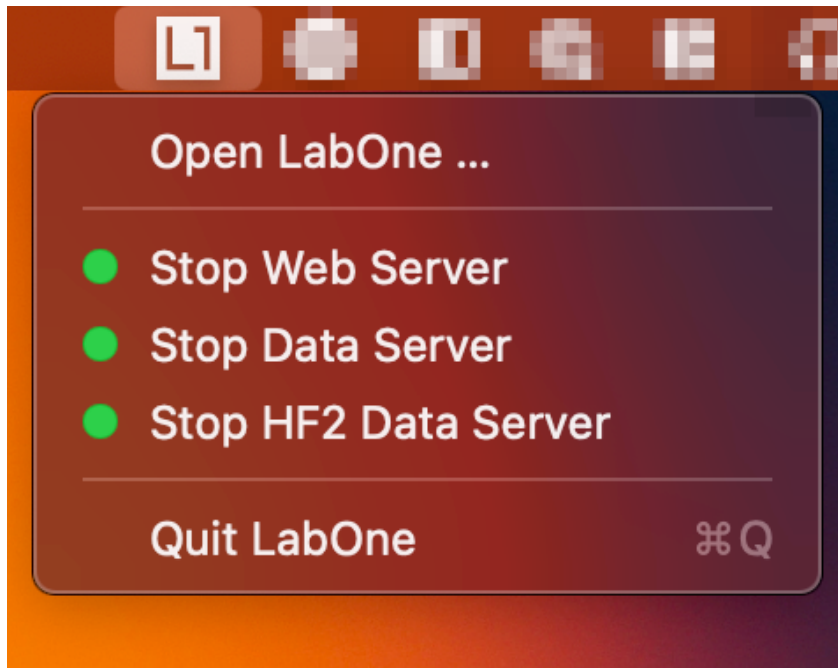
LabOne supports both Intel and ARM (M-series) architectures within a single universal disk image (DMG) file available in our Download Center.

- Download and double-click the DMG file to mount the image.



LabOne application with all services needed. - Once the application is started, a labone icon will appear in the menu bar. It allows the user to easily open a new session and shows the status of all services.

- The image contains a single



Uninstalling LabOne on macOS

To uninstall LabOne on macOS, simply drag the LabOne application to the trash bin.

Application Content

The LabOne application contains all resources available for macOS. This includes:

- The binaries for the Web Server and Data Servers.
- The binaries for the C, MATLAB, and LabVIEW APIs.
- An offline version of the user manuals.
- The latest firmware images for all instruments.

To access this content, right-click on the LabOne application and select "Show Package Contents". Then, go into Contents/Resources.

Note

Since the application name contains a space, one needs to escape it when using the command line to access the contents: `cd /Applications/LabOne\ 2X.XX.app/Contents/Resources`

Start LabOne Manually on the Command Line

To start the LabOne services like the data server and web server manually, one can use the command line.

The data server binary is called **ziDataServer** (**ziServer** for HF2 instruments) and is located at `Applications/LabOne\ 2X.XX.app/Contents/Resources/DataServer/`.

The web server binary is called **ziWebServer** and is located at `Applications/LabOne\ 2X.XX.app/Contents/Resources/WebServer/`.

Note

No special command line arguments are needed to start the LabOne services. Use the `--help` argument to see all available options.

Installing LabOne on Linux

This section describes how to additionally install the LabOne Software on Linux. This is only necessary if you wish to run the Web Server on a computer rather than on the MFIA itself.

Requirements

Ensure that the following requirements are fulfilled before trying to install the LabOne software package:

1. LabOne software supports typical modern GNU/Linux distributions (Ubuntu 14.04+, CentOS 7+, Debian 8+). The minimum requirements are glibc 2.17+ and kernel 3.10+.
2. You have administrator rights for the system.
3. The correct version of the LabOne installation package for your operating system and platform have been downloaded from the Zurich Instruments [Download Center](#):

```
LabOneLinux<arch>-<release>-<revision>.tar.gz,
```

Please ensure you download the correct architecture (x86-64 or arm64) of the LabOne installer. The `uname` command can be used in order to determine which architecture you are using, by running:

```
uname -m
```

in a command line terminal. If the command outputs **x86_64** the x86-64 version of the LabOne package is required, if it displays **aarch64** the ARM64 version is required.

Linux LabOne Installation

Proceed with the installation in a command line shell as follows:

1. Extract the LabOne tarball in a temporary directory:

```
tar xzvf LabOneLinux<arch>-<release>-<revision>.tar.gz
```

2. Navigate into the extracted directory.

```
cd LabOneLinux<arch>-<release>-<revision>
```

3. Run the install script with administrator rights and proceed through the guided installation, using the default installation path if possible:

```
sudo bash install.sh
```

The install script lets you choose between the following three modes: - Type "a" to install the Data Server program, the Web Server program, documentation and APIs. - Type "u" to install **udev** support (only necessary if HF2 Instruments will be used with this LabOne installation and not relevant for other instrument classes). - Type "ENTER" to install both options "a" and "u".

4. Test your installation by running the software as described in the next section.

Running the Software on Linux

The following steps describe how to start the LabOne software in order to access and use your instrument in the User Interface.

1. Start the LabOne Data Server program at a command prompt:

```
$ ziDataServer
```

You should be able to access your instrument. In case of problems please consult the [Troubleshooting](#) at the end of this chapter.

2. Start the Web Server program at a command prompt:

```
$ ziWebServer
```

3. Start an up-to-date web browser and enter the **127.0.0.1:8006** in the browser's address bar to access the Web Server program and start the LabOne User Interface. The LabOne Web Server installed on the PC listens by default on port number 8006 instead of 80 to minimize the probability of conflicts.
4. You can now start the LabOne User Interface as described in [LabOne Software Start-up](#) and choose an instrument to connect to via the Device Connection dialog shown in [Device Connection dialog](#).

Warning

Do not use two Data Server instances running in parallel; only one instance may run at a time.

Uninstalling LabOne on Linux

The LabOne software package copies an uninstall script to the base installation path (the default installation directory is `/opt/zi/`). To uninstall the LabOne package please perform the following steps in a command line shell:

1. Navigate to the path where LabOne is installed, for example, if LabOne is installed in the default installation path:

```
$ cd /opt/zi/
```

2. Run the uninstall script with administrator rights and proceed through the guided steps:

```
$ sudo bash uninstall_LabOne<arch>-<release>-<revision>.sh
```

2.5.2. LabOne Software Architecture

The Zurich Instruments LabOne software gives quick and easy access to the instrument. LabOne also supports advanced configurations with simultaneous access by multiple software clients (i.e., LabOne User Interface clients and/or API clients), and even simultaneous access by several users working on different computers. Here we give a brief overview of the architecture of the LabOne software. The information in this section is presented for the case when LabOne is run on a separate PC rather than on the embedded computer on the MFIA Instrument.

The software of Zurich Instruments equipment is server-based. The servers and other software components are organized in layers as shown in [Figure 2.26](#).

- The lowest layer running on the PC is the LabOne Data Server, which is the interface to the connected instrument.
- The middle layer contains the LabOne Web Server, which is the server for the browser-based LabOne User Interface.
- The graphical user interface, together with the programming user interfaces, are contained in the top layer.

The architecture with one central Data Server allows multiple clients to access a device with synchronized settings. The following sections explain the different layers and their functionality in more detail.

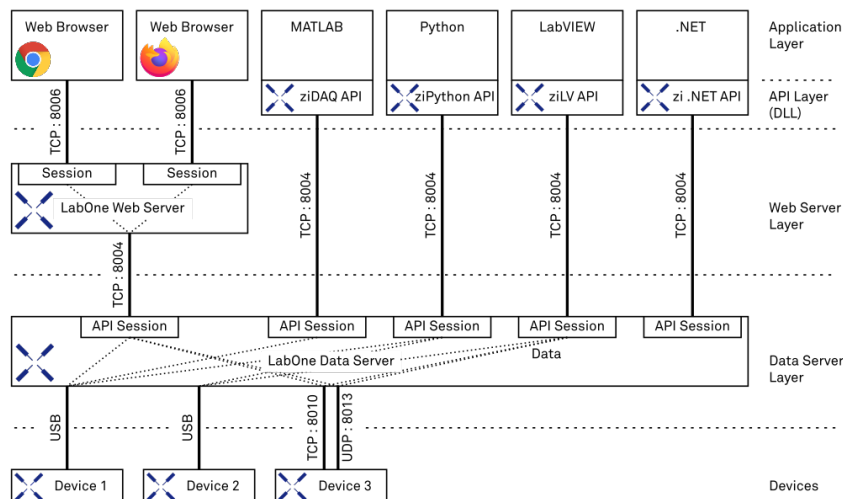


Figure 2.26: LabOne Software architecture

LabOne Data Server

The **LabOne Data Server** program is a dedicated server that is in charge of all communication to and from the device. The Data Server can control a single or also multiple instruments. It will distribute the measurement data from the instrument to all the clients that subscribe to it. It also ensures that settings changed by one client are communicated to other clients. The device settings are therefore synchronized on all clients. The Data Server is started automatically by a service when the PC is started. This service can be disabled if necessary, though the Data Server consumes only little resources when there is no active session. On a PC, only a single instance of a LabOne Data Server should be running.

LabOne Web Server

The LabOne Web Server is an application dedicated to serving up the web pages that constitute the LabOne user interface. The user interface can be opened with any device with a web browser. Since it is touch enabled, it is possible to work with the LabOne User Interface on a mobile device - like a tablet. The LabOne Web Server supports multiple clients simultaneously. This means that more than one session can be used to view data and to manipulate the instrument. A session could be running in a browser on the PC on which the LabOne software is installed. It could equally well be running in a browser on a remote machine.

With a LabOne Web Server running and accessing an instrument, a new session can be opened by typing in a network address and port number in a browser address bar. In case LabOne is run on the embedded computer of the Instrument, please refer to [Connecting to the MFIA Instrument](#) to learn what address to use to access the Web Server. In case the Web Server runs on the **same** computer as the browser, the address is the localhost address (both are equivalent):

- **127.0.0.1:8006** (only valid when LabOne is running on a separate PC)
- **localhost:8006** (only valid when LabOne is running on a separate PC)

In case the Web Server runs on a **remote** computer, the address is the IP address or network name of the remote computer:

- **192.168.x.y:8006**
- **myPC.company.com:8006**

Note that the latter is technically equivalent to the case when the Web Server runs on the embedded computer on the Instrument. In that case the IP address of the Instrument is used. The most recent versions of the most popular browsers are supported: Chrome, Firefox, Edge, Safari and Opera.

LabOne API Layer

The instrument can also be controlled via the application program interfaces (APIs) provided by Zurich Instruments. APIs are provided in the form of DLLs for the following programming environments:

- MATLAB
- Python
- LabVIEW
- .NET
- C

The instrument can therefore be controlled by an external program, and the resulting data can be processed there. The device can be concurrently accessed via one or more of the APIs and via the user interface. This enables easy integration into larger laboratory setups. See the LabOne Programming Manual for further information. Using the APIs, the user has access to the same functionality that is available in the LabOne User Interface.

2.6. LabOne Software Start-up

This section describes the start-up of the LabOne User Interface, which is used to control the MFIA Instrument. The start-up is slightly different depending on whether the LabOne software is installed on the PC (see [Running LabOne on a Separate PC](#)), or if the software on the Instrument is used. It's not required to install LabOne on the PC to operate the Instrument. In case you use the software on the instrument, the first steps to connect to the instrument and open a browser window with the LabOne start-up screen are explained in [Connecting to the MFIA Instrument](#).

In case you have installed LabOne on the PC, the LabOne User Interface start-up link can be found under the Windows 10 Start Menu (Under Windows 7 and 8, the LabOne User Interface start-up link can be found in **Start Menu → all programs / all apps → Zurich Instruments LabOne**). As shown in [Figure 2.27](#), click on **Start Menu → Zurich Instruments LabOne**. This will open the User Interface in a new tab in your default web browser and start the LabOne Data Server and LabOne Web Server programs in the background.

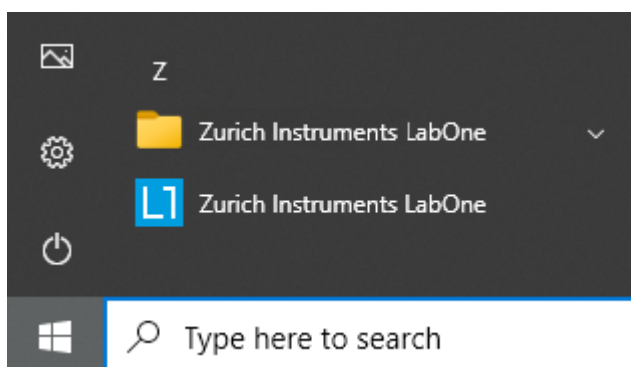


Figure 2.27: Link to the LabOne User Interface in the Windows 10 Start Menu (only when LabOne has been installed on the PC)

After starting LabOne, the Device Connection dialog [Figure 2.28](#) is shown to select the device for the session. The term "session" is used for an active connection between the user interface and the device. Such a session is defined by device settings and user interface settings. Several sessions can be started in parallel. The sessions run on a shared LabOne Web Server.

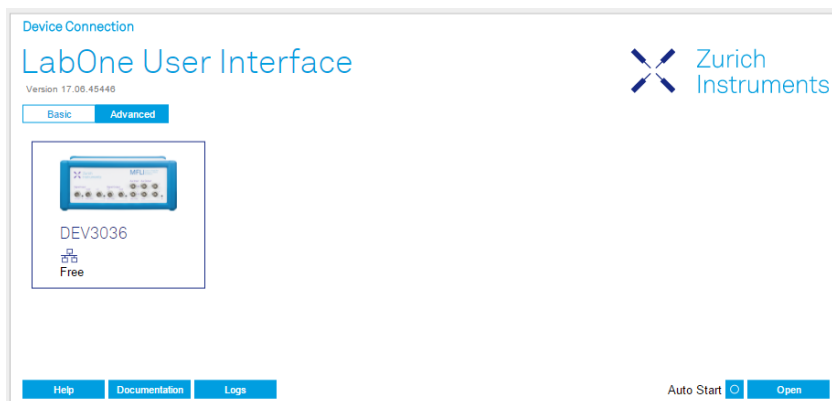


Figure 2.28: Device Connection dialog

The Device Connection dialog opens in the Basic view by default. In this view, all devices that are available for connection are represented by an icon with serial number and status information. If required, a button appears on the icon to perform a firmware upgrade. Otherwise, the device can be connected by a double click on the icon, or a click on the **Open** button at the bottom right of the dialog.

In some cases it's useful to switch to the Advanced view of the Device Connection dialog by clicking on the "Advanced" button. The Advanced view offers the possibility to select custom device and UI settings for the new session and gives further connectivity options that are particularly useful for multi-instrument setups.

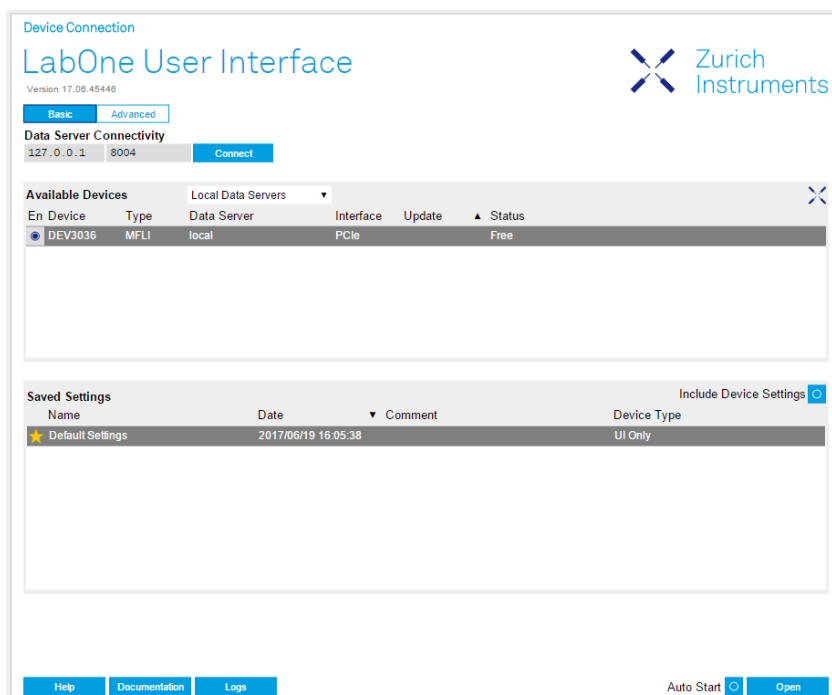


Figure 2.29: Device Connection dialog (Advanced view)

The Advanced view consists of three parts:

- Data Server Connectivity
- Available Devices
- Saved Settings

The Available Devices table has a display filter, usually set to **Default Data Server**, that is accessible by a drop-down menu in the header row of the table. When changing this to **Local Data Servers**, the Available Devices table will show only connections via the Data Server on the host PC and will contain all instruments directly connected to the host PC via USB or to the local network via 1GbE. When using the **All Data Servers** filter, connections via Data Servers running on other PCs in the network also become accessible. Once your instrument appears in the Available Devices table, perform the following steps to start a new session:

1. Select an instrument in the **Available Devices** table.
2. Select a setting file in the **Saved Settings** list unless you would like to use the Default Settings.

3. Start the session by clicking on 

Note

By default, opening a new session will only load the UI settings (such as plot ranges), but not the device settings (such as signal amplitude) from the saved settings file. In order to include the device settings, enable the **Include Device Settings** checkbox. Note that this can affect existing sessions since the device settings are shared between them.

Note

In case devices from other Zurich Instruments series (UHF, HF2, MF, HDAWG, PQSC, GHF, or SHF) are used in parallel, the list in **Available Devices** section can contain those as well.

The following sections describe the functionality of the **Device Connection** dialog in detail.

2.6.1. Data Server Connectivity

The Device Connection dialog represents a Web Server. However, on start-up the Web Server is not yet connected to a LabOne Data Server. With the **Connect/Disconnect** button the connection to a Data Server can be opened and closed.

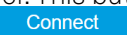
This functionality can usually be ignored when working with a single MFIA Instrument and a single host computer. Data Server Connectivity is important for users operating their instruments from a remote PC, i.e., from a PC different to the PC on which the Data Server is running or for users working with multiple instruments. The Data Server Connectivity function then gives the freedom to connect the Web Server to one of several accessible Data Servers. This includes Data Servers running on remote computers, and also Data Servers running on an MF Series instrument.

With the display filter set to "Local Data Server", all instruments accessible by a Data Server on the same computer and on an MFIA instrument connected to the PC are visible. With the filter set to "All Data Servers", all accessible Data Servers in the network are shown, including those running on other PCs.

Note

When using the filter "All Data Servers", take great care to connect to the right instrument, especially in larger local networks. Always identify your instrument based on its serial number in the form DEV0000, which can be found on the instrument back panel.

2.6.2. Available Devices

The Available Devices table gives an overview of the visible devices. A device is ready for use if either marked free or connected. The first column of the list holds the **Enable** button controlling the connection between the device and a Data Server. This button is greyed out until a Data Server is connected to the LabOne Web Server using the  button. If a device is connected to a Data Server, no other Data Server running on another PC can access this device.

The second column indicates the serial number and the third column shows the instrument type. The fourth column shows the host name of the LabOne Data Server controlling the device. The next column shows the interface type. For MF Instruments the interfaces USB or 1GbE are available and are listed if physically connected. For MF series instruments the interface is indicated as PCIe in case the Data Server is running on the instrument, even if the physical connection between PC and instrument is USB or 1GbE. PCIe corresponds to the interface between the embedded PC and the measurement unit inside the MF instrument. The LabOne Data Server will scan for the available devices and interfaces every second. If a device has just been switched on or physically connected it may take up to 20 s before it becomes visible to the LabOne Data Server.

Table 2.5: Device Status Information

Connected	The device is connected to a LabOne Data Server, either on the same PC (indicated as local) or on a remote PC (indicated by its IP address). The user can start a session to work with that device.
-----------	---

Free	The device is not in use by any LabOne Data Server and can be connected by clicking the Open button.
In Use	The device is in use by a LabOne Data Server. As a consequence the device cannot be accessed by the specified interface. To access the device, a disconnect is needed. Only applies to UHF Instruments.
Device FW upgrade required/available	The firmware of the device is out of date. Please first upgrade the firmware as described in Software Update .
Device not yet ready	The device is visible and starting up.

2.6.3. Saved Settings

Settings files can contain both UI and device settings. UI settings control the structure of the LabOne User Interface, e.g. the position and ordering of opened tabs. Device settings specify the set-up of a device. The device settings persist on the device until the next power cycle or until overwritten by loading another settings file.

The columns are described in [Table 2.6](#). The table rows can be sorted by clicking on the column header that should be sorted. The default sorting is by time. Therefore, the most recent settings are found on top. Sorting by the favorite marker or setting file name may be useful as well.

Table 2.6: Column Descriptions

	Allows favorite settings files to be grouped together. By activating the stars adjacent to a settings file and clicking on the column heading, the chosen files will be grouped together at the top or bottom of the list accordingly. The favorite marker is saved to the settings file. When the LabOne user interface is started next time, the row will be marked as favorite again.
Name	The name of the settings file. In the file system, the file name has the extension .md.
Date	The date and time the settings file was last written.
Comment	Allows a comment to be stored in the settings file. By clicking on the comment field a text can be typed in which is subsequently stored in the settings file. This comment is useful to describe the specific conditions of a measurement.
Device Type	The instrument type with which this settings file was saved.

Special Settings Files

Certain file names have the prefix "last_session_". Such files are created automatically by the LabOne Web Server when a session is terminated either explicitly by the user, or under critical error conditions, and save the current UI and device settings. The prefix is prepended to the name of the most recently used settings file. This allows any unsaved changes to be recovered upon starting a new session.

If a user loads such a last session settings file the "last_session_" prefix will be cut away from the file name. Otherwise, there is a risk that an auto-save will overwrite a setting which was saved explicitly by the user.

The settings file with the name "Default Settings" contains the default UI settings. See button description in [Table 2.7](#).

Table 2.7: Button Descriptions

Open	The settings contained in the selected settings file will be loaded. The button "Include Device Settings" controls whether only UI settings are loaded, or if device settings are included.
Include Device Settings	Controls which part of the selected settings file is loaded upon clicking on Open. If enabled, both the device and the UI settings are loaded.
Auto Start	Skips the session dialog at start-up if selected device is available. The default UI settings will be loaded with unchanged device settings.

Note

The user setting files are saved to an application-specific folder in the directory structure. The best way to manage these files is using the File Manager tab.

Note

The factory default UI settings can be customized by saving a file with the name "default_ui" in the Config tab once the LabOne session has been started and the desired UI setup has been established. To use factory defaults again, the "default_ui" file must be removed from the user setting directory using the File Manager tab.

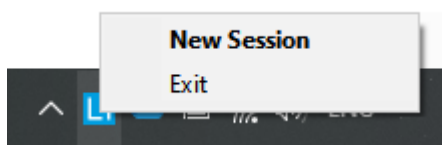
Note

Double clicking on a device row in the Available Devices table is a quick way of starting the default LabOne UI. This action is equivalent to selecting the desired device and clicking the **Open** button.

Double clicking on a row in the Saved Settings table is a quick way of loading the LabOne UI with those UI settings and, depending on the "Include Device Settings" checkbox, device settings. This action is equivalent to selecting the desired settings file and clicking the **Open** button.

2.6.4. Tray Icon

When LabOne is started, a tray icon appears by default in the bottom right corner of the screen, as shown in the figure below. By right-clicking on the icon, a new web server session can be opened quickly, or the LabOne Web and Data Servers can be stopped by clicking on Exit. Double-clicking the icon also opens a new web server session, which is useful when setting up a connection to multiple instruments, for example.

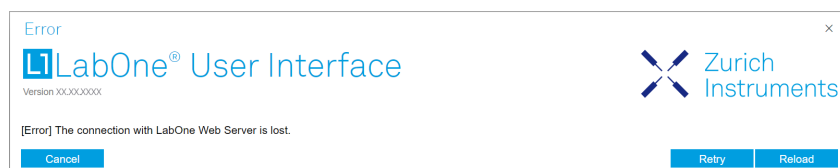


2.6.5. Messages

The LabOne Web Server will show additional messages in case of a missing component or a failure condition. These messages display information about the failure condition. The following paragraphs list these messages and give more information on the user actions needed to resolve the problem.

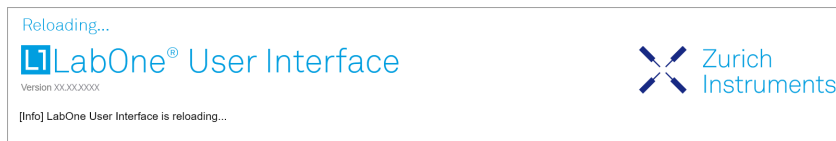
Lost Connection to the LabOne Web Server

In this case the browser is no longer able to connect to the LabOne Web Server. This can happen if the Web Server and Data Server run on different PCs and a network connection is interrupted. As long as the Web Server is running and the session did not yet time out, it is possible to just attach to the existing session and continue. Thus, within about 15 seconds it is possible with **Retry** to recover the old session connection. The **Reload** button opens the Device Connection dialog shown in [Figure 2.28](#). The figure below shows an example of the Connection Lost dialog.



Reloading...

If a session error cannot be handled, the LabOne Web Server will restart to show a new Device Connection dialog as shown in [Figure 2.28](#). During the restart a window is displayed indicating that the LabOne User Interface will reload. If reloading does not happen the same effect can be triggered by pressing F5 on the keyboard. The figure below shows an example of this dialog.



2.7. Using the LabOne Programming Interfaces

LabOne provides Application Programming Interfaces (APIs) for MATLAB, LabVIEW, Python, C, and .NET. These APIs require installation on the PC where they will be used as described in the following. The necessary installers are available on the Zurich Instruments download page, www.zhinst.com/downloads.

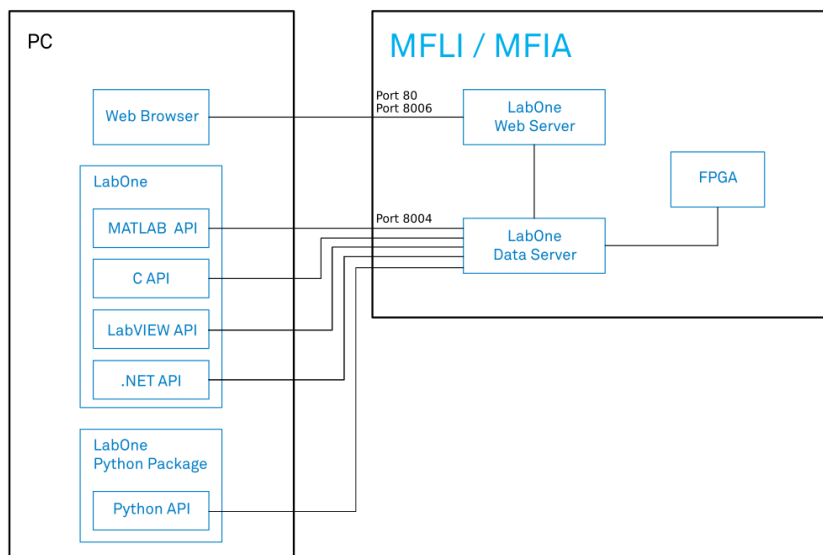


Figure 2.29: LabOne APIs installed on separate PC

The LabOne APIs communicate directly with the Data Server running on the MFIA Instrument via IP port 8004.

The device address can be `<instrument-serial>,mf-<instrument-serial>,mf-<instrument-serial>.<domain>`, or `<IP address>`. If the address is not fully qualified, means domain name is not added, the API will perform a network discovery with multicast to get the IP address. This ensures that devices connected by USB will be accessible by the same method as devices connected via 1GbE.

2.7.1. Using the MATLAB, LabVIEW APIs

There are two possibilities to install the MATLAB and LabVIEW APIs on a PC. The first option is to install the main LabOne package which includes these APIs. This is a good option if you would also like to run the Data Server on this PC as described in [Running LabOne on a Separate PC](#). Installation of the main LabOne package is described in [software installation](#). The second option is to install only the required API package. Please refer to the LabOne Programming Manual for details on installing and using the MATLAB and LabVIEW APIs.

2.7.2. Using the Python API

To use the Python API, the separate Zurich Instruments LabOne Python package needs to be installed. Please refer to the LabOne Programming Manual for details on installing and getting started with the LabOne Python API.

2.7.3. Using the .NET and C APIs

To use the .NET and C APIs, the main LabOne package needs to be installed which includes these APIs along with those for MATLAB and LabVIEW. Installation of the main LabOne package is described in [Section 2.5.1](#).

2.8. Software Update

2.9. Overview

It is recommended to regularly update the LabOne software on the MFIA Instrument to the latest version. In case the Instrument has access to the internet, this is a very simple task and can be done with a single click in the software itself, as shown in [Updating LabOne using Automatic Update Check](#). If there's no internet access or if the Automatic Update Check feature is disabled, the software can alternatively be updated by a drag-and-drop procedure, or by using a USB mass storage device (e.g. a memory stick). These methods are explained in [LabOne Update by Drag-and-Drop](#) and [LabOne Update from a USB Stick](#). If you use one of the LabOne APIs with a separate installer, don't forget to update this part of the software, too.

2.9.1. Updating LabOne using Automatic Update Check

Updating the software is done in two steps. First, LabOne is updated on the PC by downloading and installing the LabOne software from the Zurich Instruments downloads page. Second, the instrument firmware needs to be updated from the Device Connection dialog after starting up LabOne. This is shown in [Updating the Instrument Firmware](#). In case "Periodically check for updates" has been enabled during the LabOne installation and LabOne has access to the internet, a notification will appear on the Device Connection dialog whenever a new version of the software is available for download. This setting can later be changed in the Config tab of the LabOne user interface. In case automatic update check is disabled, the user can manually check for updates at any time by clicking on the button [Check For Update](#) in the Device Connection dialog. In case an update is found, clicking on the button "Update Available" shown in [Figure 2.30](#) will start a download and installation of the latest LabOne software on the MFIA instrument. In case the LabOne software runs on a separate PC (see [Running LabOne on a Separate PC](#)), clicking on "Update Available" allows you to download the latest LabOne installer for Windows or Linux, see [Figure 2.31](#). After download, proceed as explained in [software installation](#) to update LabOne. Following this update, the instrument firmware also needs to be updated as described in [Updating the Instrument Firmware](#).

Device Connection

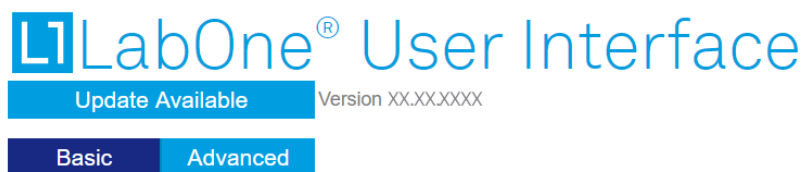


Figure 2.30: Device Connection dialog: LabOne update available

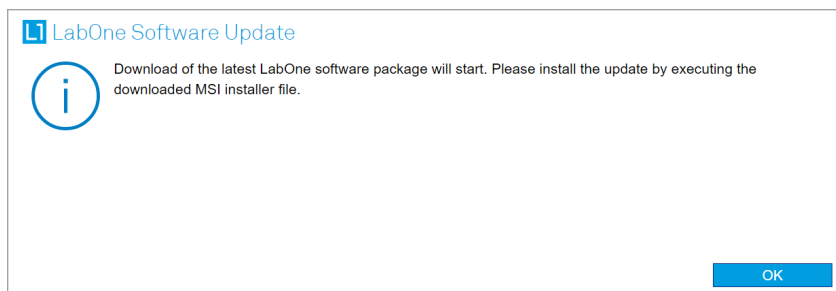


Figure 2.31: Download LabOne MSI using Automatic Update Check feature

2.9.2. LabOne Update by Drag-and-Drop

Note

Updating the LabOne software on the Instrument requires that the browser is connected to the Web Server running on the MFIA Instrument.

- Download the latest LabOne software version from the [Zurich Instruments Download Center](#). Choose the version for the MFIA Instrument. The file has the form **LabOneXX-00.00.00000.tar**.
- Open a Windows Explorer window and navigate to the location of the downloaded LabOne installation file (on Windows this is typically the Downloads folder). On Linux, the Files application or some other file manager utility supporting drag and drop can be used. We will use the term explorer window henceforth in this description.
- Start the LabOne User Interface in a web browser (connect to the Web Server running on the Instrument by typing the serial number in the address bar).
- Open the LabOne Config tab.
- Position/Re-size the explorer window and the web browser such that the dotted rectangle under the File Upload section of the Config tab is visible.

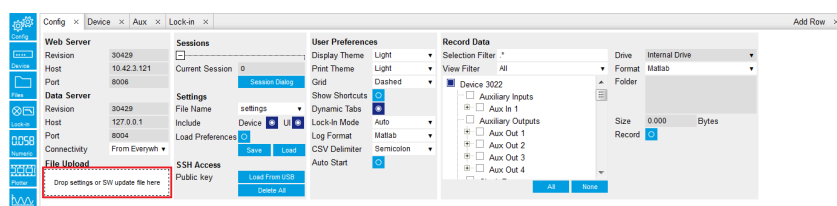


Figure 2.32: Config Tab showing drop zone for software update

- Drag and drop the downloaded LabOne installation file from the explorer window onto the dotted rectangle. An Upload pop-up window will appear indicating the progress of the upload.

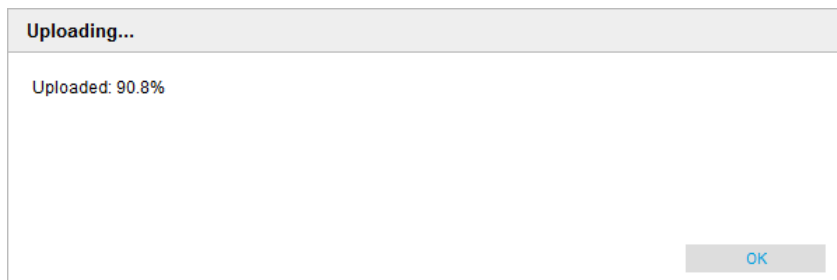


Figure 2.33: Upload pop-up window

Once uploaded, The LabOne Software Update pop-up window will appear.

```
{id=commons.getting_started.update_popup2}
```

- Click **OK** in the LabOne Software Update pop-up window. This completes the software update.

The servers running on the instrument will now be restarted necessitating a reload of the LabOne User Interface. An Error pop-up window will appear containing the text "The connection with the LabOne Web Server is lost". * Reload the LabOne User Interface by clicking on the **Reload** button.

2.9.3. LabOne Update from a USB Stick

Note

Updating the LabOne software on the Instrument requires that the browser is connected to the Web Server running on the MFIA Instrument.

—

Download the latest LabOne software version from the [Zurich Instruments Download Center](#) and copy it onto a USB mass storage device. Choose the version for the MFIA Instrument. The file has the form **LabOneMF-00.00.00000.tar**.

- Start the LabOne User Interface in a web browser (connect to the Web Server running on the Instrument by typing the serial number in the address bar).
- Insert the USB mass storage device into one of the USB sockets on the back of the MFIA Instrument.
- Open the LabOne File Manager tab.
- A folder labelled USB1 will be displayed in the File Manager tab, corresponding to the inserted USB device. Expand the USB1 branch to display its contents (double click on the folder icon).
- Right click the LabOne update .tar file and select **SW Update** to update the software.

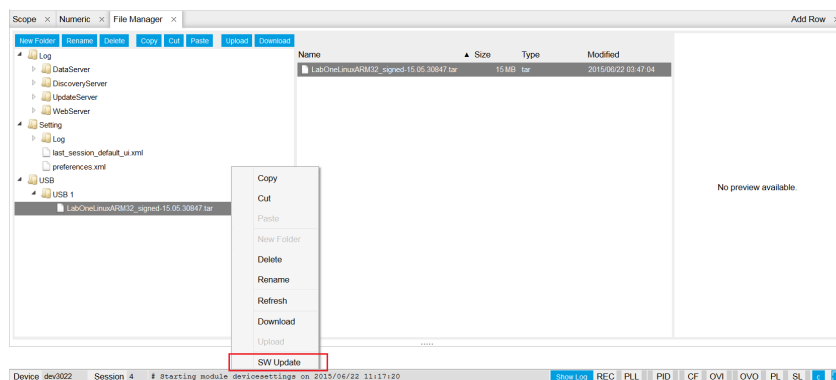


Figure 2.34: File Manager tab

An Upload pop-up window will appear indicating the progress of the upload. Once uploaded, The LabOne Software Update pop-up window will appear.



Figure 2.35: LabOne Software Update pop-up window

- Click **OK** in the LabOne Software Update pop-up window. This completes the software update. The servers running on the instrument will now be restarted necessitating a reload of the LabOne User Interface. An Error pop-up window will appear containing the text "The connection with the LabOne Web Server is lost". - Reload the LabOne User Interface by clicking on the **Reload** button.

2.9.4. Updating the Instrument Firmware

Even when running the LabOne software on a separate PC, a part of the software runs on the instrument. In order to distinguish between the two, the latter will be called firmware for the rest of this document. When upgrading to a new software release, it's also necessary to update the instrument firmware.

If the firmware needs an update, this is indicated in the Device Connection dialog of the LabOne user interface under Windows.

In the Basic view of the dialog, there will be a button "Upgrade FW" appearing together with the instrument icon as shown in [Figure 2.36](#). In the Advanced view, there will be a link "Upgrade FW" in the Update column of the Available Devices table. Click on **Upgrade FW** to open the firmware update start-up dialog shown in [Figure 2.37](#). The firmware upgrade takes approximately 2 minutes.

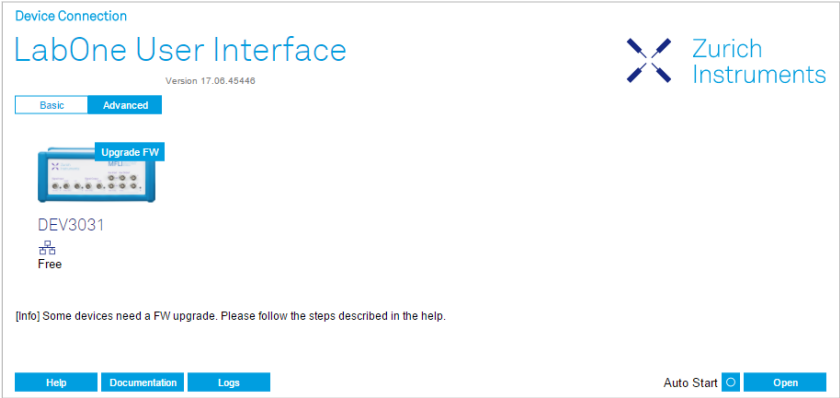


Figure 2.36: Device Connection dialog with available firmware update

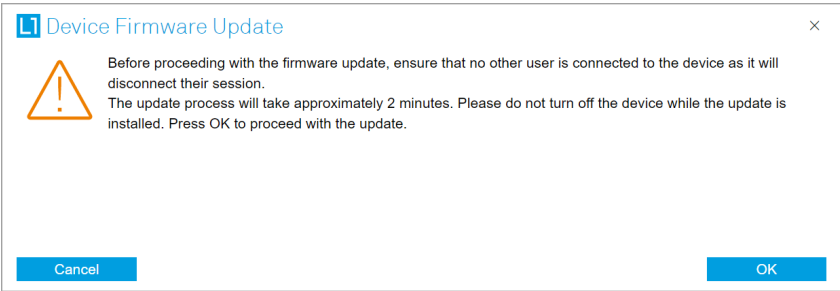


Figure 2.37: Device Firmware Update start-up dialog

Warning

Do not disconnect the USB or 1GbE cable to the Instrument or power-cycle the Instrument during a firmware update.

If you encounter any issues while upgrading the instrument firmware, please contact Zurich Instruments at support@zhinst.com.

2.10. Troubleshooting

This section aims to help the user solve and avoid problems while using the software and operating the instrument.

2.10.1. Common Problems

Your MFIA Instrument is an advanced piece of laboratory equipment which has many more features and capabilities than a traditional impedance analyzer. In order to benefit from these, the user needs access to a large number of settings in the LabOne User Interface. The complexity of the settings might overwhelm a first-time user, and even expert users can get surprised by certain combinations of settings. To avoid problems, it's good to use the possibility to save and load settings in the Config Tab. This allows one to keep an overview by operating the instrument based on known configurations. This section provides an easy-to-follow checklist to solve the most common mishaps.

Table 2.8: Common Problems

Problem	Check item
The software cannot be installed or uninstalled	Please verify you have administrator/root rights.
The software cannot be updated	Please use the Modify option in Windows Apps & Features functionality. In the software installer select Repair, then uninstall the old software version, and install the new version.

Problem	Check item
The Instrument does not turn on	Please verify the power supply connection and inspect the fuse. The fuse holder is integrated in the power connector on the back panel of the instrument.
The Instrument performs poorly in single-ended operation	the signal inputs of the instrument might be set to differential operation. Please ensure that differential input mode is turned off in the Lock-in Tab or In / Out Tab.
The Instrument has a high input noise floor (when connected to host computer by USB)	the USB cable connects the Instrument ground to computer ground, which might inject some unwanted noise to the measurements results. In this case it is recommended to use the Ethernet connection which is galvanically isolated using a UTP Cat 5 or 6 cable (UTP stands for "unshielded twisted pair").
The Instrument performs poorly at low frequencies (below 100 Hz)	the signal inputs of the instrument might be set to AC operation. Please verify to turn off the AC switch in the Lock-in Tab or In / Out Tab.
The Instrument performs poorly during operation	the demodulator filters might be set too wide (too much noise) or too narrow (slow response) for your application. Please verify if the demodulator filter settings match your frequency versus noise plan.
The Instrument performs poorly during operation	clipping of the input signal may be occurring. This is detectable by monitoring the red LEDs on the front panel of the instrument or the Input Overflow (OVI) flags on the STATUS_TAB of the user interface. It can be avoided by adding enough margin on the input range setting (for instance 50% to 70% of the maximum signal peak).
The Instrument performs strangely when working with the MF-MD Multi-demodulator Option	it is easily possible to turn on more signal generators than intended. Check the generated Signal Output with the integrated oscilloscope and check the number of simultaneously activated oscillator voltages.
The Instrument measurements are unpredictable	Please check the Status Tab to see if there is any active warning (red flag), or if one has occurred in the past (yellow flag).
The Instrument does not generate any output signal	verify that signal output switch has been activated in the Lock-in Tab or in the In / Out Tab.
The Instrument locks poorly using the digital I/O as reference	make sure that the digital input signal has a high slew rate and clean level crossings.
The Instrument locks poorly using the auxiliary analog inputs as reference	the input signal amplitude might be too small. Use proper gain setting of the input channel.
The sample stream from the Instrument to the host computer is not continuous	Check the communication (COM) flags in the status bar. The three flags indicate occasional sample loss, packet loss, or stall. Sample loss occurs when a sampling rate is set too high (the instrument sends more samples than the interface and the host computer can absorb). The packet loss indicates an important failure of the communications to the host computer and compromises the behavior of the instrument. Both problems are prevented by reducing the sample rate settings. The stall flag indicates that a setting was actively changed by the system to prevent UI crash.

Problem	Check item
The LabOne User Interface does not start (when running the LabOne on a PC)	Verify that the LabOne Data Server (ziDataServer.exe) and the LabOne Web Server (ziWebServer.exe) are running via the Windows Task Manager. The Data Server should be started automatically by ziService.exe and the Web Server should be started upon clicking "Zurich Instruments LabOne" in the Windows Start Menu. If both are running, but clicking the Start Menu does not open a new User Interface session in a new tab of your default browser then try to create a new session manually by entering 127.0.0.1:8006 in the address bar of your browser.
The user interface does not start or starts but remains idle	Verify that the Data Server has been started and is running on your host computer.
The user interface is slow and the web browser process consumes a lot of CPU power	Make sure that the hardware acceleration is enabled for the web browser that is used for LabOne. For the Windows operating system, the hardware acceleration can be enabled in Control Panel → Display → Screen Resolution . Go to Advanced Settings and then Troubleshoot. In case you use a NVIDIA graphics card, you have to use the NVIDIA control panel. Go to Manage 3D Settings, then Program Settings and select the program that you want to customize.

2.10.2. Location of the Log Files

The most recent log files of the LabOne Web and Data Server programs are most easily accessed by clicking on **Logs** in the **LabOne Device Connection dialog** of the user interface. The Device Connection dialog opens on software start-up or upon clicking on **Session Manager** in the Config tab of the user interface.

In the case that the Web and/or Data Server are running on the MFIA device (and not on the PC) the log files are physically stored on the MFIA instrument and can be accessed through the **File Manager Tab** of the LabOne User Interface in the **Log** folder. If the Web and/or Data Server are ran on a PC and not on the embedded computer of the MFIA (see **Running LabOne on a Separate PC**) the log file locations on disk are given in the sections below.

Windows

If the Web and/or Data Server are running on a Windows PC, their log files can be found in the following directories. - LabOne Data Server (**ziDataServer.exe**):

```
C:\Windows\ServiceProfiles\LocalService\AppData\Local\Temp\Zurich
Instruments\LabOne\ziDataServerLog
```

- LabOne Web Server (**ziWebServer.exe**):
C:\Users[USER]\AppData\Local\Temp\Zurich Instruments\LabOne\ziWebServerLog

Note

The **C:\Users\[USER]\AppData** folder is hidden by default under Windows. A quick way of accessing it is to enter **%AppData%\..** in the address bar of the Windows File Explorer.

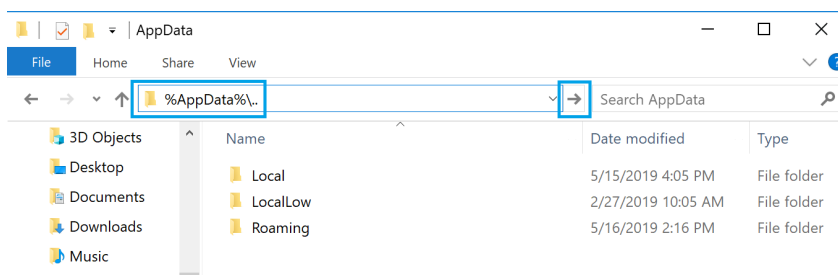


Figure 2.38: Using the

Linux and macOS

If the Web and/or Data Server are running on Linux or macOS, their log files can be found in the following directories. - LabOne Data Server (**ziDataServer**):

```
/tmp/ziDataServerLog_[USER]
```

- LabOne Web Server (**ziWebServer**):
/tmp/ziWebServerLog_[USER]

2.10.3. Prevent web browsers from sleep mode

It often occurs that an experiment requires a long-time signal acquisition; therefore, the setup including the measurement instrument and LabOne software are left unattended. By default, many web browsers go to a sleep mode after a certain idle time which results in the loss of acquired data when using the web-based user interface of LabOne for measurement. Although it is recommended to take advantage of LabOne APIs in these situations to automate the measurement process and avoid using web browsers for data recording, it is still possible to adjust the browser settings to prevent it from entering the sleep mode. Below, you will find how to modify the settings of your preferred browser to ensure a long-run data acquisition can be implemented properly.

Edge

1. Open **Settings** by typing **edge://settings** in the address bar
2. Select **System** from the icon bar.
3. Find the **Never put these sites to sleep** section of the **Optimized Performance** tab.
4. Add the IP address and the port of LabOne Webserver, e.g., **127.0.0.1:8006** or **192.168.73.98:80** to the list.

Chrome

1. While LabOne is running, open a tab in Chrome and type **chrome://discards** in the address bar.
2. In the shown table listing all the open tabs, find LabOne and disable its **Auto Discardable** feature.
3. This option avoids discarding and refreshing the LabOne tab as long as it is open. To disable this feature permanently, you can use an extension from the Chrome Webstore.

Firefox

1. Open **Advanced Preferences** by typing **about:config** in the address bar.
2. Look for **browser.tabs.unloadOnLowMemory** in the search bar.

3. Change it to **false** if it is **true**.

Opera

1. Open **Settings** by typing **opera://settings** in the address bar.
2. Locate the **User Interface** section in the **Advanced** view.
3. Disable the **Snooze inactive tabs to save memory** option and restart Opera.

Safari

1. Open **Debug** menu.
2. Go to **Miscellaneous Flags**.
3. Disable **Hidden Page Timer Throttling**.

3. Functional Overview

This chapter provides the overview of the features of the MFIA Instrument. The first section contains the graphical overview as well as the hardware and software feature list. The next sections detail the front panel and the back panel of the measurement instrument. The following section describes the MFITF Impedance Test Fixture. The last section provides product selection and ordering support.

3.1. Features

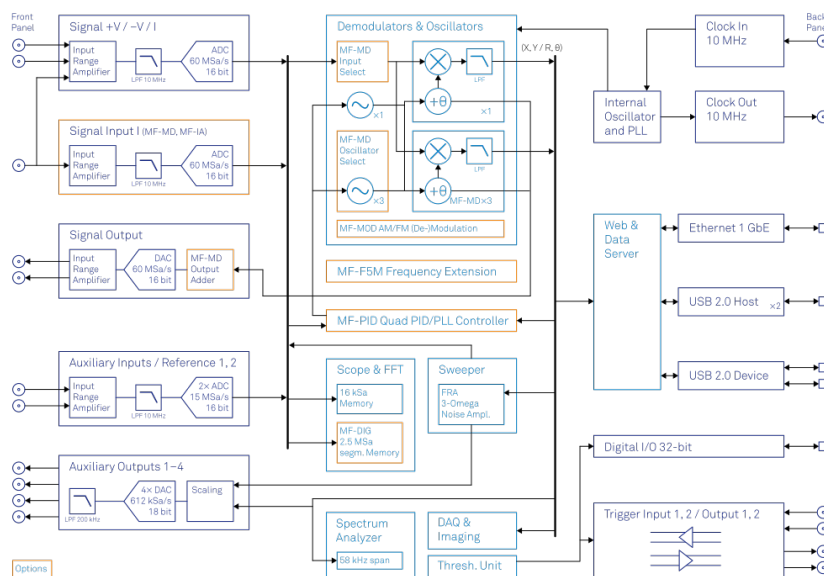


Figure 3.1: MFIA Instrument Overview

Every MF Instrument (MFLI, MFIA) described on [Figure 3.1](#) consists of several internal units (light blue color) surrounded by several interface units (dark blue color) and the front panel on the left-hand side and the back panel on the right-hand side. The orange blocks are optional units that can be either ordered at the beginning or upgraded later in the field. The arrows between the panels and the interface units indicate the physical connections and the data direction flow. Only a small subset of internal connections is depicted.

The MFIA 5 MHz Precision LCR Meter and Impedance Analyzer provides fast and accurate measurements over a wide impedance range. Unlike other impedance analyzers the MFIA measures the voltage and the current signals directly without the need of a feedback loop, e.g. balanced bridge configuration. As a result the MFIA is capable of measuring at frequencies as low as 1 Hz and to provide reliable impedance measurements up to 10 GΩ.

The device-under-test is connected to the low-noise inputs and outputs L_{CUR} , H_{CUR} , L_{POT} , H_{POT} via the MFITF Impedance Test fixture. If a different test fixture is used, the Compensation Advisor makes it easy to obtain accurate measurement data. The measurement signals are amplified to an appropriate range and digitized at very high speed. The resulting samples are fed into the digital signal processor with dual-phase demodulators for independent current and voltage measurement. The measurement signals are further processed on the embedded processor that provides the LabOne Web Server. Any device running a browser (PC, tablet or smart phone) can connect to the Web Server and no software installation is required. Both Ethernet and USB interfaces are supported. The measurement data are also available as analog signal on the auxiliary outputs on the front panel of the MF Instrument. When the MF-MD option is installed, the drive output H_{CUR} is capable of generating a multi-frequency output signal which enables impedance measurements at a second independent frequency.

The MFIA Impedance Analyzer provides full lock-in functionality equivalent to the MFLI Lock-in Amplifier. Several trigger and reference connections provide the associated functionality for purposes such as triggering demodulation, triggering oscilloscope data acquisition, or to generate external reference clocks or triggering signals to other equipment.

Impedance Analyzer features

- 2-terminal and 4-terminal measurement modes
- Multiple DUT representations (replacement circuits)
- Confidence Indicator
- Compensation Advisor
- Auto range mode
- Auto bandwidth mode
- Auto test signal level
- Dual-frequency mode (requires MF-MD option, simultaneous impedance measurement at two frequencies)

Medium Frequency Voltage Input L_{POT} , H_{POT}

- 1 low-noise MF voltage input, single-ended or differential, 5 MHz bandwidth
- Variable input range
- Switchable input impedance
- Selectable AC/DC coupling
- Selectable instrument ground or floating

Medium Frequency Current Input L_{CUR}

- 1 low-noise MF current input, single-ended, 5 MHz bandwidth
- Variable input range

Medium Frequency Signal Output H_{CUR}

- Low-distortion MF outputs, single-ended, differential, 5 MHz bandwidth
- Variable output range
- Digital and analog offset

Auxiliary Input, Outputs and Triggers

- 4 auxiliary outputs, user defined signals
- 2 auxiliary inputs, general purpose
- 2 input and 2 output trigger signals

High-speed Connectivity

- USB 2.0 Device high-speed 480 Mbit/s interface
- Dual USB 2.0 Host high-speed interface
- LAN 1 Gbit/s controller interface
- DIO: 32-bit digital input-output port
- Clock input connector (10 MHz)
- Clock output connector (10 MHz)

LabOne Toolset

- Sweeper
- Scope
- Numeric
- Spectrum
- Plotter
- Data Acquisition

Software Connectivity

- Data server with multi-client support
- API for C, LabVIEW, MATLAB, Python based instrument programming

3.2. Front Panel Tour

The front panel BNC connectors and control LEDs are arranged as shown in [Figure 3.2](#) and listed in [Table 3.1](#).

Note

Some BNC connectors on the front panel have secondary labels below the connector. These are relevant when operating the MFIA as a lock-in amplifier and are listed in brackets in [Table 3.1](#).

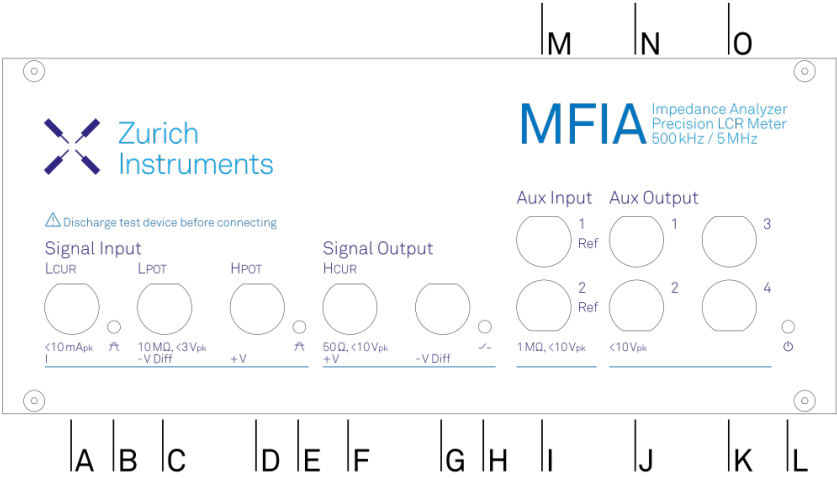


Figure 3.2: MF Instrument front panel

Table 3.1: MF Instrument front panel description

Position	Label / Name	Description
A	Signal Input L_{CUR} (I)	single-ended current input
B	Current Input Signal Over	this red LED indicates that the current input signal saturates the A/D converter and therefore the current input range must be increased or the signal must be attenuated
C	Signal Input L_{POT} (-V Diff)	voltage input
	single-ended mode: internally shorted to ground	
	differential mode: negative voltage input	
D	Signal Input H_{POT} (+V)	voltage input
	in single-ended mode: single-ended voltage input	
	differential mode: positive voltage input	
E	Voltage Input Signal Over	this red LED indicates that the voltage input signal saturates the A/D converter and therefore the voltage input range must be increased or the signal must be attenuated
F	Signal Output H_{CUR} (+V)	Drive voltage output
	single-ended mode: single-ended voltage output	
	differential mode: positive voltage output	

Position	Label / Name	Description
G	Signal Output –V Diff	voltage output
single-ended mode: internally shorted to ground		
differential mode: negative voltage output		
H	Signal Output ON	this blue LED indicates that the signal output is actively driven by the instrument
I	Aux Input 2 Ref	auxiliary input 2, can be used as external reference input supporting the full bandwidth of the device
J	Aux Output 2	auxiliary output 2, this connector provides a user defined signal, often used to output demodulated samples (X,Y) or (R,THETA)
K	Aux Output 4	auxiliary output 4, this connector provides a user defined signal, often used to output demodulated samples (X,Y) or (R,THETA)
L	Power	this LED indicates that the instrument is powered
color blue: the device has an active connection to the LabOne data server and is ready for operation		
color green blinking: the firmware is ready, waiting for LabOne data server connection. This process takes around 20 seconds.		
color red: the device is not initialized respectively is performing the internal auto calibration process.		
color purple blinking: a firmware update is in progress		
color purple: the boot process failed		
M	Aux Input 1 Ref	auxiliary input 1, can be used as external reference input supporting the full bandwidth of the instrument; the value of auxiliary input 1 can be added as offset to the signal output
N	Aux Output 1	auxiliary output 1, this connector provides a user-defined signal, often used to output demodulated samples (X,Y) or (R,Θ)
O	Aux Output 3	auxiliary output 3, this connector provides a user-defined signal, often used to output demodulated samples (X,Y) or (R,Θ)

Please refer to the troubleshooting section for further information regarding the meaning of the Power LED colors.

3.3. Back Panel Tour

The back panel is the main interface for power, control, service and connectivity to other ZI instruments. Please refer to [Figure 3.3](#) and [Table 3.2](#) for the detailed description of the items.

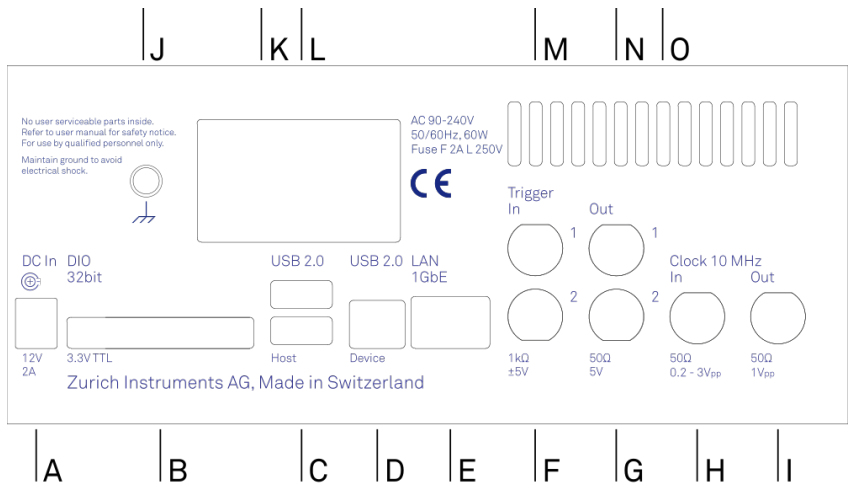


Figure 3.3: MF Instrument back panel

Table 3.2: MF Instrument back panel description

Position	Label / Name	Description
A	DC In	DC external 12 V power supply
B	DIO	32-bit digital input/output connector
C	USB 2.0 Host	universal serial bus host connector
D	USB 2.0 Device	universal serial bus device connector to computer
E	LAN 1GbE	1 Gbit Ethernet LAN connector
F	Trigger In 2	digital TTL trigger input 2
G	Trigger Out 2	digital TTL trigger output 2
H	Clk 10 MHz In	clock input (10 MHz) to be used for synchronization from external instruments
I	Clk 10 MHz Out	clock output (10 MHz) to be used for synchronization of external instruments
J	Earth ground	4 mm banana jack connector for earth ground, electrically connected to the chassis and the earth pin of the power inlet
K	USB 2.0 Host	universal serial bus host connector
L	Power inlet	power inlet with ON/OFF switch
M	Trigger In 1	digital TTL trigger input 1
N	Trigger Out 1	digital TTL trigger output 1
O	Cooling outlet	ventilator (important: keep clear from obstruction)

3.4. MFITF Impedance Test Fixture

The MFIA Impedance Analyzer is sold in combination with the MFITF Impedance Test Fixture. The test fixture shown in [Figure 3.4](#) is to be directly connected to the L_{CUR} , L_{POT} , H_{POT} , H_{CUR} BNC connectors on the MFIA front panel. The spacing of the BNC connectors is compatible with established standards for LCR meter and impedance analyzer test fixtures. The MFITF comes with a set of low-capacitive and low-loss DUT carriers for surface-mounted devices (SMD) and for through-hole components. Using the MFIA together with the MFITF and these carriers ensures the best possible accuracy.

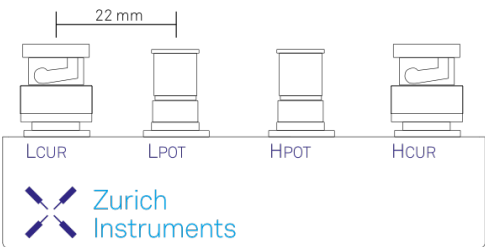


Figure 3.4: MFITF Impedance Test Fixture top view

The MFITF front panel as shown in [Figure 3.5](#) integrates an 8-pin connector (Sullins PPPC081LFBN-RC, Digikey part number S7041-ND) providing four signal contacts and four guard contacts for customized DUT holders.

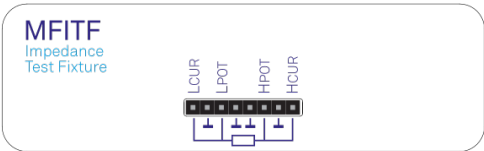
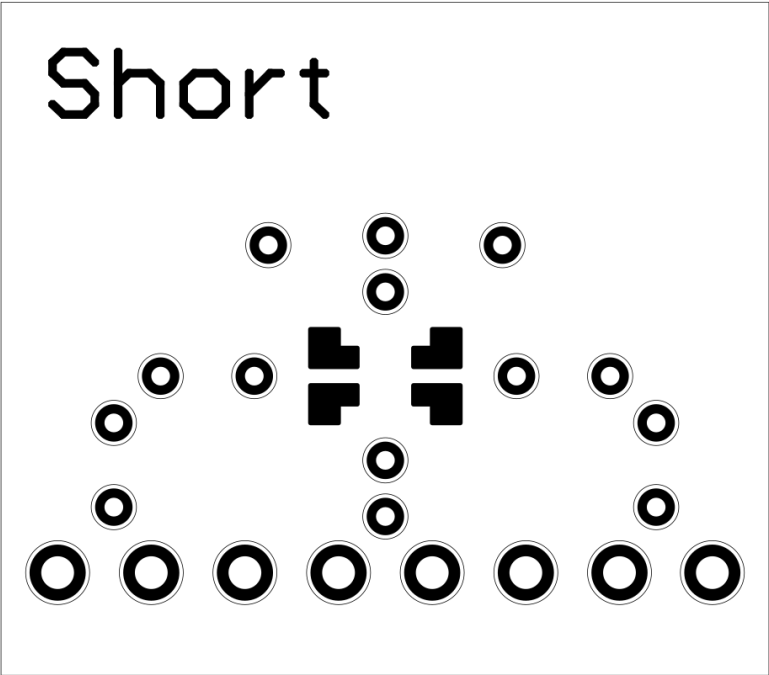
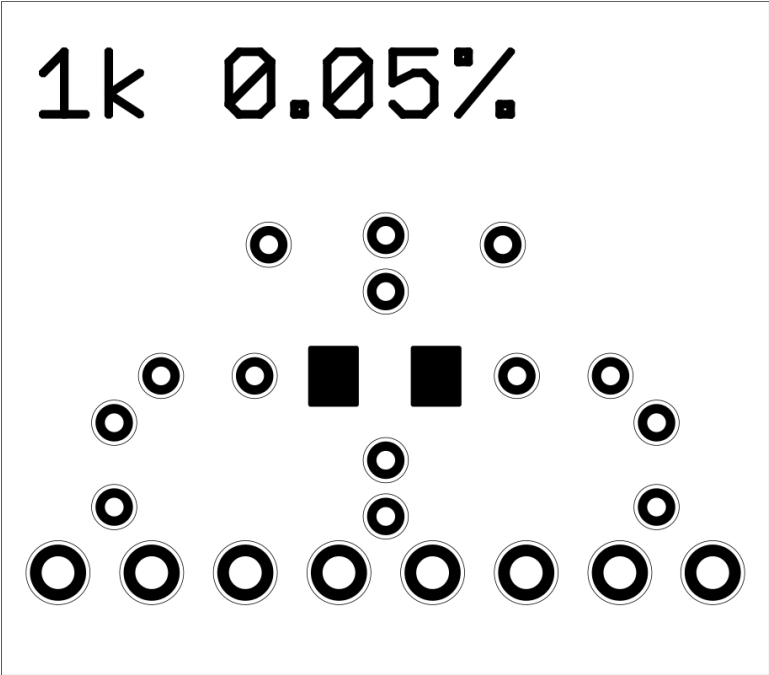


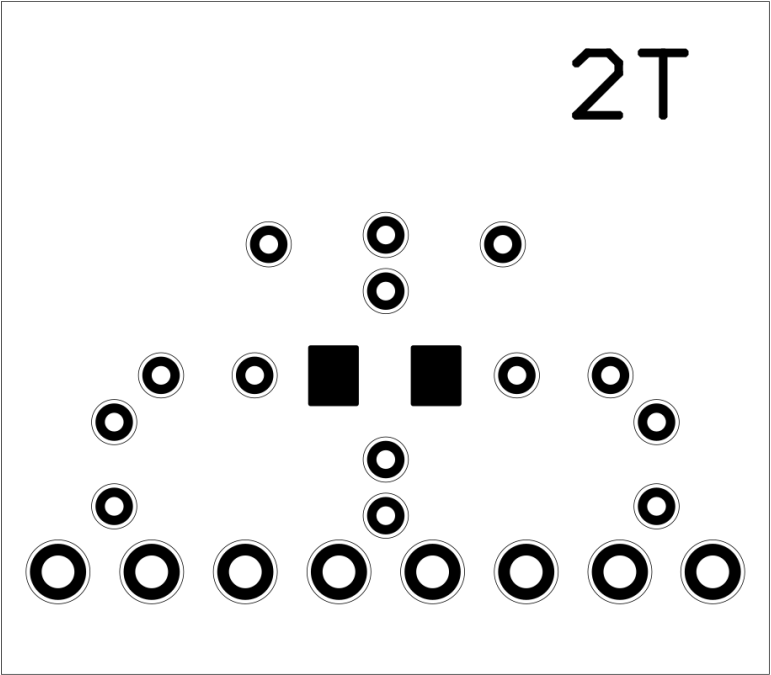
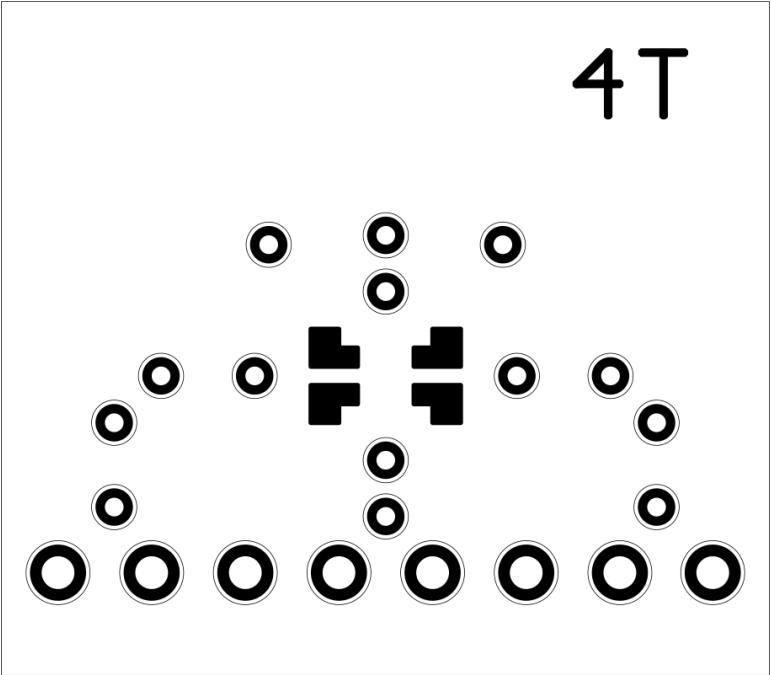
Figure 3.5: MFITF Impedance Test Fixture front view

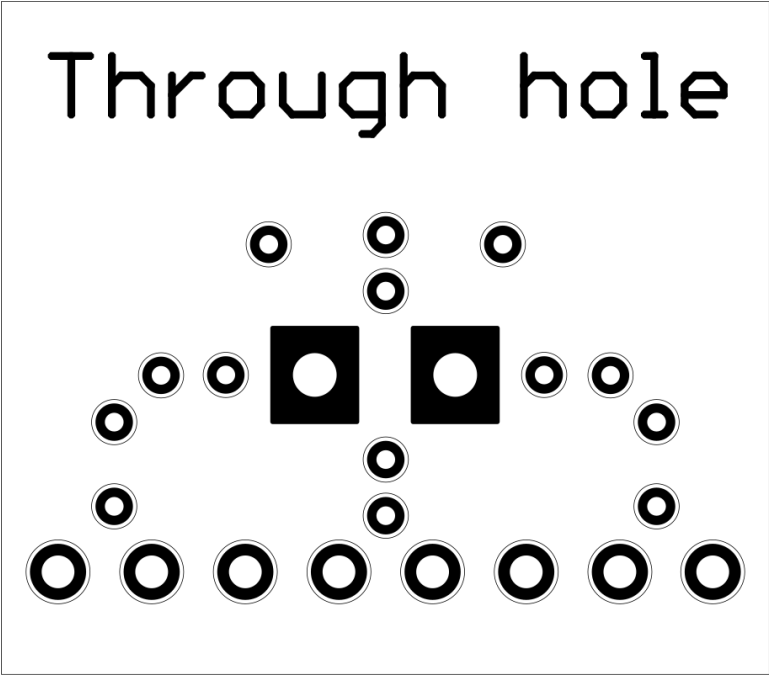
The PCB-based carriers delivered with the MFITF are described in the following table.

Table 3.3: DUT carriers for MFITF Impedance Test Fixture

Label	Image	Description	Number of units
Open		Open circuit for compensation and testing	1

Label	Image	Description	Number of units
Short		Short-circuit for compensation and testing	1
1k 0.05%		1 k Ω load with 0.05% accuracy, 10 ppm/ $^{\circ}$ C temperature coefficient, for testing and compensation	1

Label	Image	Description	Number of units
2 Pt		Carrier for SMDs with 2-terminal connection	3
4 Pt		Carrier for SMDs with 4-terminal connection	4

Label	Image	Description	Number of units
Through hole		Carrier for through-hole components	2

3.5. Ordering Guide

Table 3.4 provides an overview of the available MF products. Upgrade options can be purchased at any time without need to send the Instrument to Zurich Instruments.

Table 3.4: MF Instrument product codes for ordering

Product code	Product name	Description	Upgrade in the field possible
MFIA 500 kHz	MFIA 500 kHz Lock-in Amplifier	base product	-
MFIA 5 MHz	MFIA 5 MHz Lock-in Amplifier	bundle	-
MF-MD	MF-MD Multi-demodulator	option	yes
MF-F5M	MF-F5M Frequency Extension	option	yes
MF-DIG	MF-DIG Digitizer	option	yes
MF-IA	MF-IA Impedance Analyzer	option (includes MFITF)	yes
MFITF	Impedance Test Fixture	accessory (included with the MFIA and MF-IA)	-

The MF product line consists of the MFLI Lock-in Amplifier series and the MFIA Impedance Analyzer series. The combination MFLI 5 MHz + MF-IA is equivalent to the MFIA 5 MHz product, and the combination MFLI 500 kHz + MF-IA is equivalent to the MFIA 500 kHz product. The following table gives an overview of the features for the most important product configurations.

Table 3.5: Product selector

Feature	MFIA	MFIA +MF-MD	MFIA +MF-F5M	MFIA +MF-MD +MF-F5M
Internal reference mode	yes	yes	yes	yes
External reference mode	yes	yes	yes	yes
Auto reference mode	yes	yes	yes	yes
Impedance mode (independent measurement of voltage and current signal inputs)	-	yes	-	yes
Signal generators	1	1	1	1
Superposed output sinusoids per generator	1	up to 4	1	up to 4
Quad-harmonic mode	-	yes	-	yes
Multi-frequency mode	-	yes	-	yes
Arbitrary frequency mode	-	yes	-	yes
Number of demodulators	1	4	1	4
Simultaneous frequencies	1	4	1	4
Simultaneous harmonics	1	4	1	4
External references	1	2	1	2
Dynamic reserve	120 dB	120 dB	120 dB	120 dB
Frequency range	500 kHz	500 kHz	5 MHz	5 MHz
USB 2.0 480 Mbit/s	yes	yes	yes	yes
Ethernet 1 GbE	yes	yes	yes	yes

3.6. Rack Mount

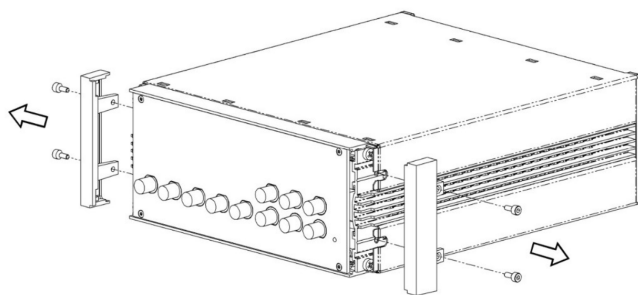
Parts for mounting the MFIA in a 19" rack are available from Zurich Instruments upon request under support@zhinst.com. Rack mount of a single MFIA as well as two MFIA instruments side-by-side is possible and explained in the following instructions.

3.6.1. Instructions for a Single Instrument

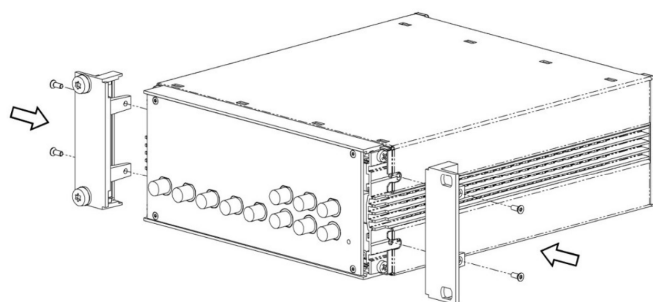
Table 3.6: Parts for rack mount of single instrument

Description	Quantity	Unit
19" Corner bracket Magic 2U RAL 9003	1	PCE
19" Adapter Plate 2U RAL 9003	1	PCE
Pan head screw M6 x 16 ISO 14583 A2 T30	4	PCE
DIN 125 A 6.4 st. zinc plated washer	4	PCE
Cage nut KM 6	4	PCE
Screw ISO 7380, M4x12, Torx, zinc plated	2	PCE
Washer DIN 9021 Ø 4.3 zinc plated	2	PCE

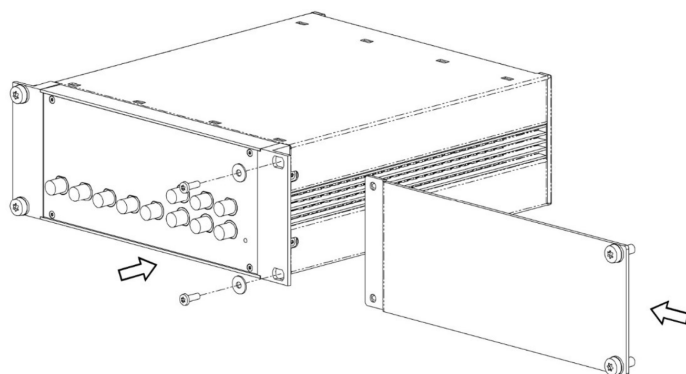
1. Remove 4x screws M3x7
2. Remove 2x trim piece



3. Attach 2x 19" Corner bracket Magic 2U RAL 9003 using the screws which were removed in step 1



4. Attach 19" Adapter plate 2U RAL 9003 using 2x screw ISO 7380 M4x12 Torx and washer DIN 9021 Ø 4.3



5. Attach 4x washer DIN 125 A 6.4 and 4x screw ISO 14583 M6x16 Torx for rack mount
 6. Attach 4x cage nuts KM6 to cabinet
 7. Attach instrument to cabinet

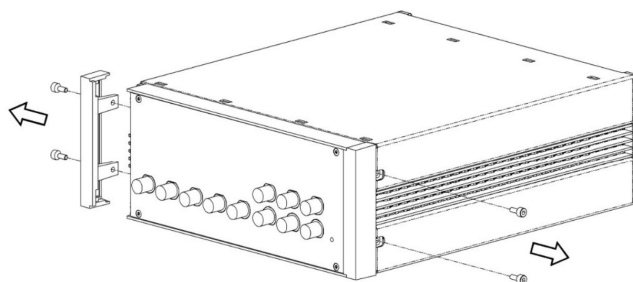
3.6.2. Instructions for Two Instruments

Table 3.7: Parts for rack mount of two instrument

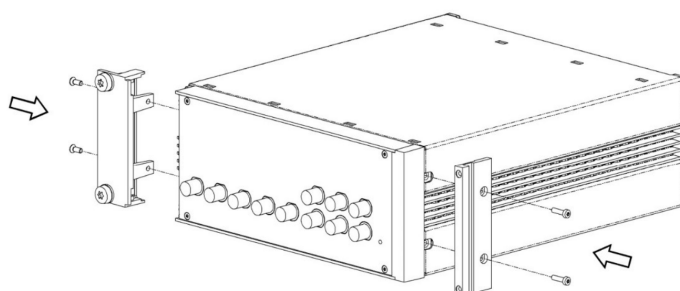
Description	Quantity	Unit
Connection Strap Magic 2U outside	1	PCE
Connection Strap Magic 2U inside	1	PCE
19" Corner bracket Magic 2U RAL 9003	2	PCE
Connection Strap Magic 2U backside	1	PCE
Screw sim. DIN 965 M2.5x10 A2 Torx lock	2	PCE
Screw sim. DIN 7985 M3x12 Torx A2 w. lock-tight	2	PCE
Pan head screw M6 x 16 ISO 14583 A2 T30	4	PCE
DIN 125 A 6.4 st. zinc plated washer	4	PCE
Cage nut KM 6	4	PCE

Assembly Instructions: Left Chassis

1. Remove 2x screws M3x7 on the left side and remove left trim piece
2. Remove 2x screws M3x7 on the right side

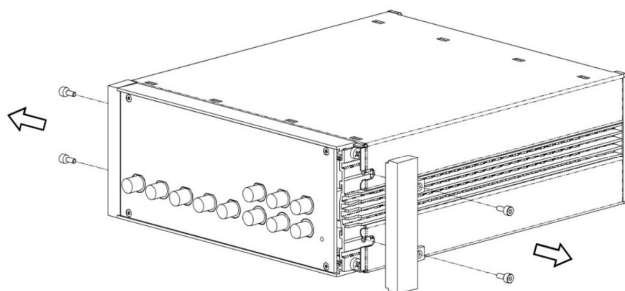


3. Attach 19" Corner bracket Magic 2U RAL 9003 to the left side with 2x screw M3x7 removed in step 1
4. Attach Connection Strap Magic 2U outside to the right side with 2x DIN 7985 screw M3x12 Torx

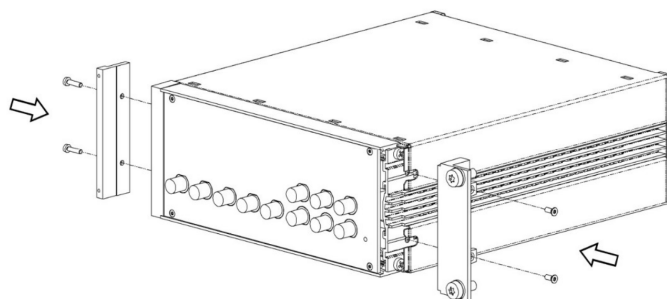


Assembly Instructions: Right Chassis

1. Remove 2x screws M3x7 on the right side and remove right trim piece
2. Remove 2x screws M3x7 left side

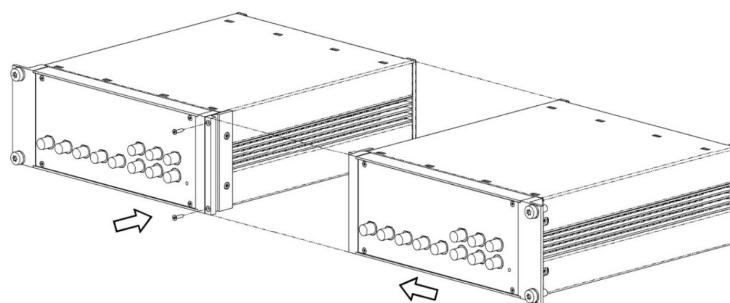


3. Attach 19" Corner bracket Magic 2U RAL 9003 to the right side with 2x screw M3x7 removed in step 1
4. Attach Connection Strap Magic 2U inside to left the side with screw Cylinder head screw M3x12 Torx

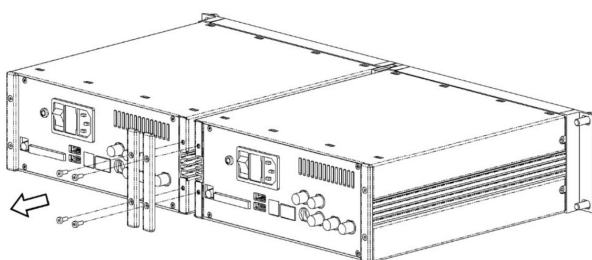


Chassis Assembly

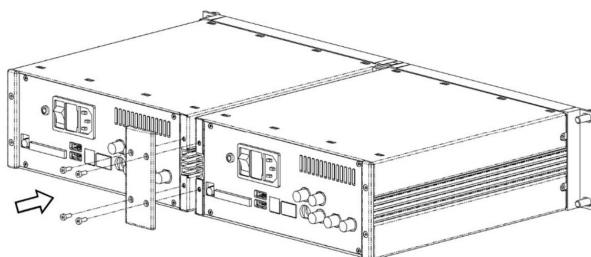
1. Attach both chassis together and fix with 2 screws DIN 965 M2.5x10 Torx through the connection strap



2. Remove 4x screws and 2x inner plastic covers



3. Attach the rear connection strap, using screws removed in step 2



4. Attach Cage nuts KM6 to cabinet

4. Tutorials

The tutorials in this chapter have been created to allow users to become more familiar with the basic technique of lock-in amplification, the operation of host-based lock-in amplifiers, the LabOne web browser based user interface, as well as some more advanced lock-in measurement techniques. In order to successfully carry out the tutorials, users are required to have certain laboratory equipment and basic equipment handling knowledge. The equipment list is given below.

Note

For all tutorials, you must have LabOne installed as described in the [Getting Started](#). We make use of a systematic nomenclature to describe the different controls and elements of the LabOne graphical user interface. This nomenclature is explained in [User Interface Overview](#).

- 1 USB or LAN cable
- 5 BNC cables
- 1 oscilloscope (optional)
- 2 BNC T-pieces
- With MF-IA Option Only
 - 1 Kelvin probe cable set and through-hole resistors 1 Ω and 10 Ω (for the Compensation tutorial)
 - 1 M Ω through-hole resistor and soldering equipment (for the Advanced Impedance Measurement tutorial)

4.1. Basic Impedance Measurement

Note

This tutorial is applicable to all MFIA Instruments and to MFLI Instruments with the MF-IA Impedance Analyzer option installed.

4.1.1. Goals and Requirements

This tutorial is for people with no or little prior experience with Zurich Instruments impedance analyzers. By using a very basic measurement setup, this tutorial shows the fundamental working principles of the MFIA Instrument and the LabOne user interface in a step-by-step hands-on approach.

The tutorial requires the MFITF Impedance Test Fixture.

4.1.2. Preparation

This tutorial guides you through the basic steps to perform a simple measurement of the 1 k Ω test resistor delivered with the MFITF Impedance Test Fixture. Start by connecting the MFITF to the BNC connectors on the front panel of the MFIA, and plug the resistor labeled "1k 0.05%" into the 8-pin connector on the front of the test fixture. [Figure 4.1](#) shows a diagram of the hardware setup.

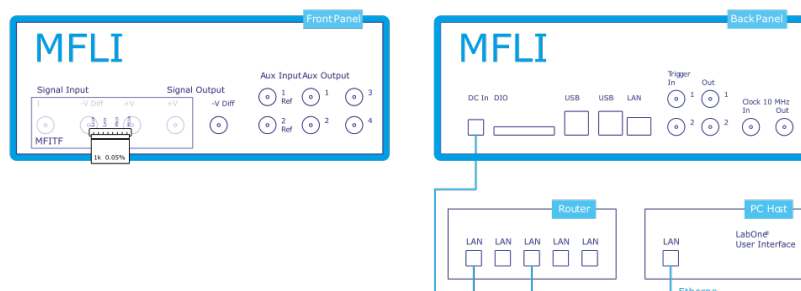


Figure 4.1: Setup for the Impedance Measurement Tutorial

Make sure that the MFIA Instrument is powered on and then connect the MFIA directly by USB to your host computer or by Ethernet to your local area network (LAN) where the host computer resides. After connecting to your Instrument through the web browser using its address, the LabOne graphical user interface is opened. Check the [Getting Started](#) for detailed instructions. The tutorial can be started with the default instrument configuration (e.g. after a power cycle) and the default user interface settings (i.e. as is after pressing F5 in the browser).

4.1.3. Configure the Impedance Analyzer

The Impedance Analyzer tab shown in [Figure 4.2](#) is open by default and provides you with the necessary settings to start measuring impedance. For a quick measurement of the 1 kΩ device-under-test (DUT), the default settings are suitable. All we have to do is to click the Enable button on top left of the Impedance Analyzer tab in order to turn on the drive signal and start the measurement. Since automatic range control is enabled by default, the current and voltage input ranges will be adjusted within a few seconds and the measurement starts.

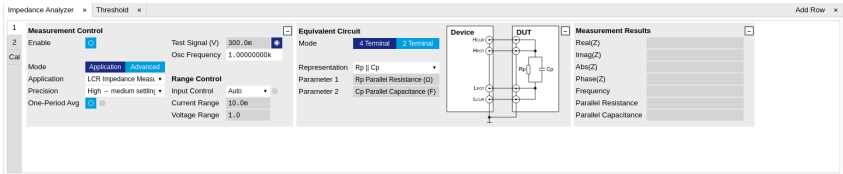


Figure 4.2: Impedance Analyzer tab (Application mode)

The measurement result is now shown in the Measurement Results section of the tab. As an alternative, the [Numeric Tab](#) provides a more configurable display of the data. Open the tab by clicking on the corresponding icon on the left of the screen. The Numeric tab shown in [Figure 4.3](#) is organized in panels which display the impedance Z_{DUT} in Cartesian and in polar form and other parameters. You can add more panels using the [Tree Selector](#) on the right if you wish to monitor other quantities such as bias voltage. Every numerical value is supported by a graphical bar scale indicator for better readability. We read an impedance absolute value $Abs(Z)$ of 999.8 Ω, and a complex phase $Phase(Z)$ of nearly 0 degrees, nicely corresponding to the 1 kΩ DUT that we plugged in.

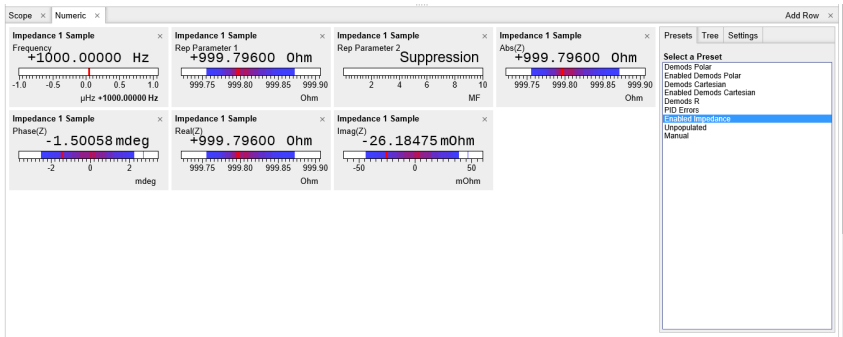


Figure 4.3: Numerical tab displaying measured impedance and circuit representation parameters

4.1.4. Measure Frequency Dependence

The frequency is a critical measurement parameter. The frequency can be changed manually by clicking on the Osc Frequency field, typing a value such as 1000 or 1k in short press either **Tab** or **Enter** on your keyboard to apply the setting. For a more systematic measurement of the frequency dependence, we'll turn to the Sweeper tool.

Open the Sweeper tab by clicking on the corresponding icon on the left side of the user interface. The Sweeper features an Application Mode providing suitable settings for common applications. An Advanced mode gives access to detailed settings in order to fully customize averaging and precision, bandwidth, sweeper speed, and so forth. By default, the Sweeper plot area contains the absolute value of the impedance, as well as the Representation Parameters 1 and 2. In the Control sub-tab you can configure the sweep parameter, the sweep range and resolution. Set the start and stop frequency to 1 kHz and 1 MHz. By default, the sweeper is set to divide this span into 100 points on a logarithmic grid. Both of these settings are configurable, as well as the Sweep Parameter. The following table summarizes the Sweeper settings we use.

Table 4.1: Settings: configure the Sweeper for frequency dependence measurement

Tab	Sub-tab	Section	#	Label	Setting / Value / State
Sweeper	Settings			Filter	Application Mode
Sweeper	Settings			Application	Impedance
Sweeper	Control	Horizontal		Sweep Param	Osc 1 Frequency
Sweeper	Control	Horizontal		Start / Stop	1k / 1M
Sweeper	Control	Horizontal		Length	100 pts

Start the frequency sweep by clicking on **Single**. The screenshot in [Figure 4.4](#) shows the measurement in the plot section of the Sweeper tab. It shows a perfectly flat curve for the real part of the impedance $\text{Abs}(Z)$ within 0.05% of 1 k Ω as is indicated with the cursor for the vertical axis. Whereas our 1 k Ω DUT has no notable frequency dependence between 1 Hz and 1 MHz, very large or small resistors will typically show a change in $\text{Abs}(Z)$ towards high frequencies due to the spurious capacitance or inductance.

The plot section features multi-trace and dual-plot functionalities for representing multiple curves with different units, such as Ohm (Ω) and Farad (F), or with very different magnitudes. The plot functionality in LabOne is described in more detail in [Plot Functionality](#). There you also find a description of the cursor and math functions for quick quantitative analysis of measurement data.

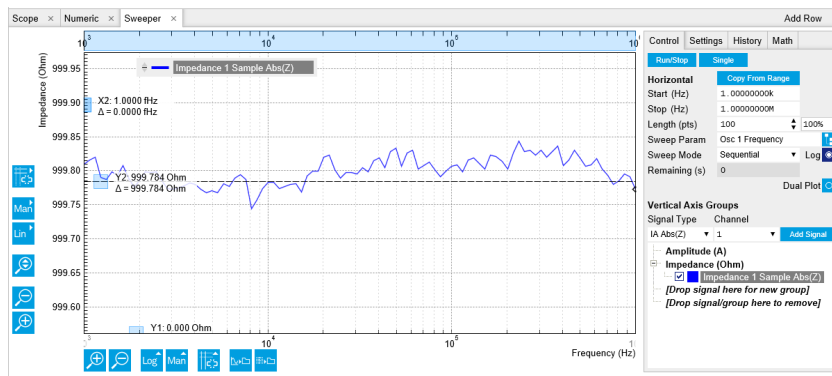


Figure 4.4: LabOne Sweeper showing the frequency dependence of a $1\text{ k}\Omega$ DUT

4.1.5. Circuit Representation and Measurement Mode

There are two settings that any user should be aware of before continuing with own measurements: the Equivalent Circuit Mode, and the DUT Representation. These settings determine the currently active measurement configuration. The configuration is graphically represented by a dynamically updated circuit diagram in the Impedance Analyzer tab.

1. The Measurement Mode (either 2 Terminal or 4 Terminal) corresponds to the way the DUT is wired to the instrument connectors. This determines where the DUT voltage V is measured: on L_{POT} and H_{POT} , or on H_{CUR} . The voltage measurement together with the measurement of the current I on L_{POT} constitutes a measurement of the impedance $Z=V/I$. For the 1 k Ω DUT, the measurement mode is 4 Terminal because all four connectors H_{CUR} , L_{POT} , H_{POT} and L_{POT} are wired to the DUT.
2. The DUT Representation determines how the measured complex-valued impedance $Z_{DUT}=V/I$ is converted into DUT circuit parameters such as resistance and capacitance. For our 1 k Ω resistor, a parallel resistor-capacitor representation $R_p \parallel C_p$ makes sense, even if the parallel capacitance is negligible.

More general information about the DUT model and the measurement mode is found in [Impedance and its Measurement](#). The DUT representation parameters (in our case Representation Parameter 1 = R_p , and Representation Parameter 2 = C_p) are available for display elsewhere in the user interface, e.g. in the Numeric or in the Sweeper tab. In the Numeric tab shown in [Figure 4.3](#), the Model Parameter 1 (R_p) reads 999.8 Ω . The panel of the Model Parameter 2 (C_p) shows no numerical reading but the message "Suppression" instead. This message of the LabOne Confidence Indicator informs you in situations when one of the Model Parameters cannot be determined reliably. In the present case, the current passing through the stray capacitance between the resistor electrodes is too small compared to the current through the resistor itself. Instead of displaying a calculated numeric value with no meaning, the "Suppression" warning prevents you from doing systematic measurement errors.

Apart from "Suppression", there are other Confidence Indicator warnings that can appear. The Confidence Indicator is for instance able to detect missing connections (e.g. when trying to measure a DUT in 4 Terminal mode when it is connected to 2 terminals only), or signal underflow and overflow conditions. The meanings of the different warnings are summarized in [List of warning messages provided by the Confidence Indicator](#). in the Functional Description chapter.

4.1.6. Time Dependence and Filtering

Next, we will have a look at the Plotter tab that allows users to observe measurement data as a function of time, e.g. to measure the long-term stability of a component impedance. Open the Plotter tab by clicking on the corresponding icon on the left, and choose the Impedance preset in the Control sub-tab in order to add a selection of useful quantities to the plot. It is possible to adjust the scaling of the graph in both directions, or make detailed measurements with 2 cursors for each direction. Try zooming in along the time dimension using the mouse wheel or the icons below the graph. When zooming in far enough, the mode in which the data are displayed will change from a min-max envelope plot to linear point interpolation depending on the density of points along the horizontal axis as compared to the number of pixels available on the screen. For a good overview, disable all traces except the resistance (Impedance 1 Representation Parameter 1) using the checkboxes in the Vertical Axis Group section.

The Plotter is a good tool to observe the noise in the measurement. Noise can be efficiently eliminated by low-pass filtering. The MFIA offers automatic bandwidth control which usually provides good noise reduction without unnecessarily slowing down the measurement. Nonetheless it can sometimes be beneficial to optimize the filtering manually, for instance in situations with strong electrical noise. For manual adjustment, disable the Bandwidth Control button in the Impedance Analyzer tab. This setting is only visible in the lower half of the IA tab after setting Mode to Advanced in the Measurement Control section, see [Figure 4.5](#). The Low-Pass Filters section now changes from a disabled (greyed-out) state to an editable (white) state. Try different filter bandwidths, e.g. 1 Hz and 100 Hz, to see the live effect on the data as seen in [Figure 4.6](#). More information about filter parameters is found in the [Signal Processing Basics](#).

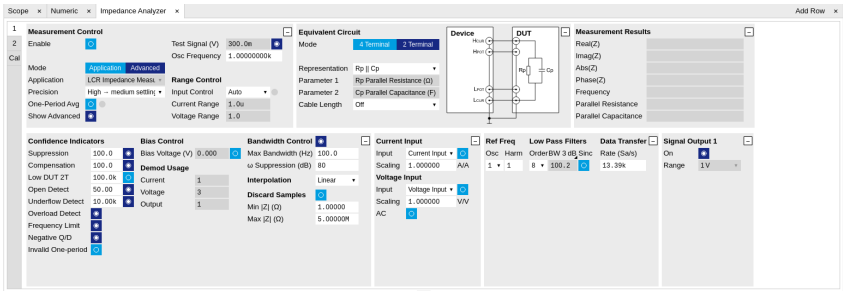


Figure 4.5: Impedance Analyzer tab (Advanced mode)

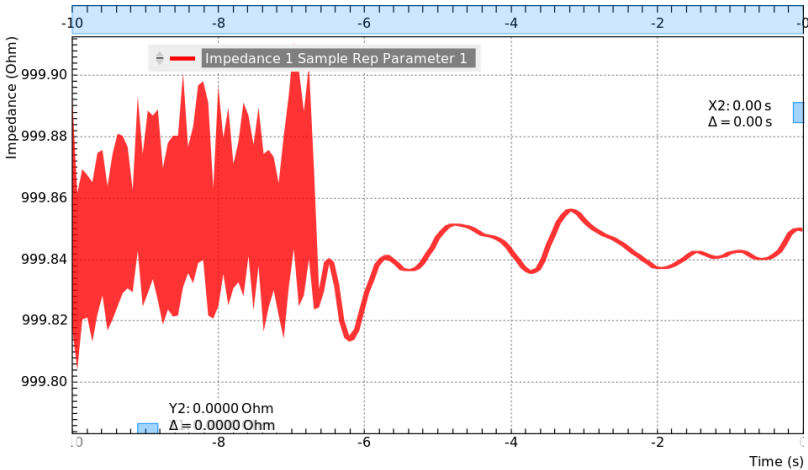


Figure 4.6: LabOne Plotter showing a resistance measurement over time. Changing the filter setting manually leads to a change in noise.

The following table summarizes the Plotter settings we use.

Table 4.2: Settings: configure the Sweeper for frequency dependence measurement

Tab	Sub-tab	Section	#	Label	Setting / Value / State
Plotter	Control	Select a Preset			Impedance

Tab	Sub-tab	Section	#	Label	Setting / Value / State
IA		Measurement Control		Mode	Advanced
IA		Bandwidth Control		Auto	OFF
IA		Low-Pass Filters		BW 3dB	1 Hz / 100 Hz

4.1.7. Measurement Accuracy

The measurement accuracy is one of the key specifications of an impedance analyzer. It relates the measured and the real DUT impedance, see [Impedance and its Measurement](#). The real impedance of the supplied 1 k Ω DUT is within 0.05% of an ideal 1 k Ω resistor in the frequency range of the MFIA, and is therefore suitable to judge the accuracy of the measurement result.

The measurement accuracy is not a universal number but depends on the frequency and on the absolute impedance $|Z_{\text{DUT}}|$. Figure 4.7 shows the accuracy across the full parameter space covered by the MFIA. This impedance accuracy chart is a handy tool not only to get the measurement accuracy, but also to quickly obtain the absolute impedance of a capacitance or inductance at a certain frequency. For 1 k Ω (vertical axis) and 1 kHz (horizontal axis), we land in a region with 0.05% accuracy.

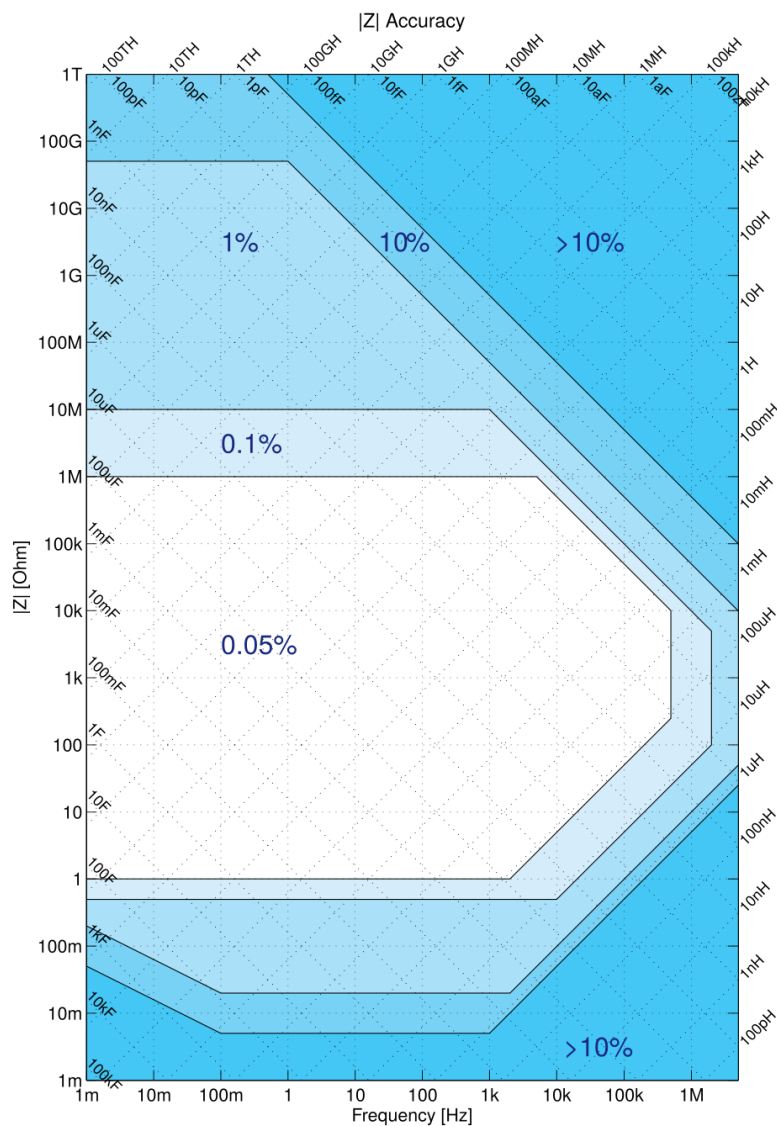


Figure 4.7: Impedance Accuracy chart

Note

The relative accuracy specification of the instrument applies to the measured total impedance. Neither does it apply to the real or imaginary parts of the impedance individually, nor to the circuit representation parameters. A 30 minute warm-up time and a 300 mV test signal are required to reach the maximum accuracy.

4.2. Advanced Impedance Measurements

Note

This tutorial is applicable to all MFIA Instruments and to MFLI Instruments with the MF-IA Impedance Analyzer option installed.

4.2.1. Goals and Requirements

The goal of this tutorial is to show some more advanced features of the LabOne user interface by guiding you through a measurement of a frequency-dependent impedance. We introduce important practical aspects about impedance measurements such as the influence of parasitics, or the effect of parameter suppression.

This tutorial is for people with some starting experience with Zurich Instruments impedance analyzers. To follow this tutorial, you need a 1 M Ω DUT fitting the MFITF Impedance Test Fixture. An easy way to get that is to solder a 1 M Ω through-hole resistor onto one of the DUT carriers delivered with the MFITF.

4.2.2. Preparation

Start by connecting the MFITF Test Fixture to the BNC connectors on the front panel of the MFIA, and plug the DUT into the 8-pin connector on the front of the test fixture. Figure 4.8 shows a diagram of the hardware setup.

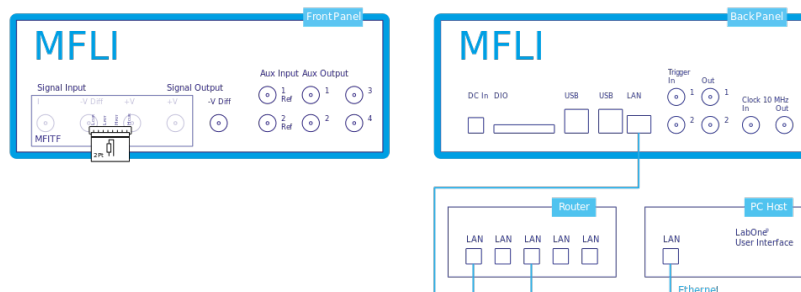


Figure 4.8: Setup for the Impedance Measurement Tutorial. The measured device is a resistor soldered onto one of the through-hole DUT carriers that come with the MFITF.

Make sure that the MFIA Instrument is powered on and then connect the MFIA directly by USB to your host computer or by Ethernet to your local area network (LAN) where the host computer resides. After connecting to your Instrument through the web browser using its address or through the MF Device Finder, the LabOne graphical user interface is opened. Check the [Getting Started](#) for detailed instructions. The tutorial can be started with the default instrument configuration (e.g. after a power cycle) and the default user interface settings (i.e. as is after pressing F5 in the browser).

4.2.3. Impedance and Representation Parameters

The real impedance of a DUT like the 1 M Ω resistor is not necessarily equal to its nominal, or ideal, value. First of all, the DUT impedance value is only specified within some accuracy, e.g. 1%. In addition to that imperfection, there are spurious impedance components such as lead inductance, or parallel capacitance. For the given 1 M Ω resistor, the real impedance is indeed close to that of a parallel R_p||C_p circuit, with a parallel capacitor C_p of some hundreds of fF. This real impedance, including device imperfections, is what the MFIA measures.

In order to measure the 1 MΩ DUT, we configure the Impedance Analyzer with a suitable Equivalent Circuit Representation of $R_p||C_p$. Since the DUT carrier we use is connected to 2 leads only, we change the measurement mode to 2 Terminal. Furthermore, we set the Range Control to Auto. This will set the input current and voltage ranges to suitable values.

Table 4.3 summarizes the instrument settings we use.

Table 4.3: Settings: configure and enable the impedance analyzer

Tab	Sub-tab	Section	#	Label	Setting / Value / State
IA		Equivalent Circuit		Mode	2 Terminal
IA		Equivalent Circuit		Representation	$R_p C_p$
IA		Measurement Control		Osc Frequency	1 k
IA		Measurement Control		Enable	ON
IA		Range Control		Input Control	Auto

4.2.4. Measure the Frequency Dependence

Open the Sweeper tab (Settings sub-tab), select the Application Mode, and Impedance as the Application. In the Control sub-tab you can configure the sweep parameter, the sweep range and resolution. We choose parameters for a logarithmic frequency sweep from 1 kHz to 1 MHz. We start with a measurement of the absolute impedance $|Z_{DUT}|$. Add this parameter to the plot by selecting "IA Abs(Z)" as a Signal Type, and click on "Add Signal". The following table summarizes the Sweeper settings we use.

Table 4.4: Settings: configure the Sweeper for frequency dependence measurement

Tab	Sub-tab	Section	#	Label	Setting / Value / State
Sweeper	Settings			Filter	Application Mode
Sweeper	Settings			Application	Impedance
Sweeper	Control	Horizontal		Sweep Param	Osc 1 Frequency
Sweeper	Control	Horizontal		Start / Stop	1k / 1M
Sweeper	Control	Horizontal		Length	100 pts

Start the frequency sweep by clicking on Single. Display the absolute impedance $|Z_{DUT}|$ by selecting "Impedance 1 Sample Abs(Z)" in the Vertical Axis Groups section. Naively we might expect this parameter to stay constant at 1 MΩ, but in fact it drops by about 50% at 500 kHz. The plot in Figure 4.9 shows the measurement of $|Z_{DUT}|$ as the red curve in the plot section of the Sweeper tab. The drop of the absolute impedance is due to the spurious parallel capacitance. At high enough frequencies, this capacitance forms a bypass for the current and therefore leads to a reduction of the impedance.

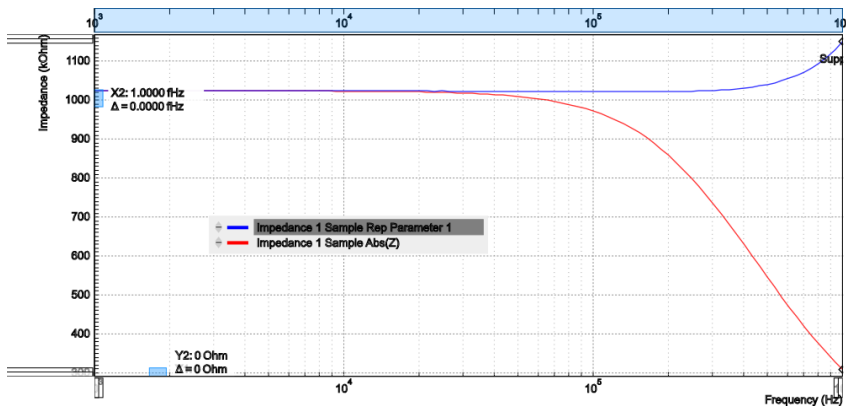


Figure 4.9: Absolute Impedance $|Z|$

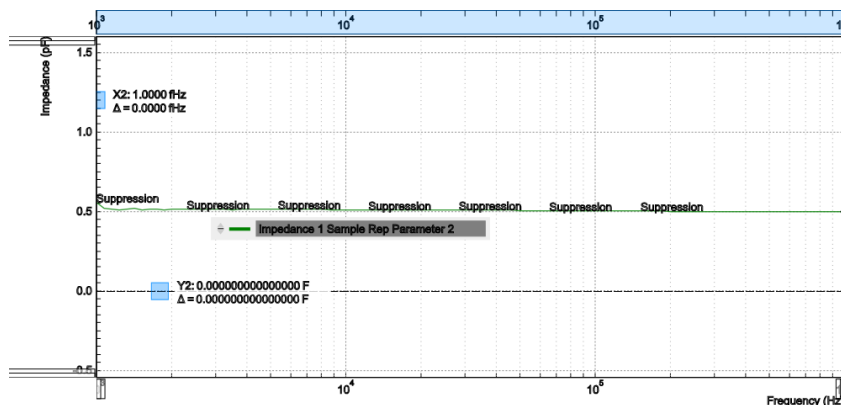


Figure 4.10: Replacement circuit capacitance of a 1 MΩ through-hole device measured with the LabOne Sweeper

The $R_p \parallel C_p$ circuit representation allows us to extract the parallel resistance R_p and capacitance C_p separately. Even after the Sweep has finished, we can add these signals to the plot by selecting IA Parameter 1 or IA Parameter 2 as Signal Type, and clicking on "Add Signal". Selecting the corresponding Impedance 1 Sample Rep Parameter 1 or 2 in the Vertical Axis Groups section will change the axis between a Farad (F) and an Ohm (Ω) scale.

Ideally, the resistance and capacitance are both frequency-independent. Looking at the blue curve in Figure 4.9 (R_p) and the green curve in Figure 4.10 (C_p), this is indeed mostly the case in the measurement: R_p stays flat at about 1 MΩ up to a few 100 kHz, and C_p stays at about 0.5 pF. However, parts of both curves are tagged by the Confidence Indicator warning "Suppression". It shows that the capacitance measurement is most reliable at high frequencies, and the resistance measurement is most reliable at low frequencies.

4.2.5. Measure a Nyquist Plot

Instead of a conventional frequency dependence with the frequency on the horizontal axis and an impedance parameter on the vertical axis, the Sweeper also features measurement of Nyquist plots. A Nyquist plot is a parametric plot of the impedance as a function of frequency with the real part of the impedance (resistance) on the horizontal axis, and the imaginary part (reactance) on the vertical axis. To obtain a Nyquist plot, please follow the instructions below.

In the Control sub-tab of the Sweeper, select the desired start frequency and stop frequency. Set XY Mode to On - Invert and select Real(Z) as the X signal from the selection tree. Select the imaginary part of Z with IA Imag(Z) as a Signal Type in the Vertical Axis Groups and add it to the plot. Start the sweep by clicking on **Single**. Scale the vertical axis by setting the axis scaling mode on the left side of the plot to Auto, see Plot Area Elements. The horizontal axis can be fixed to match the scale of the vertical axis by using the Track axis scaling mode instead of FS, Manual, or Auto. This ensures that a circle in the complex plane is displayed as a circle in the plot independent of the window aspect ratio.

The table below summarizes the settings necessary to obtain a Nyquist plot

Table 4.5: Settings: configure the Sweeper for frequency dependence measurement

Tab	Sub-tab	Section	#	Label	Setting / Value / State
Sweeper	Control			XY Mode	On - Invert
Sweeper	Control			X Signal	Impedances / Impedance 1 Sample / Real(Z)
Sweeper	Control	Vertical Axis Groups		Signal Type	IA Imag(Z)
Sweeper	Control	Vertical Axis Groups		Add Signal	Click
Sweeper		Plot Area (left)		Axis Scaling Mode	Auto
Sweeper		Plot Area (bottom)		Axis Scaling Mode	Track

4.3. Compensation

The 1 M Ω resistor connected as a DUT in the setup [Figure 4.8](#) for this tutorial has relatively unspectacular, linear Nyquist plot. To illustrate the characteristics of a Nyquist plot better, we show the measurement of an R-R-C circuit which has the shape of a semicircle. The measurement shown in [Figure 4.11](#) is easy to reproduce by soldering the sample from through-hole components as shown in [Figure 4.12](#) onto a through-hole DUT carrier. The components used here were $R_p = 6.8$ k Ω , $R_s = 2.2$ k Ω , and $C = 1$ μ F.

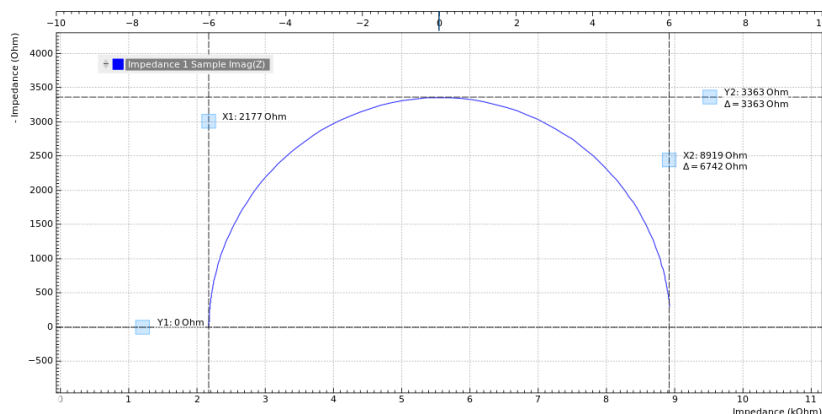


Figure 4.11: Nyquist plot of an R-R-C sample

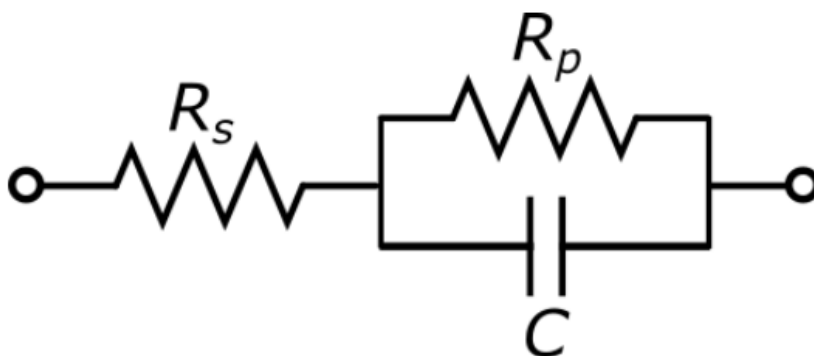


Figure 4.12: R-R-C sample circuit diagram

4.2.6. Saving and Exporting Data

Data acquired with the Sweeper can be easily saved to comma-separated value (CSV), MATLAB, and ZView format. Particularly the ZView format is attractive for further equivalent circuit modeling with specialized software tools. Such an analysis often enables precise characterization of the DUT circuit, going much beyond the extraction of a pair of representation parameters.

All past sweeps are listed in the History sub-tab of the Sweeper, up to a maximum number of list entry specified by the History Length parameter. Clicking on [Save](#) will store all the sweeps that are currently in the History list. The data are stored in a subfolder of the LabOne Data folder which can be conveniently accessed in the [File Manager Tab](#). The Enable button next to each History entry controls the plot display, but not which sweeps are saved. Find more information, for instance on how to use the Reference feature, in [Sweeper Tab](#). Select your preferred file format from the drop-down list at the bottom of the History sub-tab of the Sweeper. This setting is global and also applies to the Plotter and the DAQ tabs.

Alternatively, the buttons [Save](#) and [Save](#) in the plot area of the Sweeper tab allow you to save the selected graphical traces as a text file or a vector graphics.

4.3. Compensation

Note

This tutorial is applicable to all MFIA Instruments and to MFLI Instruments with the MF-IA Impedance Analyzer option installed.

4.3.1. Goals and Requirements

The goal of this tutorial is to demonstrate a user compensation for impedance measurements with the MFIA instrument. A compensation improves the measurement accuracy when using a custom impedance fixture.

Some starting experience with Zurich Instruments impedance analyzers is required to follow the tutorial. You also need a Kelvin probe cable set and a $1\ \Omega$ through-hole resistor, a $10\ \Omega$ resistor, and a simple piece of wire. Instead of the Kelvin probe set, you can use a set of BNC cables and crocodile clamps.

4.3.2. Preparation

Start by connecting the Kelvin probe set to the BNC connectors on the front panel of the MFIA, paying attention not to mix up the connectors. One clamp should be electrically connected to Signal Output +V and Signal Input +V, the other clamp to Signal Input I and Signal Input -V. Fix the $1\ \Omega$ resistor with the two clamps. [Figure 4.13](#) shows a diagram of the hardware setup.

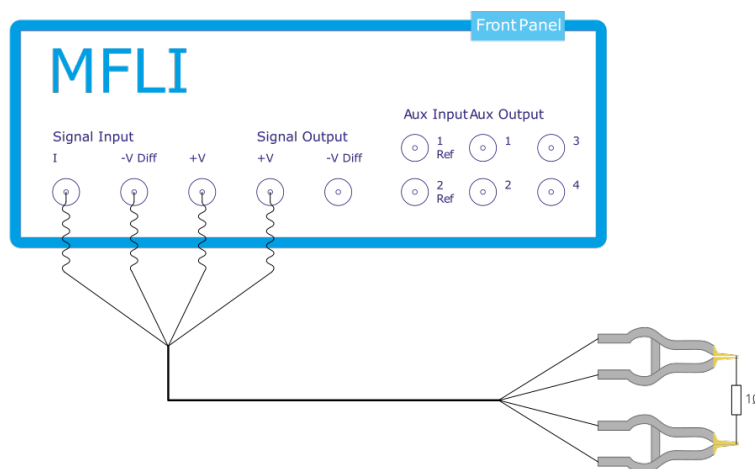


Figure 4.13: Setup for the Impedance Compensation Tutorial

Make sure that the MFIA Instrument is powered on and then connect the MFIA directly by USB to your host computer or by Ethernet to your local area network (LAN) where the host computer resides. After connecting to your Instrument through the web browser using its address, the LabOne graphical user interface is opened. Check the [Getting Started](#) for detailed instructions. The tutorial can be started with the default instrument configuration (e.g. after a power cycle) and the default user interface settings (i.e. as is after pressing F5 in the browser).

4.3.3. Configure the Impedance Analyzer

A measurement fixture like the Kelvin probe set affects the impedance measurement through its electrical properties, such as its stray capacitance, or the delay caused by its cables. A carefully executed compensation can practically eliminate the negative influence and thus improve measurement accuracy.

To get an idea of the Kelvin probe's influence, let's first make a quick measurement without compensation. We'll select the R_s+L_s Representation in the Impedance Analyzer tab, which is a good choice for small-valued resistors. The two representation parameters can be read in the Numeric tab with activated Impedance preset. We furthermore relax the threshold of the Suppression warning by setting Suppression Ratio to 100 in the Impedance Analyzer tab. This setting becomes visible in the lower half of the tab after setting the Measurement Control Mode to Advanced. The relaxed condition is deemed sufficient for the level of accuracy that we can achieve with such a cable-based setup.

Let's start by reading some measurement values at different oscillator frequencies, which we change manually in the Impedance Analyzer tab.

Table 4.6: Measurement: $1\ \Omega$ through-hole resistor without compensation

Frequency	Rep. parameter 1 (resistance Rs)	Rep. parameter 2 (inductance Ls)
1 kHz	1.001 Ω	Suppression
500 kHz	0.986 Ω	110 nH
3 MHz	0.738 Ω	112 nH
5 MHz (MF-5M option)	Suppression	111 nH

The resistance (Rep. parameter 1) at 1 kHz looks quite reasonable. The inductance (Rep. parameter 2) is not reliably measurable at this frequency and is flagged with the "Suppression" warning. At higher frequencies such as 500 kHz, the inductance is measurable, but it seems unexpectedly large: 110 nH would correspond to 10 to 15 cm of wire, but the resistor leads are much shorter. The spurious series inductance Ls of a through-hole component is typically of the order of 5 nH. At 3 MHz the measured resistance drops to 0.738 Ω which does not seem realistic. At the maximum frequency 5 MHz, the resistance measurement finally fails. The following table summarizes the settings we used.

Table 4.7: Settings: quick measurement with uncompensated Kelvin Probe set

Tab	Sub-tab	Section	#	Label	Setting / Value / State
IA		Equivalent Circuit		Representation	Rs + Ls
IA		Measurement Control		Enable	ON
IA		Measurement Control		Mode	Advanced
IA		Confidence Indicators		Suppression	100
IA		Measurement Control		Osc Frequency	1k, 500k, 3M, 5M

4.3.4. Carry out the Compensation

Some of the apparent problems seen before can be attributed to the uncompensated Kelvin probe set. In order to do a compensation, open the Cal side-tab in the Impedance Analyzer tab and select the SL (short-load) Compensation Sequence. It's necessary to specify the parameters of the reference "load" DUT, such as 10 Ω , and since we don't know precisely its stray capacitance, we'll enter a value of 0 F as Load C. The compensation steps can now be simply carried out by fixing the appropriate DUT (the wire for the S step, and the 10 Ω resistor for the L step) and clicking on the "Short" or "Load" button followed by "Compensate". For a precise calibration, try to keep the position of the clamps the same in both steps and in the measurement itself. The spurious capacitance is very sensitive to the position of the clamps and the parts of the cables which are unshielded.

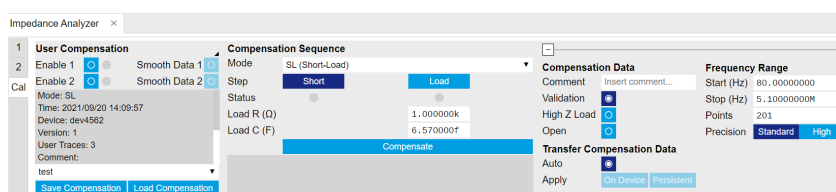


Figure 4.14: Impedance Analyzer tab: Cal side-tab

Each compensation step takes about a minute with the standard settings, and its progress and termination is reported in the text field below the "Compensate" button. The Status LEDs show which of the steps are completed. Once all three are done, the compensation data are transferred to the instrument. Compensation files can be saved and reloaded as necessary.

Table 4.8: Settings: perform a user compensation

Tab	Sub-tab	Section	#	Label	Setting / Value / State
IA	Cal	Compensation Sequence		Mode	SOL (short-load)
IA	Cal	Compensation Sequence		Load R (Ω)	10
IA	Cal	Compensation Sequence		Load C (F)	0
IA	Cal	Compensation Sequence		Step	Short / Load
IA	Cal	Compensation Sequence		Compensate	ON

The left section of the Cal side-tab displays information and contains the enable button of the current user compensation and the internal calibration. The label User Compensation (or Internal Calibration) acts as a control to switch the display (the small black triangle to the bottom right of the label indicates the control). When switched to User Compensation, the compensation can easily be activated and deactivated with the Enable button. This has an immediate effect on the measurement data displayed in the Numeric, the Plotter, or the Sweeper tab.

Note

The internal calibration should normally be enabled at all times.

4.3.5. Measure with Compensation

In order to compare the results with and without compensation, we'll perform a sweep up to the maximum available frequency as shown in the previous tutorials. In the Sweeper tab, perform a linear frequency sweep from 1 kHz to 5 MHz. Repeat that once with and once without enabled compensation.

We look at two selected quantities in the following plots in [Figure 4.15](#). The Resistance (= Rep. Parameter 1) remains flat at 1.0 Ω with compensation (green), whereas it goes down below 0.4 Ω without compensation.

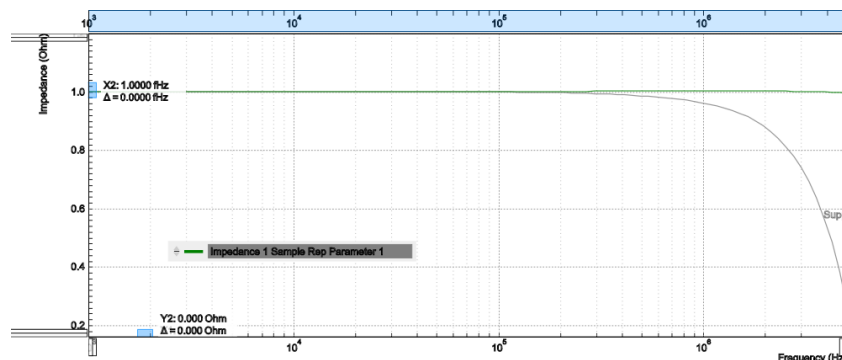


Figure 4.15: LabOne Sweeper showing the resistance of a 1 Ω DUT measured with compensation (green) and without (grey)

The Inductance (= Parameter 2) is relatively flat in frequency both with and without compensation, see [Figure 4.16](#). But its value is much too large without compensation. With compensation the measured value is less than 10 nH which is the expected order of magnitude, in any case it is too small to be measured with high precision and is flagged with the "Suppression" warning. Note that the lead inductance of the resistor is about 8 nH/cm. Clamping the DUT at a slightly different position and measuring again will show an effect on the measured inductance L_s , whereas the resistance R_s is largely insensitive to that.

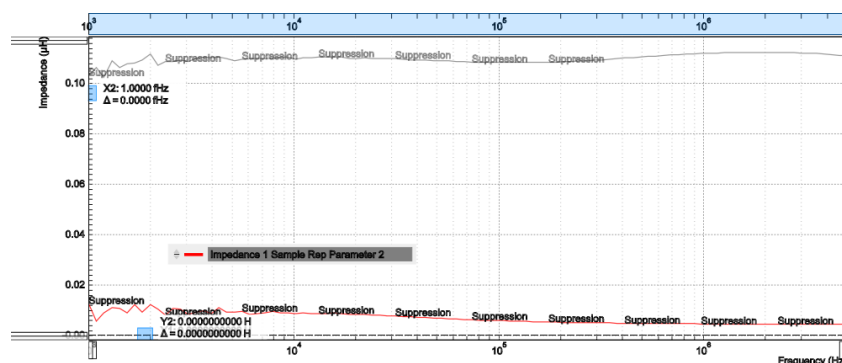


Figure 4.16: LabOne Sweeper showing the serial inductance of a 1 Ω DUT measured with compensation (red) and without (grey)

5. Functional Description LabOne User Interface

This chapter gives a detailed description of the functionality available in the LabOne User Interface (UI) for the Zurich Instruments MFIA. LabOne provides a data server and a web server to control the Instrument with any of the most common web browsers (e.g. Firefox, Chrome, Edge, etc.). This platform-independent architecture supports interaction with the Instrument using various devices (PCs, tablets, smartphones, etc.) even at the same time if needed.

On top of standard functionality like acquiring and saving data points, this UI provides a wide variety of measurement tools for time and frequency domain analysis of measurement data as well as for convenient servo loop implementation.

5.1. User Interface Overview

5.2. UI Nomenclature

This section provides an overview of the LabOne User Interface, its main elements and naming conventions. The LabOne User Interface is a browser-based UI provided as the primary interface to the MFIA instrument. Multiple browser sessions can access the instrument simultaneously and the user can have displays on multiple computer screens. Parallel to the UI, the instrument can be controlled and read out by custom programs written in any of the supported languages (e.g. LabVIEW, MATLAB, Python, C) connecting through the LabOne APIs.

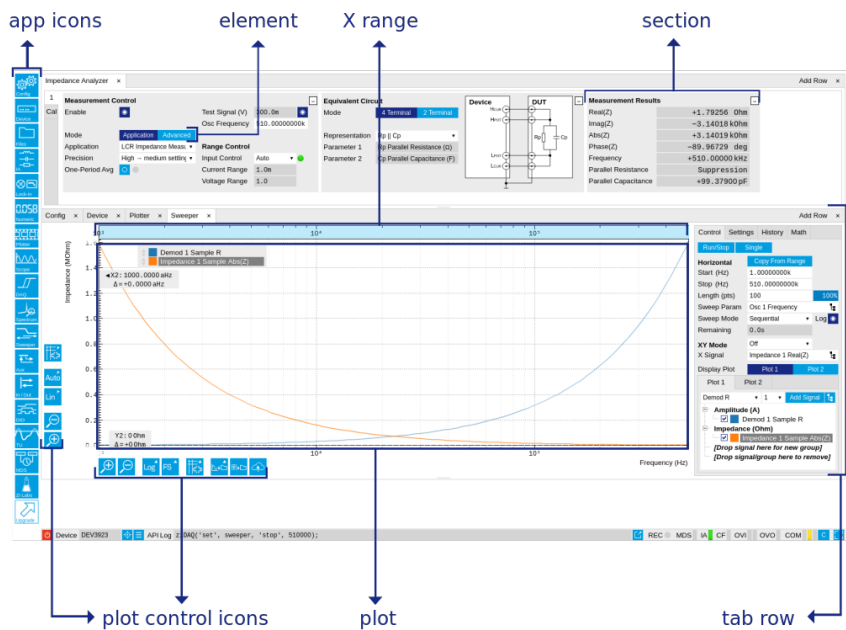


Figure 5.1: LabOne User Interface (default view)

The LabOne User Interface automatically opens some tabs by default after a new UI session has been started. At start-up, the UI is divided into two tab rows, each containing a tab structure that gives access to the different LabOne tools. Depending on display size and application, tab rows can be freely added and deleted with the control elements on the right-hand side of each tab bar. Similarly, the individual tabs can be deleted or added by selecting app icons from the side bar on the left. A click on an icon adds the corresponding tab to the display, alternatively the icon can be dragged and dropped into one of the tab rows. Moreover, tabs can be moved by drag-and-drop within a row or across rows.

Table 5.1 gives a brief descriptions and naming conventions for the most important UI items.

Table 5.1: LabOne User Interface features

Item name	Position	Description	Contains
side bar	left-hand side of the UI	contains app icons for each of the available tabs - a click on an icon adds or activates the corresponding tab in the active tab row	app icons
status bar	bottom of the UI	contains important status and warning indicators, device and session information, and access to the command log	status indicators
main area	center of the UI	accommodates all active tabs – new rows can be added and removed by using the control elements in the top right corner of each tab row	tab rows, each consisting of tab bar and the active tab area
tab area	inside of each tab	provides the active part of each tab consisting of settings, controls and measurement tools	sections, plots, sub-tabs, unit selections

Further items are highlighted in [Figure 5.2](#).

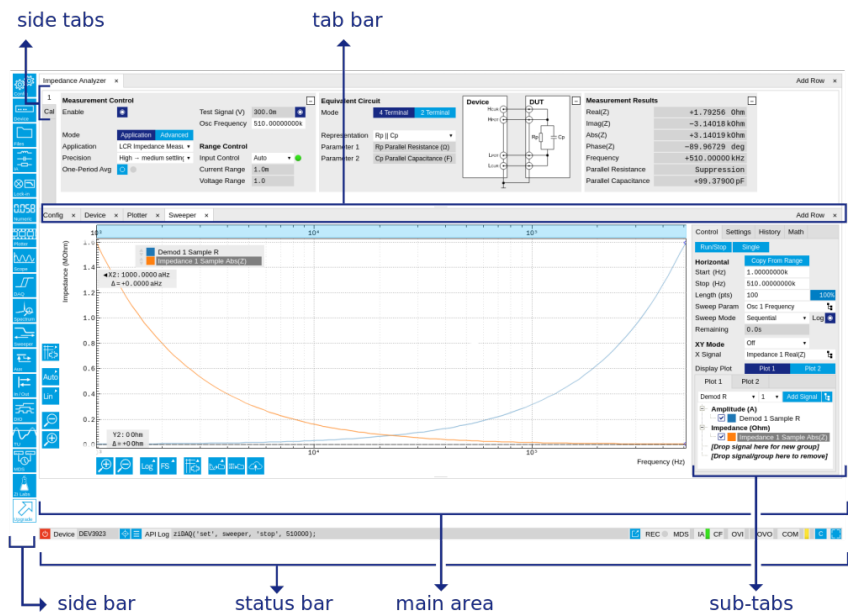


Figure 5.2: LabOne User Interface (more items)

5.2.1. Unique Set of Analysis Tools

All instruments feature a comprehensive tool set for time and frequency domain analysis for both raw and demodulated signals.

The app icons on the left side of the UI can be roughly divided into two categories: settings and tools.

Settings-related tabs are in direct connection to the instrument hardware, allowing the user to control all the settings and instrument states.

Tools-related tabs place a focus on the display and analysis of gathered measurement data.

There is no strict distinction between settings and tools, e.g. the Sweeper will change certain demodulator settings while performing a frequency sweep. Within the tools one can often further discriminate between time domain and frequency domain analysis. Moreover, a distinction can be made between the analysis of fast input signals - typical sampling rate of 60 MSa/s - and the measurement of orders of magnitude slower data - typical sampling rate of <200 kSa/s - derived for instance from demodulator outputs and auxiliary inputs. [Table 5.2](#) provides a brief classification of the tools.

Table 5.2: Tools for time domain and frequency domain analysis

	Time Domain	Frequency Domain
Fast signals (60 MSa/s)	Oscilloscope (Scope tab)	FFT Analyzer (Scope tab)
Slow signals (<200 kSa/s)	Numeric	Spectrum Analyzer (Spectrum tab)
	Plotter	Sweeper
	Data Acquisition	-

The following table gives the overview of all app icons. Note that the selection of app icons may depend on the upgrade options installed on a given instrument.

Table 5.3: Overview of app icons and short description

Control/Tool	Option/Range	Description
IA		Quick overview and access to all the settings and properties for impedance measurements.
Files		Access files on internal flash memory and USB drive.
Numeric		Access to all continuously streamed measurement data as numerical values.
Plotter		Displays various continuously streamed measurement data as traces over time (roll mode).
Scope		Displays shots of data samples in time and frequency domain (FFT) representation.
DAQ		Provides complex trigger functionality on all continuously streamed data samples and time domain display.
Spectrum		Provides FFT functionality to all continuously streamed measurement data.
Sweeper		Sweep frequencies, voltages, and other quantities over a defined range and display various response functions including statistical operations.
Lock-in		Quick overview and access to all the settings and properties for signal generation and demodulation.
Lock-in MD		Quick overview and access to all the settings and properties for signal generation and demodulation.
Aux		Controls all settings regarding the auxiliary inputs and auxiliary outputs.
In/Out		Gives access to all controls relevant for the Signal Inputs and Signal Outputs of each channel.
DIO		Gives access to all controls relevant for the digital inputs and outputs including DIO, Trigger Inputs, Trigger Outputs, and Marker Outputs.
Config		Provides access to software configuration.
Device		Provides instrument specific settings.
PID		Features all control, analysis, and simulation capabilities of the PID controllers.
MOD		Control panel to enable (de)modulation at linear combinations of oscillator frequencies.
MDS		Synchronize multiple instruments.
ZI Labs		Experimental settings and controls.

Table 5.4 provides a quick overview over the different status bar elements along with a short description.

Table 5.4: Status bar description

Control/Tool	Option/Range	Description
Command log	last command	Shows the last command. A different formatting (MATLAB, Python, ..) can be set in the config tab. The log is also saved in [User] \Documents\Zurich Instruments\LabOne\WebServer\Log
Show Log		Show the command log history in a separate browser window.
Errors	Errors	Display system errors in separate browser tab.
Device	devXXX	Indicates the device serial number.
Identify Device		When active, device LED blinks
Shutdown		Shuts down the instrument.
MDS	grey/green/red/yellow	Multiple device synchronization indicator. Grey: Nothing to synchronize - single device on the UI. Green: All devices on the UI are correctly synchronized. Yellow: MDS sync in progress or only a subset of the connected devices is synchronized. Red: Devices not synchronized or error during MDS sync.
REC	grey/red	A blinking red indicator shows ongoing data recording (related to global recording settings in the Config tab).
IA	grey/green	Impedance Analyzer - Green: indicates which of the impedance analyzer modules is enabled.
CF	grey/yellow/red	Clock Failure - Red: present malfunction of the external 10 MHz reference oscillator. Yellow: indicates a malfunction occurred in the past.
OVI	grey/yellow/red	Signal Input Overload - Red: present overload condition on the signal input also shown by the red front panel LED. Yellow: indicates an overload occurred in the past.
OVO	grey/yellow/red	Overload Signal Output - Red: present overload condition on the signal output. Yellow: indicates an overload occurred in the past.
COM	grey/yellow/red	Packet Loss - Red: present loss of data between the device and the host PC. Yellow: indicates a loss occurred in the past.
COM	grey/yellow/red	Sample Loss - Red: present loss of sample data between the device and the host PC. Yellow: indicates a loss occurred in the past.
COM	grey/red	Stall - Red: indicates that the sample transfer rates have been reset to default values to prevent severe communication failure. This is typically caused by high sample transfer rates on a slow host computer.
C		Reset status flags: Clear the current state of the status flags
MOD	grey/green	MOD - Green: indicates which of the modulation kits is enabled.
PID	grey/green	PID - Green: indicates which of the PID units is enabled. Red: indicates PID unit is in PLL or ExtRef mode but is not locked. Yellow: indicates PID unit was not locked in the past.
PLL	grey/green	PLL - Green: indicates which of the PLLs is enabled.
Full Screen		Toggles the browser between full screen and normal mode.

5.2.2. Plot Functionality

Several tools provide a graphical display of measurement data in the form of plots. These are multi-functional tools with zooming, panning and cursor capability. This section introduces some of the highlights.

Plot Area Elements

Plots consist of the plot area, the X range and the range controls. The X range (above the plot area) indicates which section of the wave is displayed by means of the blue zoom region indicators. The two ranges show the full scale of the plot which does not change when the plot area displays a zoomed view. The two axes of the plot area instead do change when zoom is applied.

The [mouse functionality](#) inside of a plot greatly simplifies and speeds up data viewing and navigation.

Table 5.5: Mouse functionality inside plots

Name	Action	Description	Performed inside
Panning	left click on any location and move around	moves the waveforms	plot area
Zoom X axis	mouse wheel	zooms in and out the X axis	plot area
Zoom Y axis	shift + mouse wheel	zooms in and out the Y axis	plot area
Window zoom	shift and left mouse area select	selects the area of the waveform to be zoomed in	plot area
Absolute jump of zoom area	left mouse click	moves the blue zoom range indicators	X and Y range, but outside of the blue zoom range indicators
Absolute move of zoom area	left mouse drag-and-drop	moves the blue zoom range indicators	X and Y range, inside of the blue range indicators
Full Scale	double click	set X and Y axis to full scale	plot area

Each plot area contains a legend that lists all the shown signals in the respective color. The legend can be moved to any desired position by means of drag-and-drop.

The X range and Y range plot controls are described in [Table 5.6](#).

Note

Plot data can be conveniently exported to other applications such as Excel or Matlab by using LabOne's Net Link functionality, see [LabOne Net Link](#) for more information.

Table 5.6: Plot control description

Control/Tool	Option/Range	Description
Axis scaling mode		Selects between automatic, full scale and manual axis scaling.
Axis mapping mode		Select between linear, logarithmic and decibel axis mapping.
Axis zoom in		Zooms the respective axis in by a factor of 2.
Axis zoom out		Zooms the respective axis out by a factor of 2.
Rescale axis to data		Rescale the foreground Y axis in the selected zoom area.
Save figure		Generates PNG, JPG or SVG of the plot area or areas for dual plots to the local download folder.
Save data		Generates a CSV file consisting of the displayed wave or histogram data (when histogram math operation is enabled). Select full scale to save the complete wave. The save data function only saves one shot at a time (the last displayed wave).
Cursor control		Cursors can be switch On/Off and set to be moved both independently or one bound to the other one.
Net Link		Provides a LabOne Net Link to use displayed wave data in tools like Excel, MATLAB, etc.

Cursors and Math

The plot area provides two X and two Y cursors which appear as dashed lines inside of the plot area. The four cursors are selected and moved by means of the blue handles individually by means of drag-and-drop. For each axis, there is a primary cursor indicating its absolute position and a secondary cursor indicating both absolute and relative position to the primary cursor.

Cursors have an absolute position which does not change upon pan or zoom events. In case a cursor position moves out of the plot area, the corresponding handle is displayed at the edge of the plot area. Unless the handle is moved, the cursor keeps the current position. This functionality is very effective to measure large deltas with high precision (as the absolute position of the other cursors does not move).

The cursor data can also be used to define the input data for the mathematical operations performed on plotted data. This functionality is available in the Math sub-tab of each tool. The [Table 5.7](#) gives an overview of all the elements and their functionality. The chosen Signals and Operations are applied to the currently active trace only.

Note

Cursor data can be conveniently exported to other applications such as Excel or MATLAB by using LabOne's Net Link functionality, see [LabOne Net Link](#) for more information.

Table 5.7: Plot math description

Control/Tool	Option/Range	Description
Source Select		Select from a list of input sources for math operations.
	Cursor Loc	Cursor coordinates as input data.
	Cursor Area	Consider all data of the active trace inside the rectangle defined by the cursor positions as input for statistical functions (Min, Max, Avg, Std).
	Tracking	Display the value of the active trace at the position of the horizontal axis cursor X1 or X2.
	Plot Area	Consider all data of the active trace currently displayed in the plot as input for statistical functions (Min, Max, Avg, Std).
	Peak	Find positions and levels of up to 5 highest peaks in the data.
	Trough	Find positions and levels of up to 5 lowest troughs in the data.
	Histogram	Display a histogram of the active trace data within the x-axis range. The histogram is used as input to statistical functions (Avg, Std). Because of binning, the statistical functions typically yield different results than those under the selection Plot Area.
	Resonance	Display a curve fitted to a resonance.
	Linear Fit	Display a linear regression curve.
Operation Select		Select from a list of mathematical operations to be performed on the selected source. Choice offered depends on the selected source.
	Cursor Loc: X1, X2, X2-X1, Y1, Y2, Y2-Y1, Y2 / Y1	Cursors positions, their difference and ratio.
	Cursor Area: Min, Max, Avg, Std	Minimum, maximum value, average, and bias-corrected sample standard deviation for all samples between cursor X1 and X2. All values are shown in the plot as well.
	Tracking: Y(X1), Y(X2), ratioY, deltaY	Trace value at cursor positions X1 and X2, the ratio between these two Y values and their difference.
	Plot Area: Min, Max, Pk Pk, Avg, Std	Minimum, maximum value, difference between min and max, average, and bias-corrected sample standard deviation for all samples in the x axis range.

Control/Tool	Option/Range	Description
	Peak: Pos, Level	Position and level of the peak, starting with the highest one. The values are also shown in the plot to identify the peak.
	Histogram: Avg, Std, Bin Size, (Plotter tab only: SNR, Norm Fit, Rice Fit)	A histogram is generated from all samples within the x-axis range. The bin size is given by the resolution of the screen: 1 pixel = 1 bin. From this histogram, the average and bias-corrected sample standard deviation is calculated, essentially assuming all data points in a bin lie in the center of their respective bin. When used in the plotter tab with demodulator or boxcar signals, there additionally are the options of SNR estimation and fitting statistical distributions to the histogram (normal and rice distribution).
	Resonance: Q, BW, Center, Amp, Phase, Fit Error	A curve is fitted to a resonator. The fit boundaries are determined by the two cursors X1 and X2. Depending on the type of trace (Demod R or Demod Phase) either a Lorentzian or an inverse tangent function is fitted to the trace. The Q is the quality factor of the fitted curve. BW is the 3dB bandwidth (FWHM) of the fitted curve. Center is the center frequency. Amp gives the amplitude (Demod R only), whereas Phase returns the phase at the center frequency of the resonance (demod Phase only). The fit error is given by the normalized root-mean-square deviation. It is normalized by the range of the measured data.
	Linear Fit: Intercept, Slope, R ²	A simple linear least squares regression is performed using a QR decomposition routine. The fit boundaries are determined by the two cursors X1 and X2. The parameter outputs are the Y-axis intercept, slope and the R ² -value, which is the coefficient of determination to determine the goodness-of-fit.
Add	Add	Add the selected math function to the result table below.
Add All	Add All	Add all operations for the selected signal to the result table below.
Clear Selected	Clear	Clear selected lines from the result table above.
Clear All	Clear All	Clear all lines from the result table above.
Copy	Copy	Copy selected row(s) to Clipboard as CSV
Unit Prefix		Adds a suitable prefix to the SI units to allow for better readability and increase of significant digits displayed.
CSV	CSV	Values of the current result table are saved as a text file into the download folder.
Net Link	Link	Provides a LabOne Net Link to use the data in tools like Excel, MATLAB, etc.
Help	Help	Opens the LabOne User Interface help.

Note

The standard deviation is calculated using the formula $\sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$ for the unbiased estimator of the sample standard deviation with a total of N samples x_i and an arithmetic average $\bar{x}$. The formula above is used as-is to calculate the standard deviation for the Histogram Plot Math tool. For large number of points (Cursor Area and Plot Area tools), the more accurate pairwise algorithm is used (Chan et al., "Algorithms for Computing the Sample Variance: Analysis and Recommendations", The American Statistician 37 (1983), 242-247).

The fitting functions used in the Resonance Plot Math tool depend on the selected signal source. The demodulator R signal is fitted with the following function:

$$R(f) = C + A \frac{f}{\sqrt{f^2 + \left(\frac{Q}{f_0}\right)^2 (f^2 - f_0^2)^2}} \quad (1)$$

where **C** accounts for a possible offset in the output, **A** is the amplitude, **Q** is the quality factor and **f₀** is the center frequency. The demodulator ϕ signal is fitted with the following function:

$$\phi(f) = \tan^{-1} \left(Q \frac{1 - \left(\frac{f}{f_0}\right)^2}{\frac{f}{f_0}} \right) \quad (2)$$

using the same parameters as above.

Tree Selector

The Tree selector allows one to access streamed measurement data in a hierarchical structure by checking the boxes of the signals that should be displayed. The tree selector also supports data selection from multiple instruments, where available. Depending on the tool, the Tree selector is either displayed in a separate Tree sub-tab, or it is accessible by a click on the  button.

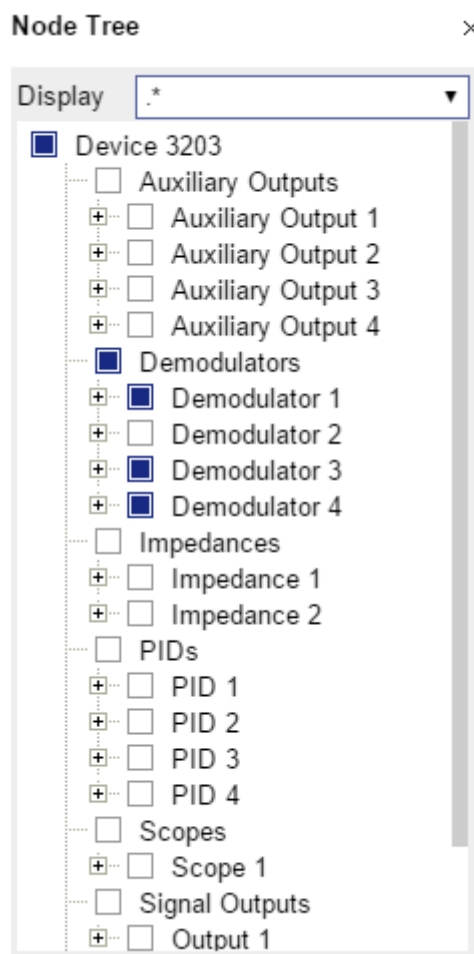


Figure 5.3: Tree selector with Display drop-down menu

Vertical Axis Groups

Vertical Axis groups are available as part of the plot functionality in many of the LabOne tools. Their purpose is to handle signals with different axis properties within the same plot. Signals with different units naturally have independent vertical scales even if they are displayed in the same plot. However, signals with the same unit should preferably share one scaling to enable quantitative comparison. To this end, the signals are assigned to specific axis group. Each axis group has its own axis system. This default behavior can be changed by moving one or more signals into a new group.

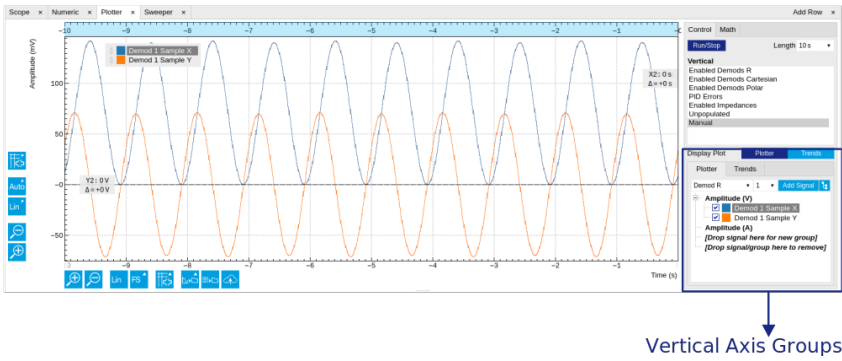


Figure 5.4: Vertical Axis Group in Plotter tool

The tick labels of only one axis group can be shown at once. This is the foreground axis group. To define the foreground group click on one of the group names in the Vertical Axis Groups box. The current foreground group gets a high contrast color.

Select foreground group

Click on a signal name or group name inside the Vertical Axis Groups. If a group is empty the selection is not performed.

Split the default vertical axis group

Use drag-and-drop to move one signal on the field **[Drop signal here to add a new group]**. This signal will now have its own axis system.

Change vertical axis group of a signal

Use drag-and-drop to move a signal from one group into another group that has the same unit.

Group separation

In case a group hosts multiple signals and the unit of some of these signals changes, the group will be split in several groups according to the different new units.

Remove a signal from the group

In order to remove a signal from a group drag-and-drop the signal to a place outside of the Vertical Axis Groups box.

Remove a vertical axis group

A group is removed as soon as the last signal of a custom group is removed. Default groups will remain active until they are explicitly removed by drag-and-drop. If a new signal is added that match the group properties it will be added again to this default group. This ensures that settings of default groups are not lost, unless explicitly removed.

Rename a vertical axis group

New groups get a default name "Group of ...". This name can be changed by double-clicking on the group name.

Hide/show a signal

Uncheck/check the check box of the signal. This is faster than fetching a signal from a tree again.

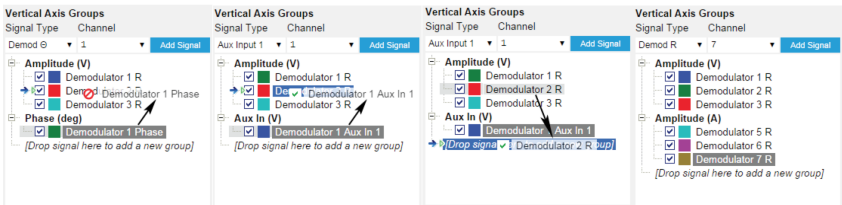


Figure 5.5: Vertical Axis Group typical drag and drop moves.

Table 5.8: Vertical Axis Groups description

Control/Tool	Option/Range	Description
Vertical Axis Group		Manages signal groups sharing a common vertical axis. Show or hide signals by changing the check box state. Split a group by dropping signals to the field [Drop signal here to add new group]. Remove signals by dragging them on a free area. Rename group names by editing the group label. Axis tick labels of the selected group are shown in the plot. Cursor elements of the active wave (selected) are added in the cursor math tab.
Signal Type	HW Trigger	Select signal types for the Vertical Axis Group.
	Demod X, Y, R, Theta	
	Frequency	
	Aux Input 1, 2	
Channel	integer value	Selects a channel to be added.
Signal	integer value	Selects signal to be added.
Add Signal	Add Signal	Adds a signal to the plot. The signal will be added to its default group. It may be moved by drag and drop to its own group. All signals within a group share a common y-axis. Select a group to bring its axis to the foreground and display its labels.
Window Length	2 s to 12 h	Window memory depth. Values larger than 10 s may cause excessive memory consumption for signals with high sampling rates. Auto scale or pan causes a refresh of the display for which only data within the defined window length are considered.

Trends

The Trends tool lets the user monitor the temporal evolution of signal features such as minimum and maximum values, or mean and standard deviation. This feature is available for the Scope, Spectrum, Plotter, and DAQ tab. Using the Trends feature, one can monitor all the parameters obtained in the [Math sub-tab](#) of the corresponding tab.

The Trends tool allows the user to analyze recorded data on a different and adjustable time scale much longer than the fast acquisition of measured signals. It saves time by avoiding post-processing of recorded signals and it facilitates fine-tuning of experimental parameters as it extracts and shows the measurement outcome in real time.

To activate the Trends plot, enable the Trends button in the Control sub-tab of the corresponding main tab. Various signal features can be added to the plot from the Trends sub-tab in the [Vertical Axis Groups](#). The vertical axis group of Trends has its own Run/Stop button and Length setting independent from the main plot of the tab. Since the Math quantities are derived from the raw signals in the main plot, the Trends plot is only shown together with the main plot. The Trends feature is only available in the LabOne user interface and not at the API level.

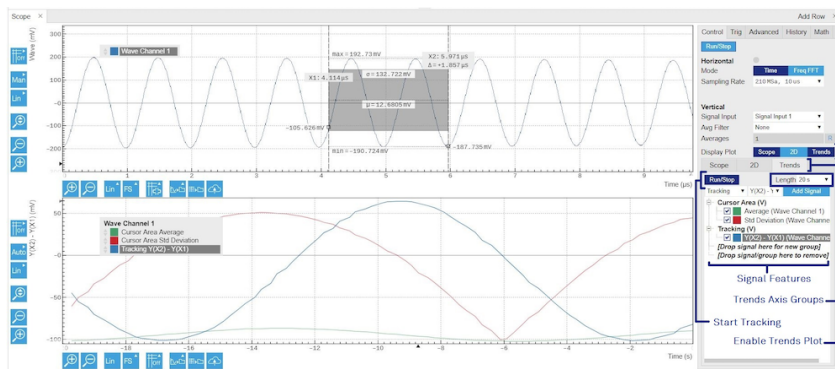


Figure 5.6: Top: main plot of the Scope tab showing the signal trace. Bottom: corresponding Trends plot tracking an average, standard deviation, and difference signal derived from the cursor positions in the main plot. The example shown is part of

the HF2LI user interface. The controls of the Trends feature and their layout are very similar in all tabs and product platforms where this feature is available.

5.3. Saving and Loading Data

5.4. Overview

In this section we discuss how to save and record measurement data with the MFIA Instrument using the LabOne user interface. In the LabOne user interface, there are 3 ways to save data:

- Saving the data that is currently displayed in a plot
- Continuously recording data in the background
- Saving trace data in the History sub-tab

Furthermore, the History sub-tab supports loading data. In the following, we will explain these methods.

5.4.1. Saving Data from Plots

A quick way to save data from any plot is to click on the Save CSV icon  at the bottom of the plot to store the currently displayed curves as a comma-separated value (CSV) file to the download folder of your web browser. Clicking on  will save a graphics file instead.

5.4.2. Recording Data

The recording functionality allows you to store measurement data continuously, as well as to track instrument settings over time. The [Config Tab](#) gives you access to the main settings for this function. The Format selector defines which format is used: HDF5, CSV, or MATLAB. The CSV delimiter character can be changed in the User Preferences section. The default option is Semicolon. The Time Zone setting allows you to adjust the time stamps of the saved data.

The node tree display of the Record Data section allows you to browse through the different measurement data and instrument settings, and to select the ones you would like to record. For instance, the demodulator 1 measurement data is accessible under the path of the form **Device 0000/Demodulators/Demod 1/Sample**. An example for an instrument setting would be the filter time constant, accessible under the path **Device 0000/Demodulators/Demod 1/Filter Time Constant**.

The storage location is selected in the Drive drop-down menu of the Record section. The default location is the internal flash memory of the MFIA. Alternatively, you can plug an external USB drive into one of the USB Device ports on the back panel of the MFIA. If you do so, an additional option USB 1 or USB 2 appears in the Drive drop-down menu. If you'd like to save data directly to the host computer, you can install and run the LabOne Web Server on the host computer as described in [Running LabOne on a Separate PC](#).

Note

Storing data on the internal flash memory is only recommended for occasional use, and for storing small amounts of data. The reason is that flash memories such as the one used in the instrument support only a finite number of write cycles and may wear out under heavy use. Storing data on the host computer (see [Running LabOne on a Separate PC](#)) or on an external USB hard drive is the safer alternative for such cases, and also offers higher speed.

Clicking on the Record checkbox will initiate the recording to the selected location. In case of demodulator and boxcar data, ensure that the corresponding data stream is enabled, as otherwise no data will be saved.

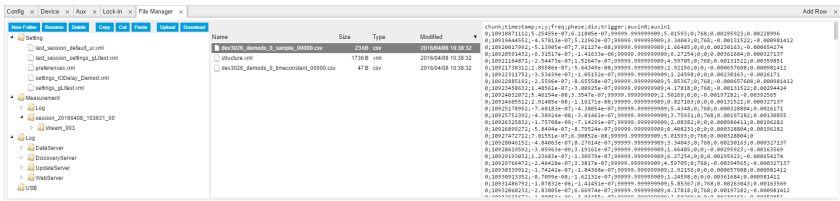


Figure 5.7: Browsing and inspecting files in the LabOne File Manager tab

In case HDF5 or MATLAB is selected as the file format, LabOne creates a single file containing the data for all selected nodes. For the CSV format, at least one file for each of the selected nodes is created from the start. At a configurable time interval, new data files are created, but the maximum size is capped at about 1 GB for easier data handling. The storage location is indicated in the Folder field of the Record Data section.

The **File Manager Tab** is a good place to inspect CSV data files. The file browser on the left of the tab allows you to navigate to the location of the data files and offers functionalities for managing files on the drive of the MF Instrument (or your computer's drive in case you run the Web Server on the host computer). In addition, you can conveniently transfer files between the MFIA and the download folder on your host computer using the Upload/Download buttons. The file viewer on the right side of the tab displays the contents of text files up to a certain size limit. [Figure 5.7](#) shows the Files tab after recording Demodulator Sample and Filter Time Constant for a few seconds. The file viewer shows the contents of the demodulator data file.

Note

The structure of files containing instrument settings and of those containing streamed data is the same. Streaming data files contain one line per sampling period, whereas in the case of instrument settings, the file usually only contains a few lines, one for each change in the settings. More information on the file structure can be found in the LabOne Programming Manual.

5.4.3. History List

Tabs with a history list such as [Sweeper Tab](#), [Data Acquisition Tab](#), [Scope Tab](#), [Spectrum Analyzer Tab](#) support feature saving, autosaving, and loading functionality. By default, the plot area in those tools displays the last 100 measurements (depending on the tool, these can be sweep traces, scope shots, DAQ data sets, or spectra), and each measurement is represented as an entry in the History sub-tab. The button to the left of each list entry controls the visibility of the corresponding trace in the plot; the button to the right controls the color of the trace. ¹Double-clicking on a list entry allows you to rename it. All measurements in the history list can be saved with **Save All**. Clicking on the **Save Sel** button (note the dropdown button ) saves only those traces that were selected by a mouse click. Use the Control or Shift button together with a mouse click to select multiple traces. The file location can be accessed by the Open Folder button . [Figure 5.10.8](#) illustrates some of these features. [Figure 5.8](#) illustrates the data loading feature.

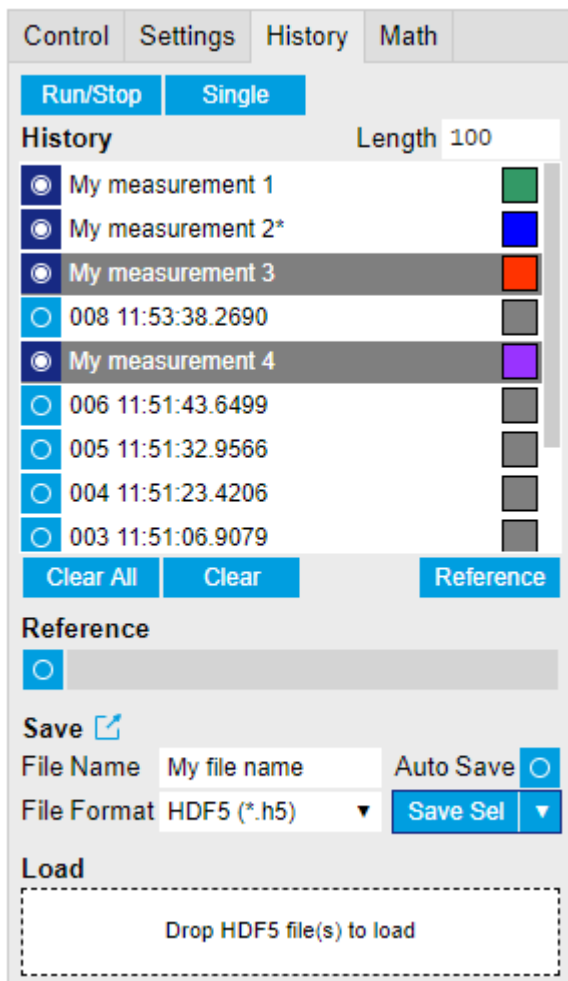


Figure 5.8: History sub-tab features. The entries "My measurement 1" etc. were renamed by the user. Measurement 1, 2, 3, 4 are currently displayed in the plot because their left-hand-side button is enabled. Clicking on Save Sel would save "My measurement 3" and "My measurement 4" to a file, because these entries were selected (gray overlay) by a Control key + mouse click action.

Which quantities are saved depends on which signals have been added to the Vertical Axis Groups section in the **Control** sub-tab. Only data from demodulators with enabled Data Transfer in the Lock-in tab can be included in the files.

The history sub-tab supports an **autosave** functionality to store measurement results continuously while the tool is running. Autosave directories are differentiated from normal saved directories by the text "autosave" in the name, e.g. sweep_autosave_000. When running a tool continuously (**Run/Stop** button) with Autosave activated, after the current measurement (history entry) is complete, all measurements in the history are saved. The same file is overwritten each time, which means that old measurements will be lost once the limit defined by the history Length setting has been reached. When performing single measurements (**Single** button) with Autosave activated, after each measurement, the elements in the history list are saved in a new directory with an incrementing count, e.g. **sweep_autosave_001**, **sweep_autosave_002**.

Data which was saved in HDF5 file format can be loaded back into the history list. Loaded traces are marked by a prefix "loaded " that is added to the history entry name in the user interface. The **createdtimestamp** information in the header data marks the time at which the data were measured.

- Only files created by the Save button in the History sub-tab can be loaded.
- Loading a file will add all history items saved in the file to the history list. Previous entries are kept in the list.
- Data from the file is only displayed in the plot if it matches the current settings in the Vertical Axis Group section the tool. Loading e.g. PID data in the Sweeper will not be shown, unless it is selected in the Control sub-tab.
- Files can only be loaded if the devices saving and loading data are of the same product family. The data path will be set according to the device ID loading the data.

Figure 5.9 illustrates the data loading feature.

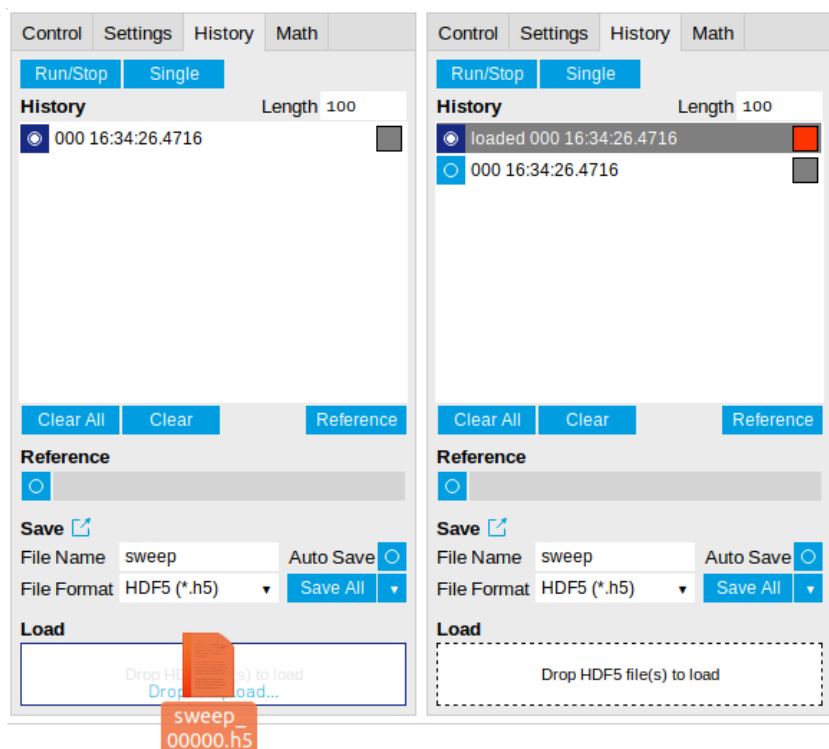


Figure 5.9: History data loading feature. Here, the file `sweep_00000.h5` is loaded by drag-and-drop. The loaded data are added to the measurements in the history list.

5.4.4. Supported File Formats

HDF5

Hierarchical Data File 5 (HDF5) is a widespread memory-efficient, structured, binary, open file format. Data in this format can be inspected using the dedicated viewer [HDFview](#). HDF5 libraries or import tools are available for Python, MATLAB, LabVIEW, C, R, Octave, Origin, Igor Pro, and others. The following example illustrates how to access demodulator data from a sweep using the `h5py` library in Python:

```
import h5py
filename = 'sweep_00000.h5'
f = h5py.File(filename, 'r')
x = f['000/dev3025/demods/0/sample/frequency']
```

The data loading feature of LabOne supports HDF5 files, while it is unavailable for other formats.

MATLAB

The MATLAB File Format (`.mat`) is a proprietary file format from MathWorks based on the open HDF5 file format. It has thus similar properties as the HDF5 format, but the support for importing `.mat` files into third-party software other than MATLAB is usually less good than that for importing HDF5 files.

SXM

SXM is a proprietary file format by Nanonis used for SPM measurements.

ZView

ZView is a comma-separated value format adapted for impedance analysis and supported by the analysis software with the same name by Scribner Associates. Since there are many variants of this

structure in use by other impedance modeling software tools, LabOne allows you to adapt the ZView format to your preferred standard using a template file. Proceed as follows:

- In the [Config Tab](#), select ZView as Format setting for data recording.
- Enable and disable the Record checkbox. This step is only needed the first time you use the ZView format and creates a default template file.
- Open the [File Manager Tab](#) and look for the file `savefile_template.txt` in the Setting folder. Use the Download button to transfer it to your PC, and modify it according to your needs (see below).
- Use the Upload button to transfer the modified file (with the same name `savefile_template.txt`) back into the Setting folder.

Any successive data recording in ZView format will adhere to this new template. The last line of the template file defines the data row format of the saved data file and is repeated for each sample. Everything above the last line defines the data file header and is written to the data file a single time. Both in the header part and in the last line, keywords of the form `${variable}` are replaced by the respective quantity or setting. Any other text in the template file is written verbatim to the data file.

The following tables list the keywords supported in the header part and in the last line of the template file.

Table 5.9: Keywords supported in the header part of the ZView template file

Keyword	Datum
<code>year,</code> <code>{month}, day,</code> <code>{hours}, minutes,</code> <code>{seconds}</code>	Respective component of the current time stamp
<code>\${month_str}</code>	Abbreviated month name (Jan, Feb, Mar, ...)
<code>\${numpoints}</code>	Number of recorded data samples
<code>\${filename}</code>	File name
<code>\${grid_columns}</code>	Number of columns in Data Acquisition Tab
<code>\${grid_rows}</code>	Number of rows in Data Acquisition Tab
<code>\${grid_mode}</code>	Mode setting in Data Acquisition Tab (nearest, linear, Lanczos)
<code>\${grid_operation}</code>	Operation setting in Data Acquisition Tab (replace, average)
<code>\${grid_scan_direction}</code>	Scan direction in Data Acquisition Tab (forward, reverse, bidirectional)
<code>\${grid_repetitions}</code>	Repetitions setting in Data Acquisition Tab

Table 5.10: Keywords supported in the last line of the ZView template file

Keyword	Datum
<code>\$</code> <code>{frequency}</code>	Measurement frequency
<code>\${phasez}</code>	Impedance complex phase
<code>\${absz}</code>	Impedance absolute value
<code>\${realz}</code>	Impedance real part
<code>\${imagz}</code>	Impedance imaginary part
<code>\${param0}</code>	Representation parameter 1
<code>\${param1}</code>	Representation parameter 2
<code>\${drive}</code>	Drive voltage amplitude
<code>\${bias}</code>	DC bias voltage
<code>\$</code> <code>{timestamp}</code>	Measurement time in units of the instrument clock period
<code>\${time_sec}</code>	Time in seconds relative to first sample

Keyword	Datum
#{count}	Successive sample count
#{flags}	Integer containing bit flags for confidence indicators and device state information (see)

Bit number	Condition
0	Internal calibration is enabled
1	User compensation is enabled
2...3	Reserved
4	Confidence indicator: overflow on voltage signal input
5	Confidence indicator: overflow on current signal input
6	Confidence indicator: underflow on voltage signal input
7	Confidence indicator: underflow on current signal input
8...10	Reserved
11	Confidence indicator: Z too low for 2-terminal
12	Confidence indicator: suppression of first representation parameter (PARAM0)
13	Confidence indicator: suppression of second representation parameter (PARAM1)
14	Reserved
15	Confidence indicator: frequency limit of current input exceeded
16	Confidence indicator: strong compensation detected on first representation parameter (PARAM0)
17	Confidence indicator: strong compensation detected on second representation parameter (PARAM1)
18...23	Reserved
24	Confidence indicator: open detected
25...31	Reserved

Bit assignment in **#{flags}** data column

5.4.5. LabOne Net Link

Measurement and cursor data can be downloaded from the browser as CSV data. This allows for further processing in any application that supports CSV file formats. As the data is stored internally on the web server it can be read by direct server access from other applications. Most up-to-date software supports data import from web pages or CSV files over the internet. This allows for automatic import and refresh of data sets in many applications. To perform the import the application needs to know the address from where to load the data. This link is supplied by the LabOne User Interface. The following chapter lists examples of how to import data into some commonly used applications.

The CSV data sent to the application is a snap-shot of the data set on the web server at the time of the request. Many applications support either manual or periodic refresh functionality.

Since tabs can be instantiated several times within the same user interface, the link is specific to the tab that it is taken from. Changing the session on the LabOne User Interface or removing tabs may invalidate the link.

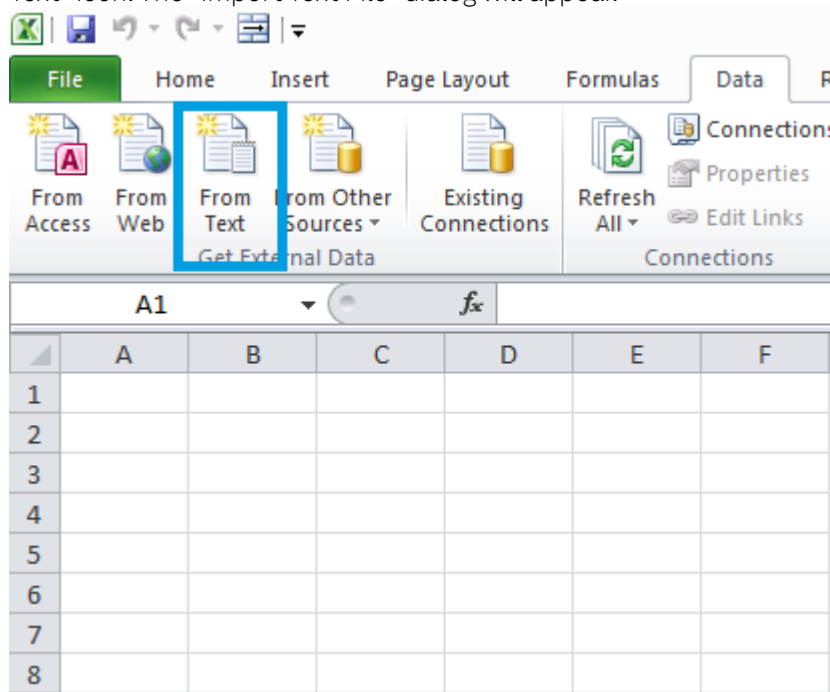
Supported applications:

- Excel
- MATLAB
- Python
- C#.NET
- Igor Pro
-

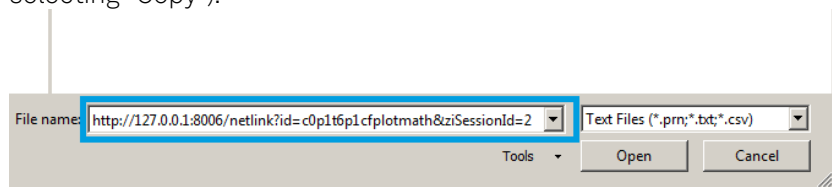
Excel

These instructions are for Excel 2010 (English). The procedure for other versions may differ.

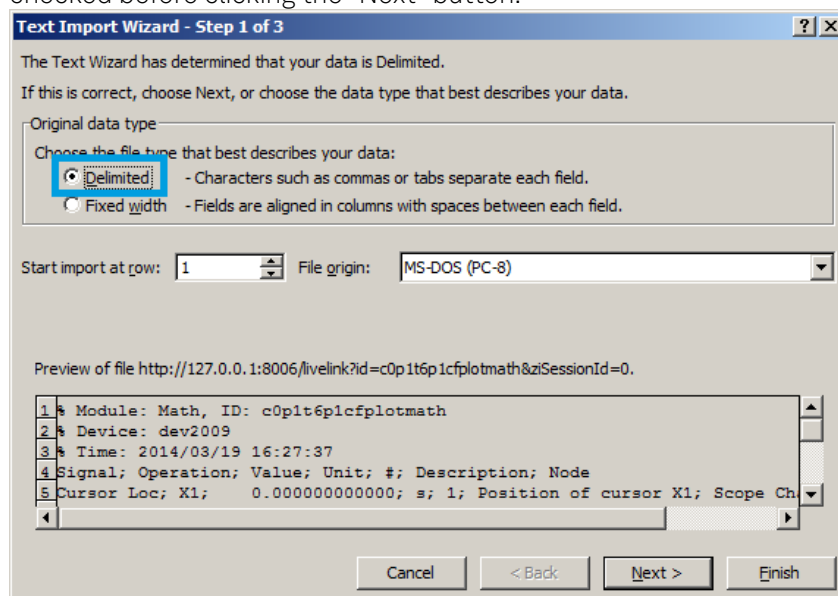
1. In Excel, click on the cell where the data is to be placed. From the Data ribbon, click the "From Text" icon. The "Import Text File" dialog will appear.



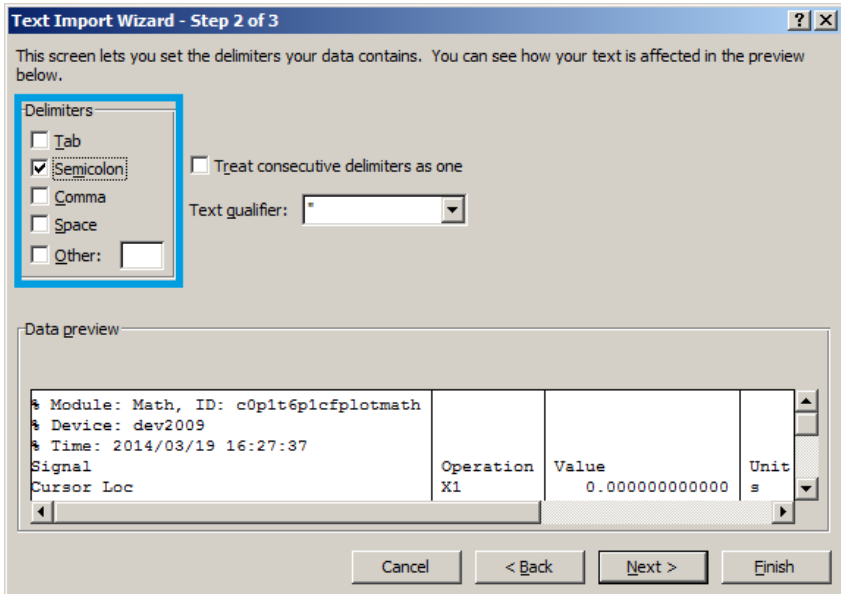
2. In LabOne, click the "Link" button of the appropriate Math tab. Copy the selected text from the "LabOne Net Link" dialog to the clipboard (either with Ctrl-C or by right clicking and selecting "Copy").



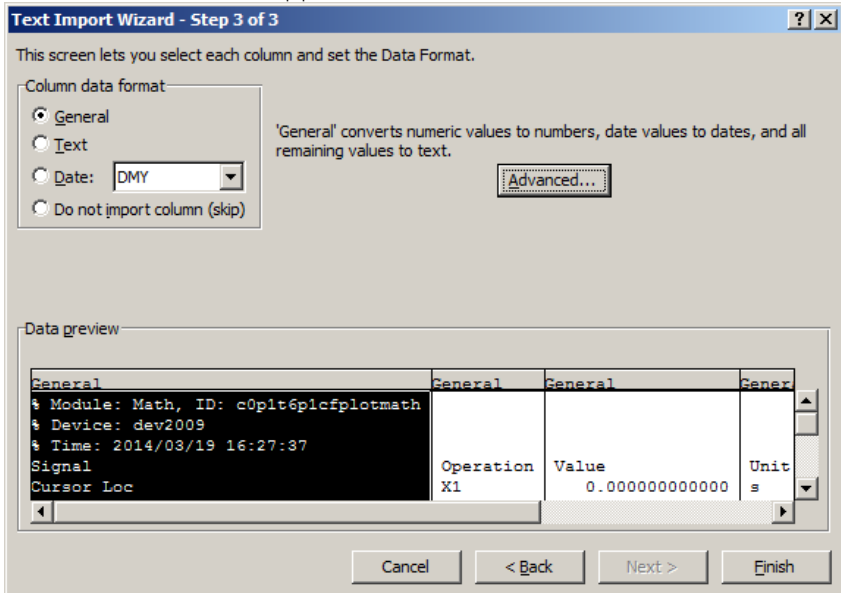
3. In Excel, paste the link into the "File name" entry field of the "Import Text File" dialog and click the "Open" button. This will start the text import wizard. Ensure that the "Delimited" button is checked before clicking the "Next" button.



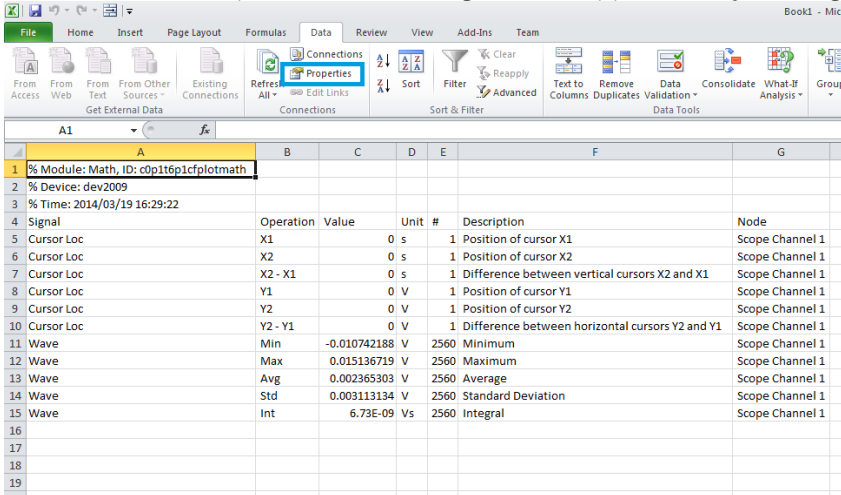
- In the next dialog, select the delimiter character corresponding to that selected in LabOne (this can be found in the "Sessions" section of the Config tab). The default is semicolon. Click the "Next" button.



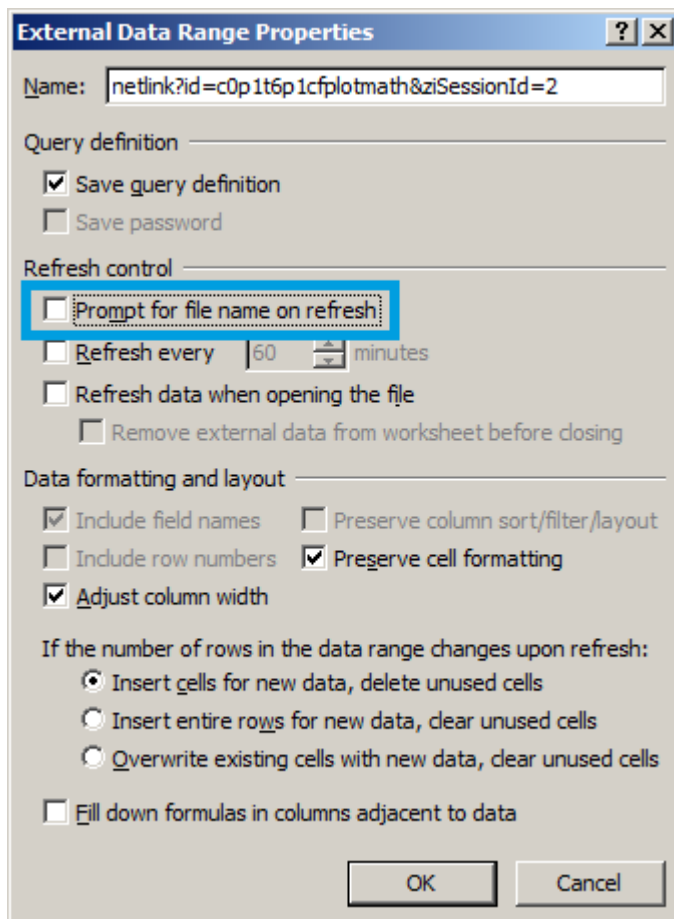
- In the next dialog, click on "Finish" and then "OK" in the "Import Data" dialog. The data from the Math tab will now appear in the Excel sheet.



- The data in the sheet can be updated by clicking the "Refresh All" icon. To make updating the data easier, the "Import text file" dialog can be suppressed by clicking on "Properties".



- Deactivate the check box "Prompt for file name on refresh".



MATLAB

By copying the link text from the "LabOne Net Link" dialog to the clipboard, the following code snippet can be used in MATLAB to read the data.

```
textscan(urlread(clipboard('paste')), '%s%f%s%d%s', 'Headerlines',
4, 'Delimiter', ';')
```

Python

The following code snippet can be used in Python 2 to read the LabOne Net Link data, where "url" is assigned to the text copied from the "LabOne Net Link" dialog.

```
import csv
import urllib2
url = "http://127.0.0.1:8006/netlink?id=c0p5t6p1cfplotmath&ziSessionId=0"
webpage = urllib2.urlopen(url)
datareader = csv.reader(webpage)
data = []
for row in datareader:
    data.append(row)
```

C#.NET

The .NET Framework offers a WebClient object which can be used to send web requests to the LabOne WebServer and download LabOne Net Link data. The string with comma separated content can be parsed by splitting the data at comma borders.

```

using System;
using System.Text;
using System.Net;

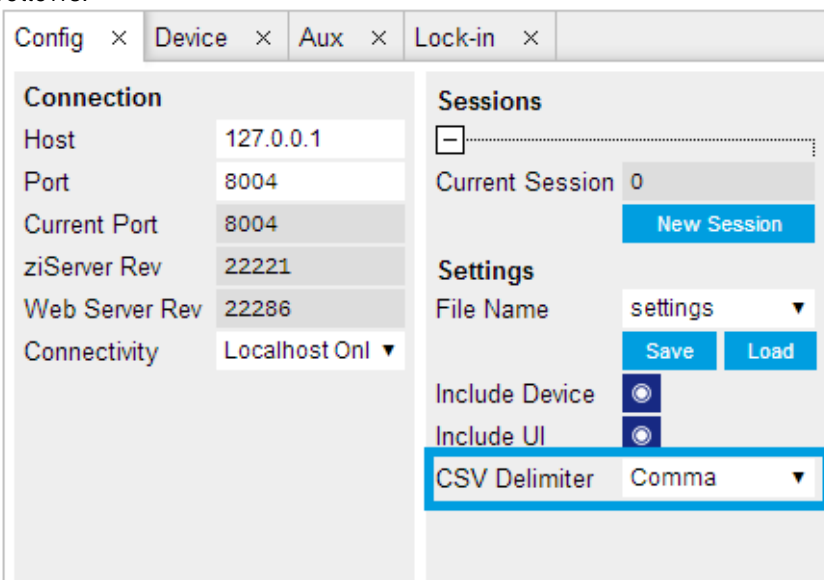
namespace ExampleCSV
{
    class Program
    {
        static void Main(string[] args)
        {
            try
            {
                WebClient wc = new WebClient();
                byte[] buffer = wc.DownloadData("http://127.0.0.1:8006/netlink?
id=c0p1t6p1cfplotmath&ziSessionId=0");
                String doc = Encoding.ASCII.GetString(buffer);
                // Parse here CSV lines and extract data
                // ...
                Console.WriteLine(doc);
            } catch (Exception e) {
                Console.WriteLine("Caught exception: " + e.Message);
            }
        }
    }
}

```

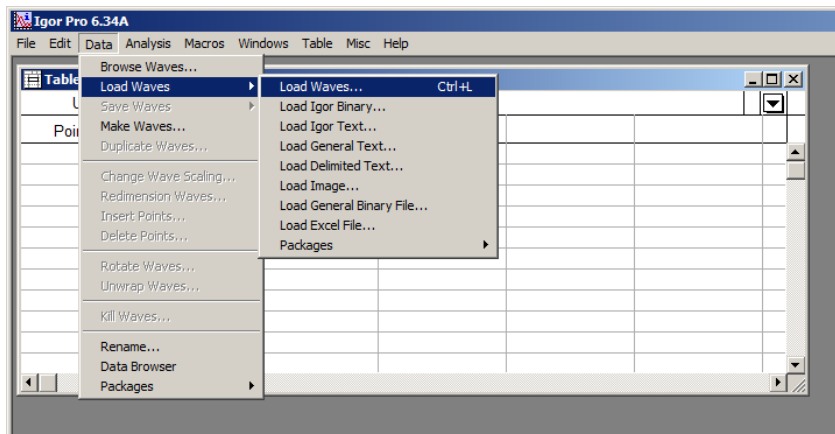
Igor Pro

These instructions are for Igor Pro 6.34A English. The procedure for other versions may differ.

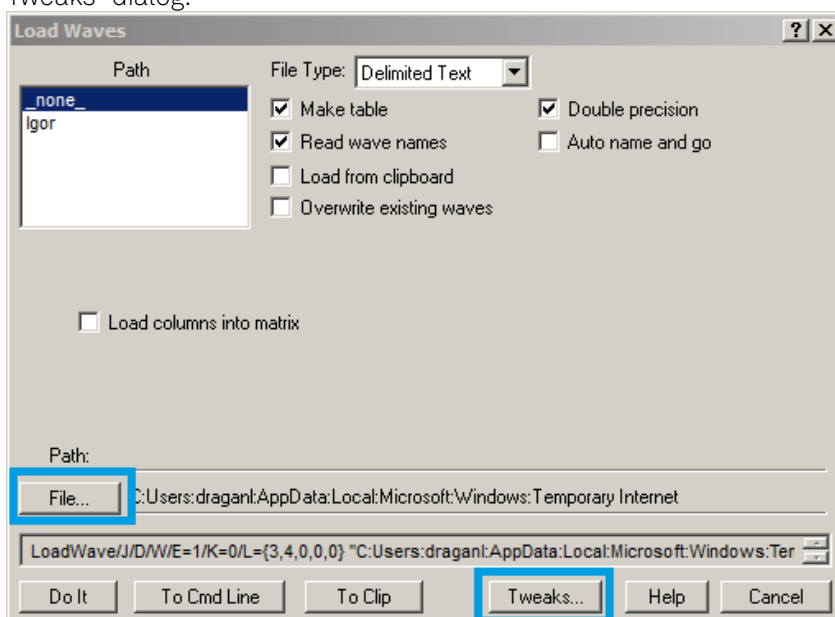
1. For Igor Pro, the CSV separator has to be the comma. Set this in the LabOne Config tab as follows:



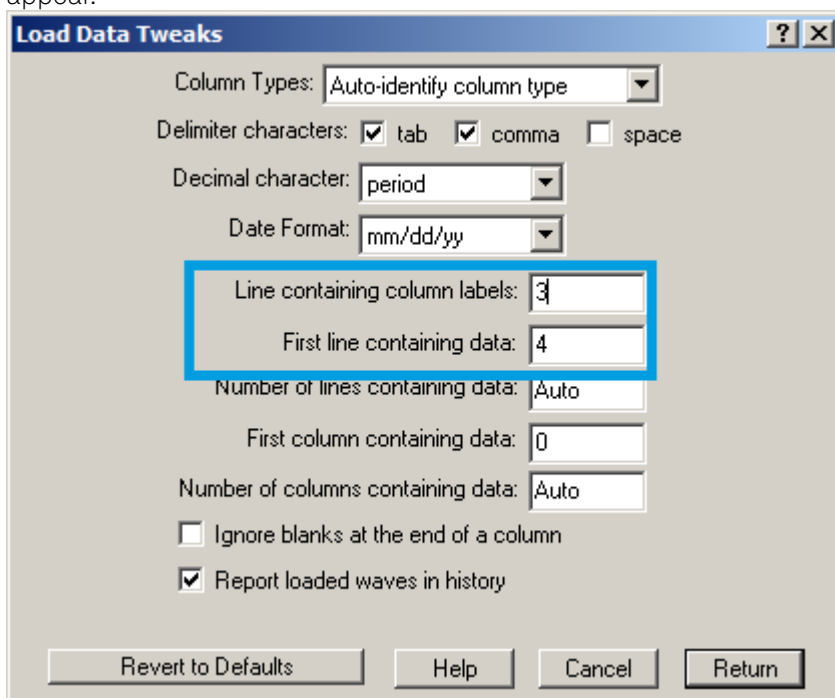
2. In Igor Pro, select the menu "Data→Load Waves→Load Waves...".



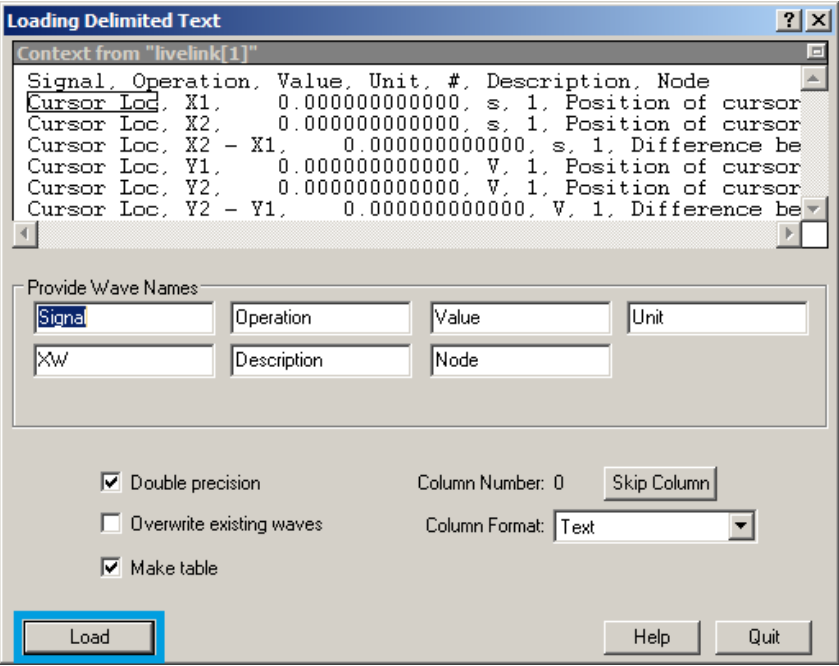
3. In the "Load Waves" dialog, click the "File..." button and paste the link text from the "LabOne Net Link" dialog into the entry field. Then click the "Tweaks..." button to open the "Load Data Tweaks" dialog.



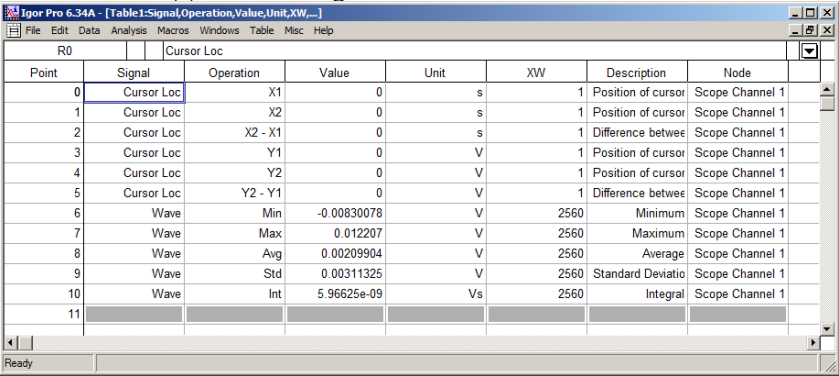
4. Adjust values as highlighted below and click "Return". The "Loading Delimited Data" dialog will appear.



5. Click the "Load" button to read the data.



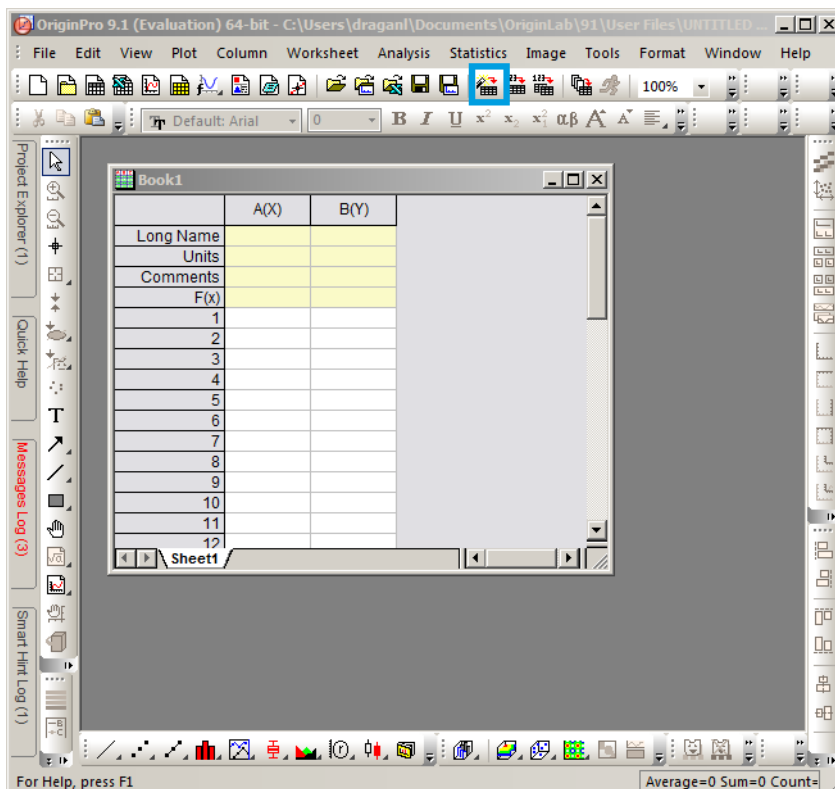
6. The data will appear in the Igor Pro main window.



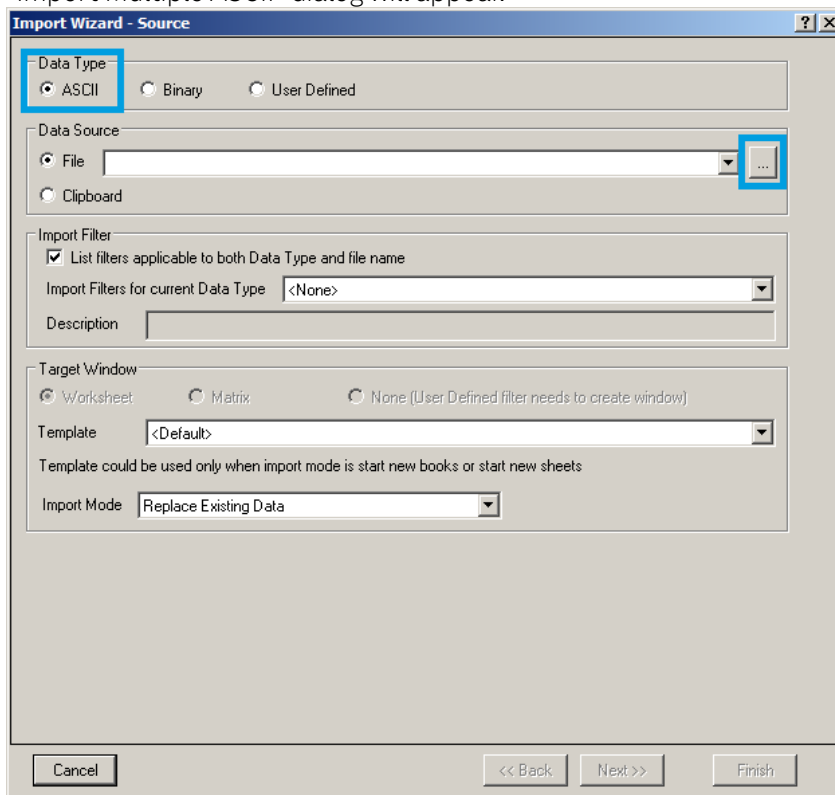
Origin

These instructions are for Origin 9.1 English. The procedure for other versions may differ.

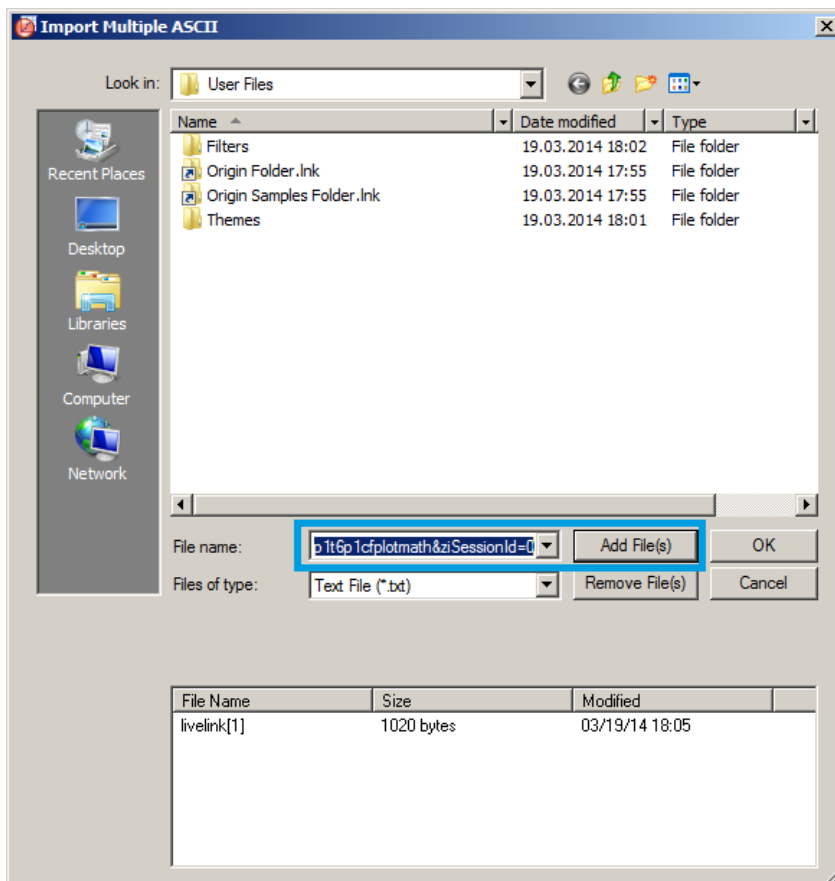
1. Open the import wizard by clicking on the icon highlighted below.



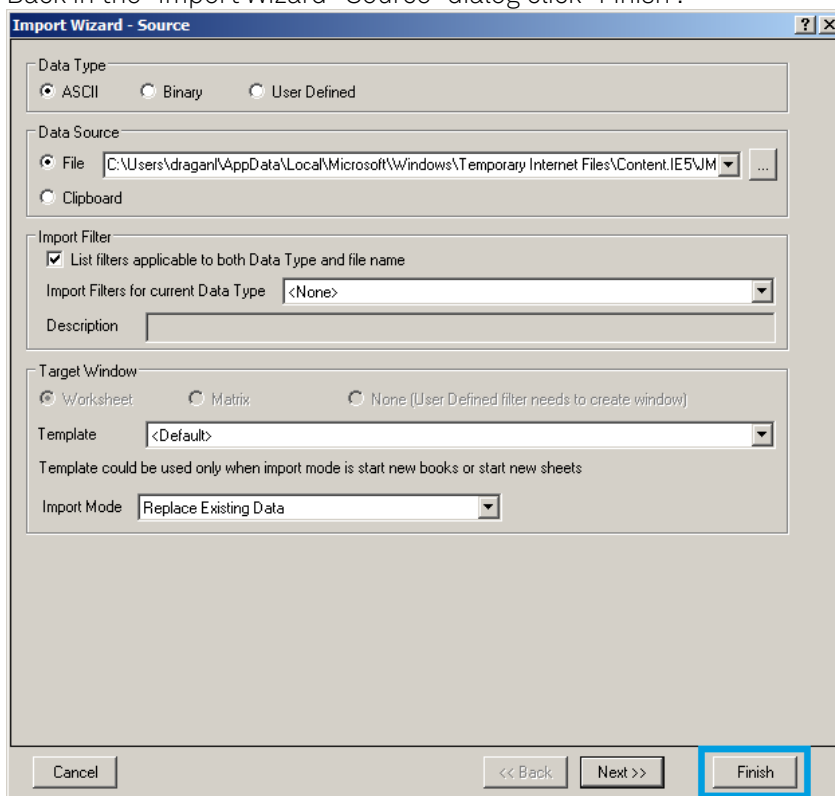
2. Ensure that the ASCII button is selected. Click the "...". See screenshot below. The "Import Multiple ASCII" dialog will appear.



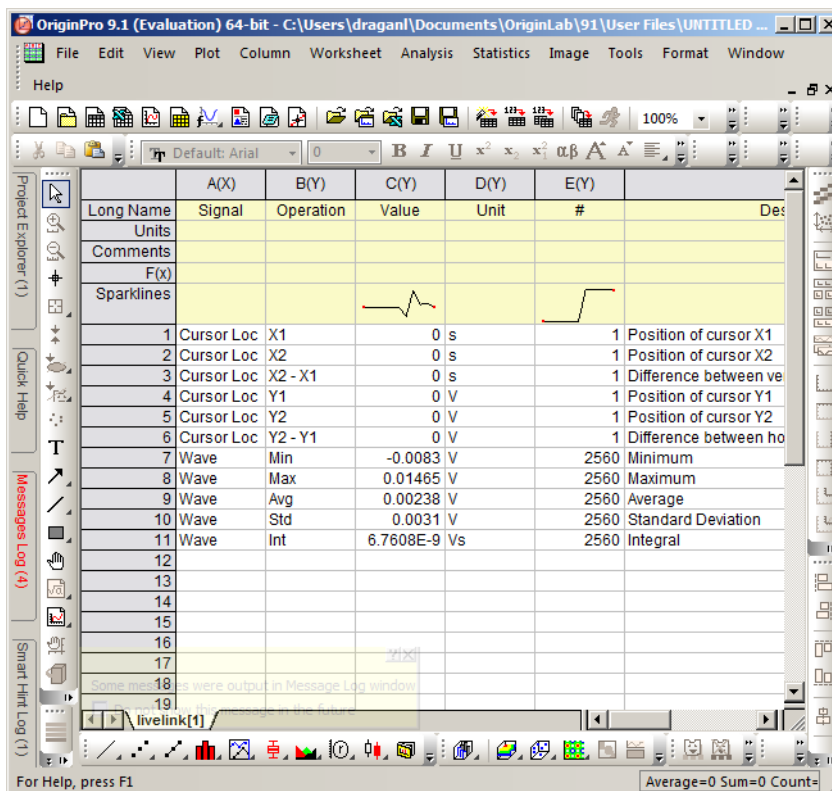
3. Paste the link text from the "LabOne Net Link" dialog into the entry field highlighted below. Then click "Add File(s)" followed by "OK".



4. Back in the "Import Wizard - Source" dialog click "Finish".



5. The data will appear in the Origin main window.



1. Among the mentioned tools, the Scope is exceptional: it displays the most recent acquisition, and its display color is fixed. However, the Persistence feature represents a more specialized functionality for multi-trace display. ↩

5.5. Impedance Analyzer Tab

The Impedance Analyzer tab is available on MFIA Impedance Analyzer instruments and on MFIA Lock-in Amplifier instruments with installed MF-IA Impedance Analyzer option (see Information section in the Device tab).

5.5.1. Features

- Control for impedance analyzer unit (2 units with MF-MD option)
- Auto ranging and auto bandwidth
- Current Zone ranging with customizable range switching frequencies
- Graphical display of measurement mode: 2- and 4-terminal configuration
- Graphical display of DUT representation: $R_p || C_p$, $R_s + C_s$, $L_s + R_s$, C , D ,...
- 0.05 % basic accuracy
- 1 mΩ to 1 TΩ measurement range
- 20 ms measurement speed for basic accuracy
- Compensation Advisor for a variety of compensation procedures
- Confidence Indicator
- Independent current and voltage measurement
- One-period averaging for low frequency impedance measurements

5.5.2. Description

The Impedance Analyzer tab is the main control panel for impedance measurements. On MFIA Impedance Analyzer instruments, the tab is open by default. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.11: App icon and short description

Control/Tool	Option/Range	Description
IA		Quick overview and access to all the settings and properties for impedance measurements.

The Impedance Analyzer tab consists of one side tab for each impedance analyzer unit, and the Cal side tab giving access to the Compensation Advisor. The numbered side-tab(s) for the impedance analyzer units (see [Figure 5.10](#)) are the main measurement interface. The main settings are accessible in the Measurement Control section on the left. The Equivalent Circuit section holds the controls for configuring the circuit representation visualized by a graphical circuit diagram. With the Mode setting, the user selects between a 2- and a 4-terminal setup which has to correspond to the physical wiring of the device-under-test (DUT). The Representation setting allows the user to convert the measured impedance Z into a parameter pair of a DUT circuit representation, such as parallel resistance and capacitance ($R_p \parallel C_p$), or serial resistance and inductance ($R_s + L_s$). The data are displayed in the Measurement Results section on the right. The equivalent circuit parameters appear as Representation Parameter 1 and 2 elsewhere in Measurement the UI, e.g. in the [Numeric Tab](#) or the [Plotter Tab](#) tab. The Enable button at the top left initiates the measurement by turning on the drive voltage and starting the current and voltage measurement.

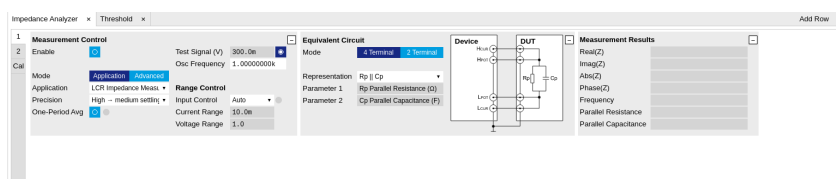


Figure 5.10: LabOne UI: Impedance Analyzer tab (Application mode)

The Measurement Control section of the tab contains settings related to measurement range, frequency, and output drive amplitude. By default, all of those parameters are under automatic control of the Impedance Analyzer providing suitable settings for most situations. Manual or Current Zone Range Control is available for some more advanced cases. E.g., disabling automatic Range Control by setting it to Manual or Current Zone can help preventing or forcing range changes in sweeper measurements. In addition, a Manual Bandwidth Control is available to help in conditions with strong noise. The measurement bandwidth should then be chosen as a compromise between speed and precision. In auto bandwidth mode, the instrument selects an appropriate bandwidth for the desired Ω suppression (see [Sinc Filtering](#)). Disabling automatic Drive Control allows the user to specify the drive amplitude when measuring an electrically sensitive DUT, for example. The numeric controls for bandwidth and drive amplitude are accessible in the [Lock-in Tab](#).

More advanced controls for signal bandwidth, digital data transfer, and more, become accessible in the lower half of the tab after switching Mode from Application to Advanced in the Measurement Control section, see [Figure 5.11](#). The Confidence Indicators offer control of a number of features assisting in the measurement. The Indicators provide warning messages in the [Sweeper Tab](#), [Plotter Tab](#), and [Numeric Tab](#) tab that prevent misinterpretation of measurement data. The different warning messages are explained in the table below. All Indicators can be individually enabled or disabled, and the numeric settings provide fine-tuning of the error thresholds. The Discard Samples feature allows the user to disregard outlier measurements using thresholds.

For impedance measurements at low frequency, the one-period averaging feature (located in the Measurement Control sub-tab) can be enabled. Once enabled, the one-period averaging feature will automatically be active below a frequency threshold (defined by the data transfer rate), and the standard measurement mode will be used above this threshold. The active state is indicated by a green LED next to the enable switch. One-period averaging uses an alternative method to calculate impedance compared with the standard measurement mode. Current and voltage are measured for one period and subsequently divided to calculate the impedance, this technique suppresses the unwanted omega components by averaging over one full period. This is in contrast to the standard measurement mode of the MFIA, where unwanted omega components are filtered out using a low pass filter. The advantage of one-period averaging is the increased measurement speed for low frequencies, as more relaxed (faster) filter settings can be used.

During one-period averaging, the Demod signals will contain significant omega components and therefore do not return reliable values. If the Demod streams are required, switch off one-period averaging. Both [Sweeper Tab](#) and [Plotter Tab](#) can be used with one-period averaging to calculate impedance. Please note, during the first period of the measurement, no data is transferred as no valid result is available during this time.

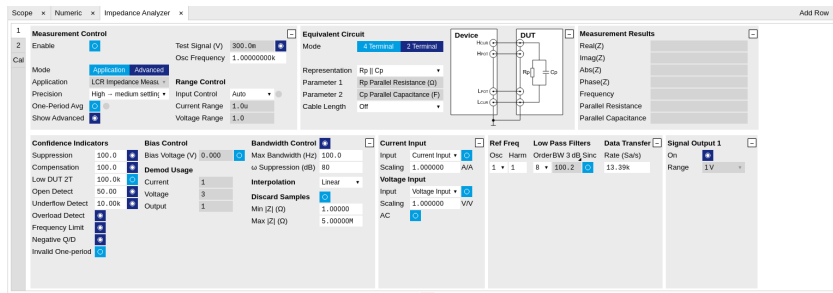


Figure 5.11: LabOne UI: Impedance Analyzer tab (Advanced mode)

Table 5.12: List of warning messages provided by the Confidence Indicator.

Confidence Indicator	Warning message	Situation	Corrective action
Suppression	Suppression	The DUT is characterized by strongly differing impedance components. The calculation of the flagged Representation Parameter is unreliable.	Change measurement frequency, select appropriate representation corresponding to the DUT
Compensation	Strong Compensation	A measurement is carried out in a parameter range in which the user compensation enforces a large correction, making the measurement unreliable.	Check whether the test fixture is suitable for the measured impedance range and frequency. Check that the chosen compensation method is suitable for the impedance range. Consider including a "load" step with a load close to the measurement range.
Open Detect	Open Detected	One of the terminals in a four-terminal is disconnected	Check the correct wiring of the DUT (e.g. 2-terminal DUT in a 4-terminal measurement). Check for broken connections.
Underflow Detect	Current Underflow, Voltage Underflow	Voltage or current reading is close to zero	Check for open connections. If connections are correctly closed, check if the selected Range is suitable for the measured impedance when using the Range Modes Manual and Impedance.
Overflow Detect	Current Overflow, Voltage Overflow	Voltage or current reading is close to the input range or exceeding it	Check if the selected Range is suitable for the measured impedance when using the Range Modes Manual and Impedance.
Frequency Limit	Freq Limit	Measurement frequency is higher than the bandwidth of the selected current input range	Select a larger current range in the range control settings or set Range Control to Auto.
Negative Q/D	Negative Q/D	The measured Q or D factor, or any measurement contributing to the displayed average, is negative	Check whether the chosen Representation corresponds well to the connected DUT, evaluate the quality of the applied compensation, and check for presence of strong noise. A negative Q or D factor mean the measured impedance can not be physically mapped to the chosen Representation.
Z too low for 2-terminal	Low DUT 2T	The measured Z is lower than the defined threshold for 2-terminal measurements	For low impedance measurements, 4-terminal measurements are recommended. The threshold for this confidence indicator can be manually adjusted or disabled in the advance section of the Impedance Tab

Confidence Indicator	Warning message	Situation	Corrective action
Different representation	Different representation	This flag highlights traces within the displayed data set which were acquired using an equivalent circuit representation other than the last set representation	Remove these traces from the recorded data using the history tab, or restart the data with the required representation

5.5.3. Compensation

The Cal side-tab (see [Figure 5.12](#)) provides user guidance through a compensation procedure. Compensation reduces the effect of test fixture parasitics and thus improves measurement accuracy. It requires a measurement of one or more reference devices under test (DUTs) with precisely known impedance values in a certain frequency range. The available compensation methods are combinations of measurements of short ("S", DUT with zero impedance) and load ("L", DUT with known impedance of arbitrary value). An Open ("O", DUT with infinite impedance) step is not recommended, but if necessary can still be found in the Advanced tab. Please have a look at [Impedance and its Measurement](#) to learn more about the background and practicalities of the compensation, and at [Compensation](#) for step-by-step instructions for a specific example. To prepare a compensation, select the appropriate Mode, connect the reference DUT and click on Compensate.

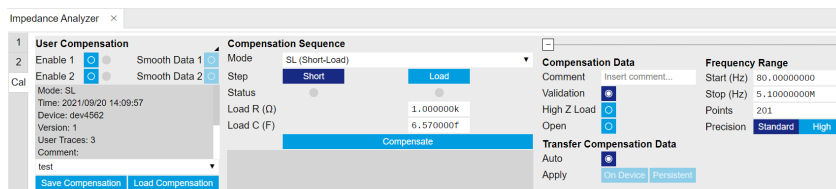


Figure 5.12: LabOne UI: Impedance Analyzer tab (Cal side tab)

5.5.4. Functional Elements

Table 5.13: Impedance Analyzer tab: Control sub-tab

Control/Tool	Option/Range	Description
Enable	ON / OFF	Enable impedance calculation for demodulator data.
One-Period Averaging	ON / OFF	Enables one-period averaging for low frequency impedance measurements.
One-Period Averaging	grey/green	The LED is green when one-period averaging is active.
Lower Frequency Limit (Hz)		Lower frequency limit where the current input range will be enabled.
Upper Frequency Limit (Hz)		Upper frequency limit where the current input range will be enabled.
Cable Length		One way length of RG58 cables used in measurement set up. This setting corrects for the phase shift introduced by an ideal cable.
Representation		Representation of the complex impedance value Z by two real values accessible as Parameter 1 and Parameter 2 on all user interface displays.
	Rp Cp	Impedance value Z is represented by a resistive element R_p in parallel with a capacitive element C_p .
	Rs + Cs	Impedance value Z is represented by a resistive element R_s in series with a capacitive element C_s .
	Rs + Ls	Impedance value Z is represented by a resistive element R_s in series with a inductive element L_s .

Control/Tool	Option/Range	Description
	G B	Impedance value Z is represented by conductance $G = \text{Real}(Y)$ and Susceptance $B = \text{Imag}(Y)$ of the admittance $Y = 1/Z$.
	D Cs	Impedance value Z is represented by a dissipation factor $D = -\text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a capacitive element.
	Q Cs	Impedance value Z is represented by a quality factor $Q = -\text{Imag}(Z)/\text{Real}(Z)$ and a capacitive element.
	D Ls	Impedance value Z is represented by a dissipation factor $D = \text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and an inductive element.
	Q Ls	Impedance value Z is represented by a quality factor $Q = \text{Imag}(Z)/\text{Real}(Z)$ and an inductive element.
	Rp Lp	Impedance value Z is represented by a resistive element Rp in parallel with a inductive element Lp.
	D Cp	Impedance value Z is represented by a dissipation factor $D = -\text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a capacitive element.
	$\epsilon_r' \epsilon_r''$	The relative dielectric constant ϵ_r is represented by the real part ϵ_r' and the frequency-normalized imaginary part ϵ_r'' ($\epsilon_r = \epsilon_r' - j \omega \epsilon_r''$).
Real(Z)		Real part of the measured impedance Z.
Imag(Z)		Imaginary part of the measured impedance Z.
Abs(Z)		Absolute value of the measured impedance Z.
Phase(Z)		Phase of the measured impedance Z.
Frequency		Frequency at which the current impedance is measured.
Parameter 1		Currently active representation parameter 1 and the corresponding measured value. A representation is based on the measured impedance and the frequency. Representations are only valid within a specific frequency or impedance region.
Parameter 2		Currently active representation parameter 2 and the corresponding measured value. A representation is based on the measured impedance and the frequency. Representations are only valid within a specific frequency or impedance region.
Thickness		Fill in the thickness of the material-under-test in mm.
Area		Fill in the area of the material-under-test in cm ² .
MUT		Representation of the relative dielectric properties of the material-under-test by the real and imaginary components.
Osc Frequency	0 to 5 MHz	Frequency control for the oscillator used for impedance measurement.
Range Control		Select input range control mode.
	Manual	In manual mode the current and voltage input ranges are adjusted manually and separately. Use this mode with care as overload will result in inaccurate impedance results.
	Auto	Dynamically adjust the input range according to the measured input signal strength. This optimizes the dynamic range and precision of impedance measurements.
	Current Zone	This ranging option allows you to manually set the switching frequency limits for all eight current input ranges.
Range Control Override	ON / OFF	Indicates disabled periodic auto range control. A running sweeper module takes over the range control and thus disables the periodic range checks.
Current Range	100 nA, 1 μ A, 10 μ A, 100 μ A, 1 mA, 10 mA	Input current range used for the impedance measurement. Small current input range have a reduced bandwidth. In the Range Control modes "Auto" and "Impedance", the current range is switched automatically to a higher range if the frequency is too high.

Control/Tool	Option/ Range	Description
Voltage Range	10 mV, 30 mV, 100 mV, 300 mV, 1 V, 3.0 V	Input voltage range used for the impedance measurement.
Drive Control	ON / OFF	If enabled, the drive voltage amplitude is controlled by the device. If disabled it can be set manually.
Test Signal (V)	numeric value	Drive amplitude on the Signal Output.
Bandwidth Control	ON / OFF	Enable automatic bandwidth control. If enabled the optimum bandwidth is calculated based on the frequency and measurement data.
Max Bandwidth (Hz)	positive numeric value	Limit of the maximum bandwidth used on the demodulator filter. Values above 1 kHz can heavily diminish measurement accuracy in the high-frequency region where the amplitude is no more constant over frequency.
ω Suppression (dB)	positive numeric value	Suppression of the omega and 2-omega components. Small omega suppression can diminish measurements of very low or high impedance because the DC component can become dominant. Large omega suppression will have a significant impact on sweep time especially for low filter orders.
Suppression Ratio	1 to 10000	Error amplification limit for which a secondary parameter is marked unreliable. Larger gain values mean larger warning tolerances. A gain value between 10 and 100 is best.
Conf. Indicators	ON / OFF	Enables/disables all confidence indicators to check the reliability of the measured data. To enable individual indicators, open the advanced tab.
Suppression Enable	ON / OFF	The Suppression Confidence Indicator indicates if one of the two parameters of a circuit representation cannot be calculated reliably from the measured impedance. This is the case if a small variation in one (dominant) representation parameter creates a strong variation of the other (suppressed) representation parameter. Such an error amplification indicates that the measurement of the secondary parameter is unreliable.
Compensation	positive numeric value	Strength of the compensation that will trigger the strong compensation warning.
Compensation Enable	ON / OFF	Enables the indication of strong compensation in the plots. A strong compensation diminishes the measurement accuracy of the parameter.
Low DUT	positive numeric value	The maximum DUT impedance for which a 2T measurement is marked unreliable.
Low DUT Enable	ON / OFF	Enables the detection of small DUTs when doing 2 terminal measurements. Use 4 terminal mode for low impedance measurements.
Open Detect	positive numeric value	Open terminal detection ratio. An open terminal is reported if the excitation calculated from current and voltage drop differs more than the specified factor from the driving voltage.
Open Detect Enable	ON / OFF	Enables the open terminal detection for 4-terminal measurements. If enabled, an open terminal is indicated in the numeric tab and in plots.
Underflow Detect	positive numeric value	The underflow condition is met if the measured amplitude is lower than the specified ratio relative to full scale.
Underflow Detect Enable	ON / OFF	Enables the underflow detection for current and voltage. If enabled, an underflow will be shown in the numeric tab and in plots.

Control/Tool	Option/Range	Description
Overload Detect Enable	ON / OFF	Enables the overload detection for current and voltage. If enabled, an overload will be shown in the numeric tab and in plots.
Frequency Limit Enable	ON / OFF	Enables the frequency limit detection based on the used current input range. It is shown in the numeric tab and in plots. Only relevant when Range Control is set to Manual.
Negative Q/D Enable	ON / OFF	Enables the detection of negative Q or D factors. Negative Q or D factors mean the measured impedance does not correspond to the chosen Representation. This can be due to an erroneous compensation, a bad choice of the Representation, or noise. A message will be shown in the numeric tab and in plots.
Invalid One-period	ON / OFF	Enables the detection of unreliable data points where data sample loss leads to an invalid one-period average. Try reducing the data transfer rate.

Table 5.14: Impedance Analyzer tab: Settings sub-tab

Control/Tool	Option/Range	Description
Mode		Switch between application-based or manually configured impedance settings. A parameter set given by an Application mode can be fine-tuned by changing to Advanced mode. Changing back to Application mode will reset the parameters.
	Application	The impedance settings are adjusted to fit best the selected application.
	Advanced	The impedance settings are manually configured.
Application		Select the Impedance application
	LCR Impedance Measurement	General-purpose settings for measuring components at finite frequency
	DC Impedance Measurement	Settings for measuring resistance at zero frequency.
Precision		Select the Impedance precision and measurement speed by automatically adjusting the filter bandwidth. If the sweeper module is in impedance application mode the precision setting will also control the sweeper measurement speed.
	Low -> fast settling	Medium accuracy/precision is optimized for fast response.
	High -> medium settling	High accuracy/precision takes more settling time.
	Very high -> slow settling	Very high accuracy/precision uses long settling time due to very low bandwidth.
Show Advanced	ON / OFF	Display the advanced settings.
Bias Control (V)	-3 V to 3 V	DC bias voltage applied across the device under test. Both positive and negative bias voltages are supported. In a 4-terminal measurement, the bias voltage is limited by the maximum common voltage input range of the device. In a 2-terminal measurement, the bias voltage can be larger because the voltage inputs are not connected.
Bias Control Enable	ON / OFF	Enables bias control. The bias is generated by the additional offset applied on the output.
Current Input	Current Input 1	Select the current input used for two- and four-terminal impedance measurements.
Current Invert Enable	ON / OFF	If enabled, the current input signal is inverted. This is useful to switch the polarity of an input signal which can be caused by additional current amplifiers.

Control/Tool	Option/Range	Description
Scaling factor for current input	numeric value	The value measured at the current input will be scaled by this value.
Voltage Input	Voltage Input 1	Select the voltage input used for a four-terminal impedance measurement.
Voltage Invert Enable	ON / OFF	If enabled, the voltage input signal is inverted.
Scaling factor for voltage input	numeric value	The value measured at the voltage input will be scaled by this value.
AC	ON / OFF	Defines the input coupling for the Signal Inputs. AC coupling inserts a high-pass filter.
Current Demodulator	demodulator index	Demodulator used for current demodulation.
Voltage Demodulator	demodulator index	Demodulator used for voltage measurement in case of a four-terminal impedance measurement.
Output Demodulator	demodulator index	Demodulator unit used to generate the excitation voltage on the Signal Output.
Interpolation		Select the interpolation method of the compensation data. The interpolation method is particularly important if the derivative changes strongly e.g at cut-off frequencies.
	Linear	The linear interpolation is fastest but may create compensation errors in between the frequency points used for compensation.
	Piecewise Cubic Hermite (PCHIP)	The piecewise cubic Hermite interpolation will result in very accurate results but requires more calculation power.
Osc	oscillator index	Oscillator used to generate the frequency of the excitation voltage on the Hcur (+V) connector.
Harm	1 to 1023	Multiplies the demodulator's reference frequency with the integer factor defined by this field.
Order		Select the filter roll-off between 6 dB/oct and 48 dB/oct. High filter orders are beneficial for impedance measurements as they help to suppress DC offsets.
	1	1st order filter 6 dB/oct
	2	2nd order filter 12 dB/oct
	3	3rd order filter 18 dB/oct
	4	4th order filter 24 dB/oct
	5	5th order filter 30 dB/oct
	6	6th order filter 36 dB/oct
	7	7th order filter 42 dB/oct
	8	8th order filter 48 dB/oct
TC/BW Select		Define the display unit of the low-pass filters: time constant (TC), noise-equivalent power bandwidth (BW NEP), 3 dB bandwidth (BW 3 dB).
	TC	Define the low-pass filter characteristic using time constant of the filter.
	BW NEP	Define the low-pass filter characteristic using the noise-equivalent power bandwidth of the filter.
	BW 3 dB	Define the low-pass filter characteristic using the cut-off frequency of the filter.
TC/BW Value	numeric value	Defines the low-pass filter characteristic in the unit defined above.
Sinc	ON / OFF	Enables the sinc filter.

Control/Tool	Option/Range	Description
Rate	0.056 Sa/s to 857 kSa/s	Impedance data streaming rate. The same data rate is applied to the demodulators that are used for the impedance measurement.
On	ON / OFF	Main switch for the Signal Output corresponding to the blue LED indicator on the instrument front panel.
Range		Defines the maximum output voltage that is generated by the corresponding Signal Output. This includes the potential multiple Signal Amplitudes and Offsets summed up. Select the smallest range possible to optimize signal quality.
	10 mV	Selects output range ± 10 mV.
	100 mV	Selects output range ± 100 mV.
	1 V	Selects output range ± 1 V.
	10 V	Selects output range ± 10 V.
Enable	ON / OFF	Discarding impedance samples outside the indicated range.
Minimum	ON / OFF	Threshold for $\text{abs}(Z)$ below which the impedance samples are discarded.
Maximum	ON / OFF	Threshold for $\text{abs}(Z)$ above which the impedance samples are discarded.

Table 5.15: Impedance Analyzer tab: Compensation sub-tab

Control/Tool	Option/Range	Description
Compensation Type Select		Displays information on either internal calibration or user compensation.
	Internal Calibration	The internal calibration matches all voltage and current ranges over the frequency range in order to achieve a high dynamic impedance range. The internal calibration is performed at Zurich instruments.
	User Compensation	The user compensation will correct for measurement setup influences like capacitive or inductive parasitics. It is applied on top of the internal calibration.
Internal	ON / OFF	Enables the internal calibration. This ensures that the input range gains match over the full frequency range. With enabled internal calibration the device fulfills the impedance accuracy specification. The internal calibration is a prerequisite to apply a user compensation.
Active	grey/green	Indicates if the internal calibration is applied to the measurement data.
Internal Calibration Info		Display of meta data saved together with the internal impedance calibration data.
User	ON / OFF	Enables the compensation of the impedance data. The user compensation is correcting parasitics and delays caused by the external setup. The user compensation is applied on top of the internal impedance calibration.
Active	grey/green	Indicates that a valid compensation is active. If active the impedance data streams deliver amplitude and phase corrected data based on the impedance compensation.
User Compensation Info		Display of meta data saved together with the user compensation.
File Name for User Compensation		Enter a new file name to save the current user compensation to a file or select an existing file to load a user compensation.
Save User Compensation		Save the current user compensation to a file.

Control/Tool	Option/ Range	Description
Load User Compensation		Load a user compensation from a file.
Mode		Select the user compensation sequence. If the sequence involves a short or open condition, it is essential to keep the noise small by using auto range on the voltage and current inputs.
	None	No compensation is performed.
	SOL (Short-Open-Load)	Compensation of user setup using a short-open-load condition. This compensation method results in very accurate results over the whole impedance range under the condition of low-noise open and short compensation. Thus, this compensation should always be used with enabled auto range.
	OL (Open-Load)	The open-load compensation is suited for measuring high impedance. Use auto range to reduce the influence of noise for the open compensation step.
	SL (Short-Load)	The short-load compensation is suited for measuring low impedance.
	L (Load)	Compensation performed with a single load. This compensation is useful if the measured DUT is close to the load used for compensation.
	SO (Short-Open)	The short-open compensation corrects for inductive and capacitive parasitics of the user setup. This compensation method is suited if no load is available for the compensation. It will not correct gain errors.
	LLL (Load-Load-Load)	The compensation based on three load values results in very accurate results for a restricted impedance range. Use this method if the noise on the open or short condition is too strong.
Load R	numeric value	Resistance value of the compensation load.
Load C	numeric value	Capacitance value of the compensation load. In case of a resistor load the capacitance value is equivalent to its capacitive parasitics.
Step	Short, Open, Load	Compensation step to record. The sequence of the steps can be freely selected. Before starting the correct device needs to be connected to the setup.
Compensation Step Short	ON / OF	Select to do a short measurement. This compensates for parasitics that are in series with your DUT.
Compensation Step Open	ON / OF	Select to do an open measurement. This compensates for parasitics that are in parallel with your DUT.
Compensation Step Load	ON / OF	Select to do a load measurement. This compensates for additional gains in your measurement setup.
Compensation Step High Load	ON / OF	Select to do a high load measurement. This compensates for additional gains in your measurement setup.
Status	grey/green	Indicates a successful compensation step. If the status indicator does not indicate green after a compensation step, check the message box to learn more about the failure.
Compensate		Start the compensation measurement for the selected condition. If a compensation measurement is completed successfully, the next pending condition is automatically selected. The message box will contain information on the quality of the completed compensation and failure messages in case of a fail. The progress of the currently active compensation run will be shown in the message box.

Control/Tool	Option/Range	Description
Message	string	The message pane displays information related to compensation. In case of a failed compensation, information on the root cause is reported. In case of successful compensation, representative parameters are reported to judge the quality of the compensation and measurement setup.
Comment	string	User comment that will be saved together with the compensation data. The comment string can also be changed between the different compensation steps.
Validation	ON / OFF	Perform a sanity check of the compensation result to detect outlier measurement results, overload, or other error conditions. Validation should only be disabled for compensations performed in special conditions.
High Z Load	ON / OFF	Two different loads are measured during user compensation. Improves compensation data.
Auto Transfer	ON / OFF	If enabled, a successful compensation will be immediately transferred to the device, activated, and stored permanently. If automatic transfer is disabled, the transfer needs to be initiated manually.
On Device	On Device	Transfer and activate the recorded compensation manually.
Persistent	Persistent	Store the compensation data persistently on the device. The compensation will be loaded automatically on every power-up.
Start (Hz)	0 to 5 MHz	Lower frequency limit used for calibration. For measurements at lower frequencies than this value, an extrapolation is performed.
Stop (Hz)	0 to 5 MHz	Upper frequency limit used for calibration. This upper limit is essential for correct measurements.
Points	positive integer value	Number of frequency points used for compensation. For frequencies between two compensation frequencies the data will be interpolated.
Precision		Precision of measurement during compensation. Using high precision can improve the quality of the compensation data in case the measurement is limited by noise.
	Standard	Standard filter and averaging settings are applied during compensation measurement
	High	Lower filter bandwidth and more averaging are used during compensation measurement

5.6. Numeric Tab

The Numeric tab provides a powerful time domain based measurement display as introduced in [Unique Set of Analysis Tools](#). It is available on all MFIA instruments.

5.6.1. Features

- Display of demodulator output data and other streamed data, e.g. auxiliary inputs, demodulator frequencies
- Graphical and numerical range indicators
- Polar and Cartesian formats
- Support for Input Scaling and Input Units

5.6.2. Description

The Numeric tab serves as the main numeric overview display of multiple measurement data. The display can be configured by both choosing the values displayed and also rearrange the display tiles

by drag-and-drop. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.16: App icon and short description

Control/Tool	Option/Range	Description
Numeric		Access to all continuously streamed measurement data as numerical values.

The numeric tab (see [Figure 5.13](#)) is divided into a display section on the left and a configuration section on the right. The configuration section is further divided into a number of sub-tabs.

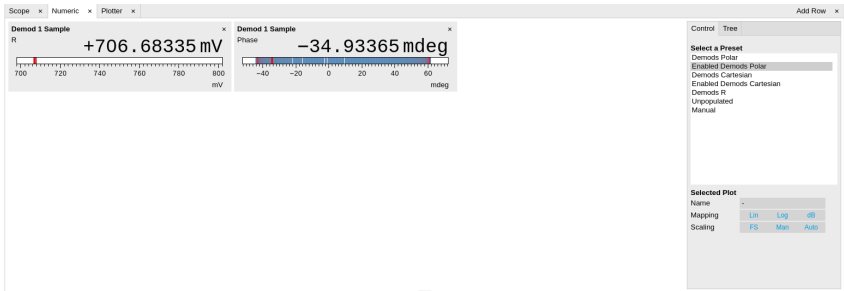


Figure 5.13: LabOne UI: Numeric tab

The numeric tab can be deployed to display the impedance parameters and demodulated current and voltage signals, phase, frequency as well as the signal levels at the auxiliary inputs. By default, the user can display the demodulated data either in polar coordinates (R, θ) or in Cartesian coordinates (X, Y) which can be toggled using the presets. To display other measurement quantities as available from any of the presets simply click on the tree tab next to the preset tab. The desired display fields can be selected under each demodulator's directory tree structure.

5.6.3. Functional Elements

Table 5.17: Numeric tab: Presets sub-tab

Control/Tool	Option/Range	Description
Select a Preset		Select numerical view based on a preset. Alternatively, the displayed value may also selected based on tree elements.
	Demods Polar	Shows R and Phase of all demodulators.
	Enabled Demods Polar	Shows R and Phase of enabled demodulators.
	Demods Cartesian	Shows X and Y of all demodulators.
	Enabled Demods Cartesian	Shows X and Y of enabled demodulators.
	Demods R	Shows R of all demodulators.
	Enabled Impedances	Shows impedance, model parameters, and frequency.
	PID Errors	Shows error of all PID.
	Unpopulated	Shows no signals.
	Manual	If additional signals are added or removed the active preset gets manual.

For the Tree sub-tab please see [the section called "Tree Selector"](#).

Table 5.18: Numeric tab: Settings sub-tab

Control/Tool	Option/Range	Description
Name	text label	Name of the selected plot(s). The default name can be changed to reflect the measured signal.

Control/Tool	Option/Range	Description
Mapping		Mapping of the selected plot(s)
	Lin	Enable linear mapping.
	Log	Enable logarithmic mapping.
	dB	Enable logarithmic mapping in dB.
Scaling	Manual/Full Scale	Scaling of the selected plot(s)
Zoom To Limits		Adjust the zoom to the current limits of the displayed histogram data.
Start Value	numeric value	Start value of the selected plot(s). Only visible for manual scaling.
Stop Value	numeric value	Stop value of the selected plot(s). Only visible for manual scaling.

5.7. Plotter Tab

The Plotter is one of the powerful time-domain measurement tools as introduced in [Unique Set of Analysis Tools](#) and is available on all MFIA instruments.

5.7.1. Features

- Vertical axis grouping for flexible axis scaling
- Polar and Cartesian data format
- Histogram and Math functionality for data analysis
- 4 cursors for data analysis
- Support for Input Scaling and Input Units

5.7.2. Description

The Plotter serves as graphical display for time domain data in a roll mode, i.e. continuously without triggering. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.19: App icon and short description

Control/Tool	Option/Range	Description
Plotter		Displays various continuously streamed measurement data as traces over time (roll mode).

The Plotter tab (see [Figure 5.14](#)) is divided into a display section on the left and a configuration section on the right.

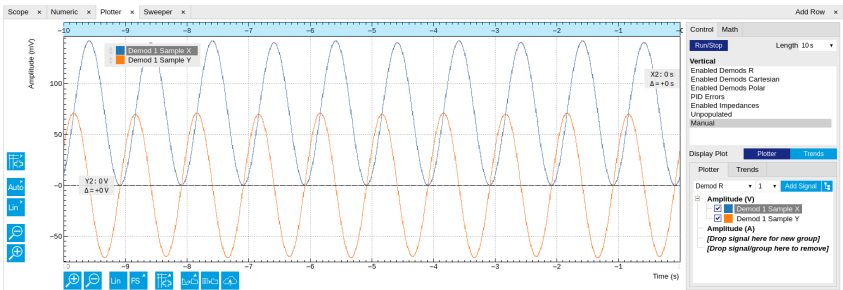


Figure 5.14: LabOne UI: Plotter tab

The Plotter can be used to monitor the evolution of demodulated data and other streamed data continuously over time. Just as in the numeric tab any continuously streamed quantity can be displayed, for instance R, θ , X, Y, frequency, and others. New signals can be added by either using the presets in the Control sub-tab or by going through the tree and selecting the signals of interest in the tree structure. The vertical and horizontal axis can be displayed in Lin, Log or dB scale. The

Plotter display can be zoomed in and out with the magnifier symbols, or through Man (Manual), Auto (Automatic) and FS (Full Scale) button settings (see also [Plot Functionality](#)). The maximum duration data is kept in the memory can be defined through the window length parameter in the Settings sub-tab. The window length also determines the file size for the Record Data functionality.

Note

Setting the window length to large values when operating at high sampling rates can lead to memory problems at the computer hosting the data server.

The sampling rate of the demodulator data is determined by the Rate value in Sa/s set in the Lock-in tab. The Plotter data can be continuously saved to disk by clicking the record button in the Config tab which will be indicated by a green Recording (REC) LED in the status bar. See [Saving and Loading Data](#) for more information on data saving.

5.7.3. Functional Elements

Table 5.20: Plotter tab: Control sub-tab

Control/Tool	Option/Range	Description
Run/Stop		Start and stop continuous data plotting (roll mode)
Select a Preset		Select a pre-defined group signals. A signal group is defined by a common unit and signal type. They should have the same scaling behavior as they share a y-axis. Split a group if the signals have different scaling properties.
	Enabled Demods R	Selects the amplitude of all enabled demodulators.
	Enabled Demods Cartesian	Selects X and Y of all enabled demodulators.
	Enabled Demods Polar	Selects amplitude and phase of all enabled demodulators.
	Enabled Impedances	Shows impedance, model parameters, and frequency.
	PID Errors	Selects the error of all PID.
	Unpopulated	Shows no signals.
	Manual	Selects the signals as defined in the tree sub-tab.

For the Vertical Axis Groups, please see [the table "Vertical Axis Groups description"](#) in the section called ["Vertical Axis Groups"](#).

For the Math sub-tab please see [the table "Plot math description"](#) in the section called ["Cursors and Math"](#).

5.8. Scope Tab

The Scope is a powerful time domain and frequency domain measurement tool as introduced in [Unique Set of Analysis Tools](#) and is available on all MFIA instruments.

5.8.1. Features

- One input channel with 16 kSa of memory; upgradable to two channels with 2.5 MSa memory per channel (MF-DIG option)
- 16 bit nominal resolution
- Segmented recording (requires MF-DIG option)
- Colorscale display for imaging (requires MF-DIG option)
- 24 bit high-definition mode (MF-DIG option)
-

Fast Fourier Transform (FFT): up to 60 MHz span, spectral density and power conversion, choice of window functions

- Sampling rates from 1.83 kSa/s to 60 MSa/s; up to 270 μ s acquisition time at 60 MSa/s or 8.9 s at 1.83 kSa/s
- 8 signal sources including Signal Inputs (I/V) and Trigger Inputs; up to 8 trigger sources and 2 trigger methods
- Independent hold-off, hysteresis, pre-trigger and trigger level settings`
- Support for Input Scaling and Input Units

5.8.2. Description

The Scope tab serves as the graphical display for time domain data. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.21: App icon and short description

Control/Tool	Option/Range	Description
Scope		Displays shots of data samples in time and frequency domain (FFT) representation.

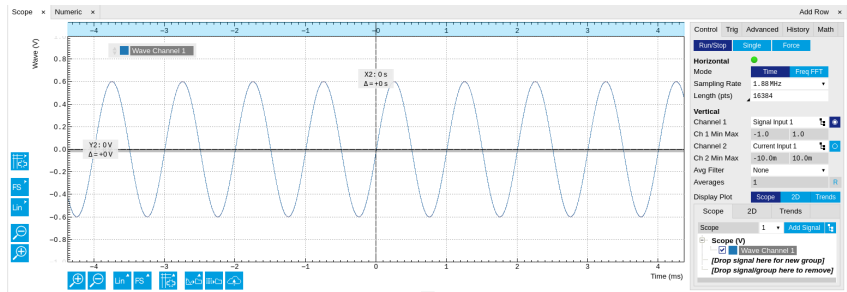


Figure 5.15: LabOne UI: Scope tab - Time domain

The Scope tab consists of a plot section on the left and a configuration section on the right. The configuration section is further divided into a number of sub-tabs. It gives access to a single-channel oscilloscope that can be used to monitor a choice of signals in the time or frequency domain. Hence the X axis of the plot area is time (for time domain display, [Figure 5.15](#)) or frequency (for frequency domain display, [Figure 5.17](#)). It is possible to display the time trace and the associated FFT simultaneously by opening a second instance of the Scope tab. The Y axis displays the selected signal that can be modified and scaled using the arbitrary input unit feature found in the Lock-in tab.

The Scope records data from a single channel at up to 60 MSa/s. The channel can be selected among the two Signal Inputs, Auxiliary Inputs, Trigger Inputs and Demodulator Oscillator Phase. The Scope records data sets of up to 16 kSa samples in the standard configuration, which corresponds to an acquisition time of 270 μ s at the highest sampling rate.

The product of the inverse sampling rate and the number of acquired points (Length) determines the total recording time for each shot. Hence, longer time intervals can be captured by reducing the sampling rate. The Scope can perform sampling rate reduction either using decimation or BW Limitation as illustrated in [Figure 5.16](#). BW Limitation is activated by default, but it can be deactivated in the Advanced sub-tab. The figure shows an example of an input signal at the top, followed by the Scope output when the highest sample rate of 60 MSa/s is used. The next signal shows the Scope output when a rate reduction by a factor of 4 (i.e. 15 MSa/s) is configured and the rate reduction method of decimation is used. For decimation, a rate reduction by a factor of N is performed by only keeping every Nth sample and discarding the rest. The advantage of this method is its simplicity, but the disadvantage is that the signal is undersampled because the input filter bandwidth of the MFIA instrument is fixed at 7 MHz. As a consequence, the Nyquist sampling criterion is no longer satisfied and aliasing effects may be observed. The default rate reduction mechanism of BW Limitation is illustrated by the lowermost signal in the figure. BW Limitation means that for a rate reduction by a factor of N, each sample produced by the Scope is computed as the average of N samples acquired at the maximum sampling rate. The effective signal bandwidth is thereby reduced and aliasing effects are largely suppressed. As can be seen from the figure, with a rate reduction by a factor of 4, every output sample is simply computed as the average of 4 consecutive samples acquired at 60 MSa/s.

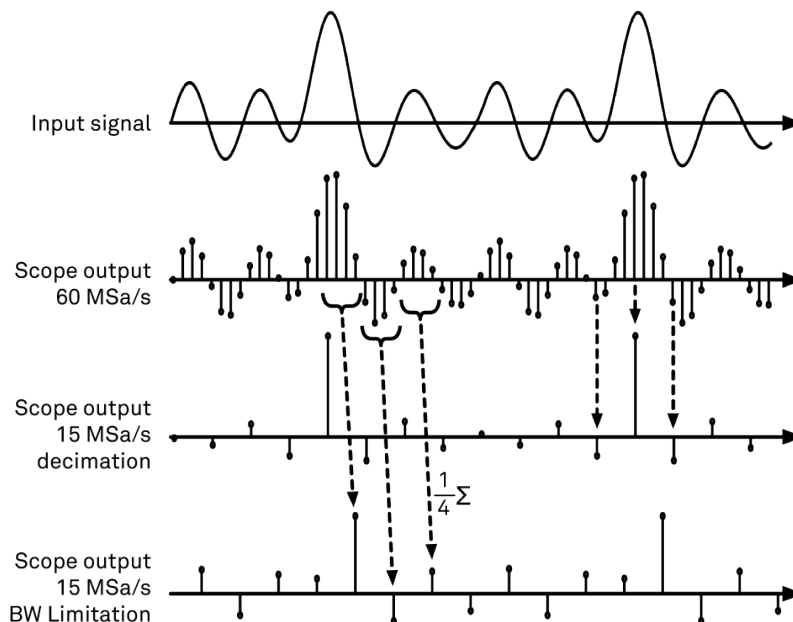


Figure 5.16: Illustration of how the Scope output is generated in BW Limitation and decimation mode when the sampling rate is reduced from the default of 60 MSa/s to 15 MSa/s

The frequency domain representation is activated in the Control sub-tab by selecting Freq Domain FFT as the Horizontal Mode. It allows the user to observe the spectrum of the acquired shots of samples. All controls and settings are shared between the time domain and frequency domain representations.

The Scope supports averaging over multiple shots. The functionality is implemented by means of an exponential moving average filter with configurable filter depth. Averaging helps to suppress noise components that are uncorrelated with the main signal. It is particularly useful in combination with the Frequency Domain FFT mode where it can help to reveal harmonic signals and disturbances that might otherwise be hidden below the noise floor.

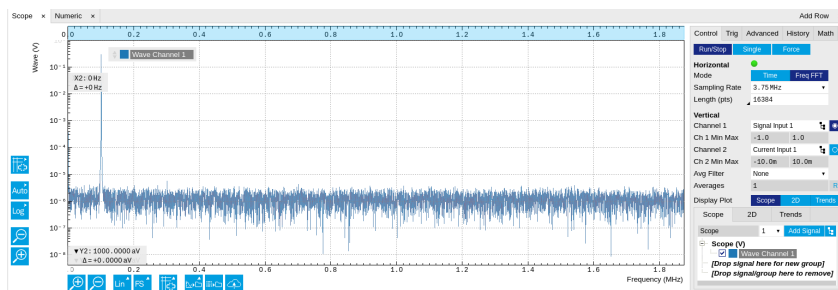


Figure 5.17: LabOne UI: Scope tab - Frequency domain

The Trigger sub-tab offers all the controls necessary for triggering on different signal sources. When the trigger is enabled, then oscilloscope shots are acquired whenever the trigger conditions are met. Trigger and Hysteresis levels can be indicated graphically in the plot. A disabled trigger is equivalent to continuous oscilloscope shot acquisition.

Digitizer upgrade option

The MF-DIG Digitizer option greatly enhances the performance of the Scope with the addition of the following features

- Simultaneous recording of two Scope channels
- Memory depth of 5 MSa for both Scope channels
- Additional input signal sources (Demodulator X, Y, R, and Θ)
- Segmented recording
- Imaging support with colorscale display
- XY display of two channels
- Trigger gating
- Resolution increase up to 24 bit with high-definition (HD) mode
- Additional trigger input sources that allow for cross-domain triggering

- Additional trigger/marker output sources based on the state of the Scope
- Continuous scope data streaming

This additional functionality can be enabled on any MFIA instrument with an option key. Please contact Zurich Instruments to get more information. The following sections explain the Digitizer features in more detail.

Two channels and extended memory depth

The MF-DIG option enables simultaneous dual-channel recording. This allows for very exact relative timing measurements. With the XY display, it's possible to plot two signals against each other, e.g. for quick visualization of phase offsets or characteristic curves. Each channel can be assigned a different signal source. Trigger settings, sampling rate, and recording length settings are shared between both channels. An increased shot length of up to 5 MSa compared to the standard 16 kSa allows for longer recording times and FFTs with finer frequency resolution for the same frequency span.

Additional input sources

Besides the Signal Input, Trigger Input, Auxiliary Input, and Oscillator Phase the MF-DIG option also allows for recording of Demodulator X, Y, R, and Θ signals. This functionality is very powerful in that it allows short bursts to be recorded with very high sampling rates. In order to optimally use the vertical resolution, the upper and lower limit of these input signals should be chosen appropriately. Before sampling, a scaling and an offset are applied to the input signal in order to get 16 bit resolution between the lower and upper limit. The applied scaling and offset values are transferred together with the scope data, which allows for recovery of the original physical signal strength in absolute values. For directly sampled input signals like the Signal Inputs or Trigger Inputs, the limits are read-only values and reflect the selected input range.

Segmented data recording and imaging

The segmented data recording mode allows for intermediate storage of a burst of up to 1024 scope shots, called segments, in the scope memory. In this way, the Scope does not have to wait after each shot until the data transfer from the scope memory to the internal computer is completed. In segmented data recording mode, the Scope provides a two-dimensional color scale display of the data particularly useful for imaging applications. When used over the API, the data of each shot will contain information on the segment number.

High-definition mode

Sampling rate reduction with BW Limitation brings the advantage of increasing the available vertical resolution beyond the 16 bit resolution of the ADC effectively up to 23.5 bit (high-definition mode). [Table 5.22](#) shows the nominal scope resolution depending on the sampling rate.

Table 5.22: Scope resolution increase with MF-DIG Digitizer option

Scope Sampling Frequency	Scope vertical resolution (with MF-DIG option)
60 MSa/s	16 bits
30 MSa/s	16.5 bits
15 MSa/s	17 bits
7.5 MSa/s	17.5 bits
3.75 MSa/s	18 bits
1.88 MSa/s	18.5 bits
938 kSa/s	19 bits

Scope Sampling Frequency	Scope vertical resolution (with MF-DIG option)
469 kSa/s	19.5 bits
234 kSa/s	20 bits
117 kSa/s	20.5 bits
58.6 kSa/s	21 bits
29.3 kSa/s	21.5 bits
14.6 kSa/s	22 bits
7.32 kSa/s	22.5 bits
3.66 kSa/s	23 bits
1.83 kSa/s	23.5 bits

Trigger gating

With the MF-DIG option installed the user can make full use of the Trigger Engine. If trigger gating is enabled, a trigger event will only be accepted if the gating input is active.

Additional trigger input sources

By using a Demodulator signal as trigger source, the Scope can be used in a cross-domain triggering mode. This allows, for example, for time domain signals to be recorded in a synchronous fashion triggered by the result from analyzing a signal in the frequency domain by means of a demodulator.

Note

Choose a negative delay (pre-trigger) to compensate for the delay introduced by the Demodulator.

Scope state output on Trigger Output

The MF-DIG option extends the list of available Trigger Outputs by the six elements: Scope Trigger, Scope Armed, Scope Active and their logically inverse signals. The Trigger Output signals are controlled on the DIO tab ([DIO Tab](#)). [Figure 5.18](#) shows an illustration of the signal that will be generated on the Trigger Output when one of the six new Scope-related sources is selected. An example input signal is shown at the top of the figure. It is assumed that the Scope is configured to trigger on this input signal on a rising edge crossing the level indicated by the stippled line.

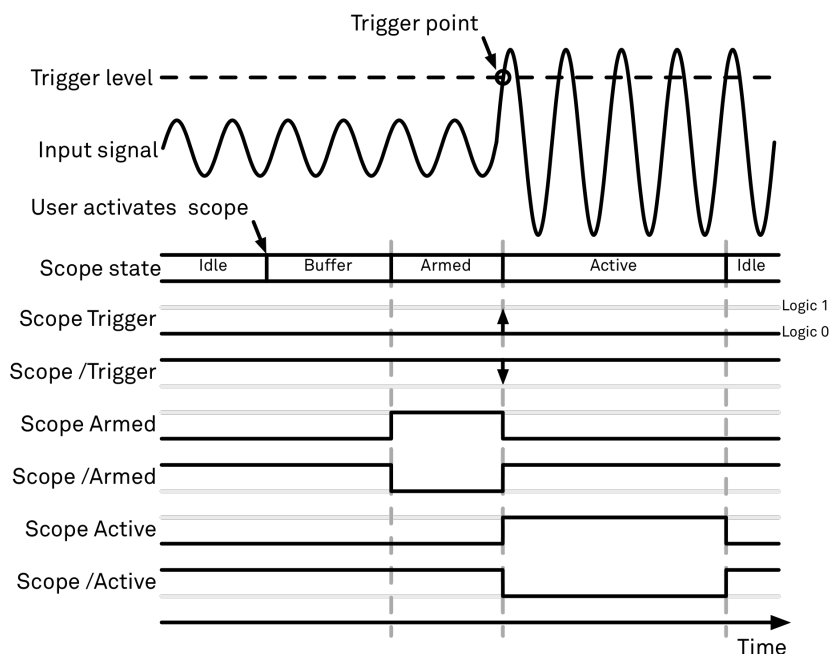


Figure 5.18: Illustration of the signal that will appear on the Trigger Output when one of the six Scope related sources is selected.

The Scope can be thought of as having a state, which changes over time. The state is shown below the input signal in the figure. When the Scope is completely inactive, it is said to be in the Idle state. When the user then activates the Scope, it will transition into a Buffer state. In this state the Scope will start to record the input signal. It will remain in this state until sufficient data has been recorded to fulfill the user requirement for recording data prior to the trigger point as controlled by the trigger Reference and Delay fields in the user interface. Once sufficient data has been recorded, the Scope will transition to the Armed state. In this state the Scope is ready to accept the trigger signal. Note that the Scope will continue to record data for as long as it is in the Armed state, and that if no trigger is defined, the Scope will simply pass straight through the Armed state. Once the input signal passes the Trigger level the Scope will trigger, and at the same time its state will change from Armed to Active. The Scope will remain in the Active state, where it also records data, until sufficient data has been recorded to fulfill the Length requirement configured in the user interface. Once enough data has been acquired, the Scope will transition back into the Idle state where it will wait for the time configured with the Hold-off time before it either starts the next measurement automatically (in case Run is active) or waits for the user to reactivate it.

The trigger source selector allows information about the Scope state to be reproduced on the Trigger Output in a number of ways. The signal that will appear on the output is shown with the six bottommost traces in the figure. Note that these traces are shown as digital signals with symbolic values of logic 0 and 1. These values will of course be actual voltages when measured on the device itself.

First, if Scope Trigger is selected then the trigger output will have a signal that is asserted, which means that it goes high, when the scope triggers, i.e. changes from the Armed to the Active state. The signal will normally have a very short duration and, therefore, it is shown with an arrow in the figure. The duration can be increased by means of the Width input field, which can be found next to the Output Signal selector on the DIO tab. If Scope/Trigger is selected, then the same signal will appear on the output, but it will simply be inverted logically.

Next, if the Scope Armed source is selected, the trigger output will be asserted as long as the Scope is in the Armed state. Again, this means that the Scope has recorded enough data to proceed with the acquisition and is waiting for the trigger condition to become satisfied. In this example, since a rising edge trigger is defined, the trigger condition becomes satisfied when the input signal goes from below the trigger level to above the trigger level.

Similarly, if Scope /Armed is selected, the trigger output will be asserted (i.e. at logic 1) whenever the Scope is in a state different from the Armed state. The same explanation holds for the remaining two configuration options, except here the trigger output is asserted when the Scope is in the Active state or when it is not in the Active state.

5.8.3. Functional Elements

Table 5.23: Scope tab: Control sub-tab

Control/Tool	Option/Range	Description
Run/Stop		Runs the scope/FFT continuously.
Single		Acquires a single shot of samples.
Force		Force a trigger event.
Mode	Freq Domain (FFT)	Switches between time and frequency domain display.
	Time Domain	
Sampling Rate	916 Sa/s to 60 MSa/s	Defines the sampling rate of the scope. The numeric values are rounded for display purposes. The exact values are equal to the base sampling rate divided by 2^n , where n is an integer.
Sampling Rate		Defines the sampling rate of the scope. The numeric values are rounded for display purposes. The exact values are equal to the base sampling rate divided by 2^n , where n is an integer. Warning: Due to the lack of sample averaging feature, reduced sampling rates can cause aliasing and thus artifacts in the signal spectrum. Currently, the Scope tool only supports sample decimation, but in the future it will also offer sample averaging.
Length Mode		Switches between length and duration display.
	Length (pts)	The scope shot length is defined in number of samples. The duration is given by the number of samples divided by the sampling rate. The MF-DIG option greatly increases the available length.
	Duration (s)	The scope shot length is defined as a duration. The number of samples is given by the duration times the sampling rate.
Length (pts) or Duration (s)	numeric value	Defines the length of the recorded scope shot. Use the Length Mode to switch between length and duration display.
Channel 1/2		Selects the signal source for the corresponding scope channel. Navigate through the tree view that appears and click on the required signal. Note: Channel 2 requires the DIG option.
Min	numeric value	Lower limit of the scope full scale range. For demodulator, PID, Boxcar, and AU signals the limit should be adjusted so that the signal covers the specified range to achieve optimal resolution.
Max	numeric value	Upper limit of the scope full scale range. For demodulator, PID, Boxcar, and AU signals the limit should be adjusted so that the signal covers the specified range to achieve optimal resolution.
Enable	ON / OFF	Activates the display of the corresponding scope channel. Note: Channel 2 requires the DIG option.
Average Filter		Enable Exponential Moving Average (EMA) filter that is applied when the average of several scope shots is computed and displayed. Depending on the mode, the source data for averaging is either the Time or the Freq FFT trace.
	Off	Averaging is turned off.
	On	Consecutive scope shots are averaged with an exponential weight.
Averages	integer value	The number of shots required to reach 63% settling. Twice the number of shots yields 86% settling.
Averages	integer value	The number of shots to average on the device before returning the data.
Reset		Resets the averaging filter.

For the Vertical Axis Groups, please see [the table "Vertical Axis Groups description" in the section called "Vertical Axis Groups"](#).

Table 5.24: Scope tab: Trigger sub-tab

Control/Tool	Option/Range	Description
Trigger	grey/green/yellow	When flashing, indicates that new scope shots are being captured and displayed in the plot area. The Trigger must not necessarily be enabled for this indicator to flash. A disabled trigger is equivalent to continuous acquisition. Scope shots with data loss are indicated by yellow. Such an invalid scope shot is not processed.
Enable	ON / OFF	When triggering is enabled scope data are acquired every time the defined trigger condition is met. If disabled, scope shots are acquired continuously.
Signal		Selects the trigger source signal. Navigate through the tree view that appears and click on the required signal.
Slope	Rising or falling edge trigger	Select the signal edge that should activate the trigger.
	None	
	Rising edge trigger	
	Falling edge trigger	
Level (V)	trigger signal range (negative values permitted)	Defines the trigger level.
Hysteresis Mode		Selects the mode to define the hysteresis strength. The relative mode will work best over the full input range as long as the analog input signal does not suffer from excessive noise.
	Hysteresis (V)	Selects absolute hysteresis.
	Hysteresis (%)	Selects a hysteresis relative to the adjusted full scale signal input range.
Hysteresis (V)	trigger signal range (positive values only)	Defines the voltage the source signal must deviate from the trigger level before the trigger is rearmed again. Set to 0 to turn it off. The sign is defined by the Edge setting.
Hysteresis (%)	numeric percentage value (positive values only)	Hysteresis relative to the adjusted full scale signal input range. A hysteresis value larger than 100% is allowed.
Show Level	ON / OFF	If enabled shows the trigger level as grey line in the plot. The hysteresis is indicated by a grey box. The trigger level can be adjusted by drag and drop of the grey line.
Trigger Gating		Select the signal source used for trigger gating if gating is enabled. This feature requires the MF-DIG option.
	Trigger In 1 High	Only trigger if the Trigger Input 1 is at high level.
	Trigger In 1 Low	Only trigger if the Trigger Input 1 is at low level.
	Trigger In 2 High	Only trigger if the Trigger Input 2 is at high level.
	Trigger In 2 Low	Only trigger if the Trigger Input 2 is at low level.
Trigger Gating Enable	ON / OFF	If enabled the trigger will be gated by the trigger gating input signal. This feature requires the MF-DIG option.
Holdoff Mode		Selects the holdoff mode.
	Holdoff (s)	Holdoff is defined as time.
	Holdoff (events)	Holdoff is defined as number of events.
Holdoff (s)	numeric value	Defines the time before the trigger is rearmed after a recording event.

Control/Tool	Option/Range	Description
Holdoff (events)	1 to 1048575	Defines the trigger event number that will trigger the next recording after a recording event. A value one will start a recording for each trigger event.
Reference (%)	percent value	Trigger reference position relative to the plot window. Default is 50% which results in a reference point in the middle of the acquired data.
Delay (s)	numeric value	Trigger position relative to reference. A positive delay results in less data being acquired before the trigger point, a negative delay results in more data being acquired before the trigger point.
Enable	ON / OFF	Enable segmented scope recording. This allows for full bandwidth recording of scope shots with a minimum dead time between individual shots. This functionality requires the DIG option.
Segments	1 to 32768	Specifies the number of segments to be recorded in device memory. The maximum scope shot size is given by the available memory divided by the number of segments. This functionality requires the DIG option.
Shown Trigger	integer value	Displays the number of triggered events since last start.
Plot Type		Select the plot type.
	None	No plot displayed.
	2D	Display defined number of grid rows as one 2D plot.
	Row	Display only the trace of index defined in the Active Row field.
	2D + Row	Display 2D and row plots.
Active Row	integer value	Set the row index to be displayed in the Row plot.
Track Active Row	ON / OFF	If enabled, the active row marker will track with the last recorded row. The active row control field is read-only if enabled.
Palette	Solar	Select the colormap for the current plot.
	Viridis	
	Inferno	
	Balance	
	Turbo	
	Grey	
Colorscale	ON / OFF	Enable/disable the colorscale bar display in the 2D plot.
Mapping		Mapping of colorscale.
	Lin	Enable linear mapping.
	Log	Enable logarithmic mapping.
	dB	Enable logarithmic mapping in dB.
Scaling	Full Scale/ Manual/Auto	Scaling of colorscale.
Clamp To Color	ON / OFF	When enabled, grid values that are outside of defined Min or Max region are painted with Min or Max color equivalents. When disabled, Grid values that are outside of defined Min or Max values are left transparent.
Start	numeric value	Lower limit of colorscale. Only visible for manual scaling.
Stop	numeric value	Upper limit of colorscale. Only visible for manual scaling.

Table 5.25: Scope tab: Advanced sub-tab

Control/ Tool	Option/ Range	Description
FFT Window	Cosine squared (ring-down)	Several different FFT windows to choose from. Each window function results in a different trade-off between amplitude accuracy and spectral leakage. Please check the literature to find the window function that best suits your needs.
	Rectangular	
	Hann	
	Hamming	
	Blackman Harris	
	Flat Top	
	Exponential (ring-down)	
	Cosine (ring-down)	
Resolution (Hz)	mHz to Hz	Spectral resolution defined by the reciprocal acquisition time (sample rate, number of samples recorded).
Absolute Frequency	ON / OFF	Shifts x-axis labeling to show the absolute frequency in the center as opposed to 0 Hz, when turned off.
Spectral Density	ON / OFF	Calculate and show the spectral density. If power is enabled the power spectral density value is calculated. The spectral density is used to analyze noise.
Power	ON / OFF	Calculate and show the power value. To extract power spectral density (PSD) this button should be enabled together with Spectral Density.
Persistence	ON / OFF	Keeps previous scope shots in the display. The color scheme visualizes the number of occurrences at certain positions in time and amplitude by a multi-color scheme.
BW Limit		Selects between sample decimation and sample averaging. Averaging avoids aliasing, but may conceal signal peaks. Channel 2 requires the DIG option.
	OFF	Selects sample decimation for sample rates lower than the maximal available sampling rate.
	ON	Selects sample averaging for sample rates lower than the maximal available sampling rate.
Rate	1.83 kHz to 3.75 MHz	Streaming rate of the scope channels. The streaming rate can be adjusted independent from the scope sampling rate. The maximum rate depends on the interface used for transfer. Note: scope streaming requires the DIG option.
Enable	ON / OFF	Enable scope streaming for the specified channel. This allows for continuous recording of scope data on the plotter and streaming to disk. Note: scope streaming requires the DIG option.
X Axis		Select the x-axis for xy-plot display mode.
	Time/Freq	The xy-plot mode is off. The x-axis is either time or frequency.
	Channel 1	The xy-plot is enabled with the first channel used for the x-axis.
	Channel 2	The xy-plot is enabled with the second channel used for the x-axis.

Table 5.26: Scope tab: History sub-tab

Control/ Tool	Option/ Range	Description
History	History	Each entry in the list corresponds to a single trace in the history. The number of traces displayed in the plot is limited to 20. Use the toggle buttons to hide or show individual traces. Use the color picker to change the color of a trace in the plot. Double click on a list entry to edit its name.

Control/Tool	Option/Range	Description
Length	integer value	Maximum number of records in the history. The number of entries displayed in the list is limited to the 100 most recent ones.
Clear All		Remove all records from the history list.
Clear		Remove selected records from the history list.
Load file		Load data from a file into the history. Loading does not change the data type and range displayed in the plot, this has to be adapted manually if data is not shown.
Name		Enter a name which is used as a folder name to save the history into. An additional three digit counter is added to the folder name to identify consecutive saves into the same folder name.
Auto Save		Activate autosaving. When activated, any measurements already in the history are saved. Each subsequent measurement is then also saved. The autosave directory is identified by the text "autosave" in the name, e.g. "sweep_autosave_001". If autosave is active during continuous running of the module, each successive measurement is saved to the same directory. For single shot operation, a new directory is created containing all measurements in the history. Depending on the file format, the measurements are either appended to the same file, or saved in individual files. For HDF5 and ZView formats, measurements are appended to the same file. For MATLAB and SXM formats, each measurement is saved in a separate file.
File Format		Select the file format in which to save the data.
Save		Save the traces in the history to a file accessible in the File Manager tab. The file contains the signals in the Vertical Axis Groups of the Control sub-tab. The data that is saved depends on the selection from the pull-down list. Save All: All traces are saved. Save Sel: The selected traces are saved.

For the Math sub-tab please see [the table "Plot math description"](#) in the section called "Cursors and Math".

5.9. Data Acquisition Tab

The Data Acquisition tool is one of the powerful time domain measurement tools as introduced in [Unique Set of Analysis Tools](#) and is available on all MFIA instruments. This tab used to be named Software Trigger tab in previous versions of the LabOne software.

5.9.1. Features

- Time-domain and frequency domain display for all continuously streamed data
- Capture and color scale display of imaging data
- Frame averaging and pixel interpolation
- Automatic trigger level determination
- Display of multiple traces
- Adjustable record history
- Mathematical toolkit for signal analysis

5.9.2. Description

The Data Acquisition tab features display and recording of shot-wise and imaging data sets upon a trigger event. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.27: App icon and short description

Control/ Tool	Option/ Range	Description
DAQ		Provides complex trigger functionality on all continuously streamed data samples and time domain display.

The Data Acquisition tab (see [Figure 5.19](#)) is divided into a display section on the left and a configuration section on the right. The configuration section is further divided into a number of sub-tabs.

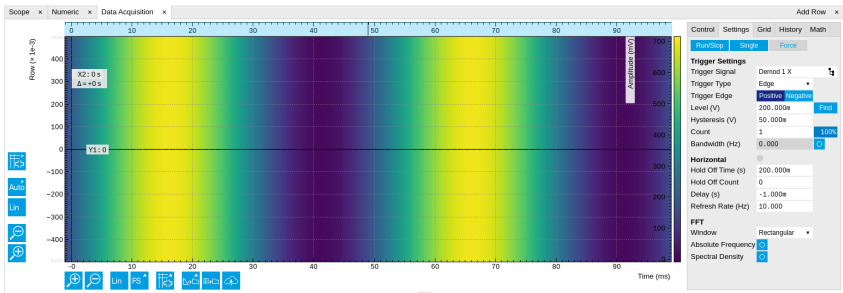


Figure 5.19: LabOne UI: Data Acquisition tab

The Data Acquisition tool brings the trigger functionality of a scope with FFT capability to the demodulator signals and other streamed data. The user can choose between a variety of different trigger and display options in the time and frequency domain.

Use the **Control** sub-tab to configure which signals are measured, both in time and in frequency domain. Measurement signals can be added to the Vertical Axis Groups section as described in [Vertical Axis Groups](#). There is one vertical axis group for each the time domain and the frequency domain data.

The trigger condition is configured in the **Settings** sub-tab. Among the selection of Trigger Types provided here, Edge and Pulse are applicable to analog trigger sources such as demodulator data, auxiliary voltages, or oscillator frequencies. The trigger time resolution is enhanced above the sampling rate of the analog data by using interpolation. Instead of manually setting a Trigger Level, you can click on **Find** to have LabOne find a value by analyzing the data stream. In case of noisy trigger sources, both the Bandwidth and the Hysteresis setting can help preventing false trigger events. The Bandwidth setting provides a configurable low-pass filter applied to the trigger source. When enabling this function, be sure to choose a sufficiently high bandwidth to resolve the signal feature that should be triggered upon, i.e., the signal edge or pulse. The Bandwidth setting does not affect the recorded data.

For trigger sources with a slowly varying offset, the Tracking Edge and Tracking Pulse Trigger Types provide continuous adjustment of the Level and Hysteresis. In Tracking mode, the Bandwidth setting plays a different role than for the Edge and Pulse trigger types. Here, the Bandwidth should be chosen sufficiently low to filter out all fast features and only let pass the slow offset. The Trigger Types HW Trigger and Digital are used for TTL signals on the Trigger Inputs or on the DIO lines. Using the Bits and Bit Mask setting, complex multi-bit trigger conditions on the DIO lines can be defined. The timing resolution for digital triggers is given by the demodulator sample rate because the state of the DIO line is transferred together with demodulator data.

The Horizontal section of the Settings sub-tab contains the settings for shot Duration and Delay (negative delays correspond to pre-trigger time). Also minimum and maximum pulse width for the Pulse and Tracking Pulse trigger types are defined here.

The **Grid** sub-tab provides imaging functionality to capture and display two-dimensional data sets organized in frames consisting of rows and columns. By default, the number of rows is 1, which means the Data Acquisition tool operates similar to a scope. With a Rows setting larger than 1, every newly captured shot of data is assigned to a row until the number of rows is reached and the frame is complete. After completion of a full frame, the new data either replace the old or averaging is performed, according to the selected Operation and Repetitions setting. On the horizontal axis, the Duration of a shot is divided into a number of samples specified with the Columns setting. The Mode settings provides the functionality for post-processing of the streamed data for interpolation, resampling, and alignment with the trigger event. This is particularly helpful when capturing data from several sources, e.g. demodulators and PID controllers. As illustrated in [Figure 5.20](#), in such situation the streamed data don't lie on the same temporal grid by default. This can be changed by setting Mode to Linear or Nearest. In these modes, the streams from several sources will be up-sampled to match the sampling rate and temporal grid of the fastest data stream. This means data processing after saving becomes more convenient, however note that the actual streamed data rate is not increased, and the data don't gain in time resolution. A two-dimensional color scale image of

the data can be enabled and controlled in the Display section. The display features configurable scaling, range, and color scale.

With enabled grid mode, the data of a completed frame after averaging appear as a list entry in the **History** sub-tab. See [History List](#) for more details on how data in the history list can be managed and stored.

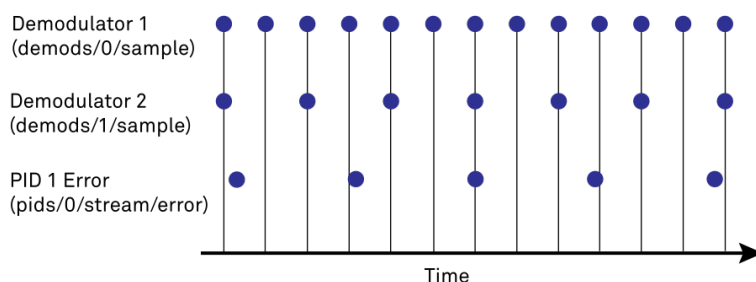


Figure 5.20: Samples from different sources configured with different rates: demodulator 1 at 2N kSa/s, demodulator 2 at N kSa/s and PID Error 1 at M kSa/s (N not divisible by M). Although each stream consists of equidistantly spaced samples in time, the sample timestamps from different streams are not necessarily aligned due to the different sampling rates

5.9.3. Functional Elements

Table 5.28: DAQ tab: Control sub-tab

Control/Tool	Option/Range	Description
Run/Stop		Start and stop the Data Acquisition tool
Single		Run the Data Acquisition tool once (record Count trigger events)
Force		Forces a single trigger event.
Triggered	grey/green	When green, indicates that new trigger shots are being captured and displayed in the plot area.

For the Vertical Axis Groups, please see [the table "Vertical Axis Groups description" in the section called "Vertical Axis Groups"](#).

Table 5.29: DAQ tab: Settings sub-tab

Control/Tool	Option/Range	Description
Trigger Signal		Source signal for trigger condition. Navigate through the tree view that appears and click on the required signal.
Trigger Type		Select the type of trigger to use. Selectable options depend on the selected trigger signal.
	Continuous	Continuous triggering.
	Edge	Analog edge triggering based on high and low level. Hysteresis on the levels and low-pass filtering can be used to reduce the risk of wrong trigger for noisy trigger signals.
	Digital	Digital triggering on the 32-bit DIO lines. The bit value defines the trigger condition. The bit mask controls the bits that are used for trigger evaluation. When using a Positive Edge trigger setting, a trigger event occurs as soon as the equality $(DIO\ Value) \text{ AND } (Bit\ Mask) = (Bits) \text{ AND } (Bit\ Mask)$ is fulfilled (and was not previously fulfilled). In order to trigger on DIO0 set bit value to 1 and bit mask to 1; to trigger on DIO1 set bit value to 2 and bit mask to 2.
	Pulse	Triggers if a pulse on an analog signal is within the min and max pulse width. Pulses can be defined as either low to high then high to low (positive), the reverse (negative) or both.

Control/Tool	Option/Range	Description
	Tracking Edge	Edge triggering with automatic adjustment of trigger levels to compensate for drifts. The tracking speed is controlled by the bandwidth of the low-pass filter. For this filter noise rejection can only be achieved by level hysteresis.
	HW Trigger	Trigger on one of the four trigger inputs. Ensure that the trigger level and the trigger coupling is correctly adjusted. The trigger input state can be monitored on the plotter.
	Tracking Pulse	Pulse triggering with automatic adjustment of trigger levels to compensate for drifts. The tracking speed is controlled by the bandwidth of the low-pass filter. For this filter noise rejection can only be achieved by level hysteresis.
Pulse Type	Positive/ Negative/ Both	Select between negative, positive or both pulse forms in the signal to trigger on.
Trigger Edge	Positive/ Negative/ Both	Triggers when the trigger input signal is crossing the trigger level from either high to low, low to high or both. This field is only displayed for trigger type Edge, Tracking Edge and Event Count.
Bits	0 to $2^{32}-1$	Specify the value of the DIO to trigger on. All specified bits have to be set in order to trigger. This field is only displayed for trigger type Digital.
Bit Mask	0 to $2^{32}-1$	Specify a bit mask for the DIO trigger value. The trigger value is bits AND bit mask (bitwise). This field is only displayed for trigger type Digital.
Level	full signal range	Specify the trigger level value.
Find		Automatically find the trigger level based on the current signal.
Hysteresis	full signal range	The hysteresis is important to trigger on the correct edge in the presence of noise. The hysteresis is applied below the trigger level for positive trigger edge selection. It is applied above for negative trigger edge selection, and on both sides for triggering on both edges.
Count	integer number	Number of trigger events to record (in Single mode)
Trigger progress	0% to 100%	The percentage of triggers already acquired (in Single mode)
Bandwidth (Hz)	0 to $0.5 * \text{Sampling Rate}$	Bandwidth of the low-pass filter applied to the trigger signal. For edge and pulse trigger use a bandwidth larger than the signal sampling rate divided by 20 to keep the phase delay. For tracking filter use a bandwidth smaller than signal sampling frequency divided by 100 to just track slow signal components like drifts.
Enable	ON / OFF	Enable low-pass filtering of the trigger signal.
Hold Off Time (s)	positive numeric value	Hold off time before the trigger is rearmed. A hold off time smaller than the duration will lead to overlapping trigger frames.
Hold Off Count	integer value	Number of skipped triggers until the next trigger is recorded again.
Delay (s)	-Duration to Duration	Time delay of trigger frame position (left side) relative to the trigger edge. For delays smaller than 0, trigger edge inside trigger frame (pre trigger). For delays greater than 0, trigger edge before trigger frame (post trigger)
Refresh Rate	100 mHz to 10 Hz	Set the maximum refresh rate for plot updates. The actual refresh rate depends on other factors such as the hold-off time and duration.
Pulse Min (s)	0 to Duration	Minimum pulse width to trigger on.
Pulse Max (s)	0 to Duration	Maximum pulse width to trigger on.

Control/Tool	Option/Range	Description
Window	Cosine squared (ring-down)	Several different FFT windows to choose from. Depending on the application it makes a huge difference which of the provided window function is used. Please check the literature to find out the best trade off for your needs.
	Rectangular	
	Hann	
	Hamming	
	Blackman Harris	
	Flat Top	
	Exponential (ring-down)	
	Cosine (ring-down)	
Absolute Frequency	ON / OFF	Shifts x-axis labeling to show the demodulation frequency in the center as opposed to 0 Hz, when turned off.
Spectral Density	ON / OFF	Calculate and show the spectral density. If power is enabled the power spectral density value is calculated. The spectral density is used to analyze noise.

Table 5.30: DAQ tab: Grid sub-tab

Control/Tool	Option/Range	Description
Mode		Select resampling method for two-dimensional data recording.
	Off	Two-dimensional data recording is disabled.
	Nearest	Resampling is performed using substitution by closest data point.
	Linear	Resampling is performed using linear interpolation.
	Exact (on-grid)	Adjust the duration so that the grid distance matches the maximal sampling rate of the selected signals. This allows for on-grid sampling of measurement data. If a signal uses lower sampling rate it will be up-sampled by linear interpolation.
On Grid Sampling	Green or yellow	When green, indicates that all the captured data is aligned to the grid. When yellow, indicates that some data is not aligned to the grid and is interpolated. This can happen when one or more data sources have different sampling rates, or when a sampling rate changes.
Operation		Select row update method.
	Replace	New row replaces old row.
	Average	The data for each row is averaged over a number of repetitions.
	Std	The data for each row is the standard deviation over a number of repetitions.
Columns	numeric value	Number of columns. The data along the horizontal axis are resampled to a number of samples defined by this setting.
Duration	up to 1000 s	Recording length for each triggered data set. In exact sampling mode the duration is a read-only field. The duration is then defined by the maximal sampling rate and column size.
Rows	numeric value	Number of rows
Scan Direction		Select the scan direction and mode
	Forward	Scan direction from left to right
	Reverse	Scan direction from right to left
	Bidirectional	Alternate scanning in both directions

Control/Tool	Option/Range	Description
Repetitions	numeric value	Number of repetitions used for averaging
Row-wise repetition	ON / OFF	Enable row-wise repetition. With row-wise repetition, each row is calculated from successive repetitions before starting the next row. With grid-wise repetition, the entire grid is calculated with each repetition.
Waterfall	ON / OFF	Enable to show the 2D plot in waterfall mode. It will always update the last line.
Overwrite	ON / OFF	Enable to overwrite the grid in continuous mode. History will not be collected. A history element will only be created when the analysis is stopped.
Plot Type		Select the plot type.
	None	No plot displayed.
	2D	Display defined number of grid rows as one 2D plot.
	Row	Display only the trace of index defined in the Active Row field.
	2D + Row	Display 2D and row plots.
Active Row	integer value	Set the row index to be displayed in the Row plot.
Track Active Row	ON / OFF	If enabled, the active row marker will track with the last recorded row. The active row control field is read-only if enabled.
Palette	Solar	Select the colormap for the current plot.
	Viridis	
	Inferno	
	Balance	
	Turbo	
	Grey	
Colorscale	ON / OFF	Enable/disable the colorscale bar display in the 2D plot.
Mapping		Mapping of colorscale.
	Lin	Enable linear mapping.
	Log	Enable logarithmic mapping.
	dB	Enable logarithmic mapping in dB.
Scaling	Full Scale/ Manual/Auto	Scaling of colorscale.
Clamp To Color	ON / OFF	When enabled, grid values that are outside of defined Min or Max region are painted with Min or Max color equivalents. When disabled, Grid values that are outside of defined Min or Max values are left transparent.
Start	numeric value	Lower limit of colorscale. Only visible for manual scaling.
Stop	numeric value	Upper limit of colorscale. Only visible for manual scaling.

Table 5.31: DAQ tab: History sub-tab

Control/Tool	Option/Range	Description
History	History	Each entry in the list corresponds to a single trace in the history. The number of traces displayed in the plot is limited to 20. Use the toggle buttons to hide or show individual traces. Use the color picker to change the color of a trace in the plot. Double click on a list entry to edit its name.

Control/Tool	Option/Range	Description
Length	integer value	Maximum number of records in the history. The number of entries displayed in the list is limited to the 100 most recent ones.
Clear All		Remove all records from the history list.
Clear		Remove selected records from the history list.
Load file		Load data from a file into the history. Loading does not change the data type and range displayed in the plot, this has to be adapted manually if data is not shown.
Name		Enter a name which is used as a folder name to save the history into. An additional three digit counter is added to the folder name to identify consecutive saves into the same folder name.
Auto Save		Activate autosaving. When activated, any measurements already in the history are saved. Each subsequent measurement is then also saved. The autosave directory is identified by the text "autosave" in the name, e.g. "sweep_autosave_001". If autosave is active during continuous running of the module, each successive measurement is saved to the same directory. For single shot operation, a new directory is created containing all measurements in the history. Depending on the file format, the measurements are either appended to the same file, or saved in individual files. For HDF5 and ZView formats, measurements are appended to the same file. For MATLAB and SXM formats, each measurement is saved in a separate file.
File Format		Select the file format in which to save the data.
Save		Save the traces in the history to a file accessible in the File Manager tab. The file contains the signals in the Vertical Axis Groups of the Control sub-tab. The data that is saved depends on the selection from the pull-down list. Save All: All traces are saved. Save Sel: The selected traces are saved.

For the Math sub-tab please see [the table "Plot math description"](#) in the section called "Cursors and Math".

5.10. Spectrum Analyzer Tab

The Spectrum Analyzer is one of the powerful frequency domain measurement tools as introduced in [Unique Set of Analysis Tools](#) and is available on all MFIA instruments.

5.10.1. Features

- Fast, high-resolution FFT spectrum analyzer
- Signals: demodulated data ($X+iY$, R , θ , f and $d\theta/dt/(2\pi)$), PID, Boxcar, Auxiliary Inputs, and more
- Variable center frequency, frequency resolution and frequency span
- Auto bandwidth
- Waterfall display
- Choice of 4 different FFT window functions
- Continuous and block-wise acquisition with different types of averaging
- Detailed noise power analysis
- Support for Input Scaling and Input Units
- Mathematical toolbox for signal analysis

5.10.2. Description

The Spectrum Analyzer provides frequency domain analysis of demodulator data. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.32: App icon and short description

Control/Tool	Option/Range	Description
Spectrum		Provides FFT functionality to all continuously streamed measurement data.

The Spectrum tab (see [Figure 5.21](#)) is divided into a display section on the left and a configuration section on the right. The configuration section is further divided into a number of sub-tabs.

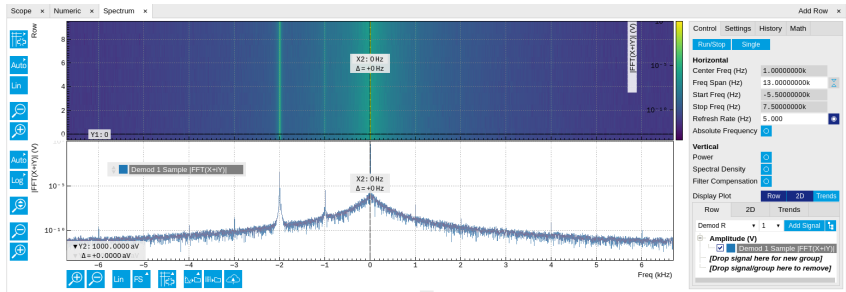


Figure 5.21: LabOne UI: Spectrum analyzer tab

The Spectrum Analyzer allows for spectral analysis of all the demodulator data by performing the fast Fourier transform (FFT) on the complex demodulator data samples $X+iY$ (with i as the imaginary unit). The result of this FFT is a spectrum centered around the demodulation frequency, whereas applying a FFT directly on the raw input data would produce a spectrum centered around zero frequency. The latter procedure corresponds to the Frequency Domain operation in the [Scope Tab](#). The main difference between the two is that the Spectrum Analyzer tool can acquire data for a much longer periods of time and therefore can achieve very high frequency resolution around the demodulation frequency. By default, the spectrum is displayed centered around zero. Sometimes however it is convenient to shift the frequency axis by the demodulation frequency which allows one to identify the frequencies on the horizontal axis with the physical frequencies at the signal inputs. This can be done by activating Absolute Frequency on the Settings sub-tab.

By default, the display section contains a line plot of the spectrum together with a color waterfall plot of the last few acquired spectra. The waterfall plot makes it easier to see the evolution of the spectrum over time. The display layout as well as the number of rows in the color plot can be configured in the Settings sub-tab.

Data shown in the Spectrum tab have passed a low-pass filter with a well-defined order and bandwidth. This is most clearly noted by the shape of the noise floor. One has to take care that the selected frequency span, which equals the demodulator sampling rate, is 5 to 10 times higher than the filter bandwidth in order to prevent measurement errors due to aliasing. The Auto Bandwidth button  adjusts the sampling rate so that it suits the filter settings. The Spectrum tab features FFT display of a selection of data available in the Signal Type drop-down list in addition to the complex demodulator samples $X+iY$. Looking at the FFT of polar demodulator values R and Θ allows one to discriminate between phase noise components and amplitude noise components in the signal. The FFT of the phase derivative $d\Theta/dt$ provides a quantitative view of the spectrum of demodulator frequencies. That is particularly useful in conjunction with the PLL or the ExtRef functionalities. The FFT of the frequency samples then provide a quantitative view of what frequency noise components are present in the reference signal and also helps to find the optimal PLL bandwidth to track the signal. Note that many of the signals in the Signal Type list are real-valued, rather than complex-valued. Their spectra are single-sided with minimum frequency of 0 Hz.

5.10.3. Functional Elements

Table 5.33: Spectrum tab: Settings sub-tab

Control/Tool	Option/Range	Description
Run/Stop		Run the FFT spectrum analysis continuously
Single		Run the FFT spectrum analysis once
Center Freq (Hz)	numeric value	Demodulation frequency of the selected demodulator used as input for the spectrum. For complex FFT($X+iY$) the demodulation frequency defines the center frequency of the displayed FFT.

Control/Tool	Option/Range	Description
Frequency Span (Hz)	numeric value	Set the frequency span of interest for the complex FFT. A FFT based on real input data will display half of the frequency span up to the Nyquist frequency.
Auto Bandwidth		Automatic adjustment of the demodulator bandwidths to obtain optimal alias rejection for the selected frequency span which is equivalent to the sampling rate. The functionality is only available if the spectrum is enabled.
Start Frequency (Hz)	numeric value	Indicates the start frequency of the FFT.
Stop Frequency (Hz)	numeric value	Indicates the end frequency of the FFT.
Refresh Rate (Hz)	numeric value	Set the maximum plot refresh rate. The actual refresh rate also depends on other parameters such as FFT length. In overlapped mode the refresh rate defines the amount of overlapping.
Overlapped FFT	ON / OFF	Enable overlapped FFTs. If disabled, FFTs are performed on distinct abutting data sets. If enabled, the data sets of successive FFTs overlap based on the defined refresh rate.
Power	ON / OFF	Calculate and show the power value. To extract power spectral density (PSD) this button should be enabled together with spectral density.
Spectral Density	ON / OFF	Calculate and show the spectral density. If power is enabled the power spectral density value is calculated. The spectral density is used to analyze noise.
Filter Compensation	ON / OFF	Spectrum is corrected by demodulator filter transfer function. Allows for quantitative comparison of amplitudes of different parts of the spectrum.
Absolute Frequency	ON / OFF	Shifts x-axis labeling to show the demodulation frequency in the center as opposed to 0 Hz, when turned off.
FFT length	numeric value	The number of samples used for the FFT. Values entered that are not a binary power are truncated to the nearest power of 2.
Sampling Progress	0% to 100%	The percentage of the FFT buffer already acquired. The progress includes the number of rows and averages.
FFT Duration (s)	numeric value	Indicates the length in time of the samples used for a single FFT.
Window	Cosine squared (ring-down)	Several different FFT windows to choose from. Depending on the application it makes a huge difference which of the provided window function is used. Please check the literature to find out the best trade off for your needs.
	Rectangular	
	Hann	
	Hamming	
	Blackman Harris	
	Flat Top	
	Exponential (ring-down)	
	Cosine (ring-down)	
Resolution (Hz)	mHz to Hz	Spectral resolution defined by the reciprocal acquisition time (sample rate, number of samples recorded).
Rows	numeric value	Number of rows
Averages	numeric value	Number of FFT averaged for each row. Setting the value to 1 will disable any averaging.
Waterfall	ON / OFF	Enable to show the 2D plot in waterfall mode. It will always update the lowest line.

Control/Tool	Option/Range	Description
Overwrite	ON / OFF	Enable to overwrite the grid in continuous mode. History will not be collected. A history element will only be created when the analysis is stopped.
Plot Type		Select the plot type.
	None	No plot displayed.
	2D	Display defined number of grid rows as one 2D plot.
	Row	Display only the trace of index defined in the Active Row field.
	2D + Row	Display 2D and row plots.
Active Row	integer value	Set the row index to be displayed in the Row plot.
Track Active Row	ON / OFF	If enabled, the active row marker will track with the last recorded row. The active row control field is read-only if enabled.
Palette	Solar	Select the colormap for the current plot.
	Viridis	
	Inferno	
	Balance	
	Turbo	
	Grey	
Colormap	ON / OFF	Enable/disable the colormap bar display in the 2D plot.
Mapping		Mapping of colormap.
	Lin	Enable linear mapping.
	Log	Enable logarithmic mapping.
	dB	Enable logarithmic mapping in dB.
Scaling	Full Scale/ Manual/Auto	Scaling of colormap.
Clamp To Color	ON / OFF	When enabled, grid values that are outside of defined Min or Max region are painted with Min or Max color equivalents. When disabled, Grid values that are outside of defined Min or Max values are left transparent.
Start	numeric value	Lower limit of colormap. Only visible for manual scaling.
Stop	numeric value	Upper limit of colormap. Only visible for manual scaling.

For the Math sub-tab please see [the table "Plot math description"](#) in the section called "Cursors and Math".

5.11. Sweeper Tab

The Sweeper is a highly versatile measurement tool available on all MFIA instruments. The Sweeper enables scans of an instrument parameter over a defined range and simultaneous measurement of a selection of continuously streamed data. In the special case where the sweep parameter is an oscillator frequency, the Sweeper offers the functionality of a frequency response analyzer (FRA), a well-known class of instruments.

5.11.1. Features

- Full-featured parametric sweep tool for frequency, phase shift, output amplitude, DC output voltages, etc.

- Simultaneous display of data from different sources (Demodulators, PIDs, auxiliary inputs, and others)
- Different application modes, e.g. Frequency response analyzer (Bode plots), noise amplitude sweeps, etc.
- Different sweep types: single, continuous (run / stop), bidirectional, binary
- Persistent display of previous sweep results
- XY Mode for Nyquist plots or I-V curves
- Normalization of sweeps
- Auto bandwidth, averaging and display normalization
- Support for Input Scaling and Input Units
- Phase unwrap
- Full support of sinc filter

5.11.2. Description

The Sweeper supports a variety of experiments where a parameter is changed stepwise and numerous measurement data can be graphically displayed. Open the tool by clicking the corresponding icon in the UI side bar. The Sweeper tab (see Figure 5.22) is divided into a plot section on the left and a configuration section on the right. The configuration section is further divided into a number of sub-tabs.

Table 5.34: App icon and short description

Control/Tool	Option/Range	Description
Sweeper		Sweep frequencies, voltages, and other quantities over a defined range and display various response functions including statistical operations.

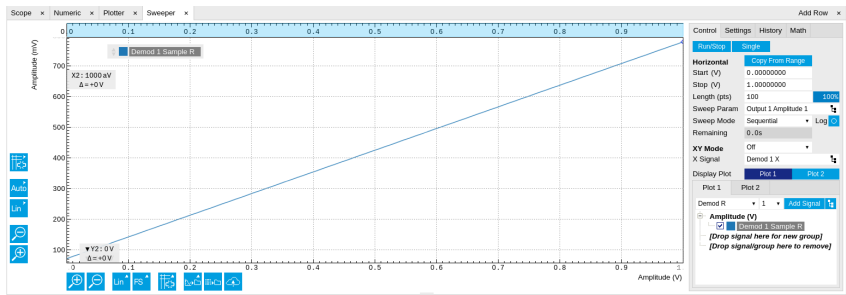


Figure 5.22: LabOne UI: Sweeper tab

The **Control** sub-tab holds the basic measurement settings such as Sweep Parameter, Start/Stop values, and number of points (Length) in the Horizontal section. Measurement signals can be added in the Vertical Axis Groups section. A typical use of the Sweeper is to perform a frequency sweep and measure the response of the device under test in the form of a Bode plot. As an example, AFM and MEMS users require to determine the resonance frequency and the phase delay of their oscillators. The Sweeper can also be used to sweep parameters other than frequency, for instance signal amplitudes and DC offset voltages. A sweep of the Auxiliary Output offset can for instance be used to measure current-voltage (I-V) characteristics. The XY Mode allows one to use a measured signal, rather than the sweep parameter, on the horizontal axis. This is useful to obtain Nyquist plots in impedance measurements, or to display an I-V curve in a four-probe measurement of a nonlinear device.

For frequency sweeps, the sweep points are distributed logarithmically, rather than linearly, between the start and stop values by default. This feature is particularly useful for sweeps over several decades and can be disabled with the Log checkbox. The Sweep Mode setting is useful for identifying measurement problems caused by hysteretic sample behavior or too fast sweeping speed. Such problems would cause non-overlapping curves in a bidirectional sweep.

Note

The Sweeper actively modifies the main settings of the demodulators and oscillators. So in particular for situations where multiple experiments are served maybe even from different control computers great care needs to be taken so that the parameters altered by the sweeper module do not have unwanted effects elsewhere.

The Sweeper offers two operation modes differing in the level of detail of the accessible settings: the Application Mode and the Advanced Mode. Both of them are accessible in the **Settings** sub-tab.

The Application Mode provides the choice between six measurement approaches that should help to quickly obtain correct measurement results for a large range of applications. Users who like to be in control of all the settings can access them by switching to the Advanced Mode.

In the Statistics section of the Advanced Mode one can control how data is averaged at each sweep point: either by specifying the Sample count, or by specifying the number of filter time constants (TC). The actual measurement time is determined by the larger of the two settings, taking into account the demodulator sample rate and filter settings. The Algorithm settings determines the statistics calculated from the measured data: the average for general purposes, the deviation for noise measurements, or the mean square for power measurements. The Phase Unwrap features ensures continuity of a phase measurement curve across the PM180 degree boundary. Enabling the Sinc Filter setting means that the demodulator Sinc Filter gets activated for sweep points below 50 Hz in Auto and Fixed mode. This speeds up measurements at small frequencies as explained in the [Sinc Filtering](#).

In the Settling section one can control the waiting time between a parameter setting and the first measurement. Similarly to the Statistics setting, one has the choice between two different representations of this waiting time. The actual settling time is the maximum of the values set in units of absolute time and a time derived from the demodulator filter and a desired inaccuracy (e.g. 1 m for 0.1%). Let's consider an example. For a 4th order filter and a 3 dB bandwidth of 100 Hz we obtain a step response that attains 90% after about 4.5 ms. This can be easily measured by using the Data Acquisition tool as indicated in [Figure 5.23](#). It is also explained in [Discrete-Time Filters](#). In case the full range is set to 1 V this means a measurement has a maximum error caused by imperfect settling of about 0.1 V. However, for most measurements the neighboring values are close compared to the full range and hence the real error caused is usually much smaller.

In the Filter section of the Advanced mode, the Bandwidth Mode setting determines how the filters of the activated demodulators are configured. In Manual mode, the current setting (in the Lock-in tab) remains unchanged by the Sweeper. In Fixed mode, the filter settings can be controlled from within the Sweeper tab. In Auto mode, the Sweeper determines the filter bandwidth for each sweep point based on a desired ω suppression. The ω suppression depends on the measurement frequency and the filter steepness. For frequency sweeps, the bandwidth will be adjusted for every sweep point within the bound set by the Max Bandwidth setting. The Auto mode is particularly useful for frequency sweeps over several decades, because the continuous adjustment of the bandwidth considerably reduces the overall measurement time.

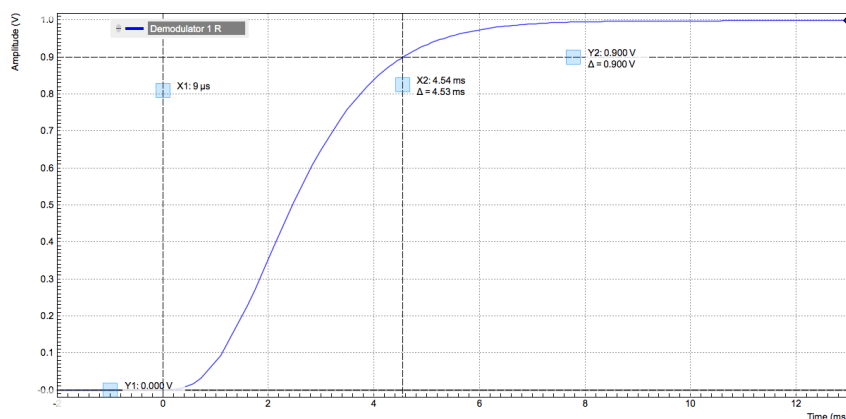


Figure 5.23: Demodulator settling time and inaccuracy: measurement carried out with the Data Acquisition tool to illustrate the settling time for a 4

By default the plot area keeps the memory and display of the last 100 sweeps represented in a list in the **History** sub-tab. See [History List](#) for more details on how data in the history list can be managed and stored. With the Reference feature, it is possible to divide all measurements in the history by a reference measurement. This is useful for instance to eliminate spurious effects in a frequency response sweep. To define a certain measurement as the reference, mark it in the list and click on [Reference](#). Then enable the Reference mode with the checkbox below the list to update the plot display. Note that the Reference setting does not affect data saving: saved files always contain raw data.

Note

The Sweeper can get stuck whenever it does not receive any data. A common mistake is to select to display demodulator data without enabling the data transfer of the associated demodulator in the Lock-in tab.

Note

Once a sweep is performed the sweeper stores all data from the enabled demodulators and auxiliary inputs even when they are not displayed immediately in the plot area. These data can be accessed at a later point in time simply by choosing the corresponding signal display settings (Input Channel).

5.11.3. Functional Elements

Table 5.35: Sweeper tab: Control sub-tab

Control/Tool	Option/Range	Description
Run/Stop		Runs the sweeper continuously.
Single		Runs the sweeper once.
Copy From X-Axis		Takes over start and stop value from the X-axis.
Copy From X-Cursors		Takes over start and stop value from X-cursors. Button is disabled when one or both X cursors are not visible.
Start (unit)	numeric value	Start value of the sweep parameter. The unit adapts according to the selected sweep parameter.
Stop (unit)	numeric value	Stop value of the sweep parameter. The unit adapts according to the selected sweep parameter.
Length	integer value	Sets the number of measurement points.
Progress	0 to 100%	Reports the sweep progress as ratio of points recorded.
Sweep Param		Selects the parameter to be swept. Navigate through the tree view that appears and click on the required parameter. The available selection depends on the configuration of the device.
Sweep Mode		Select the scanning type, default is sequential (incremental scanning from start to stop value)
	Sequential	Sequential sweep from Start to Stop value
	Binary	Non-sequential sweep continues increase of resolution over entire range
	Bidirectional	Sequential sweep from Start to Stop value and back to Start again
	Reverse	Reverse sweep from Stop to Start value
X Distribution	Linear / Logarithmic	Selects between linear and logarithmic distribution of the sweep parameter.
Remaining	numeric value	Reporting of the remaining time of the current sweep. A valid number is only displayed once the sweeper has been started. An undefined sweep time is indicated as NaN.
Invert Y Axis	ON / OFF	The xy-plot is displayed with inverted y-axis. This mode is used for Nyquist plots that allow for displaying $-\text{imag}(z)$ on the y-axis and $\text{real}(z)$ on the x-axis.
X Signal		Selects the signal that defines the x-axis for xy-plots. The available selection depends on the configuration of the device. Displaying the selected signal source will result in a diagonal straight line.

For the Vertical Axis Groups, please see [the table "Vertical Axis Groups description" in the section called "Vertical Axis Groups"](#).

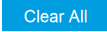
Table 5.36: Sweeper tab: Settings sub-tab

Control/Tool	Option/Range	Description
Filter		Application Mode: preset configuration. Advanced Mode: manual configuration.

Control/Tool	Option/Range	Description
	Application Mode	The sweeper sets the filters and other parameters automatically.
	Advanced Mode	The sweeper uses manually configured parameters.
Application		Select the sweep application mode
	Parameter Sweep	Only one data sample is acquired per sweeper point.
	Parameter Sweep Averaged	Multiple data samples are acquired per sweeper point of which the average value is displayed.
	Noise Amplitude Sweep	Multiple data samples are acquired per sweeper point of which the standard deviation is displayed (e.g. to determine input noise). For accurate noise measurement, the signal amplitude R is replaced by its quadrature components X and Y.
	Freq Response Analyzer	Narrow band frequency response analysis. Averaging is enabled.
	3-Omega Sweep	Optimized parameters for 3-omega application. Averaging is enabled.
	FRA (Sinc Filter)	The sinc filter helps to speed up measurements for frequencies below 50 Hz in FRA mode. For higher frequencies it is automatically disabled. Averaging is off.
	Impedance	This application mode uses narrow bandwidth filter settings to achieve high calibration accuracy.
Precision		Choose between a high speed scan speed or high precision and accuracy.
	Low -> fast sweep	Medium accuracy/precision is optimized for sweep speed.
	High -> standard speed	Medium accuracy/precision takes more measurement time.
	Very high -> slow sweep	High accuracy/precision takes more measurement time.
Bandwidth Mode		Automatically is recommended in particular for logarithmic sweeps and assures the whole spectrum is covered.
	Auto	All bandwidth settings of the chosen demodulators are automatically adjusted. For logarithmic sweeps the measurement bandwidth is adjusted throughout the measurement.
	Fixed	Define a certain bandwidth which is taken for all chosen demodulators for the course of the measurement.
	Manual	The sweeper module leaves the demodulator bandwidth settings entirely untouched.
Time Constant/ Bandwidth Select		Defines the display unit of the low-pass filter to use for the sweep in fixed bandwidth mode: time constant (TC), noise equivalent power bandwidth (NEP), 3 dB bandwidth (3 dB).
	TC	Defines the low-pass filter characteristic using time constant of the filter.
	Bandwidth NEP	Defines the low-pass filter characteristic using the noise equivalent power bandwidth of the filter.
	Bandwidth 3 dB	Defines the low-pass filter characteristic using the cut-off frequency of the filter.
Time Constant/ Bandwidth	numeric value	Defines the measurement bandwidth for Fixed bandwidth sweep mode, and corresponds to either noise equivalent power bandwidth (NEP), time constant (TC) or 3 dB bandwidth (3 dB) depending on selection.

Control/Tool	Option/Range	Description
Order	numeric value	Selects the filter roll off to set on the device in Fixed and Auto bandwidth modes. It ranges from 1 (6 dB/octave) to 8 (48 dB/octave).
Max Bandwidth (Hz)	numeric value	Maximal bandwidth used in auto bandwidth mode. The effective bandwidth will be calculated based on this max value, the frequency step size, and the omega suppression.
BW Overlap	ON / OFF	If enabled the bandwidth of a sweep point may overlap with the frequency of neighboring sweep points. The effective bandwidth is only limited by the maximal bandwidth setting and omega suppression. As a result, the bandwidth is independent of the number of sweep points. For frequency response analysis bandwidth overlap should be enabled to achieve maximal sweep speed.
Omega Suppression (dB)	numeric value	Suppression of the omega and 2-omega components. Large suppression will have a significant impact on sweep time especially for low filter orders.
Min Settling Time (s)	numeric value	Minimum wait time in seconds between a sweep parameter change and the recording of the next sweep point. This parameter can be used to define the required settling time of the experimental setup. The effective wait time is the maximum of this value and the demodulator filter settling time determined from the Inaccuracy value specified.
Inaccuracy	numeric value	Demodulator filter settling inaccuracy defining the wait time between a sweep parameter change and recording of the next sweep point. The settling time is calculated as the time required to attain the specified remaining proportion [1e-13, 0.1] of an incoming step function. Typical inaccuracy values: 10 m for highest sweep speed for large signals, 100 u for precise amplitude measurements, 100 n for precise noise measurements. Depending on the order the settling accuracy will define the number of filter time constants the sweeper has to wait. The maximum between this value and the settling time is taken as wait time until the next sweep point is recorded.
Settling Time (TC)	numeric value	Calculated wait time expressed in time constants defined by the specified filter settling inaccuracy.
Algorithm		Selects the measurement method.
	Averaging	Calculates the average on each data set.
	Standard Deviation	Calculates the standard deviation on each data set.
	Average Power	Calculates the electric power based on a 50 Ω input impedance.
Count (Sa)	integer number	Sets the number of data samples per sweeper parameter point that is considered in the measurement. The maximum between samples, time and number of time constants is taken as effective calculation time.
Count (s)	numeric value	Sets the time during which data samples are processed. The maximum between samples, time and number of time constants is taken as effective calculation time.
Count (TC)	0/5/15/50/100 TC	Sets the effective measurement time per sweeper parameter point that is considered in the measurement. The maximum between samples, time and number of time constants is taken as effective calculation time.
Phase Unwrap	ON / OFF	Allows for unwrapping of slowly changing phase evolutions around the +/-180 degree boundary.
Spectral Density	ON / OFF	Selects whether the result of the measurement is normalized versus the demodulation bandwidth.
Sinc Filter	ON / OFF	Enables sinc filter if sweep frequency is below 50 Hz. Will improve the sweep speed at low frequencies as omega components do not need to be suppressed by the normal low-pass filter.

Table 5.37: Sweeper tab: History sub-tab

Control/Tool	Option/Range	Description
History	History	Each entry in the list corresponds to a single trace in the history. The number of traces displayed in the plot is limited to 20. Use the toggle buttons to hide or show individual traces. Use the color picker to change the color of a trace in the plot. Double click on a list entry to edit its name.
Length	integer value	Maximum number of records in the history. The number of entries displayed in the list is limited to the 100 most recent ones.
Clear All		Remove all records from the history list.
Clear		Remove selected records from the history list.
Load file		Load data from a file into the history. Loading does not change the data type and range displayed in the plot, this has to be adapted manually if data is not shown.
Name		Enter a name which is used as a folder name to save the history into. An additional three digit counter is added to the folder name to identify consecutive saves into the same folder name.
Auto Save		Activate autosaving. When activated, any measurements already in the history are saved. Each subsequent measurement is then also saved. The autosave directory is identified by the text "autosave" in the name, e.g. "sweep_autosave_001". If autosave is active during continuous running of the module, each successive measurement is saved to the same directory. For single shot operation, a new directory is created containing all measurements in the history. Depending on the file format, the measurements are either appended to the same file, or saved in individual files. For HDF5 and ZView formats, measurements are appended to the same file. For MATLAB and SXM formats, each measurement is saved in a separate file.
File Format		Select the file format in which to save the data.
Save		Save the traces in the history to a file accessible in the File Manager tab. The file contains the signals in the Vertical Axis Groups of the Control sub-tab. The data that is saved depends on the selection from the pull-down list. Save All: All traces are saved. Save Sel: The selected traces are saved.
Reference		Use the selected trace as reference for all active traces.
Reference On	ON / OFF	Enable/disable the reference mode.
Reference name	name	Name of the reference trace used.

For the Math sub-tab please see [the table "Plot math description"](#) in the section called "Cursors and Math".

5.12. Auxiliary Tab

The Auxiliary tab provides access to the settings of the Auxiliary Inputs and Auxiliary Outputs; it is available on all MFIA instruments.

5.12.1. Features

- Monitor signal levels of auxiliary input connectors
- Monitor signal levels of auxiliary output connectors
- Auxiliary output signal sources: Demodulators and manual setting
- Define Offsets and Scaling for auxiliary output values
- Control auxiliary output range limitations

5.12.2. Description

The Auxiliary tab serves mainly to monitor and control the auxiliary inputs and outputs. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.38: App icon and short description

Control/Tool	Option/Range	Description
Aux		Controls all settings regarding the auxiliary inputs and auxiliary outputs.

The Auxiliary tab (see [Figure 5.24](#)) is divided into three sections. The Aux Input section gives two graphical and two numerical monitors for the signal amplitude applied to the auxiliary inputs on the back panel. In the middle of the tab the Aux Output section allows to associate any of the measured signals to one of the 4 auxiliary outputs on the instrument front panel. With the action buttons next to the Preoffset and Offset values the effective voltage on the auxiliary outputs can be automatically set to zero. The analog output voltages can be limited to a certain range in order to avoid damaging the parts connected to the outputs.

Note

Please note the change of units of the scaling factor depending on what measurement signal is chosen.

Two Aux Output Levels on the right provides 4 graphical and 4 numerical indicators to monitor the voltages currently set on the auxiliary outputs.

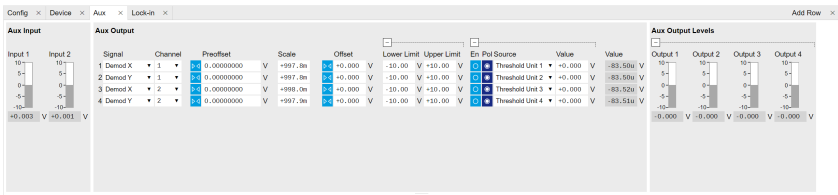


Figure 5.24: LabOne UI: Auxiliary tab

5.12.3. Functional Elements

Table 5.39: Auxiliary tab

Control/Tool	Option/Range	Description
Auxiliary Input Voltage	-10 V to 10 V	Voltage measured at the Auxiliary Input.
Signal		Select the signal source to be represented on the Auxiliary Output.
	X	Select the demodulator X component for auxiliary output.
	Y	Select the demodulator Y component for auxiliary output.
	R	Select the demodulator magnitude component for auxiliary output.
	Θ	Select the demodulator phase component for auxiliary output.
	PID Out	Select one of the PID controllers output. MF-PID option needs to be installed.
	PID Shift	Select one of the PID controllers' shift signal. MF-PID option needs to be installed.
	PID Error	Select one of the PID controllers' error signal. MF-PID option needs to be installed.

Control/Tool	Option/Range	Description
	TU Filtered Value	Select one of the Threshold unit input signals after filtering for auxiliary output.
	TU Output Value	Select one of the Threshold unit output signals for auxiliary output.
	Manual	Manually define an auxiliary output voltage using the offset field.
Channel	index	Select the channel according to the selected signal source.
Preoffset	numerical value in signal units	Add a pre-offset to the signal before scaling is applied. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset
Auto-zero		Automatically adjusts the Pre-offset to set the Auxiliary Output Value to zero.
Scale	numerical value	Multiplication factor to scale the signal. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset
Auto-zero		Automatically adjusts the Offset to set the Auxiliary Output Value to zero.
Offset	numerical value in Volts	Add the specified offset voltage to the signal after scaling. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset
Lower Limit	-10 V to 10 V	Lower limit for the signal at the Auxiliary Output. A smaller value will be clipped.
Upper Limit	-10 V to 10 V	Upper limit for the signal at the Auxiliary Output. A larger value will be clipped.
Enable	ON / OFF	Enable Tip Protect unit.
Polarity	ON / OFF	Select the active level of Source signal.
Source		Select the input source for Tip Protect.
	Threshold Unit 1	Output value of the TU logic unit 1.
	Threshold Unit 2	Output value of the TU logic unit 2.
	Threshold Unit 3	Output value of the TU logic unit 3.
	Threshold Unit 4	Output value of the TU logic unit 4.
Value		Signal value written to the Aux Output port when Tip Protect is active. The value overwrites Aux Output Internal value and the Lower and Upper Limits.
Value	-10 V to 10 V	Voltage present on the Auxiliary Output port. The value is obtained by clamping the calculated signal (Signal+Preoffset)*Scale + Offset based on Lower and Upper Limits. It is overwritten when Tip Protect is active.

5.13. Inputs/Outputs Tab

The In / Out tab provides access to the settings of the Instrument's main Signal Inputs and Signal Outputs. It is available on all MFIA instruments.

5.13.1. Features

- Signal input configuration
- Signal output configuration

5.13.2. Description

The In / Out tab gives access to the same settings as do the left-most and the right-most sections of the Lock-in tab. It is mainly intended to be used on small screens that can not show the entire the Lock-in tab at once. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.40: Overview of app icons and short description

Control/Tool	Option/Range	Description
Lock-in		Quick overview and access to all the settings and properties for signal generation and demodulation.
Lock-in MD		Quick overview and access to all the settings and properties for signal generation and demodulation.
Files		Access files on internal flash memory and USB drive.
Numeric		Access to all continuously streamed measurement data as numerical values.
Plotter		Displays various continuously streamed measurement data as traces over time (roll mode).
Scope		Displays shots of data samples in time and frequency domain (FFT) representation.
DAQ		Provides complex trigger functionality on all continuously streamed data samples and time domain display.
Spectrum		Provides FFT functionality to all continuously streamed measurement data.
Sweeper		Sweep frequencies, voltages, and other quantities over a defined range and display various response functions including statistical operations.
Aux		Controls all settings regarding the auxiliary inputs and auxiliary outputs.
In/Out		Gives access to all controls relevant for the Signal Inputs and Signal Outputs of each channel.
DIO		Gives access to all controls relevant for the digital inputs and outputs including DIO, Trigger Inputs, Trigger Outputs, and Marker Outputs.
Config		Provides access to software configuration.
Device		Provides instrument specific settings.
PID		Features all control, analysis, and simulation capabilities of the PID controllers.
MOD		Control panel to enable (de)modulation at linear combinations of oscillator frequencies.
IA		Quick overview and access to all the settings and properties for impedance measurements.
MDS		Synchronize multiple instruments.
ZI Labs		Experimental settings and controls.

The In / Out tab contains one section for the signal inputs and one for the signal outputs. All of the corresponding connectors are placed on the instrument front panel. The In / Out tab looks differently depending on whether the MF-MD Multi-demodulator option is installed or not.

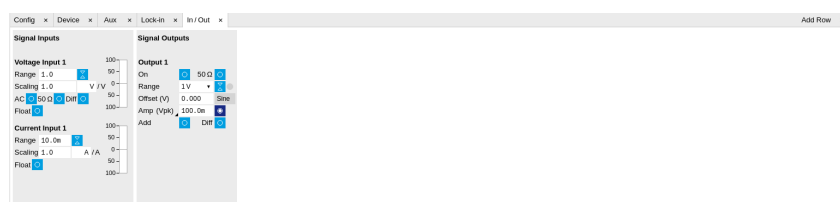


Figure 5.25: LabOne UI: Inputs/Outputs tab (base configuration)

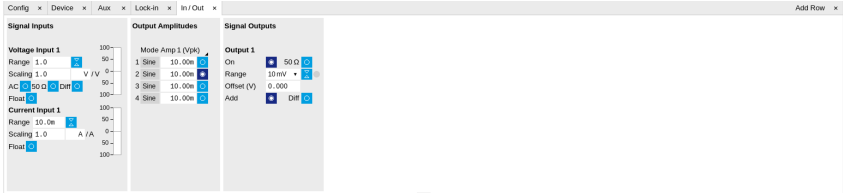


Figure 5.26: LabOne UI: Inputs/Outputs tab (with MF-MD Multi-demodulator option)

5.13.3. Functional Elements

All functional elements are equivalent to the ones on the Lock-in tab. See [the Lock-in Tab](#) or [Lock-in MF Tab](#) for a detailed description of the functional elements.

5.14. DIO Tab

The DIO tab provides access to the settings and controls of the digital I/O as well as the Trigger channels and is available on all MFIA instruments.

5.14.1. Features

- Monitor and control of digital I/O connectors
- Control settings for external reference and triggering

5.14.2. Description

The DIO tab is the main panel to control the digital inputs and outputs as well as the trigger levels and external reference channels . Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.41: App icon and short description

Control/ Tool	Option/ Range	Description
DIO		Gives access to all controls relevant for the digital inputs and outputs including DIO, Trigger Inputs, Trigger Outputs, and Marker Outputs.

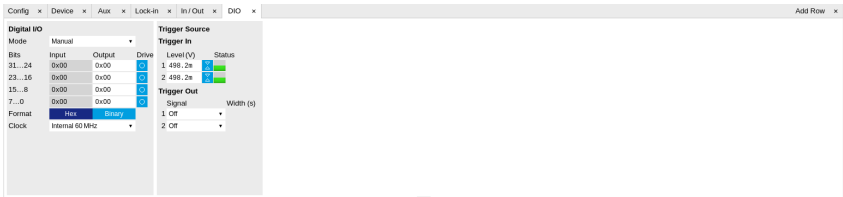


Figure 5.27: LabOne UI: DIO tab

The Digital I/O section provides numerical monitors to observe the states of the digital inputs and outputs. Moreover, with the values set in the Output column and the Drive button activated the states can also be actively set in different numerical formats.

The Trigger section shows the settings for the 2 trigger inputs and 2 trigger outputs on the instrument back panel. The trigger outputs can be referenced to the demodulator frequencies and output a TTL signal with a 50% duty cycle.

Note

The Input Level determines the trigger threshold for trigger state discrimination. Also a 100 mV hysteresis is applied that cannot be adjusted such that a minimum amplitude of more than 100 mV is needed for the Trigger inputs to work reliably.

5.14.3. Functional Elements

Table 5.42: Digital input and output channels, reference and trigger

Control/Tool	Option/Range	Description
DIO mode		Select DIO mode
	Manual	Enables manual control of the DIO output bits.
	Threshold Outputs	Enables setting of DIO output values by the threshold unit.
QA Result Overflow	grey/yellow/red	Red: present overflow condition on the DIO interface during readout. Yellow: indicates an overflow occurred in the past. An overflow can happen if readouts are triggered faster than the maximum possible data-rate of the DIO interface.
DIO bits	label	Partitioning of the 32 bits of the DIO into 4 buses of 8 bits each. Each bus can be used as an input or output.
DIO input	numeric value in either Hex or Binary format	Current digital values at the DIO input port.
DIO output	numeric value in either hexadecimal or binary format	Digital output values. Enable drive to apply the signals to the output.
DIO drive	ON / OFF	When on, the corresponding 8-bit bus is in output mode. When off, it is in input mode.
Format		Select DIO view format.
	Hexadecimal	DIO view format is hexadecimal.
	Binary	DIO view format is binary.
Clock		Select DIO internal or external clocking.
	Internal 60 MHz	The DIO is internally clocked with a fixed frequency of 60 MHz.
	Clk Pin 68	The DIO is externally clocked with a clock signal connected to DIO Pin 68. Available frequency range 1 Hz to 60 MHz.
Trigger level	-5 V to 5 V	Trigger voltage level at which the trigger input toggles between low and high. Use 50% amplitude for digital input and consider the trigger hysteresis.
Auto Threshold	Press once	Automatically adjust the trigger threshold. The level is adjusted to fall in the center of the applied transitions.
Trigger Input Low status		Indicates the current low level trigger state.
	Off	A low state is not being triggered.
	On	A low state is being triggered.
Trigger Input High status		Indicates the current high level trigger state.
	Off	A high state is not being triggered.
	On	A high state is being triggered.
Trigger output signal		Select the signal assigned to the trigger output.
	Off	The output trigger is disabled.
	Osc Phase Demod 2/4	Oscillator phase of demod 2 (trigger output channel 1) or demod 4 (trigger output channel 2). Trigger event is output for each zero crossing of the oscillator phase.
	Scope Trigger	Trigger output is asserted when the scope trigger condition is satisfied.
	Scope /Trigger	Trigger output is deasserted when the scope trigger condition is satisfied.

Control/Tool	Option/Range	Description
	Scope Armed	Trigger output is asserted when the scope is waiting for the trigger condition to become satisfied.
	Scope /Armed	Trigger output is deasserted when the scope is waiting for the trigger condition to become satisfied.
	Scope Active	Trigger output is asserted when the scope has triggered and is recording data.
	Scope /Active	Trigger output is deasserted when the scope has triggered and is recording data.
	Threshold 1	Threshold Logic Unit 1 output.
	Threshold 2	Threshold Logic Unit 2 output.
	Threshold 3	Threshold Logic Unit 3 output.
	Threshold 4	Threshold Logic Unit 4 output.
	MDS Sync Out	Trigger output is driven by the multi-device synchronisation signal.
Delay (s)		This delay adds an offset that acts only on the trigger/marker output. The total delay to the trigger/marker output is the sum of this value and the value of the output delay node.
Width	0 s to 0.149 s	Defines the minimal pulse width for the case of Scope events written to the trigger outputs of the device.

5.15. Lock-in Tab

This tab is the main lock-in amplifier control panel. Users with instruments with MF-MD Multi-demodulator option installed are kindly referred to [Lock-in Tab \(MF-MD option\)](#)

5.15.1. Features

- Functional block diagram with access to main input, output and demodulator controls
- Parameter table with main input, output and demodulator controls
- Control elements for 1 configurable demodulator
- Auto ranging, scaling, arbitrary input units for both input channels
- Control for 1 oscillator
- Settings for main signal inputs and signal outputs
- Flexible choice of reference source, trigger options and data transfer rates

5.15.2. Description

The Lock-in tab is the main control center of the instrument and open after start up by default. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.43: App icon and short description

Control/Tool	Option/Range	Description
Lock-in		Quick overview and access to all the settings and properties for signal generation and demodulation.

The default view of the Lock-in tab is the parameter table view. It is accessible under the side tab labeled All and provides controls for all demodulators in the instrument. Moreover, for each individual demodulator there is a functional block diagram available. It is accessible under the side tab labeled with the corresponding demodulator number.

Parameter Table

The parameter table (see Figure 5.28) consists of 4 sections: Signal Inputs, Oscillators, Demodulators and Signal Outputs. The Demodulator section consists of two rows where the upper row offers access to all the settings of the dual phase demodulator, the second row represents the phase detector for the PLL used for external reference. The user can obtain and change the filter settings but not transfer the data to the data server.

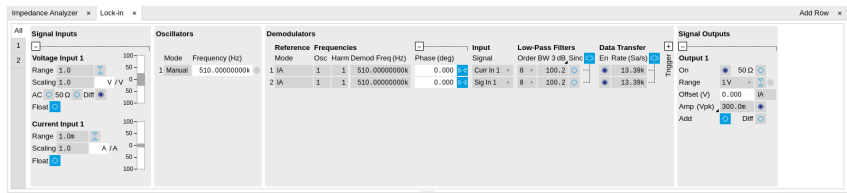


Figure 5.28: LabOne User Interface Lock-in tab - Parameter table (All)

The **Signal Inputs** section allows the user to define all relevant settings specific to the signal entered as for example input coupling, range, etc. Some of the available options like phase adjustment and the trigger functionality are collapsed by default. It takes one mouse click on the "+" icon in order to expand those controls. On the right-hand side of the Lock-in tab the Signal Outputs section allows defining signal amplitudes, offsets and range values.

The Scaling field below the Range field can be used to multiply the Signal Input data for instance to account for the gain of an external amplifier. In case there is a transimpedance gain of 10 V/A applied to the input signal externally, then the Scaling field can be set to 0.1 and the Units field can be set to A in order to show the actual current readings through the entire user interface. Below the Scaling field there is the AC/DC button and the 50 Ω / 10 M Ω . The AC/DC button sets the coupling type: AC coupling has a high-pass cutoff frequency that can be used to block large DC signal components to prevent input signal saturation during amplification. The 50 Ω / 10 M Ω button toggles the input impedance between low (50 Ω) and high (approx. 10 M Ω) input impedance. With 50 Ω input impedance, one will expect a reduction of a factor of 2 in the measured signal if the signal source also has an output impedance of 50 Ω .

Note

The Signal Inputs can be set to float which means that the BNC connector shield is no longer connected to the instrument ground. This setting affects both the current input and the voltage input. It is recommended to discharge the test device before connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.

The **Oscillator** section indicates the frequency of the internal oscillator . Where the Mode indicator shows Manual, the user can define the oscillator frequency manually defined by typing a frequency value in the field. In case the oscillator is referenced to an external source, the Mode indicator will show ExtRef and the frequency field is set to read-only. External reference requires a PLL to do the frequency mapping onto an internal oscillator. Successful locking is indicated by a green light right next to the frequency field.

In the following, we discuss the **Demodulators settings** in more detail. The block diagram displayed in Figure 5.29 indicates the main demodulator components and their interconnection. The understanding of the wiring is essential for successfully operating the instrument.

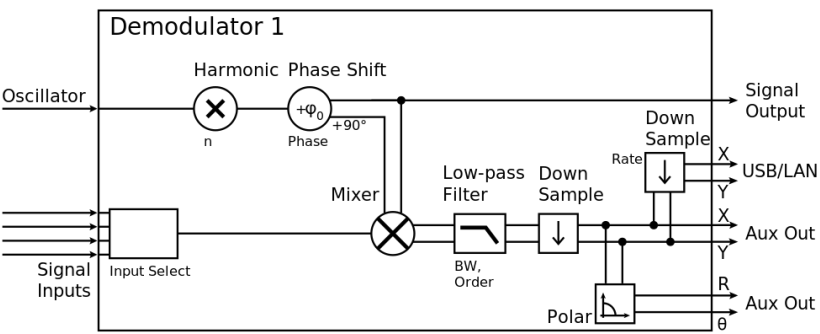


Figure 5.29: Demodulator block diagram without MF-MD Multi-demodulator option.

The first line in the Demodulators section represents the demodulator available for measurements. The Mode column is read-only set to internal reference (Demod). The second line represents an additional Demodulator that is reserved for the exclusive use as a phase detector when the mode is switched to external reference (ExtRef). The user can select from a number of different inputs to be used as external reference signals and the filter settings provide the user with an idea of the PLL speed. However, this second demodulator does not produce any output data that could be used for measurements. In the Input Signal column one defines the signal that is taken as input for a given demodulator. A wide choice of signals can be selected: Signal Inputs, the Trigger Inputs, the Auxiliary Inputs and Auxiliary Outputs. This allows using the instrument for many different measurement topologies. For each demodulator an additional phase shift can be introduced to the associated oscillator by entering the phase offset in the Phase column. This phase is added both to the reference channel and to the output of the demodulator. Hence, when the frequency is generated and detected using the same demodulator, signal phase and reference phase change by the same amount and no change will be visible in the demodulation result. Demodulation of frequencies that are integer multiples of any of the oscillator frequencies is achieved by entering the desired factor in the Harm column. The result of the demodulation, i.e. the amplitude and phase can be read e.g. using the Numeric tab which is described in [Numeric Tab](#).

In the middle of the Lock-in tab is the Low-Pass Filters section where the filter order can be selected in the drop-down list for each demodulator and the filter bandwidth (BW 3dB) can be chosen by typing a numerical value. Alternatively, the time constant of the filter (TC) or the noise equivalent power filter bandwidth (BW NEP) can be chosen by clicking on the column's header. For example, setting the filter order to 4 corresponds to a roll off of 24 dB/oct or 80 dB/dec i.e. an attenuation of 10^4 for a tenfold frequency increase. If the Low-Pass Filter bandwidth is comparable to or larger than the demodulation frequency, the demodulator output may contain frequency components at the frequency of demodulation and its higher harmonics. In this case, the additional Sinc Filter should be enabled. It attenuates those unwanted harmonic components in the demodulator output. The Sinc Filter is useful when measuring at low frequencies, since it allows one to apply a Low-Pass Filter bandwidth closer to the demodulation frequency, thus speeding up the measurement time.

The data transfer of demodulator outputs is activated by the En button in the Data Transfer section where also the sampling rate (Rate) for each demodulator can be defined.

The Trigger section next to the Data Transfer allows for setting trigger conditions in order to control and initiate data transfer from the Instrument to the host PC by the application of logic signals (e.g. TTL) to either Trigger Input 1 or 2 on the instrument back panel.

In the **Signal Outputs** section the On buttons are used to activate the Signal Output. The Range drop-down list is used to select the proper output range setting. On the Signal Output an offset voltage (Offset) can be defined. The maximum output signal permitted is $\pm V$.

Block Diagram

The block diagram view of the main instrument functions is also sometimes called the "Graphical Lock-in Tab". A set of indexed side tabs in the Lock-in Tab give access to a block diagram for each demodulator. The block diagrams are fully functional and provide the user with a visual feedback of what is going on inside the instrument. All control elements that are available in the Parameter Table detailed in the previous section are also present in the graphical representation.

The block diagram in [Figure 5.30](#) shows the signal path through the instrument for the case when the internal oscillator is used as reference. The Signal Inputs and Reference/Internal Frequency are shown on the left-hand side. The actual demodulation, i.e. the mixing and low-pass filtering is represented in the center of the tab. On the bottom right the user can set Signal Output parameters. On the top right there are the settings related to the output of the measurement data, either by digital means (PC Data Transfer) or by analog means (Auxiliary Outputs 1 to 4).

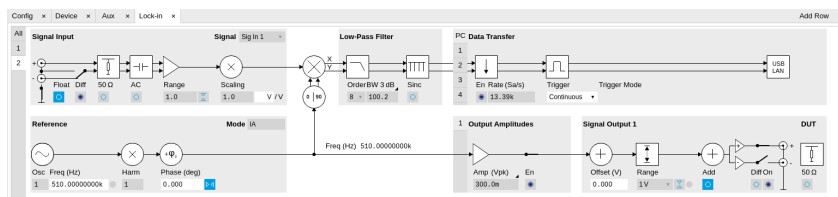


Figure 5.30: LabOne User Interface Lock-in tab - Graphical Lock-in tab in Internal Reference mode

The block diagram in [Figure 5.31](#) shows the signal path through the instrument for the case when an external reference is used. This setting is only available for demodulator 2. In order to map an external frequency to oscillator 1 go to the Reference section of demodulator 2 and change the mode to ExtRef. This demodulator will then be used as a phase detector within the phase locked

loop. The software will choose the appropriate filter settings according to the frequency and properties of the reference signal. Once demodulator 2 is used to map an external frequency on to one of the internal oscillators, it is no longer available for other measurements.

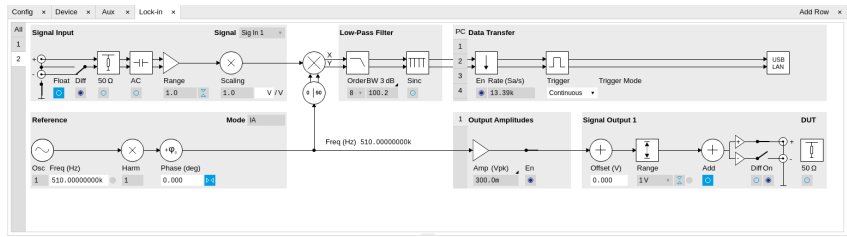


Figure 5.31: LabOne User Interface Lock-in tab - Graphical Lock-in tab in External Reference mode

5.15.3. Functional Elements

Table 5.44: Lock-in tab

Control/Tool	Option/Range	Description
Range	3.0 mV, 10 mV, 30 mV, 100 mV, 300 mV, 1 V, 3.0 V	Defines the gain of the analog input amplifier. The range should exceed the incoming signal by roughly a factor two including a potential DC offset. The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used.
On	ON / OFF	Enable Signal Input.
Auto		Automatic adjustment of the Range to about two times the maximum signal input amplitude measured over about 100 ms.
Scaling	numeric value	Applies an arbitrary scale factor to the input signal.
Measurement Unit	unit acronym	Defines the physical unit of the input signal. Use *, / and ^ operators, e.g., m or m/s^2. The value in this field modifies the readout of all measurement tools in the user interface. Typical uses of this field is to make measurements in the unit before the sensor/transducer, e.g. to take an transimpedance amplifier into account and to directly read results in Ampere instead of Volts.
Coupling	OFF: DC coupling ON: AC coupling	Defines the input coupling for the Signal Inputs. AC coupling inserts a high-pass filter.
50 Ω	OFF: 10 MΩ ON: 50 Ω	Switches between 50 Ω (ON) and 10 MΩ (OFF).
Diff	ON: Differential voltage input OFF: Single ended voltage input	Switches between single ended (OFF) and differential (ON) measurements.
Float	OFF: GND connected ON: Floating	Switches between floating (ON) and connected grounds (OFF). This setting applies both to the voltage and the current input. It is recommended to discharge the test device before connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.

Control/Tool	Option/ Range	Description
Range	10 mA	Defines the gain of the current input amplifier. The range should exceed the incoming signal by roughly a factor two including a potential DC offset.
	1 nA (*)	
	10 nA	The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used. Ranges marked with an asterisk (*) are only available on instruments with serial numbers MF-DEV3200 and higher.
	100 nA (*)	
	1 μ A	
	10 μ A (*)	
	100 μ A	
	1 mA (*)	
Auto		Automatic adjustment of the Range to about two times the maximum current input amplitude measured over about 100 ms.
Scaling	numeric value	Applies an arbitrary scale factor to the current input.
Measurement Unit	unit acronym	Defines the physical unit of the current input. Use *, / and ^ operators, e.g., m or m/s^2 The value in this field modifies the readout of all measurement tools in the user interface.
Mode		Indicates how the frequency of the corresponding oscillator is controlled (manual, external reference). Read only flag.
	Manual	The user setting defines the oscillator frequency.
	ExtRef	An external reference is mapped onto the oscillator frequency.
Frequency (Hz)	0 to 5 MHz	Frequency control for each oscillator.
Locked	ON / OFF	Oscillator locked to external reference when turned on.
Mode		Select the reference mode (manual or external reference) or indicate the unit that uses the demodulator (e.g. PLL).
	Manual	Default lock-in operating mode with manually set reference frequency.
	ExtRef	The demodulator is used for external reference mode and tracks the frequency of the selected reference input. The demodulator bandwidth is set automatically to adapt to the signal properties.
	ExtRef Low BW	The demodulator is used for external reference mode and tracks the frequency of the selected reference input. The demodulator bandwidth is fixed at low value. Use when automatic bandwidth adjustment interferes with a stable lock at a fixed frequency.
	ExtRef High BW	The demodulator is used for external reference mode and tracks the frequency of the selected reference input. The demodulator bandwidth is fixed at a high value. Use when automatic bandwidth adjustment interferes with tracking a strongly fluctuating frequency.
	PLL	The demodulator is used in PLL mode for frequency tracking of the signal. This function requires the MF-PID option.
	IA	The demodulator is used by the impedance analyzer.
Osc	oscillator index	Connects the selected oscillator with the demodulator corresponding to this line. Number of available oscillators depends on the installed options.
Harm	1 to 1023	Multiplies the demodulator's reference frequency with the integer factor defined by this field. If the demodulator is used as a phase detector in external reference mode (PLL), the effect is that the internal oscillator locks to the external frequency divided by the integer factor.
Harm	1 to 1023	Divides the demodulator's reference frequency by an integer factor in external reference mode.

Control/Tool	Option/ Range	Description
Demod Freq (Hz)	0 to 5 MHz	Indicates the frequency used for demodulation and for output generation. The demodulation frequency is calculated with oscillator frequency times the harmonic factor. When the MF-MOD option is used linear combinations of oscillator frequencies including the harmonic factors define the demodulation frequencies.
Phase (deg)	-180° to 180°	Phase shift applied to the reference input of the demodulator.
Zero		Adjust the phase of the demodulator reference automatically in order to read zero degrees at the demodulator output. This action maximizes the X output, zeros the Y output, zeros the Θ output, and leaves the R output unchanged.
Signal		Selects the signal source to be associated to the demodulator.
	Sig In 1	Signal Input 1 is connected to the corresponding demodulator.
	Curr In 1	Current Input 1 is connected to the corresponding demodulator.
	Trigger 1	Trigger 1 is connected to the corresponding demodulator.
	Trigger 2	Trigger 2 is connected to the corresponding demodulator.
	Aux Out 1	Internal value of Auxiliary Output 1 is applied to the input of the corresponding demodulator.
	Aux Out 2	Internal value of Auxiliary Output 2 is applied to the input of the corresponding demodulator.
	Aux Out 3	Internal value of Auxiliary Output 3 is applied to the input of the corresponding demodulator.
	Aux Out 4	Internal value of Auxiliary Output 4 is applied to the input of the corresponding demodulator.
	Aux In 1	Auxiliary Input 1 is connected to the corresponding demodulator.
	Aux In 2	Auxiliary Input 2 is connected to the corresponding demodulator.
	Constant	Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).
Order		Selects the filter roll off between 6 dB/oct and 48 dB/oct.
	1	1st order filter 6 dB/oct
	2	2nd order filter 12 dB/oct
	3	3rd order filter 18 dB/oct
	4	4th order filter 24 dB/oct
	5	5th order filter 30 dB/oct
	6	6th order filter 36 dB/oct
	7	7th order filter 42 dB/oct
	8	8th order filter 48 dB/oct
TC/BW Select		Defines the display unit of the low-pass filters: time constant (TC) in seconds, noise equivalent power bandwidth (BW NEP) in Hz, 3 dB bandwidth (BW 3 dB) in Hz.
	TC	Defines the low-pass filter characteristic using time constant (s) of the filter.

Control/Tool	Option/ Range	Description
	BW NEP	Defines the low-pass filter characteristic using the noise equivalent power bandwidth (Hz) of the filter.
	BW 3 dB	Defines the low-pass filter characteristic using the 3 dB cut-off frequency (Hz) of the filter.
TC/BW Value	numeric value	Defines the low-pass filter characteristic in the unit defined above.
Sinc	ON / OFF	Enables the sinc filter. When the filter bandwidth is comparable to or larger than the demodulation frequency, the demodulator output may contain frequency components at the frequency of demodulation and its higher harmonics. The sinc is an additional filter that attenuates these unwanted components in the demodulator output.
Filter Lock		Makes all demodulator filter settings equal (order, time constant, bandwidth). Enabling the lock copies the settings from demodulator 1 to all other demodulators. With locked filters, any modification to a filter setting is applied to all other filters, too. Releasing the lock does not change any setting.
Enable Streaming	ON / OFF	Enables the data acquisition and streaming of demodulated samples to the host computer for the corresponding demodulator. The streaming rate is defined in the field on the right hand side. Enabling a stream activates a corresponding element in the numeric tab and allows for demodulated samples to be visualized and analyzed in any of the LabOne measurement tools. Note: increasing number of active demodulators increases load on physical connection to the host computer.
Rate (Sa/s)	0.056 Sa/s to 857 kSa/s	Defines the demodulator sampling rate, the number of samples that are sent to the host computer per second. A rate of about 7-10 higher as compared to the filter bandwidth usually provides sufficient aliasing suppression. This is also the rate of data received by LabOne Data Server and saved to the computer hard disk. This setting has no impact on the sample rate on the auxiliary outputs connectors. Note: the value inserted by the user may be approximated to the nearest value supported by the instrument.
Demodulator Sampling Rate Lock		Makes all demodulator sampling rates equal. Enabling the lock copies the settings from demodulator 1 to all other demodulators. With locked sampling rates, any modification to a sampling rate is applied to all other sampling rate fields, too. Releasing the lock does not change any setting.
Trigger		Selects the acquisition mode of demodulated samples. Continuous trigger means data are streamed to the host computer at the Rate indicated.
	Continuous	Selects continuous data acquisition mode. The demodulated samples are streamed to the host computer at the Rate indicated on the left hand side. In continuous mode the numerical and plotter tools are continuously receiving and display new values.
	Trigger 1	Selects external triggering by means of the Trigger 1 connector. Demodulated samples are sent to the host computer for each event defined in the Trig Mode field. When edge trigger is selected the rate field is greyed out and has no meaning.
	Trigger 2	Selects external triggering by means of the Trigger 2 connector. Demodulated samples are sent to the host computer for each event defined in the Trig Mode field. When edge trigger is selected the rate field is greyed out and has no meaning.
	Trigger 1\ 2	Same functionality as above, but triggering is based on a logical OR function of Trigger 1 and Trigger 2.

Control/Tool	Option/ Range	Description
Trigger Mode		Defines the edge or level trigger mode for the selected Trigger input. Note: this field only appears when a non-continuous trigger is selected in the Trigger field.
	Rising	Selects triggered sample acquisition mode on rising edge of the selected Trigger input.
	Falling	Selects triggered sample acquisition mode on falling edge of the selected Trigger input.
	Both	Selects triggered sample acquisition mode on both edges of the selected Trigger input.
	High	Selects continuous sample acquisition mode on high level of the selected Trigger input. In this selection, the sample rate field determines the frequency in which demodulated samples are sent to the host computer.
	Low	Selects continuous sample acquisition mode on low level of the selected Trigger input. In this selection, the sample rate field determines the frequency in which demodulated samples are sent to the host computer.
Amplitude Unit	Vpk, Vrms, dBm	Select the unit of the displayed amplitude value. The dBm value is only valid for a system with 50 Ω termination.
Amplitude Mode		Indicates the type or source of the waveform being generated. 'Sine' indicates a sinusoidal waveform from the internal oscillator.
Amplitude Enable	ON / OFF	Enables individual output signal amplitude. When the MF-MD option is used, it is possible to generate signals being the linear combination of the available demodulator frequencies.
On	ON / OFF	Main switch for the Signal Output corresponding to the blue LED indicator on the instrument front panel.
50 Ω	ON / OFF	Select the load impedance between 50 Ω and HiZ. The impedance of the output is always 50 Ω . For a load impedance of 50 Ω the displayed voltage is half the output voltage to reflect the voltage seen at the load.
Range		Defines the maximum output voltage that is generated by the corresponding Signal Output. This includes the potential multiple Signal Amplitudes and Offsets summed up. Select the smallest range possible to optimize signal quality. This setting ensures that no levels or peaks above the setting are generated, and therefore it limits the values that can be entered as output amplitudes. Therefore selected output amplitudes are clipped to the defined range and the clipping indicator turns on. If 50 Ω target source or differential output is enabled the possible maximal output range will be half.
	10 mV	Selects output range ± 10 mV.
	100 mV	Selects output range ± 100 mV.
	1 V	Selects output range ± 1 V.
	10 V	Selects output range ± 10 V.
Auto Range		Selects the most suited output range automatically.
Output Clipping	grey/red	Indicates that the specified output amplitude(s) exceeds the range setting. Signal clipping occurs and the output signal quality is degraded. Adjustment of the range or the output amplitudes is required.
Offset	-range to range	Defines the DC voltage that is added to the dynamic part of the output signal.
Add	ON / OFF	The signal supplied to the Aux Input 1 is added to the signal output. For differential output the added signal is a common mode offset.

Control/Tool	Option/Range	Description
Diff	ON / OFF	Switch between single-ended output (OFF) and differential output (ON). In differential mode the signal swing is defined between Signal Output +V / -V.
Output	-range to range	Defines the output amplitude as rms or peak-to-peak value. A negative amplitude value is equivalent to a phase change of 180 degree.

5.16. Lock-in Tab (MF-MD option)

This tab is the main lock-in amplifier control panel for MFIA Instruments with the MF-MD Multi-demodulator option installed. Users with instruments without this option installed are kindly referred to [Lock-in Tab](#).

5.16.1. Features

- Functional block diagram with access to main input, output and demodulator controls
- Parameter table with main input, output and demodulator controls
- Controls for 4 individually configurable demodulators
- Auto ranging, scaling, arbitrary input units for both input channels
- Control for 4 oscillators
- Settings for main signal inputs and signal outputs
- Choice of reference source, trigger options and data transfer rates

5.16.2. Description

The Lock-in tab gives access to the lock-in amplifier functionality of the instrument and provides some of the advanced settings for impedance measurements. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.45: App icon and short description

Control/Tool	Option/Range	Description
Lock-in MD		Quick overview and access to all the settings and properties for signal generation and demodulation.

The default view of the Lock-in tab is the parameter table view. It is accessible under the side tab labeled All and provides controls for all demodulators in the instrument. Moreover, for each individual demodulator there is a functional block diagram available. It is accessible under the side tab labeled with the corresponding demodulator number.

Parameter Table

The parameter table (see [Figure 5.32](#)) consists of 5 sections: Signal Inputs, Oscillators, Demodulators, Output Amplitudes and Signal Outputs. The Demodulator section contains 4 rows each of them providing access to the settings of one dual phase demodulator. Demodulators 2 and 4 can be used for external referencing. Every demodulator can be connected to any of the possible inputs and oscillators.

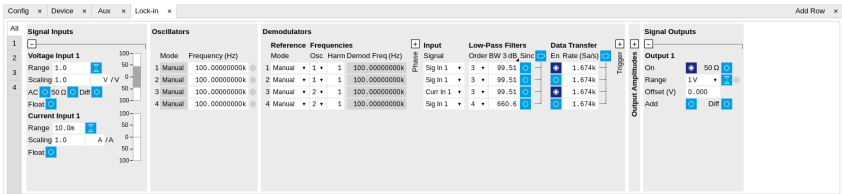


Figure 5.32: LabOne User Interface Lock-in tab with MF-MD Multi-demodulator option.

The **Signal Inputs** section allows the user to define all relevant settings specific to the signal entered as for example input coupling, range, etc. Some of the available options like phase adjustment and the trigger functionality are collapsed by default. It takes one mouse click on the "+" icon in order to expand those controls. On the right-hand side of the Lock-in tab the Signal Outputs section allows to define signal amplitudes, offsets and range values.

The Scaling field below the Range field can be used to multiply the Signal Input data for instance to account for the gain of an external amplifier. In case there is a transimpedance gain of 10 V/A applied to the input signal externally, then the Scaling field can be set to 0.1 and the Units field can be set to A in order to show the actual current readings through the entire user interface.

There are two buttons below the Scaling field that can be toggled: the AC/DC button and the 50 Ω /10 M Ω . The AC/DC button sets the coupling type: AC coupling has a high-pass cutoff frequency that can be used to block large DC signal components to prevent input signal saturation during amplification. The 50 Ω /10 M Ω button toggles the input impedance between low (50 Ω) and high (approx. 10 M Ω) input impedance. With 50 Ω input impedance, one will expect a reduction of a factor of 2 in the measured signal if the signal source also has an output impedance of 50 Ω .

Note

The Signal Inputs can be set to float which means that the BNC connector shield is no longer connected to the instrument ground. This setting affects both the current input and the voltage input. It is recommended to discharge the test device before connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.

The **Oscillator** section indicates the frequency of the 4 internal oscillators. Where the Mode indicator shows Manual the user can define the oscillator frequency manually defined by typing a frequency value in the field. In case the oscillator is referenced to an external source the Mode indicator will show ExtRef and the frequency field is set to read-only. External reference requires a PLL to do the frequency mapping onto an internal oscillator. Successful locking is indicated by a green light right next to the frequency field.

The next section contains the **Demodulators settings**. The block diagram displayed in [Figure 5.33](#) indicates the main demodulator components and their interconnection. The understanding of the wiring is essential for successfully operating the instrument.

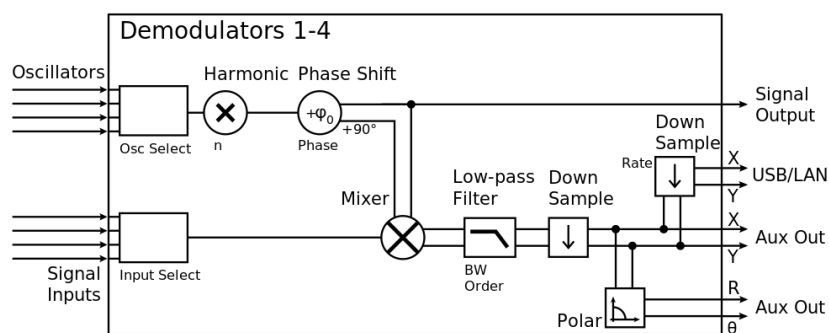


Figure 5.33: Demodulator block diagram with MF-MD Multi-demodulator option.

Every line in the Demodulators section represents one demodulator. The Mode column is read-only for all demodulators except 2 and 4, which can be set to either internal reference (Demod) or external reference mode (ExtRef). When internal reference mode is selected, it is possible to demodulate the input signal with 4 demodulators simultaneously at 4 independent frequencies and using different filter settings. For external reference mode, one demodulator is used for the reference recovery and a few settings are greyed-out, and therefore 3 demodulators remain for simultaneous measurements. In the Input Signal column one defines the signal that is taken as input for the demodulator. A wide choice of signals can be selected: Signal Inputs, the Trigger Inputs, the Auxiliary Inputs and Auxiliary Outputs. This allows to use the instrument for many different measurement topologies.

For each demodulator an additional phase shift can be introduced to the associated oscillator by entering the phase offset in the Phase column. This phase is added both, to the reference channel and the output of the demodulator. Hence, when the frequency is generated and detected using the same demodulator, signal phase and reference phase change by the same amount and no change will be visible in the demodulation result. Demodulation of frequencies that are integer multiples of any of the oscillator frequencies is achieved by entering the desired factor in the Harm column. The demodulator readout can be obtained using the Numeric tab which is described in [Numeric Tab](#).

In the middle of the Lock-in tab is the Low-Pass Filters section where the filter order can be selected in the drop down list for each demodulator and the filter bandwidth (BW 3dB) can be chosen by typing a numerical value. Alternatively the time constant of the filter (TC) or the noise equivalent power filter bandwidth (BW NEP) can be chosen by clicking on the column's header. For example, setting the filter order to 4 corresponds to a roll off of 24 dB/oct or 80 dB/dec i.e. an attenuation of 10^4 for a tenfold frequency increase. If the Low-Pass Filter bandwidth is comparable to or larger than the demodulation frequency, the demodulator output may contain frequency components at the frequency of demodulation and its higher harmonics. In this case, the additional Sinc Filter can be enabled. It attenuates those unwanted harmonic components in the demodulator output. The Sinc Filter is also useful when measuring at low frequencies, since it allows to apply a Low-Pass Filter bandwidth closer to the demodulation frequency, thus speeding up the measurement time.

The data transfer of demodulator outputs is activated by the En button in the Data Transfer section where also the sampling rate (Rate) for each demodulator can be defined.

The Trigger section next to the Data Transfer allows for setting trigger conditions in order to control and initiate data transfer from the Instrument to the host PC by the application of logic signals (e.g. TTL) to either Trigger Input 1 or 2 on the back panel.

The **Output Amplitudes** section is only available for Instruments with the MF-MD option installed and allows for the flexible adjustment of output amplitudes of different demodulators and their summation on the Signal Output. In order to avoid signal clipping the sum of amplitudes of each signal output needs to be smaller than the range defined in the Signal Outputs section on the right. By clicking the headline of each column one can switch between amplitude definitions in terms of root mean square values, peak-to-peak values or even units of dBm, when the 50 Ω option in the Signal Output section is activated. In the **Signal Outputs** section the On buttons allow to activate the Signal Output of the front panel. The Range drop down list is used to select the proper output range setting. On the Signal Output a digital offset voltage (Offset) can be defined. The maximum output signal permitted is ± 10 V.

Block Diagram

The block diagram view of the main instrument functions is also sometimes called the "Graphical Lock-in Tab". Depending on how many demodulators are available in the instrument a set of numbered side tabs are available giving access to a Graphical Lock-in Tab for each demodulator. The block diagrams are fully functional and provide the user with a visual feedback of what is going on inside the instrument. All control elements that are available in the Parameter Table detailed in the previous section are also present in the graphical representation.

The block diagram in [Figure 5.34](#) describes the signal path throughout the instrument for the case when the internal oscillator is used as reference. The Signal Inputs and Reference/Internal Frequency are described on the left side, the core of demodulation with the mixer and low-pass filter is located in the center of the tab and the Signal Outputs, the Auxiliary Outputs as well as the data transfer to the PC is sketched on the right.

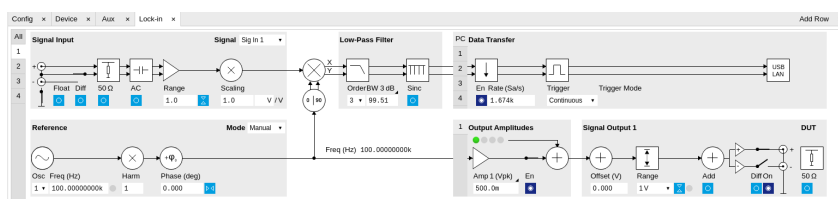


Figure 5.34: LabOne User Interface Lock-in tab - Graphical Lock-in tab in Internal Reference mode

The block diagram in [Figure 5.35](#) describes the signal path throughout the instrument for the case when an external reference is used. This setting is only available for demodulator 2/4. In order to map an external frequency to any of the oscillators, go to the Reference section of demodulator 2/4 and change the mode to ExtRef. This demodulator will then be used as a phase detector within the phase-locked loop. The software will choose the appropriate filter settings according to the frequency and properties of the reference signal. Once a demodulator is used to map an external frequency on to one of the internal oscillators, it is no longer available for other measurements.

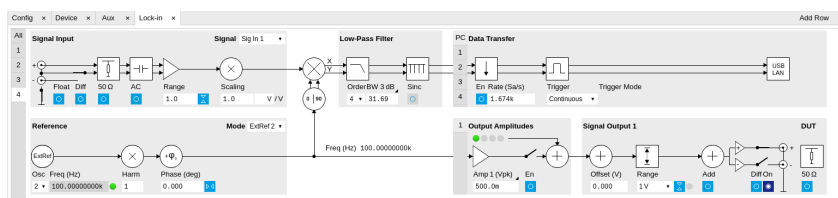


Figure 5.35: LabOne User Interface Lock-in tab - Graphical Lock-in tab in External Reference mode

5.16.3. Functional Elements

Table 5.46: Lock-in MF tab

Control/Tool	Option/Range	Description
Range	3.0 mV, 10 mV, 30 mV, 100 mV, 300 mV, 1 V, 3.0 V	Defines the gain of the analog input amplifier. The range should exceed the incoming signal by roughly a factor two including a potential DC offset. The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used.
On	ON / OFF	Enable Signal Input.
Auto		Automatic adjustment of the Range to about two times the maximum signal input amplitude measured over about 100 ms.
Scaling	numeric value	Applies an arbitrary scale factor to the input signal.
Measurement Unit	unit acronym	Defines the physical unit of the input signal. Use *, / and ^ operators, e.g., m or m/s^2. The value in this field modifies the readout of all measurement tools in the user interface. Typical uses of this field is to make measurements in the unit before the sensor/transducer, e.g. to take an transimpedance amplifier into account and to directly read results in Ampere instead of Volts.
Coupling	OFF: DC coupling	Defines the input coupling for the Signal Inputs. AC coupling inserts a high-pass filter.
	ON: AC coupling	
50 Ω	OFF: 10 M Ω	Switches between 50 Ω (ON) and 10 M Ω (OFF).
	ON: 50 Ω	
Diff	ON: Differential voltage input	Switches between single ended (OFF) and differential (ON) measurements.
	OFF: Single ended voltage input	
Float	OFF: GND connected	Switches between floating (ON) and connected grounds (OFF). This setting applies both to the voltage and the current input. It is recommended to discharge the test device before connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.
	ON: Floating	
Range	10 mA	Defines the gain of the current input amplifier. The range should exceed the incoming signal by roughly a factor two including a potential DC offset. The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used. Ranges marked with an asterisk (*) are only available on instruments with serial numbers MF-DEV3200 and higher.
	1 nA (*)	
	10 nA	
	100 nA (*)	
	1 μ A	
	10 μ A (*)	
	100 μ A	
	1 mA (*)	
Auto		Automatic adjustment of the Range to about two times the maximum current input amplitude measured over about 100 ms.
Scaling	numeric value	Applies an arbitrary scale factor to the current input.

Control/Tool	Option/ Range	Description
Measurement Unit	unit acronym	Defines the physical unit of the current input. Use *, / and ^ operators, e.g., m or m/s^2 The value in this field modifies the readout of all measurement tools in the user interface.
Mode		Indicates how the frequency of the corresponding oscillator is controlled (manual, external reference). Read only flag.
	Manual	The user setting defines the oscillator frequency.
	ExtRef	An external reference is mapped onto the oscillator frequency.
Frequency (Hz)	0 to 5 MHz	Frequency control for each oscillator.
Locked	ON / OFF	Oscillator locked to external reference when turned on.
Mode		Select the reference mode (manual or external reference) or indicate the unit that uses the demodulator (e.g. PLL).
	Manual	Default lock-in operating mode with manually set reference frequency.
	ExtRef	The demodulator is used for external reference mode and tracks the frequency of the selected reference input. The demodulator bandwidth is set automatically to adapt to the signal properties.
	ExtRef Low BW	The demodulator is used for external reference mode and tracks the frequency of the selected reference input. The demodulator bandwidth is fixed at low value. Use when automatic bandwidth adjustment interferes with a stable lock at a fixed frequency.
	ExtRef High BW	The demodulator is used for external reference mode and tracks the frequency of the selected reference input. The demodulator bandwidth is fixed at a high value. Use when automatic bandwidth adjustment interferes with tracking a strongly fluctuating frequency.
	PLL	The demodulator is used in PLL mode for frequency tracking of the signal. This function requires the MF-PID option.
	IA	The demodulator is used by the impedance analyzer.
Osc	oscillator index	Connects the selected oscillator with the demodulator corresponding to this line. Number of available oscillators depends on the installed options.
Harm	1 to 1023	Multiplies the demodulator's reference frequency with the integer factor defined by this field. If the demodulator is used as a phase detector in external reference mode (PLL), the effect is that the internal oscillator locks to the external frequency divided by the integer factor.
Harm	1 to 1023	Divides the demodulator's reference frequency by an integer factor in external reference mode.
Demod Freq (Hz)	0 to 5 MHz	Indicates the frequency used for demodulation and for output generation. The demodulation frequency is calculated with oscillator frequency times the harmonic factor. When the MF-MOD option is used linear combinations of oscillator frequencies including the harmonic factors define the demodulation frequencies.
Phase (deg)	-180° to 180°	Phase shift applied to the reference input of the demodulator.
Zero		Adjust the phase of the demodulator reference automatically in order to read zero degrees at the demodulator output. This action maximizes the X output, zeros the Y output, zeros the Θ output, and leaves the R output unchanged.
Signal		Selects the signal source to be associated to the demodulator.
	Sig In 1	Signal Input 1 is connected to the corresponding demodulator.

Control/Tool	Option/ Range	Description
	Curr In 1	Current Input 1 is connected to the corresponding demodulator.
	Trigger 1	Trigger 1 is connected to the corresponding demodulator.
	Trigger 2	Trigger 2 is connected to the corresponding demodulator.
	Aux Out 1	Internal value of Auxiliary Output 1 is applied to the input of the corresponding demodulator.
	Aux Out 2	Internal value of Auxiliary Output 2 is applied to the input of the corresponding demodulator.
	Aux Out 3	Internal value of Auxiliary Output 3 is applied to the input of the corresponding demodulator.
	Aux Out 4	Internal value of Auxiliary Output 4 is applied to the input of the corresponding demodulator.
	Aux In 1	Auxiliary Input 1 is connected to the corresponding demodulator.
	Aux In 2	Auxiliary Input 2 is connected to the corresponding demodulator.
	Constant	Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).
Order		Selects the filter roll off between 6 dB/oct and 48 dB/oct.
	1	1st order filter 6 dB/oct
	2	2nd order filter 12 dB/oct
	3	3rd order filter 18 dB/oct
	4	4th order filter 24 dB/oct
	5	5th order filter 30 dB/oct
	6	6th order filter 36 dB/oct
	7	7th order filter 42 dB/oct
TC/BW Select		Defines the display unit of the low-pass filters: time constant (TC) in seconds, noise equivalent power bandwidth (BW NEP) in Hz, 3 dB bandwidth (BW 3 dB) in Hz.
	TC	Defines the low-pass filter characteristic using time constant (s) of the filter.
	BW NEP	Defines the low-pass filter characteristic using the noise equivalent power bandwidth (Hz) of the filter.
	BW 3 dB	Defines the low-pass filter characteristic using the 3 dB cut-off frequency (Hz) of the filter.
TC/BW Value	numeric value	Defines the low-pass filter characteristic in the unit defined above.
Sinc	ON / OFF	Enables the sinc filter.
		When the filter bandwidth is comparable to or larger than the demodulation frequency, the demodulator output may contain frequency components at the frequency of demodulation and its higher harmonics. The sinc is an additional filter that attenuates these unwanted components in the demodulator output.

Control/Tool	Option/ Range	Description
Filter Lock		Makes all demodulator filter settings equal (order, time constant, bandwidth). Enabling the lock copies the settings from demodulator 1 to all other demodulators. With locked filters, any modification to a filter setting is applied to all other filters, too. Releasing the lock does not change any setting.
Enable Streaming	ON / OFF	Enables the data acquisition and streaming of demodulated samples to the host computer for the corresponding demodulator. The streaming rate is defined in the field on the right hand side. Enabling a stream activates a corresponding element in the numeric tab and allows for demodulated samples to be visualized and analyzed in any of the LabOne measurement tools. Note: increasing number of active demodulators increases load on physical connection to the host computer.
Rate (Sa/s)	0.056 Sa/s to 857 kSa/s	Defines the demodulator sampling rate, the number of samples that are sent to the host computer per second. A rate of about 7-10 higher as compared to the filter bandwidth usually provides sufficient aliasing suppression. This is also the rate of data received by LabOne Data Server and saved to the computer hard disk. This setting has no impact on the sample rate on the auxiliary outputs connectors. Note: the value inserted by the user may be approximated to the nearest value supported by the instrument.
Demodulator Sampling Rate Lock		Makes all demodulator sampling rates equal. Enabling the lock copies the settings from demodulator 1 to all other demodulators. With locked sampling rates, any modification to a sampling rate is applied to all other sampling rate fields, too. Releasing the lock does not change any setting.
Trigger		Selects the acquisition mode of demodulated samples. Continuous trigger means data are streamed to the host computer at the Rate indicated.
	Continuous	Selects continuous data acquisition mode. The demodulated samples are streamed to the host computer at the Rate indicated on the left hand side. In continuous mode the numerical and plotter tools are continuously receiving and display new values.
	Trigger 1	Selects external triggering by means of the Trigger 1 connector. Demodulated samples are sent to the host computer for each event defined in the Trig Mode field. When edge trigger is selected the rate field is greyed out and has no meaning.
	Trigger 2	Selects external triggering by means of the Trigger 2 connector. Demodulated samples are sent to the host computer for each event defined in the Trig Mode field. When edge trigger is selected the rate field is greyed out and has no meaning.
	Trigger 1\ 2	Same functionality as above, but triggering is based on a logical OR function of Trigger 1 and Trigger 2.
Trigger Mode		Defines the edge or level trigger mode for the selected Trigger input. Note: this field only appears when a non-continuous trigger is selected in the Trigger field.
	Rising	Selects triggered sample acquisition mode on rising edge of the selected Trigger input.
	Falling	Selects triggered sample acquisition mode on falling edge of the selected Trigger input.
	Both	Selects triggered sample acquisition mode on both edges of the selected Trigger input.
	High	Selects continuous sample acquisition mode on high level of the selected Trigger input. In this selection, the sample rate field determines the frequency in which demodulated samples are sent to the host computer.

Control/Tool	Option/ Range	Description
	Low	Selects continuous sample acquisition mode on low level of the selected Trigger input. In this selection, the sample rate field determines the frequency in which demodulated samples are sent to the host computer.
Amplitude Unit	Vpk, Vrms, dBm	Select the unit of the displayed amplitude value. The dBm value is only valid for a system with 50 Ω termination.
Amplitude Mode		Indicates the type or source of the waveform being generated. 'Sine' indicates a sinusoidal waveform from the internal oscillator.
Amplitude Enable	ON / OFF	Enables individual output signal amplitude. When the MF-MD option is used, it is possible to generate signals being the linear combination of the available demodulator frequencies.
Amplitude (V)	-range to range	Defines the output amplitude for each demodulator frequency as rms or peak-to-peak value. A negative amplitude value is equivalent to a phase change of 180 degree. Linear combination of multiple amplitude settings on the same output are clipped to the range setting. Note: the value inserted by the user may be approximated to the nearest value supported by the Instrument.
On	ON / OFF	Main switch for the Signal Output corresponding to the blue LED indicator on the instrument front panel.
50 Ω	ON / OFF	Select the load impedance between 50 Ω and HiZ. The impedance of the output is always 50 Ω . For a load impedance of 50 Ω the displayed voltage is half the output voltage to reflect the voltage seen at the load.
Range		Defines the maximum output voltage that is generated by the corresponding Signal Output. This includes the potential multiple Signal Amplitudes and Offsets summed up. Select the smallest range possible to optimize signal quality. This setting ensures that no levels or peaks above the setting are generated, and therefore it limits the values that can be entered as output amplitudes. Therefore selected output amplitudes are clipped to the defined range and the clipping indicator turns on. If 50 Ω target source or differential output is enabled the possible maximal output range will be half.
	10 mV	Selects output range ± 10 mV.
	100 mV	Selects output range ± 100 mV.
	1 V	Selects output range ± 1 V.
	10 V	Selects output range ± 10 V.
Auto Range		Selects the most suited output range automatically.
Output Clipping	grey/red	Indicates that the specified output amplitude(s) exceeds the range setting. Signal clipping occurs and the output signal quality is degraded. Adjustment of the range or the output amplitudes is required.
Offset	-range to range	Defines the DC voltage that is added to the dynamic part of the output signal.
Add	ON / OFF	The signal supplied to the Aux Input 1 is added to the signal output. For differential output the added signal is a common mode offset.
Diff	ON / OFF	Switch between single-ended output (OFF) and differential output (ON). In differential mode the signal swing is defined between Signal Output +V / -V.

5.17. Config Tab

The Config tab provides access to all major LabOne settings and is available on all MFIA instruments.

5.17.1. Features

- define instrument connection parameters
- browser session control
- define UI appearance (grids, theme, etc.)
- store and load instrument settings and UI settings
- configure data recording

5.17.2. Description

The Config tab serves as a control panel for all general LabOne settings and is opened by default on start-up. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.47: App icon and short description

Control/Tool	Option/Range	Description
Config		Provides access to software configuration.

The Config tab (see [Figure 5.36](#)) is divided into four sections to control connections, sessions, settings, user interface appearance and data recording.

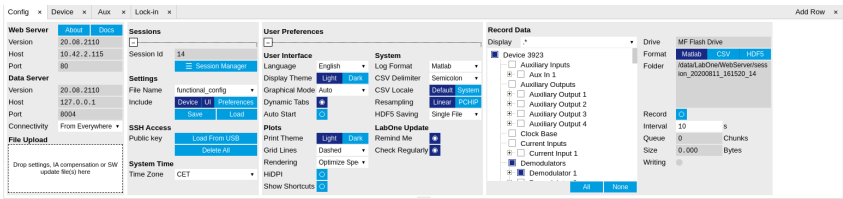


Figure 5.36: LabOne UI: Config tab

The **Connection** section provides information about connection and server versions. Access from remote locations can be restricted with the connectivity setting.

The **Session** section provides the session number which is also displayed in the status bar. Clicking on Session Dialog opens the session dialog window (same as start up screen) that allows one to load different settings files as well as to connect to other instruments.

The **Settings** section allows one to load and save instrument and UI settings. The saved settings are later available in the session dialog.

The **User Preferences** section contains the settings that are continuously stored and automatically reloaded the next time an MFIA instrument is used from the same computer account.

For low ambient light conditions the use of the dark display theme is recommended (see [Figure 5.37](#)).

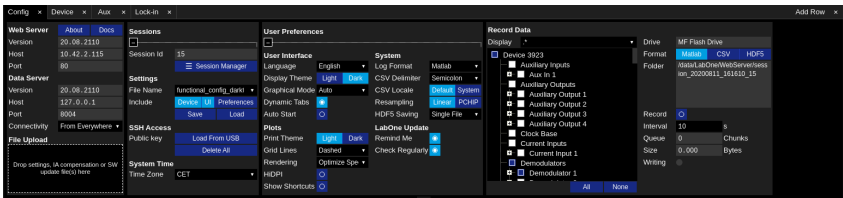


Figure 5.37: LabOne UI: Config tab - dark theme

The **Record Data** section contains all settings necessary to obtain hard copies of measurement data. The tree structure (see [Tree Selector](#) section) provides access to a large number of signals and instrument settings. Use the View Filter in order to reduce the tree structure to the most commonly used nodes such as the demodulator sample nodes. Whenever the Record button is enabled, all selected nodes get saved continuously in MATLAB, comma-separated value (CSV), or other

supported file formats. For each selected node at least one file gets generated, but the data may be distributed over several files during long recordings. The files are normally saved to the internal storage of the MFIA or on a mass storage device connected to one of the instrument's USB ports. In order to save data directly on the host computer, please install and run the LabOne Web Server on the host computer as described in [Running LabOne on a Separate PC](#). See [Saving and Loading Data](#) for more information on data saving. The quickest way to inspect the files after recording is to use the File Manager tab described in [File Manager Tab](#). Apart from the numerical data and settings, the files contain timestamps. These integer numbers encode the measurement time in units of the instrument clock period $1/(60 \text{ MHz})$. The timestamps are universal within one instrument and can be used to align the data from different files.

5.17.3. Functional Elements

Table 5.48: Config tab

Control/Tool	Option/Range	Description
About	About	Get information about LabOne software.
Web Server Version and Revision	string	Web Server version and revision number
Host	default is localhost: 127.0.0.1	IP-Address of the LabOne Web Server
Port	4 digit integer	LabOne Web Server TCP/IP port
Data Server Version and Revision	string	Data Server version and revision number
Host	default is localhost: 127.0.0.1	IP-Address of the LabOne Data Server
Port	default is 8004	TCP/IP port used to connect to the LabOne Data Server.
Connect/Disconnect		Connect/disconnect the LabOne Data Server of the currently selected device. If a LabOne Data Server is connected only devices that are visible to that specific server are shown in the device list.
Status	grey/green	Indicates whether the LabOne User Interface is connected to the selected LabOne data server. Grey: no connection. Green: connected. Red: error while connecting.
Connectivity	From Everywhere	Forbid/Allow to connect to this Data Server from other computers.
	Localhost Only	
File Upload	drop area	Drag and drop files in this box to upload files. Clicking on the box opens a file dialog for file upload. Supported files: Settings (*.xml), software update (LabOne*.tar). Uploading software updates will automatically trigger the update process if the file is valid and has a different revision than the currently installed software.
Session Id	integer number	Session identifier. A session is a connection between a client and LabOne Data Server.
Session Manager	Session Manager	Open the session manager dialog. This allows for device or session change. The current session can be continued by pressing cancel.
File Name	selection of available file names	Save/load the device and user interface settings to/from the selected file on the internal flash drive. The setting files can be downloaded/uploaded using the Files tab.
Include		Enable Save/Load of particular settings.
	No Include Settings	Please enable settings type to be included during Save/Load.
	Include Device	Enable Save/Load of Device settings.

Control/Tool	Option/Range	Description
	Include UI	Enable Save/Load of User Interface settings.
	Include UI and Device	Enable Save/Load of User Interface and Device settings.
	Include Preferences	Enable loading of User Preferences from settings file.
	Include UI, Device and Preferences	Enable Save/Load of User Interface, Device and User Preferences.
Save		Save the user interface and device setting to a file.
Load		Load the user interface and device setting from a file.
Load From USB		Load public SSH keys from the USB mass storage connected to the device. Uploading of SSH keys enables SSH access to the device.
Delete All		Delete all public SSH keys on the device.
Time Zone	Timezone offset from UTC (Coordinated Universal Time).	Select the time zone in which the device is used. Settings, data and log files receive timestamps from this time zone. Requires a power cycle for changes to become effective.
Language		Choose the language for the tooltips.
Display Theme	Dark	Choose theme of the user interface.
	Light	
Plot Print Theme	Dark	Choose theme for printing SVG plots.
	Light	
Plot Grid	None	Select active grid setting for all SVG plots.
	Dashed	
	Solid	
Plot Rendering		Select rendering hint about what tradeoffs to make as the browser renders SVG plots. The setting has impact on rendering speed and plot display for both displayed and saved plots.
	Auto	Indicates that the browser shall make appropriate tradeoffs to balance speed, crisp edges and geometric precision, but with geometric precision given more importance than speed and crisp edges.
	Optimize Speed	The browser shall emphasize rendering speed over geometric precision and crisp edges. This option will sometimes cause the browser to turn off shape anti-aliasing.
	Crisp Edges	Indicates that the browser shall attempt to emphasize the contrast between clean edges of artwork over rendering speed and geometric precision. To achieve crisp edges, the user agent might turn off anti-aliasing for all lines and curves or possibly just for straight lines which are close to vertical or horizontal.
	Geometric Precision	Indicates that the browser shall emphasize geometric precision over speed and crisp edges.
Resampling Method		Select the resampling interpolation method. Resampling corrects for sample misalignment in subsequent scope shots. This is important when using reduced sample rates with a time resolution below that of the trigger.
	Linear	Linear interpolation
	PCHIP	Piecewise Cubic Hermite Interpolating Polynomial
Show Shortcuts	ON / OFF	Displays a list of keyboard and mouse wheel shortcuts for manipulating plots.

Control/Tool	Option/Range	Description
Dynamic Tabs	ON / OFF	If enabled, sections inside the application tabs are collapsed automatically depending on the window width.
Graphical Mode	Collapsed	Select the display mode for the graphical elements. Auto format will select the format which fits best the current window width.
	Auto	
	Expanded	
Log Format	.NET	Choose the command log format. See status bar and [User] \Documents\Zurich Instruments\LabOne\WebServer\Log
	MATLAB	
	Python	
CSV Delimiter	Tab	Select which delimiter to insert for CSV files.
	Comma	
	Semicolon	
CSV Locale	System locale. Use the symbols set in the language and region settings of the computer	Select the locale used for defining the decimal point and digit grouping symbols in numeric values in CSV files. The default locale uses dot for the decimal point and no digit grouping, e.g. 1005.07. The system locale uses the symbols set in the language and region settings of the computer.
	Default locale. Dot for the decimal point and no digit grouping, e.g. 1005.07	
HDF5 Saving	Multiple files. Each measurement goes in a separate file	For HDF5 file format only: Select whether each measurement should be stored in a separate file, or whether all measurements should be saved in a single file.
	Single file. All measurements go in one file	
Auto Start	ON / OFF	Skip session manager dialog at start-up if selected device is available. In case of an error or disconnected device the session manager will be reactivated.
Update Reminder	ON / OFF	Display a reminder on start-up if the LabOne software wasn't updated in 180 days.
Update Check	ON / OFF	Periodically check for new LabOne software over the internet.
Drive		Select the drive for data saving.
	MF Flash Drive	Internal flash mass storage device of the MFLI/MFIA.
	USB 1/2	Mass storage device connected on USB.
Format	SXM (Nanonis)	File format of recorded and saved data.
	MATLAB	
	CSV	
	ZView	
Open Folder		Open recorded data in the system File Explorer.
Folder	path indicating file location	Folder containing the recorded data.
Save Interval	Time in seconds	Time between saves to disk. A shorter interval means less system memory consumption, but for certain file formats (e.g. MATLAB) many small files on disk. A longer interval means more system memory consumption, but for certain file formats (e.g. MATLAB) fewer, larger files on disk.
Queue	integer number	Number of data chunks not yet written to disk.

Control/Tool	Option/Range	Description
Size	integer number	Accumulated size of saved data in the current session.
Record	ON / OFF	Start and stop saving data to disk as defined in the selection filter. Length of the files is determined by the Window Length setting in the Plotter tab.
Writing	grey/green	Indicates whether data is currently written to disk.
Display	filter or regular expression	Display specific tree branches using one of the preset view filters or a custom regular expression.
Tree	ON / OFF	Click on a tree node to activate it.
All		Select all tree elements.
None		Deselect all tree elements.

For more information on the tree functionality in the Record Data section, please see [Tree Selector](#).

5.18. Device Tab

The Device tab is the main settings tab for the connected instrument and is available on all MFIA instruments.

5.18.1. Features

- Option and upgrade management
- External clock referencing (10 MHz)
- Instrument connectivity parameters
- Device monitor

5.18.2. Description

The **Device tab** serves mainly as a control panel for all settings specific to the instrument that is controlled by LabOne in this particular session. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.49: App icon and short description

Control/Tool	Option/Range	Description
Device		Provides instrument specific settings.

The Device tab (see [Figure 5.38](#)) is divided into five sections: general instrument information, configuration, communication parameters, device presets, and a device monitor.

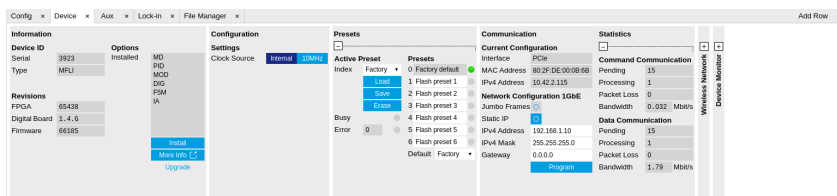


Figure 5.38: LabOne UI: Device tab

The **Information** section provides details about the instrument hardware and indicates the installed upgrade options. This is also the place where new options can be added by entering the provided option key.

The **Configuration** section allows one to change the reference from the internal clock to an external 10 MHz reference. The reference is to be connected to the Clock Input on the instrument back panel.

The **Presets** section allows you to define a custom instrument start-up configuration different from the factory default. This configuration is stored in the instrument itself and are applied

independently of the control PC. This saves time in cases where the control PC is not routinely needed, for instance when using only analog interfaces the instrument configuration is fixed.

The **Communication** section offers access to the instruments TCP/IP settings.

The **Statistics** section gives an overview on communication statistics. In particular the current data rate (Bandwidth) that is consumed.

Note

Packet loss on data streaming over UDP or USB: data packets may be lost if total bandwidth exceeds the available physical interface bandwidth. Data may also be lost if the host computer is not able to handle high-bandwidth data.

Note

Packet loss on command streaming over TCP or USB: command packets should never be lost as it creates an invalid state.

The **Device Monitor** section is collapsed by default and generally only needed for servicing. It displays vitality signals of some of the instrument's hardware components.

5.18.3. Functional Elements

Table 5.50: Device tab

Control/Tool	Option/Range	Description
Serial	1-4 digit number	Device serial number
Type	string	Device type
FPGA	integer number	HDL firmware revision.
Digital Board	version number	Hardware revision of the FPGA base board.
Firmware	integer number	Revision of the device internal controller software.
Installed Options	short names for each option	Options that are installed on this device.
Install		Click to install options on this device. Requires a unique feature code and a power cycle after entry.
More Information		Display additional device information in a separate browser tab.
Upgrade Device Options		Display available upgrade options.
Clock Source		10MHz reference clock source.
	Internal	The internal 10MHz clock is used as the frequency and time base reference.
	Clk 10MHz	An external 10MHz clock is intended to be used as the frequency and time base reference. Provide a clean and stable 10MHz reference to the appropriate back panel connector.
Index		Select between factory preset or presets stored in internal flash memory.
	Factory	Select factory preset.
	Flash 1-6	Select one of the presets stored in internal flash memory 1-6.
Load		Load the selected preset.

Control/Tool	Option/Range	Description
Save		Save the actual setting as preset.
Erase		Erase the selected preset.
Busy	grey/green	Indicates that the device is busy with either loading, saving or erasing a preset.
Error		Returns a 0 if the last preset operation was successfully completed or 1 if the last preset operation was illegal.
	0	Last preset operation was successfully completed.
	1	Last preset operation was illegal.
Error LED	grey/red	Turns red if the last operation was illegal.
Valid LED	grey/green	Turns green if a valid preset is stored at the respective location.
Presets		Shows a list of available presets including factory preset.
	0	Factory default preset. The name of the factory default preset is given and can not be edited.
	1	Flash preset 1. The name of this preset can be edited.
	2	Flash preset 2. The name of this preset can be edited.
	3	Flash preset 3. The name of this preset can be edited.
	4	Flash preset 4. The name of this preset can be edited.
	5	Flash preset 5. The name of this preset can be edited.
	6	Flash preset 6. The name of this preset can be edited.
Default		Indicates the preset which is used as default preset at start-up of the device.
	Factory	Select factory preset as default preset.
	Flash 1-6	Select one of the presets stored in internal flash memory 1-6 as default preset.
Interface	1. PCIe	Active interface between device and data server. In case multiple options are available, the priority as indicated on the left applies.
MAC Address	80:2F:DE:xx:xx:xx	MAC address of the device. The MAC address is defined statically, cannot be changed and is unique for each device.
IPv4 Address	default 192.168.1.10	Current IP address of the device. This IP address is assigned dynamically by a DHCP server, defined statically, or is a fall-back IP address if the DHCP server could not be found (for point to point connections).
Jumbo Frames	ON / OFF	Enable jumbo frames for this device and interface as default.
Static IP	ON / OFF	Enable this flag if the device is used in a network with fixed IP assignment without a DHCP server.
IPv4 Address	default 192.168.1.10	Static IP address to be written to the device.
IPv4 Mask	default 255.255.255.0	Static IP mask to be written to the device.
Gateway	default 192.168.1.1	Static IP gateway
Program		Click to program the specified IPv4 address, IPv4 Mask and Gateway to the device.
Pending	integer value	Number of buffers ready for receiving command packets from the device.
Processing	integer value	Number of buffers being processed for command packets. Small values indicate proper performance. For a TCP/IP interface, command packets are sent using the TCP protocol.

Control/Tool	Option/Range	Description
Packet Loss	integer value	Number of command packets lost since device start. Command packets contain device settings that are sent to and received from the device.
Bandwidth	numeric value	Command streaming bandwidth usage on the physical network connection between device and data server.
Pending	integer value	Number of buffers ready for receiving data packets from the device.
Processing	integer value	Number of buffers being processed for data packets. Small values indicate proper performance. For a TCP/IP interface, data packets are sent using the UDP protocol.
Packet Loss	integer value	Number of data packets lost since device start. Data packets contain measurement data.
Bandwidth	numeric value	Data streaming bandwidth usage on the physical network connection between device and data server.
Program		Program the specified WiFi settings to the device.
FW Load	numeric value	Indicates the CPU load on the processor where the firmware is running.
CPU Load	numeric value	Indicates the total CPU load on the machine where the data server is running.
CPU Temperature	numeric value	Indicates the temperature of the MF-internal CPU where the data server is running.
Memory Usage	numeric value	Indicates the total memory usage of the machine where the data server is running.

5.19. File Manager Tab

The File Manager tab provides a quick access to the files stored on the instrument flash drive as well as any mass storage attached to one of the USB connectors.

5.19.1. Features

- Download measurement data, instruments settings and log files to a local device
- Manage file structure (browse, copy, rename, delete) on instrument flash drive and attached USB mass storage devices
- Update instrument from USB mass storage
- File preview for settings files and log files

5.19.2. Description

The File Manager tab provides standard tools to see and organize the files on the instrument flash drive as well as on attached USB storage devices. Files can be conveniently copied, renamed, downloaded and deleted. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.51: App icon and short description

Control/Tool	Option/Range	Description
Files		Access files on internal flash memory and USB drive.

The Files tab (see [Figure 5.39](#)) provides three windows for exploring. The left window allows one to browse through the directory structure, the center window shows the files of the folder selected in the left window, and the right window displays the content of the file selected in the center window, e.g. a settings file or log file.

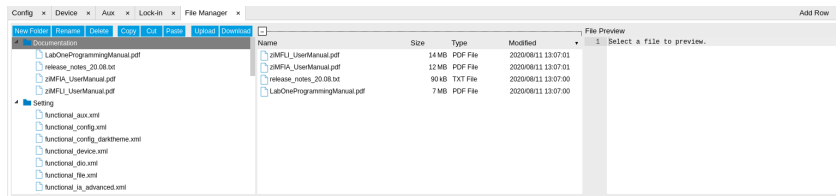


Figure 5.39: LabOne UI: File Manager tab

5.19.3. Functional Elements

Table 5.52: File tab

Control/Tool	Option/Range	Description
New Folder	New Folder	Create new folder at current location.
Rename	Rename	Rename selected file or folder.
Delete	Delete	Delete selected file(s) and/or folder(s).
Copy	Copy	Copy selected file(s) and/or folder(s) to Clipboard.
Cut	Cut	Cut selected file(s) and/or folder(s) to Clipboard.
Paste	Paste	Paste file(s) and/or folder(s) from Clipboard to the selected directory.
Upload	Upload	Upload file(s) and/or folder(s) to the selected directory.
Download	Download	Download selected file(s) and/or folder(s).

5.20. PID / PLL Tab

The PID / PLL tab is only available if the MF-PID Quad PID/PLL Controller option is installed on the MFIA Instrument (the installed options are displayed in the Device tab).

Note

The feedback controllers provide general-purpose PID functionality, phase-locked loop (PLL) functionality, and External Reference functionality. When the user sets one of the demodulators to ExtRef mode (see Lock-in tab, Demodulators section, Mode column), one of the PID controllers will be reserved for that purpose.

Note

Some settings in the PID / PLL tab are interdependent with settings that are accessible from other tabs. If the PID output controls a certain variable, e.g. Signal Output Offset, this variable will be shown as read-only where it appears in other tabs (i.e. in the Lock-in tab for this case).

5.20.1. Features

- Four fully programmable proportional, integral, derivative (PID) controllers
- Two fully programmable 5 MHz / 500 kHz phased-locked loops
- PID / PLL Advisor with multiple DUT models, transfer function, and step function modeling
- Auto Tune: Automatic minimization of the amplitude of the PID error signal
- High speed operation with up to 50 kHz loop filter bandwidth
- Input parameters: demodulator data, auxiliary inputs, auxiliary outputs
- Output parameters: output amplitudes, oscillator frequencies, demodulator phase, auxiliary outputs, signal output offsets
- Phase unwrap for demodulator θ data ($\pm 1024 \pi$), e.g. for optical phase-locked loops
- Bandwidth limit for the derivative (D) feedback component
- Programmable PLL center frequency and phase setpoint
- Programmable PLL phase detector filter settings
- Auto-zero functions for PLL center frequency and setpoint

- Generation of sub-multiple frequencies by use of harmonic multiplication factor

5.20.2. Description

The PID / PLL tab is the main control center for the feedback loop controllers in the instrument. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.53: App icon and short description

Control/Tool	Option/Range	Description
PID		Features all control, analysis, and simulation capabilities of the PID controllers.

The PID / PLL tab (see [Figure 5.40](#)) consists of four identical side-tabs, each of them providing access to the functionality of one of the four PID / PLL controllers and the associated PID Advisor.

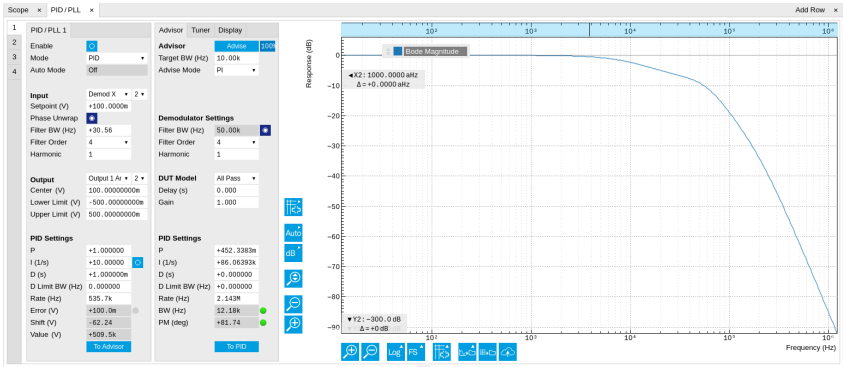


Figure 5.40: LabOne UI: PID / PLL tab

With their variety of different input and output connections, the LabOne PID controllers are extremely versatile and can be used in a wide range of different applications including laser locking or high-speed SPM. [Figure 5.41](#) shows a block diagram of all PID controller components, their interconnections and the variables to be specified by the user.

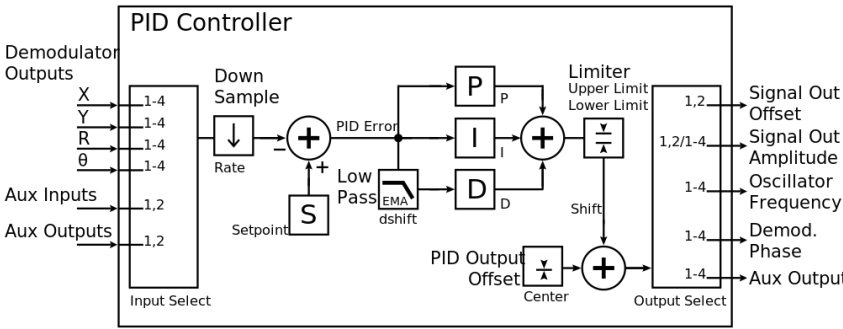


Figure 5.41: PID controller block diagram

Setting up a Control Loop

Depending on the application there are a number of ways to set up a control loop. Let's consider a few different approaches and see how the Advisor can help to reduce the effort and improve on the result and understanding of the setup.

Manual Setup

In cases where the transfer function of the device under test (DUT) is unknown and only little noise couples into the system from the environment, a manual approach is often the quickest way to get going. For manual configuration of a new control loop it is recommended to start with a small value for P and set the other parameters (I, D, D Limit) to zero. By enabling the controller one will then immediately see if the sign of P is correct and if the feedback is acting on the correct output parameter for instance by checking the numbers (Error, Shift, Out) displayed in the PID / PLL tab. A stepwise increase of the integral gain I will then help to zero the PID error signal completely. Enabling the derivative gain D can increase the speed of the feedback loop, but it can also cause an instable feedback loop behavior. Monitoring the PID error in the [Plotter Tab](#) in parallel can be a great help at this stage. The math tools offered by the Plotter allow us to display the standard deviation and the average value of the error. These values should be minimized by tweaking the PID parameters and the associated histogram should have a symmetric (ideally Gaussian) envelope.

In order to characterize the feedback loop quantitatively, you can measure the step response in the [Data Acquisition Tab](#). To do that measurement, change the PID setpoint manually after you have configured the DAQ Trigger level half way in between the old and new setpoint. DAQ Delay and Duration are chosen to roughly match the expected bandwidth. For a step response curve with fine time resolution, the PID data rate should be high enough.

PID Advisor

For many experimental situations the external device or DUT can be well approximated by a simple model. The LabOne PID Advisor allows you to simulate the behavior of a number of different DUT types in a feedback loop and choose feedback gain parameters based on the simulation. The DUTs are characterized by a model function with a number of parameters found on the Advisor sub-tab. All models include a setting for the delay that occurs outside the instrument. Depending on the targeted servo bandwidth, the external delay can often be the limiting factor and should be sensibly chosen.

Note

The delay specified for each model is the earliest possible response to a stepwise change of the instrument output to be seen on the instrument input. It describes the causality of the system and does not affect the shape of the DUT transfer function. Standard coaxial cables cause a signal delay of about 5 ns/m.

The most simple approach to modeling is to assume a DUT with a unity transfer function by using All Pass. The low-pass filters allow for limiting the bandwidth, to set an overall gain and a damping for the second order filter. With a Gain set to 1 and a Delay set to 0, All Pass can be used to model the PID controller independent of the external device. Resonator Frequency is a model that applies well in situations with a passive external component, e.g. a AFM cantilever or a quartz resonator, whose frequency should be tracked by a PLL over time. In cases where the amplitude of the resonator signal needs to be stabilized with a second control loop (automatic gain control), the Resonator Amplitude model is the right choice. Setting the resonance frequency and the Q factor, both can be obtained before by a frequency scan over the resonance using the [Sweeper Tab](#), allows the Advisor to estimate the gain and low-pass behavior of the resonator. Internal PLL is used whenever an external oscillating signal is provided that shall be followed by one of the internal oscillators. The VCO setting describes a situation where the input variable of the DUT is a voltage and the output is a frequency. The gain parameter specifies how much voltage change on the input causes how much frequency shift on the VCO output. In case the frequency of the VCO can be tracked by using the external reference mode, one can easily measure this gain with the [Sweeper Tab](#) by scanning the Auxiliary Output voltage and displaying the resulting oscillator frequency. The gain is given by the slope of the resulting line at the frequency of interest.

With a model and parameters set to best describe the actual measurement situation, one can now continue by defining a target bandwidth for the entire control loop and the Advise Mode, i.e. the feedback gain parameters that shall be used for the control operation. Whenever the input signal is derived from one of the demodulators it is convenient to activate the box next to target bandwidth. With that in place the Advise algorithm will automatically adjust the demodulator bandwidth to a value about 5 times higher than the target bandwidth in order to avoid to be limited by demodulation speed. The Advisor algorithm will now calculate a target step response function that it will try to achieve by adjusting the feedback gain parameters in the next step. Before doing so in case of a newly set up DUT model, the algorithm will first try to estimate the PID parameters by

using the Ziegler-Nichols method. When there has been a previous run, the user can also change the parameters in the model manually which will be used as new start parameters of the next Advise run. Starting from the initial parameters, the Advisor will then perform a numerical optimization in order to achieve a least-squares fit of the calculated step response to a target step response determined from the Target Bandwidth. The result is numerically characterized by an achieved bandwidth (BW) and a phase margin (PM). Moreover, the large plot area on the right can be used to characterize the result by displaying transfer functions, magnitude and phase, and step responses between different signal nodes inside the loop. Once the modeling is completed one can copy the resulting parameters to the physical PID by clicking on [To PID](#).

Table 5.54: DUT transfer functions

Name	Function	Parameters
All pass	$H(s) = g$	1. Gain g
Low-pass 1st	$H(s) = g \frac{1}{t_c s + 1} = g \frac{\omega_n}{s + \omega_n}$	1. Gain g 2. Filter bandwidth (BW) $f_{-3dB} = \omega_n / 2\pi$
Low-pass 2nd	$H(s) = g \frac{\omega_n^2}{s^2 + 2\omega_n \zeta s + \omega_n^2}$	1. Gain g 2. Resonance frequency $f_{res} = \omega_n / 2\pi$ 3. Damping ratio ζ with $f_{-3dB} = 2\zeta f_{res}$
Resonator frequency	$H(s) = -360^\circ \frac{t_c}{t_c s + 1}$ with $t_c = \frac{1}{2\pi BW} = \frac{2Q}{2\pi f_{res}}$	1. Resonance frequency f_{res} 2. Quality factor Q
Resonator amplitude	$H(s) = g \frac{\omega/(2Q)}{s + \omega/(2Q)}$ with $\omega = 2\pi f_{res}$	1. Gain g 2. Resonance frequency f_{res} 3. Quality factor Q
Internal PLL	$H(s) = -\frac{360^\circ}{s}$	
VCO	$H(s) = g \frac{360^\circ}{s(t_c s + 1)}$ with $t_c = \frac{1}{2\pi f_{res}}$	1. Gain g (Hz/V) 2. Bandwidth (BW) f_{-3dB}

Note

It is recommended to use the Advisor in a stepwise approach where one increases the free parameters from P to PI, to PID. This can save time because it prevents optimizing into local minima. Also it can be quite illustrative to see which of the feedback parameters leads to which effect in the feedback behavior.

Note

The low-pass filter in the differential part is implemented as an exponential moving average filter described by $y_t = (1 - \alpha) \cdot y_{t-1} + \alpha x_t$ with $\alpha = 2^{-dshift}$, x_t the filter input, and y_t the filter output. The default value for dshift is 0 which corresponds to a disabled filter. On the UI the filter properties can be changed in units of bandwidth or a time constant.

In case the feedback output is a voltage applied to sensitive external equipment it is recommended to make use of the center value and the upper and lower limit values. This will guarantee that the output stays in the defined range even when the lock fails and the integrator goes into saturation.

Auto Tune

The Auto Tune feature found on the Tuner sub-tab can now help to minimize the residual error signal. Auto Tune will vary the feedback gain parameters, as selected in the Advise Mode field in the Advisor sub-tab, in order to minimize the root mean square of the PID error signal. Being based on measurement, Auto Tune can often improve on the results of the model-based PID Advisor because it can take into account the real experimental noise and device transfer function. For Auto Tune to deliver good results, it should be applied in the actual operating conditions of the PID loop, as otherwise the PID bandwidth may end up too low. E.g. it makes no sense to apply Auto Tune to a PLL on a lifted AFM cantilever when the PLL later is used to track the cantilever during scanning.

The transfer function of the chosen PID settings can always be checked by copying the values to the Advisor by clicking the **To Advisor** button and selecting the Advanced Mode in the Display sub-tab. With the Response In set to Setpoint, the Response Out set to PID Output and with Closed-Loop deactivated one can visualize the Bode Magnitude of the PID controller's transfer function. This graph is what is usually shown in textbooks and is independent of the model function chosen in the DUT section. However, in order to simulate step responses or to calculate a bandwidth, a suitable model for the entire loop is required.

Setting up a phase-locked loop (PLL)

The PID controllers 1 and 2 can be set to PLL mode which the Mode selector. Changing to PLL mode sets the PID controller input to a demodulator phase, the PID controller output to the frequency of an internal oscillator, and changes some of the parameters to appropriate default values. [Figure 5.42](#) shows a block diagram of a PLL with its components, their interconnections and the variables to be specified by the user. The demodulator and the PID controller are slightly simplified for this sketch. Their full detailed block diagrams are given in [Demodulator block diagram without MF-MD Multi-demodulator option](#), [Demodulator block diagram with MF-MD Multi-demodulator option](#), and [Figure 5.41](#) respectively.

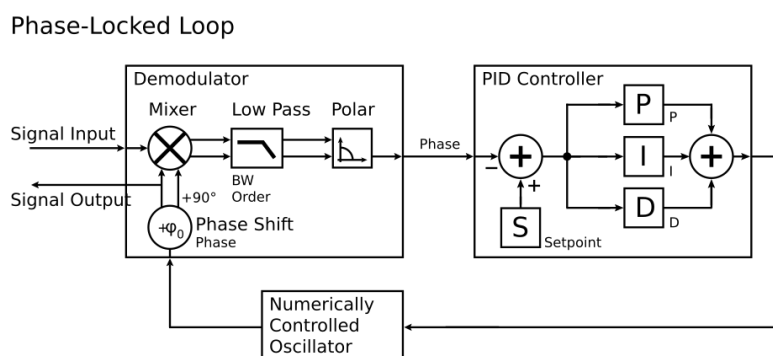


Figure 5.42: Phase-Locked Loop block diagram (components simplified)

In a typical procedure to set up a PLL one would first define the center frequency, frequency limits, and the phase setpoint in the left section. If the frequency is not known beforehand, it can often be determined using the Sweeper or Spectrum tool. Then one would set a target bandwidth in the Advisor section and click on the Advise button. The feedback parameters calculated by the Advisor will be shown in the fields just below. A graphical representation of the calculated transfer function is shown in the plot on the right-hand side. Once satisfied with the result, one can transfer the values to the instrument by clicking the **To PID** button, and then enable the PLL. If the Error/ PLL Lock field now displays very small values, the phase lock was successful. One can now iterate the process and e.g. play with the target bandwidth in the PLL Advisor to calculate a new set of feedback parameters. Displaying the oscillator frequency in the Plotter along with a Histogram and Math function (e.g. standard deviation) can help to quantify the residual phase error and further improve the lock performance by manual tweaking.

Note

The span set by the PLL frequency limits should exceed the target bandwidth by a factor of 5 to 10 or more.

Note

In the PID/PLL tab you select which of the demodulators you use as a phase detector. Open the Lock-in tab to check if the right Signal Input is associated with the demodulator in use.

5.20.3. Functional Elements

Table 5.55: PID tab: PID section

Control/Tool	Option/Range	Description
Enable	ON / OFF	Enable the PID controller
Mode		Operation mode of the PID module.
	PID	The PID is used for a general application.
	PLL	The PID is used to control an internal oscillator.
	ExtRef	The PID is used by the external reference to control an internal oscillator.
Auto Mode		This defines the type of automatic adaptation of parameters in the PID.
	Off	No automatic adaptation.
	PID Coeff	The coefficients of the PID controller are automatically set.
	Coeff + BW (low)	The PID coefficients, the filter bandwidth and the output limits are automatically set using a low bandwidth.
	Coeff + BW (high)	The PID coefficients, the filter bandwidth and the output limits are automatically set using a high bandwidth.
	Adaptive	All parameters of the PID including the center frequency are adapted.
Input		Select input source of PID controller
	Demodulator X	Demodulator cartesian X component
	Demodulator Y	Demodulator cartesian Y component
	Demodulator R	Demodulator magnitude component
	Demodulator Theta	Demodulator phase
	Aux Input	Auxiliary Input
	Aux Output	Internal value of Auxiliary Output
	TU Output Value	Output value of TU logic unit
Input Channel	index	Select input channel of PID controller.
Setpoint	numeric value	PID controller setpoint
Phase Unwrap	ON / OFF	Enables the phase unwrapping to track phase errors past the +/-180 degree boundary and increase PLL bandwidth.
Filter BW	numeric value	Bandwidth of the demodulator filter used as an input.
Filter Order		Selects the filter roll off between 6 dB/oct and 48 dB/oct of the current demodulator.
	1	1st order filter 6 dB/oct
	2	2nd order filter 12 dB/oct
	3	3rd order filter 18 dB/oct
	4	4th order filter 24 dB/oct
	5	5th order filter 30 dB/oct
	6	6th order filter 36 dB/oct

Control/Tool	Option/Range	Description
	7	7th order filter 42 dB/oct
	8	8th order filter 48 dB/oct
Harmonic	1 to 1023	Multiplier of the for the reference frequency of the current demodulator.
Output		Select output of the PID controller
	Sig Out 1 Amplitude	Feedback to the main signal output amplitude 1
	Sig Out 2 Amplitude	Feedback to the main signal output amplitude 2
	Oscillator Frequency	Feedback to any of the internal oscillator frequencies
	Demodulator Phase	Feedback to any of the 8 demodulator phase set points
	Aux Output Offset	Feedback to any of the 4 Auxiliary Output's Offset
	Signal Output Offset	Feedback to the main Signal Output offset adjustment
Output Channel	index	Select output channel of PID controller.
Center	numeric value	After adding the Center value to the PID output, the signal is clamped to Center + Lower Limit and Center + Upper Limit.
Lower Limit	numeric value	After adding the Center value to the PID output, the signal is clamped between Center - Lower Limit, and Center + Upper Limit.
Upper Limit	numeric value	After adding the Center value to the PID output, the signal is clamped between Center - Lower Limit, and Center + Upper Limit.
P (Hz/deg)	numeric value	PID proportional gain P
I (Hz/deg/s)	numeric value	PID integral gain I
D (Hz/deg*s)	numeric value	PID derivative gain D
D Limit TC/BW 3 dB	673 ns to 15 ms/ 10 Hz to 236 kHz	The cutoff of the low-pass filter for the D limitation, shown as either the filter time constant or the 3 dB cutoff frequency, depending on the selected TC mode. When set to 0, the low-pass filter is disabled.
Rate	3.7 Hz to 469 kHz	PID sampling rate and update rate of PID outputs. Needs to be set substantially higher than the targeted loop filter bandwidth. The numerical precision of the controller is influenced by the loop filter sampling rate. If the target bandwidth is below 1 kHz is starts to make sense to adjust this rate to a value of about 100 to 500 times the target bandwidth. If the rate is set too high for low bandwidth applications, integration inaccuracies can lead to non linear behavior.
Error	numeric value	Error = Set point - PID Input
Lock LED	grey/green	Indicates when the PID (configured as PLL) is locked. The PLL error is sampled at 5 Sa/s and its absolute value is calculated. If the result is smaller than 5 degrees the loop is considered locked. Only works if mode is PLL or ExtRef.
Shift	numeric value	Difference between the current output value Out and the Center. $\text{Shift} = P \cdot \text{Error} + I \cdot \text{Int}(\text{Error}, dt) + D \cdot d\text{Error}/dt$
Value	numeric value	Current output value
To Advisor	To Advisor	Copy the current PID settings to the PID Advisor.

Table 5.56: PID tab: Advisor sub-tab

Control/Tool	Option/Range	Description
Advise	Advise	Calculate the PID coefficients based on the used DUT model and the given target bandwidth. If optimized values can be found the coefficients are updated and the response curve is updated on the plot. Only PID coefficients specified with the advise mode are optimized. The Advise mode can be used incremental, means current coefficients are used as starting point for the optimization unless other model parameters are changed in-between.
Progress		The percentage of design algorithm already done when the Advisor is in progress.
Target BW (Hz)	numeric value	Target bandwidth for the closed loop feedback system which is used for the advising of the PID parameters. This bandwidth defines the trade-off between PID speed and noise.
Advise Mode		Select the PID coefficients that are optimized. The other PID coefficients remain unchanged but are used during optimization. This enables keeping selected coefficients at a fixed value while optimizing the rest. The advise time will increase significantly with the number of parameters to be optimized.
	P	Only optimize the proportional gain.
	I	Only optimize the integral gain.
	PI	Only optimize the proportional and the integral gain.
	PID	Optimize the proportional, integral, and derivative gains.
	PIDF	Optimize the proportional, integral, and derivative gains. Also the derivative gain bandwidth will be optimized.
Filter BW	numeric Value	Defines the low-pass filter characteristic of the selected demodulator input.
Auto Bandwidth	ON / OFF	Adjusts the demodulator bandwidth to fit best to the specified target bandwidth of the full system. If disabled, a demodulator bandwidth too close to the target bandwidth may cause overshoot and instability. In special cases the demodulator bandwidth can also be selected smaller than the target bandwidth.
Filter Order		Selects the filter roll off between 6 dB/oct and 48 dB/oct of the modelled demodulator.
	1	1st order filter 6 dB/oct
	2	2nd order filter 12 dB/oct
	3	3rd order filter 18 dB/oct
	4	4th order filter 24 dB/oct
	5	5th order filter 30 dB/oct
	6	6th order filter 36 dB/oct
	7	7th order filter 42 dB/oct
	8	8th order filter 48 dB/oct
Harmonic	1 to 1023	Multiplier of the for the reference frequency of the modelled demodulator.
DUT Model		Type of model used for the external device to be controlled by the PID. A detailed description of the transfer function for each model is found in the previous section.
	All Pass	The external device is modelled by an all pass filter. Parameters to be configured are delay and gain.

Control/ Tool	Option/ Range	Description
	LP 1st	The external device is modelled by a first-order low-pass filter. Parameters to be configured are delay, gain and filter bandwidth.
	LP 2nd	The external device is modelled by a second-order low-pass filter. Parameters to be configured are delay, gain, resonance frequency and damping ratio.
	Resonator Frequency	The external device is modelled by a resonator. Parameters to be configured are delay, center frequency and quality factor.
	Internal PLL	The DUT is the internal oscillator locked to an external signal through a phase-locked loop. The parameter to be configured is the delay.
	VCO	The external device is modelled by a voltage controlled oscillator. Parameters to be configured are delay, gain and bandwidth.
	Resonator Amplitude	The external device is modelled by a resonator. Parameters to be configured are delay, gain, center frequency and quality factor.
Delay	numeric value	Parameter that determines the earliest response for a step change. This parameter does not affect the shape of the DUT transfer function.
Gain	numeric value	Parameter that determines the gain of the DUT transfer function.
BW (Hz)	numeric value	Parameter that determines the bandwidth of the first-order low-pass filter respectively the bandwidth of the VCO.
Damping Ratio	numeric value	Parameter that determines the damping ratio of the second-order low-pass filter.
Res Freq	numeric value	Parameter that determines the resonance frequency of the of the modelled resonator.
Q	numeric value	Parameter that determines the quality factor of the modelled resonator.
P (Hz/deg)	numeric value	Proportional gain P coefficient used for calculation of the response of the PID model. The parameter can be optimized with PID advise or changed manually. The parameter only gets active on the PID after pressing the button To PLL.
I (Hz/deg/s)	numeric value	Integral gain I coefficient used for calculation of the response of the PID model. The parameter can be optimized with PID advise or changed manually. The parameter only gets active on the PID after pressing the button To PLL.
D (Hz/deg*s)	numeric value	Derivative gain D coefficient used for calculation of the response of the PID model. The parameter can be optimized with PID advise or changed manually. The parameter only gets active on the PID after pressing the button To PLL.
D Limit TC/ BW 3 dB	numeric value	The cutoff of the low-pass filter for the D limitation, shown as either the filter time constant or the 3 dB cutoff frequency, depending on the selected TC mode. When set to 0, the low-pass filter is disabled.
Rate	3.7 Hz to 469 kHz	PID sampling rate used for simulation. The advisor will update the rate to match with the specified target bandwidth. A sampling rate close to the target bandwidth and excessive higher bandwidth will results in a simulation mismatch.
BW (Hz)	numeric value	Simulated bandwidth of the full close loop model with the current PID settings. This value should be larger than the target bandwidth.
Target BW LED	green/red	Green indicates that the target bandwidth can be achieved. For very high PID bandwidth the target bandwidth might be only achieved using marginal stable PID settings. In this case, try to lower the bandwidth or optimize the loop delays of the PID system.
PM (deg)	numeric value	Simulated phase margin of the PID with the current settings. The phase margin should be greater than 45 deg for internal PLL and 60 deg for all other DUT for stable conditions. An Infinite value is shown if no unity gain crossing is available to determine a phase margin.

Control/Tool	Option/Range	Description
Stable LED	green/red	Green indicates that the phase margin is fulfilled and the PID system should be stable.
To PID	To PID	Copy the PID Advisor settings to the PID.

Table 5.57: PID tab: Tuner sub-tab

Control/Tool	Option/Range	Description
Auto Tune	Run/Stop	Optimize the PID parameters so that the noise of the closed-loop system gets minimized. The tuning method needs a proper starting point for optimization (away from the limits). The tuning process can be interrupted and restarted. The tuning will try to match the PID bandwidth with the loop bandwidth of the DUT, signal input (demodulator), and signal output.
Mode		Select the PID coefficients that are optimized. The other PID coefficients remain unchanged but are used during optimization. This allows to force coefficients to a value while optimizing the rest.
	P	Only optimize the proportional gain.
	I	Only optimize the integral gain.
	PI	Only optimize the proportional and the integral gain.
	PID	Optimize the proportional, integral, and derivative gains.
	PIDF	Optimize the proportional, integral, and derivative gains. Also the derivative gain bandwidth will be optimized.
Advanced Mode	ON / OFF	Enables manual selection of tuner averaging and setpoint toggling.
Iteration Time	numeric Value	Set the length for a tuner optimization iteration.
Enable	ON / OFF	Enables the setpoint toggle
Alt Setpoint	numeric Value	Defines the second setpoint used for the setpoint toggle.
Toggle Rate	numeric Value	Defines the rate of setpoint toggling. Note that possible values are logarithmically spaced with a factor of 4 between values.

Table 5.58: PID tab: Display sub-tab

Control/Tool	Option/Range	Description
Advanced Mode	ON / OFF	Enables manual selection of display and advice properties. If disabled the display and advice settings are automatically with optimized default values.
Display		Select the display mode used for rendering the system frequency or time response.
	Bode Magnitude	Display the Bode magnitude plot.
	Bode Phase	Display the Bode phase plot.
	Step Resp	Display the step response plot.
Start (Hz)	numeric value	Start frequency for Bode plot display. For disabled advanced mode the start value is automatically derived from the system properties and the input field is read-only.
Stop (Hz)	numeric value	Stop frequency for Bode plot display. For disabled advanced mode the stop value is automatically derived from the system properties and the input field is read-only.

Control/Tool	Option/Range	Description
Start (s)	numeric value	Start time for step response display. For disabled advanced mode the start value is zero and the field is read-only.
Stop (s)	numeric value	Stop time for step response display. For disabled advanced mode the stop value is automatically derived from the system properties and the input field is read-only.
Transfer Function Selector		Selection of the displayed transfer function of the loop. 2 presets and a manual selection are possible. In closed loop configuration all elements from output to input will be included as feedback elements.
	System	From Setpoint to System Output.
	PID	From Setpoint to PID Output.
	Manual	Any transfer function in the open or closed loop can be visualized.
Response In		Start point for the plant response simulation for open or closed loops. In closed loop configuration all elements from output to input will be included as feedback elements.
	Demod Input	Start point is at the demodulator input.
	Setpoint	Start point is at the setpoint in front of the PID.
	PID Output	Start point is at PID output.
	Instrument Output	Start point is at the instrument output.
	DUT Output	Start point is at the DUT output and instrument input.
Response Out		End point for the plant response simulation for open or closed loops. In closed loop configuration all elements from output to input will be included as feedback elements.
	PID Output	End point is at PID output.
	Instrument Output	End point is at the instrument output.
	DUT Output	End point is at the DUT output and instrument input.
	Demod Input	End point is at the demodulator input.
	System Output	End point is at the output of the controlled system.
Closed-Loop	ON / OFF	Switch the display of the system response between closed or open loop.
TC Mode	ON / OFF	Enables time constant representation of PID parameters.
Set Limits	ON / OFF	Switch the writing of PID limits when 'To PID' is pressed. Only applies in case of internal PLL.
Advisor Link		Automatically copy cursor values displayed below to the PID advisor. To enable cursor helpers, switch Advanced Mode on and set Display to Bode Magnitude with PID Transfer Function. Cursors will be displayed in Log and dB axis scale combinations.
P		Cursor value representing PID proportional gain P. Drag the plot cursor with the mouse pointer or directly insert numerical value here.
I		Cursor value representing PID integral gain I. Drag the plot cursor with the mouse pointer or directly insert numerical value here.
D		Cursor value representing PID derivative gain D. Drag the plot cursor with the mouse pointer or directly insert numerical value here.
Max Rate (Sa/s)	1 to 469 kSa/s	Target Rate for PID output data sent to PC. This value defines the applied decimation for sending data to the PC. It does not affect any other place where PID data are used.

Control/ Tool	Option/ Range	Description
Data Rate (Sa/s)		Actual data transfer rate of the PID stream data sent to PC. It is defined based on Max Rate which can be adjusted in Display tab of PID Advisor.
Data Rate (Sa/s)		Actual data transfer rate of the PID stream data sent to PC. It is defined based on Max Rate.
Stream Overflow Indicator		Indicates an overflow in the stream.

5.21. Threshold Tab

The Threshold Unit tab allows the user to define discrimination conditions and logical operations on a selection of input signals and output the result as TTL signals in real time. This functionality and tab is available on all MF instruments.

5.21.1. Features

- 4 threshold units for state detection
- analog input signals: demodulator data (X, Y, R, Θ), PID data (error, output, shift; requires MF-PID Quad PID/PLL Controller option option)
- digital input signals: 32 DIO channels, input and output overflows, etc.
- threshold detection: above, below, inside, outside, rising edge, falling edge
- absolute value and low-pass filtering can be applied prior to threshold detection
- configurable activation and deactivation times
- combination of up to 3 logical signals on one output with NOT, AND, OR, and XOR operation
- minimum length, hold, and invert functionality

5.21.2. Description

The Threshold tab enables real-time logical analysis of measurement data in order to detect special conditions or failures, and trigger a reaction via a TTL output. Typical applications are AFM tip protection or microfluidics cell sorting. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.59: App icon and short description

Control/ Tool	Option/ Range	Description
Threshold		Configure the threshold unit and the logical combinations to generate digital output signals.

The Threshold tab (see [Figure 5.43](#)) is divided in a Thresholds section on the left, and a Logic units section on the right. The rows in the Thresholds section represent the input signals of the unit and the discretization action that converts them into logical signals. The rows in the Logic units section represent the output signals of the unit and the logical operations performed to generate these signals.

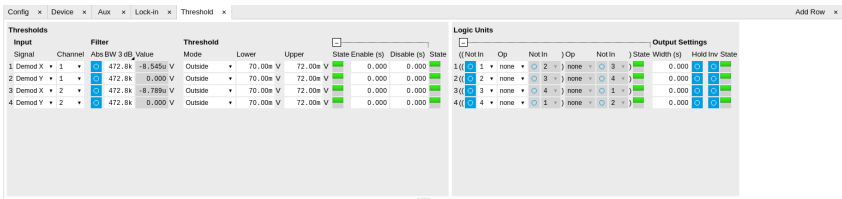


Figure 5.43: LabOne UI: Threshold tab

The available input signals include analog signals such as demodulator R, but also digital inputs or error flags of the instrument such as Signal Input overload. An analog input signal can be low-pass filtered which helps to avoid false alarms when short glitches occur. After filtering the signal is discretized by comparing it to a user-defined threshold or range. Configurable minimum activation times after crossing a threshold allow further tuning of the time-domain behavior. The filtered

threshold signal can be displayed in the [Scope Tab](#) which helps in defining meaningful threshold conditions. When digital signals are selected as inputs, the activation and deactivation times can be set the same way as for the analog signals. The figures below illustrate and detail the signal processing in the Threshold Unit using flow diagrams and signal graphs.

In order to output these signals on the Trigger outputs, the [DIO Tab](#) Mode setting needs to be set to Threshold units, and the Trigger output Drive needs to be enabled. By default, the four logical signals are routed as is to the four Trigger outputs. To set up more complex state detection, up to three input signals can be combined on one output using logical operations. The configuration of the logical operations in the Logic units section is collapsed by default. Base logical operators (AND, OR, XOR) as well as logical inversion (NOT) are available and grouping of the operations is indicated by brackets in the user interface.

Thresholds

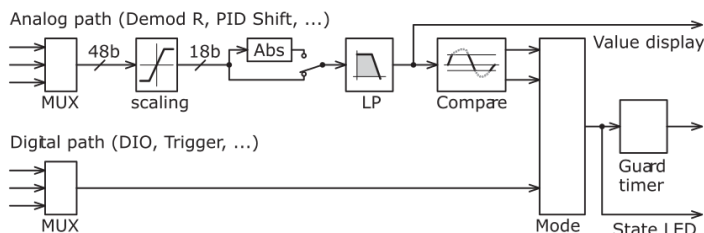


Figure 5.44: Flow diagram representing the Thresholds section of the Threshold tab.

Guard timer

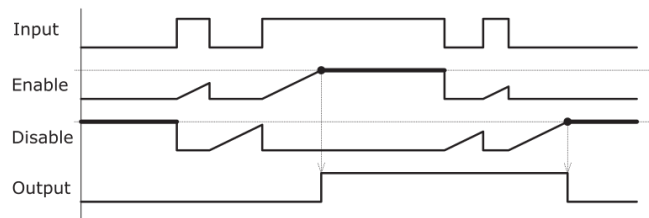


Figure 5.45: Signal graph showing an example of the signal processing that occurs in the Guard Timer. The Guard Timer block corresponds to the Enable/Disable settings in the Thresholds section of the Threshold tab and is represented in

Logic Units

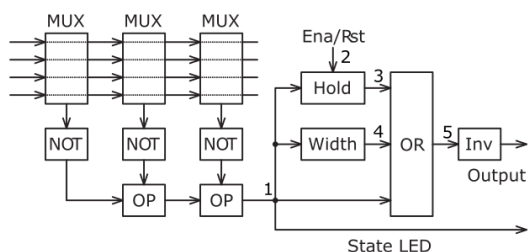


Figure 5.46: Flow diagram for the Logic Units section of the Threshold tab.

Hold and Width functionality

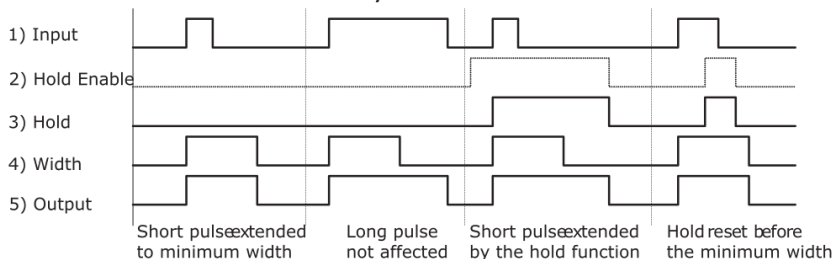


Figure 5.47: Signal graph showing an example of the signal processing by the Hold and Width functionality of the Threshold Unit. The represented signals 1 to 5 are indicated in

5.21.3. Functional Elements

Table 5.60: Threshold tab

Control/Tool	Option/ Range	Description
Signal		Select the signal source to be used in the threshold unit.
	X	Select the demodulator X component as input.
	Y	Select the demodulator Y component as input.
	R	Select the demodulator magnitude component as input.
	Θ	Select the demodulator phase component as input.
	PID Out	Use the PID controller's output signal as input. PID option needs to be installed.
	PID Shift	Use the PID controller's shift signal. PID option needs to be installed.
	PID Error	Use the PID controller's error as input. PID option needs to be installed.
	DIO	Select one of the DIO channels as input.
	Trigger In	Select a trigger input as input.
	Trigger Out	Select a trigger output as input.
	Input Overload (V)	Use voltage input overload as input.
	Input Overload (I)	Use current input overload as input.
	Output Overload	Use signal output overload as input.
	Aux Input Overload	Use aux input overload as input.
	Aux Output Overload	Use aux output overload as input.
	PID Output Overload	Use PID output overload as input.
	TU Output Value	Use TU output value as input.
Channel	index	Select the channel according to the selected signal source.
Abs	ON / OFF	Takes the absolute value of an analog input signal.
TC/BW Value	numeric value	Defines the characteristic time constant of a low-pass filter applied to an analog input signal.
Value	numeric value	Shows the value after the low-pass filter.
Source Value Lower Saturation Status		Indicates whether the source saturates at the lower bound.
	Off	The source value does not saturate at the lower bound.
	On	The source value saturates at the lower bound.
Source Value Upper Saturation Status		Indicates whether the source saturates at the upper bound.
	Off	The source value does not saturate at the upper bound.
	On	The source value saturates at the upper bound.
Mode		Selects the analysis mode defining the output signal.
	Above	Enable if the Signal is above Upper Threshold.
	Below	Enable if the Signal is below Lower Threshold.
	Outside	Enable if the Signal is outside the range [Lower Threshold, Upper Threshold].

Control/Tool	Option/Range	Description
	Rising Edge	Enable if the Signal crosses Upper Threshold from below Lower Threshold. The difference between Upper and Lower Threshold defines the threshold hysteresis.
	Falling Edge	Enable if the Signal crosses Lower Threshold from above Upper Threshold. The difference between Upper and Lower Threshold defines the threshold hysteresis.
Lower	numeric value	Lower threshold used to generate the output. In Rising Edge mode, this parameter defines the hysteretic behavior as the output state is reset only when the signal crosses Lower Threshold from above.
Upper	numeric value	Upper threshold used to generate the output. In Falling Edge mode, this parameter defines the hysteretic behavior as the output state is reset only when the signal crosses Upper Threshold from below.
State	low / high / toggling	Indicates the current output of the threshold unit before the Minimum Time analysis.
Enable	numeric value	Minimum duration of threshold violation needed before the output is activated. This can be used as a glitch filter.
Disable	numeric value	Minimum duration of threshold compliance needed before the output is deactivated.
State	low / high / toggling	Indicates the current output of the threshold after the Minimum Time analysis.
Not	ON / OFF	Inverts the input signal.
In	index	Selects threshold channel to be used as input.
Op		Logical operation applied to the signals to the left and right of the control.
	none	No logical operation selected. No additional signals will be used to generate the output.
	AND	Use a logical AND operation.
	OR	Use a logical OR operation.
	XOR	Use a logical XOR operation.
State	low / high / toggling	Indicates the current output of the threshold after the logical combination.
Width	numeric value	Select a minimum pulse width for the generated output signal. The width of pulses shorter than this time will be extended.
Hold	ON / OFF	Hold the output state indefinitely once it has changed to the activated state.
Inv	ON / OFF	Changes the output signal to low-active.
State	low / high / toggling	Final output state.

5.22. MOD Tab

The MOD tab provides access to the settings of the amplitude and frequency modulation units. This tab is only available when the MF-MOD AM/FM Modulation option is installed on the Instrument (see Information section in the Device tab).

Note

The MF-MOD AM/FM Modulation option requires the MF-MD Multi-demodulator option.

5.22.1. Features

- Phase coherently add and subtract oscillator frequencies and their multiples
- Control for AM and FM demodulation
- Control for AM and narrow-band FM generation
- Direct analysis of higher order carrier frequencies and sidebands

5.22.2. Description

The MOD tab offers control in order to phase coherently add and subtract the frequencies of multiple numerical oscillators. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.61: App icon and short description

Control/Tool	Option/Range	Description
MOD		Control panel to enable (de)modulation at linear combinations of oscillator frequencies.

The MOD tab (see [Figure 5.48](#)) is divided into two horizontal sections, one for each modulation unit.

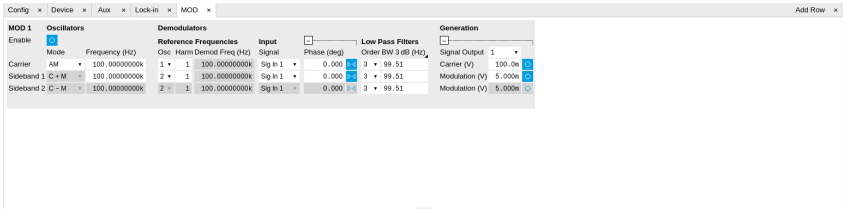


Figure 5.48: LabOne UI: MOD tab

The modulation units are designed for experiments involving multiple frequencies. For many of such experiments the associated spectrum reveals a dominant center frequency, often called the carrier, and one or multiple sidebands symmetrically placed around the carrier. Typical examples are amplitude modulated (AM) signals with one carrier and two sidebands separated from the carrier by the AM modulation frequency. Another example is frequency modulation (FM) where multiple sidebands to the left and right of the carrier can appear. The relative amplitude of the sideband for both AM and FM depends on the modulation depth, which is often expressed by the modulation index.

The classical approach of analyzing such signals (in particular when only analog instruments are available) is to use a configuration called tandem demodulation. This is essentially the serial cascading of lock-in amplifiers. The first device is referenced to the carrier frequency and outputs the in-phase component. This is then fed into the subsequent lock-in amplifiers in order to extract the different sideband components. There are several downsides to this scheme:

- The quadrature component of the first lock-in tuned to the carrier has to be continuously zeroed out by adjusting the reference phase. Otherwise a serious part of the signal power is lost for the analysis which usually leads to a drop in SNR.
- The scheme scales badly in terms of the hardware resources needed, in particular if multiple sideband frequencies need to be extracted.
- Every time a signal enters or exits an instrument the SNR gets smaller (e.g. due to the instrument inputs noise). Multiple such steps can deteriorate signal quality significantly.

All these shortcomings are nicely overcome by providing the ability to generate linear combinations of oscillator frequencies and use these combinations as demodulation references.

The MOD unit is linked to demodulators 1, 2, and 3. It can make use of up to 3 oscillators, which can be even referenced to an external source by using ExtRef or a PLL. [Figure 5.49](#) gives an overview of the different components involved and their interconnections.

MF-MOD Option

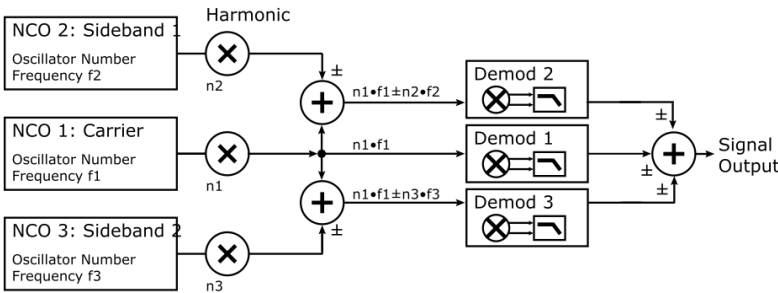


Figure 5.49: Modulation Option block diagram

For convenience the UI provides access to presets for AM and FM in the Mode column. In Manual Mode all settings can be chosen freely.

Note

Whenever a MOD unit is enabled, all the settings in the Lock-in tab that are controlled by this unit will be set to read-only.

On top of signal analysis the MF-MOD AM/FM Modulation option can also be utilized for signal generation. The Generation section provides all the necessary controls to adjust the carrier and sideband amplitudes.

Note

FM signals are generated by coherent superposition of the carrier signal with two sideband frequencies on either side that have the same amplitudes but opposite phases. The phase shift is achieved by using negative amplitudes as displayed in the Lock-in tab. This FM generation method approximates true FM as long as the modulation index is well below 1, i.e. higher-order sidebands can be neglected. For a modulation index of 1 true FM provides more than 13% of signal power in the second and higher order sidebands.

More details regarding AM and FM signal analysis and generation can be found on the Zurich Instruments web page.

5.22.3. Functional Elements

Table 5.62: MOD tab

Control/Tool	Option/Range	Description
Enable	ON / OFF	Enable the modulation
Mode	AM/FM/manual	Select the modulation mode.
Mode		Enabling of the first sideband and selection of the position of the sideband relative to the carrier frequency for manual mode.
	Off	First sideband is disabled. The sideband demodulator behaves like a normal demodulator.
	C + M	First sideband to the right of the carrier
	C - M	First sideband to the left of the carrier
Mode		Enabling of the second sideband and selection of the position of the sideband relative to the carrier frequency for manual mode.
	Off	Second sideband is disabled. The sideband demodulator behaves like a normal demodulator.

Control/Tool	Option/Range	Description
	C + M	Second sideband to the right of the carrier
	C - M	Second sideband to the left of the carrier
Frequency (Hz)	0 to 5 MHz	Sets the frequency of the carrier.
Frequency (Hz)	0 to 5 MHz	Frequency offset to the carrier from the first sideband.
Frequency (Hz)	0 to 5 MHz	Frequency offset to the carrier from the second sideband.
Carrier	oscillator index	Select the oscillator for the carrier signal.
Sideband 1	oscillator index	Select the oscillator for the first sideband.
Sideband 2	oscillator index	Select the oscillator for the second sideband.
Harm	1 to 1023	Set harmonic of the carrier frequency. 1=Fundamental
Harm	1 to 1023	Set harmonic of the first sideband frequency. 1 = fundamental
Harm	1 to 1023	Set harmonic of the second sideband frequency. 1 = fundamental
Demod Freq (Hz)	0 to 5 MHz	Carrier frequency used for the demodulation and signal generation on the carrier demodulator.
Demod Freq (Hz)	0 to 5 MHz	Absolute frequency used for demodulation and signal generation on the first sideband demodulator.
Demod Freq (Hz)	0 to 5 MHz	Absolute frequency used for demodulation and signal generation on the second sideband demodulator.
Channel	Signal Input, Current Input, Ref / Trigger, Auxiliary Inputs, Auxiliary Outputs	Select Signal Input for the carrier demodulation
Channel	Signal Input, Current Input, Ref / Trigger, Auxiliary Inputs, Auxiliary Outputs	Select Signal Input for the sideband demodulation
Phase	-180° to 180°	Phase shift applied to the reference input of the carrier demodulator and also to the carrier signal on the Signal Outputs
Phase	-180° to 180°	Phase shift applied to the reference input of the sideband demodulator and also to the sideband signal on the Signal Outputs
Zero		Adjust the carrier demodulator's reference phase automatically in order to read zero degrees at the demodulator output. This action maximizes the X output, zeros the Y output, zeros the Θ output, and leaves the R output unchanged.
Zero		Adjust the sideband demodulator's reference phase automatically in order to read zero degrees at the demodulator output. This action maximizes the X output, zeros the Y output, zeros the Θ output, and leaves the R output unchanged.
Order	1 to 8	Filter order used for carrier demodulation
Order	1 to 8	Filter order used for sideband demodulation
TC/BW Value	numeric value	Defines the low-pass filter characteristic in the unit defined above for the carrier demodulation
TC/BW Value	numeric value	Defines the low-pass filter characteristic in the unit defined above for the sideband demodulation

Control/Tool	Option/Range	Description
Signal Output	1, 2 or both	Select Signal Output 1, 2 or none
Carrier (V)	-range to range	Set the carrier amplitude
Modulation (V)	-range to range	Set the amplitude of the first sideband component.
Modulation (V)	-range to range	Set the amplitude of the second sideband component.
Index	-range to range	In FM mode, set modulation index value. The modulation index equals peak deviation divided by modulation frequency.
Peak Dev (Hz)	-range to range	In FM mode, set peak deviation value.
Enable FM Peak Mode	ON / OFF	In FM mode, choose to work with either modulation index or peak deviation. The modulation index equals peak deviation divided by modulation frequency.
Enable	ON / OFF	Enable the signal generation for the first sideband
Enable	ON / OFF	Enable the signal generation for the second sideband
Enable	ON / OFF	Enable the carrier signal

5.23. Multi Device Sync Tab

The Multi Device Sync (MDS) tab gives access to the automatic timing synchronization of measurement data from multiple MF instruments. This functionality and tab is available on all MF instruments.

5.23.1. Features

- Automatic timing synchronization across instruments
- Periodic check of synchronization
- Selectable instrument subgroup
- Status display

5.23.2. Description

The Multi Device Sync tab contains the controls and status information for synchronized measurements on multiple instruments. Whenever the tab is closed or an additional one of the same type is needed, clicking the following icon will open a new instance of the tab.

Table 5.63: App icon and short description

Control/Tool	Option/Range	Description
MDS		Synchronize multiple instruments.

The Multi Device Sync tab shown in [Figure 5.50](#) consists of the Available Devices section, a Status section, and a wiring diagram.

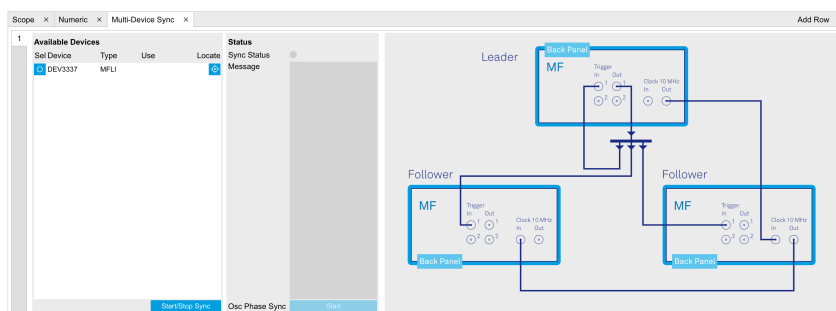


Figure 5.50: LabOne UI: Multi Device Sync tab

The Multi Device Synchronization feature provides an automated functionality to remove the clock offset of separate MF instruments. This enables a correct simultaneous display of their data in the [Plotter Tab](#) and helps when analyzing recorded data. In multi-channel applications that require sub-microsecond timing precision, the user can therefore benefit from having synchronized data from the start, rather than having to manually measure and compensate the clock offset in post-processing.

The first prerequisite for automatic synchronization is that all instruments are connected to the same LabOne Data Server (see [LabOne Software Start-up](#)). This is only possible when running the LabOne software on a separate PC as explained in [Running LabOne on a Separate PC](#).

To make these connections, click on [Session Manager](#) in the [Config Tab](#) to open the Device Connection dialog. Go to the Advanced view of this dialog and click on the Enable checkbox next to the corresponding entries in the Available Devices list. Once all instruments are connected, they are selectable in the [Tree Selector](#) of a newly opened Plotter tab allowing you to visualize their data simultaneously, though by default these data are not synchronized yet. The settings of multiple instruments can be accessed in parallel by opening a new Web Server session for each of them. This is done by opening a new browser tab and connecting to `localhost:8006` or `127.0.0.1:8006`, respectively, and then double-clicking the respective instrument entry in the Available Devices list. With multiple instruments connected to the same Data Server, tabs that are available for several instruments will feature a device selector as shown in [Figure 5.51](#).



Figure 5.51: Example of the device selector for the Device tab

The second prerequisite for automatic synchronization is correct cabling of the instruments explained in the diagram in [Figure 5.52](#). The instruments should share the same 10 MHz reference clock and communicate via the Ref / Trigger connectors for absolute timing synchronization. The 10 MHz clock signal is distributed in a daisy chain arrangement, where Clock 10 MHz Out of a leading instrument is connected to Clock 10 MHz In of the next instrument, and so forth. On all following instruments, the Clock Source in the [Device Tab](#) needs to be set to Clk 10 MHz rather than Internal. The Trigger Out 1 signal of the leading instrument should be distributed to the Trigger In 1 connector of all following instruments **and** of the leading instrument. This star-shaped connection should be made with a 1-to-N power divider and cables of equal length.

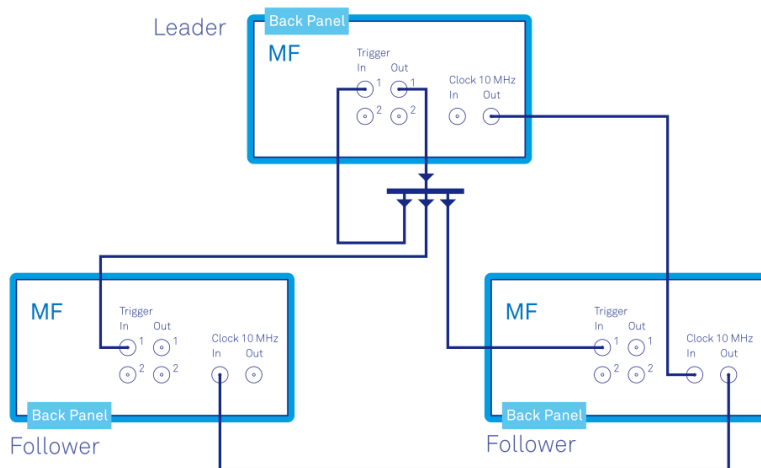


Figure 5.52: Cabling for automatic synchronization of multiple MF instruments

Once the cabling and the connectivity is set up correctly, automatic synchronization is started in the Multi Device Sync tab by checking the Enable button on the instruments in the Available Devices list, and then clicking on [Start/Stop Sync](#). The sequence assignment of the instruments (Leader, Follower 1, Follower 2,...) can be defined by the order in which the Enable button is clicked. This assignment has to agree with the way the cabling is made. The Message display on the right will then report on the progress, and the Sync Status LED will turn green if the synchronization was successful. In that case, visualizing a time-dependent measurement of multiple instruments in the Plotter will demonstrate the timing synchronization.

5.23.3. Functional Elements

Table 5.64: Multi Device Sync tab

Control/Tool	Option/Range	Description
Start Sync		Start the automatic synchronization of the selected devices.
Sync Status		Indicates the status of the synchronization within this group. Green: synchronization successful. Yellow: synchronization in progress. Red: error (see message).
Message		Displays a status message of the synchronization group.
Cabling		This image shows how to connect the devices for device synchronization.
Phase Synchronization		Reset phases of all oscillators on all synchronized devices.
Identify Device		Make device's front LED blink

5.24. ZI Labs Tab

The ZI Labs tab contains experimental LabOne functionalities added by the ZI development team. The settings found here are often relevant to special applications, but have not yet found their definitive place in one of the other LabOne tabs. Naturally this tab is subject to frequent changes, and the documentation of the individual features would go beyond the scope of this user manual. Clicking the following icon will open a new instance of the tab.

Table 5.65: App Icon and short description

Control/Tool	Option/Range	Description
ZI Labs		Experimental settings and controls.

5.25. Upgrade Tab

The Upgrade tab serves as a source of information about the possible upgrade options for the instrument in use. The tab has no functional purpose but provides the user with a quick link to further information about the upgrade options online.

6. Signal Processing Basics

This chapter provides background knowledge about impedance measurements with a focus on the methods used by Zurich Instruments impedance analyzers. The first section explains the basic concepts behind impedance and quickly turns to the practical discussion of compensation. The remaining sections discuss subjects related to the digital lock-in technology used by the MFIA Instrument. This part of the document is of interest to users that wish to better understand signal demodulation and filtering in order to optimize their measurements for noise suppression or speed. In particular, the sinc filtering technique for low-frequency measurements is explained.

6.1. Impedance and its Measurement

6.1.1. Introduction

The impedance determines the electrical current flowing through a component when a voltage is applied. In general the current can be alternating at a frequency $f = \omega/2\pi$ and is characterized by an amplitude and a phase, and so is the voltage. The current I and voltage V can be conveniently described using complex notation, and the impedance is defined as the complex ratio $Z = Z(\omega) = V/I$.

A measurement of a component's impedance requires simultaneous phase-sensitive measurement of the current and voltage at the component, usually called the device under test (DUT). Zurich Instruments products employ one of two basic measurement circuits for this task: a two-terminal setup advantageous for high impedances, and a four-terminal circuit which is suitable for most situations and advantageous for low impedances. Figure 6.1 shows the four-terminal circuit: a drive voltage is applied to one side of the DUT. On the other side, the DUT is shorted to ground passing through a current meter. Two additional leads are connected to either side of the DUT for measuring the voltage drop across. All the current flows through the L_{CUR} and H_{CUR} connectors, whereas L_{POT} and H_{POT} are current-free and serve as non-invasive probes for the electrical potential. The four-terminal setup is advantageous for measuring small impedances, since it is insensitive to the effect of series impedances in the cables, connectors, soldering points, etc.

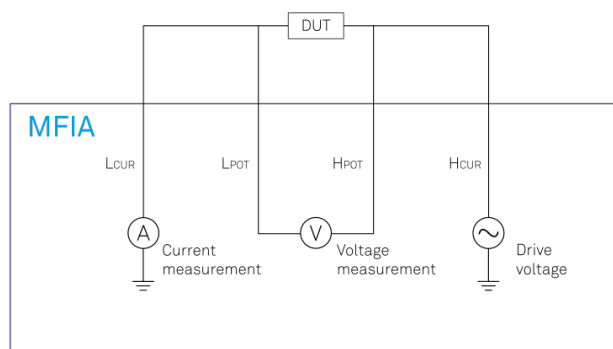


Figure 6.1: Four-terminal circuit for impedance measurement. The impedance measurement value is given by the measured voltage divided by the measured current.

Figure 6.2 shows the two-terminal circuit: it is identical to the four-terminal circuit except that the voltage measurement terminals are missing. The impedance is calculated as the drive voltage divided by the measured current. This assumes that the voltage at the DUT is equal to the drive voltage, which is the case for high-impedance DUTs. The two-terminal method is advantageous in these cases because it eliminates unnecessary stray capacitances and leakage currents parallel to the DUT.

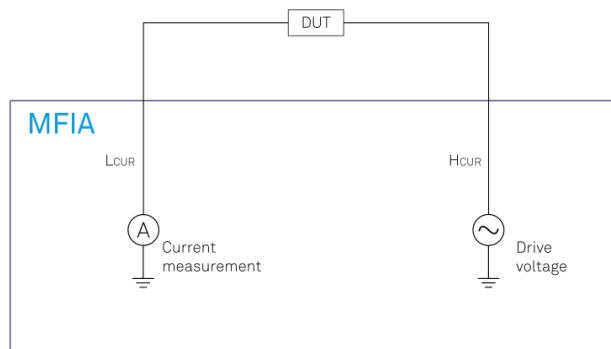


Figure 6.2: Two-terminal circuit for impedance measurement. The impedance measurement value is given by the bias voltage divided by the measured current.

The phase sensitivity of the current and voltage measurements in Zurich Instruments impedance analyzers is accomplished by using lock-in detection. This method offers superior noise performance, large dynamic range, and high precision in phase measurements. It is explained in more detail in [Principles of Lock-in Detection](#).

The real and imaginary parts of the impedance Z are called resistance R and reactance X , so we have $Z=R+jX$. Sometimes it's more natural to express things in terms of the reciprocal of the impedance, the admittance $Y=I/V=1/Z$. Its real and imaginary parts are called conductance G and susceptance B , so we have $Y=G+jB$.

Note

In the context of lock-in amplifier measurements, it's common to use the symbols for euclidean or polar coordinates to specify a measured quantity. For instance, a voltage V would be denoted as $V=R \cdot \exp(j\theta)$, or as $V=X+jY$. With Zurich Instruments impedance analyzers, V and I can be displayed directly in these terms. Consider these conventions to avoid confusion with resistance, reactance, and admittance which are denoted with the same symbols.

6.1.2. Ideal, Real, Measured Values

When talking about impedance, one often has in mind the cases of a resistance R , inductance L , capacitance C . The corresponding impedances are R , $j\omega L$, $1/(j\omega C)$. Components typically have nominal or **ideal** impedance values of a pure R , L , or C . But the impedance of any real-world component will show deviations from its ideal value. Mostly these deviations can be explained by so-called spurious impedance components. For instance, the impedance of a large resistor has a contribution due to the stray capacitance between its leads, and the contribution will be the more important the higher is the frequency. This stray capacitance lies parallel to the resistance, in this configuration it's common to name the capacitance C_p and the resistance R_p . Recalling the law for parallel combination of impedances $Z_{\text{tot}}=1/(1/Z_{p1}+1/Z_{p2})$, the **real** impedance of the resistor will be close to $Z_{\text{real}}=1/(1/R_p+j\omega C_p)$. The real impedance is furthermore not identical to the **measured** impedance Z_{meas} . The latter is what an impedance analyzer delivers, and the better the accuracy of the impedance analyzer, the closer are Z_{meas} and Z_{real} .

Circuit representations such as the parallel $C_p \parallel R_p$ play an important role in practical impedance measurements. Knowing the impedance Z_{meas} of a DUT a priori does not tell us much about the internal structure of the DUT. In order to get a deeper insight, we have to propose a realistic representation for the equivalent circuit of the DUT. Based on that, Z_{meas} can be split into its resistive and reactive parts, and the values of the principle and spurious component can be calculated. The main models are: - $C_p \parallel R_p$ (examples: large resistor with stray capacitor, capacitor with leakage current) - $C_s + R_s$ (example: large capacitor with resistive leads) - $L_s + R_s$ (examples: small resistor with lead inductance, inductor with wire resistance) - $L_p \parallel R_p$ (example: inductor with core loss)

Other, equivalent terminologies involve the dimensionless dissipation factor (loss tangent) D or the quality factor $Q=1/D$. A parallel combination of reactive components (such as $C_p \parallel L_p$) can not be split up in an impedance measurement at a single frequency. This is why such combinations don't appear in the list of basic circuit representations. Frequency dependence measurements allow for a detailed analysis of such circuits and other, more complex ones. Extracting the circuit type and parameters is possible by carrying out a model parameter fit of the data using specialized software such as ZView.

6.1.3. Test Fixtures and Compensation

When an impedance analyzer is shipped it contains an internal calibration that ensures accurate measurements at the plane of the output and input connectors. But measurements are always carried out with some sort of test fixture or cable linking the connectors to the DUT. A test fixture has its own impedance and therefore would normally distort or even invalidate the measurement of the DUT impedance. If the test fixture's electrical characteristics are known, its spurious effect can however be calculated and largely eliminated from the measurement data. This process is called compensation. The basis for a compensation is a characterization of the test fixture through a series of standard measurements.

One generic model for a test fixture is the L/R/C/G circuit shown in Figure 6.3. It is a good approximation to many compact test fixtures, and it helps in getting an intuitive understanding of the properties of a test fixture and the purpose of the compensation measurement. It contains terms due to the inductance and resistance of the leads, and terms due to the stray capacitance and leakage current between the two sides of the DUT.

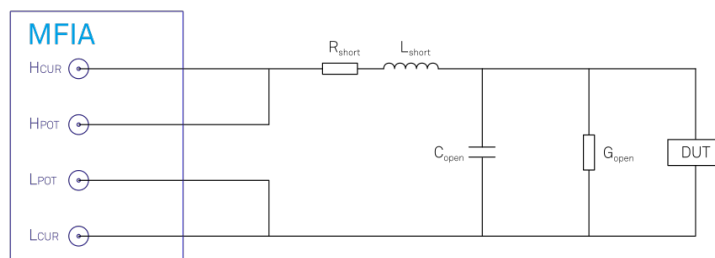


Figure 6.3: L/R/C/G circuit representation of a test fixture

The compensation measurements usually consist of one or several of the following steps: - An "open" (O) step, which is a measurement of a device with infinite impedance – an open circuit - A "short" (S) step, which is a measurement of a device with zero impedance – a short circuit - A "load" (L) step, which is a measurement of a device with known impedance ideally in the desired measurement range

The "open" compensation step yields C_{open} and G_{open} . The "short" step yields the values of L_{short} and R_{short} . The "load" step is required to compensate for effects that go beyond the L/R/C/G circuit. Among these effects are cable delays, or inductive cross-talk due to the mutual inductance M between the current leads (H_{CUR} , L_{CUR}) and the voltage (H_{POT} , L_{POT}) leads.

6.1.4. Optimizing the Setup

Having in mind the equivalent L/R/C/G circuit, the user setup should be optimized or chosen in view of what impedance and frequency is used. User setup here means any custom-built or commercial test fixture, or any custom-built device carrier that is plugged into the test fixture delivered with the instrument. In the regime of small impedances $|Z_{DUT}|$, the series L_{short} and R_{short} and the mutual inductance M are most relevant. This regime applies to small resistors, capacitors at high frequencies, and inductors at low frequencies. In the regime of large impedances $|Z_{DUT}|$, the parallel C_{open} and G_{open} are most relevant. This regime applies to large resistors, capacitors at low frequencies, and inductors at high frequencies. The cable delay is always relevant when measuring at high frequencies, no matter what is the impedance. - Minimize C_{open} by electrostatic guarding: enclose the conductors as much as possible by grounded electrostatic guards. The corresponding "guard" connection pins are marked with a ground symbol on the MFITF test fixture. Coaxial cables provide good electrostatic guarding. For measurements of very large impedances, consider placing a guarding plate perpendicular to the axis of the DUT. - Minimize the effect of R_{short} : use 4-terminal connections where the current leads (H_{CUR} , L_{CUR}) and the voltage leads (H_{POT} , L_{POT}) join only as close to the DUT as possible, ideally after the last soldering point. Avoid using very narrow PCB strips for current leads. - Minimize L_{short} and M by magnetic guarding: allow for inductive currents to flow parallel to the current leads on the full length of the current path. Ensure a continuous guard connection along the path.

6.1.5. Which Compensation Method?

There are a number of different compensation methods in use, such as short-open-load (SOL), short-open (SO), or load-load-load (LLL). Independent of the combination used, some specific

conditions and requirements apply to the S, O, and L steps. These are summarized in the following table.

Compensation steps:

S ("short")

- Requires "short" impedance much smaller than the DUT impedance
- sensitive to noise (small measured voltages)

O ("open")

- Requires "open" impedance much larger than the DUT impedance
- sensitive to noise (small measured currents)
- position of the electrodes identical to position with DUT

L ("load")

- "load" impedance of similar magnitude as DUT impedance recommended
- robust against noise
- accurate and stable "load" impedance required: capacitor or resistor, no inductor

The MFIA supports SOL, SO, LLL, OL, SL, L, compensation ('O' can be found in the Advanced tab). Which procedure is best suited depends first of all on the desired impedance measurement range – e.g. when aiming at measuring small impedances, a method involving a short (S) step is recommended. In contrast, an open (O) step is often not necessary. Second, it depends on the time the user is willing to spend in order to improve measurement accuracy, as every step requires a manual change of the test impedance. And third, it depends on the availability and accuracy of the test impedances, as a user compensation can only be as accurate as the available short, open, or load test impedances.

Table 6.1: Compensation methods with properties and user cases

Compensation method	Properties and use cases
SO	The short-open compensation corrects for inductive and capacitive parasitics of the user setup. This compensation method is in place if no accurate load is available, but its accuracy is limited by the fact that gain errors and cable delays can't be compensated.
SL	The short-load method is a modification of the short-open method for low impedances. Gain errors and cable delays can't be compensated.
OL	The open-load method is a modification of the short-open method for high impedances. Gain errors and cable delays can't be compensated.
SOL	The short-open-load compensation provides very good accuracy and is recommended if an accurate load device is available. It is required if gain errors or delays need to be compensated. Non-unity gain is typically caused by cable attenuation, amplification, or attenuation by additional circuits.
LLL	The load-load-load method ensures the best possible accuracy in the impedance range covered by the three loads. Similarly to SOL, it allows for gain and delay compensation. It can improve on the results of the SOL method but depends on the availability of accurate load devices.
L	A simple compensation method offering accuracy only close to the load impedance.

6.2. Principles of Lock-in Detection

Lock-in demodulation is a technique to measure the amplitude A_s and the phase θ of a periodic signal with the frequency $\omega_s = 2\pi f_s$ by comparing the signal to a reference signal. This technique is also called phase-sensitive detection. By averaging over time the signal-to-noise ratio (SNR) of a signal can be increased by orders of magnitude, allowing very small signals to be detected with a high accuracy making the lock-in amplifier a tool often used for signal recovery. For both signal recovery and phase-sensitive detection, the signal of interest is isolated with narrow band-pass filtering therefore reducing the impact of noise in the measured signal.

Figure 6.4 shows a basic measurement setup: a reference V_r signal is fed to the device under test. This reference signal is modified by the generally non-linear device with attenuation, amplification,

phase shifting, and distortion, resulting in a signal $V_s = A_s \cos(\omega_s t + \theta_s)$ plus harmonic components.

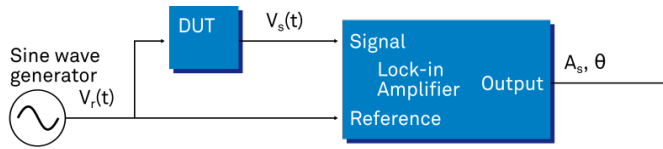


Figure 6.4: Basic measurement setup incorporating a lock-in amplifier

For practical reasons, most lock-in amplifiers implement the band-pass filter with a mixer and a low-pass filter (depicted in Figure 6.5): the mixer shifts the signal of interest into the baseband, ideally to DC, and the low-pass filter cuts all unwanted higher frequencies.

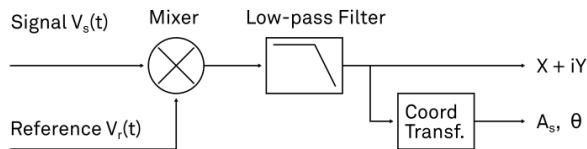


Figure 6.5: Mixing and low-pass filtering performed by the lock-in amplifier

The input signal $V_s(t)$ is multiplied by the reference signal $V_r(t) = \sqrt{2}e^{-i\omega_r t}$, where $\omega_r = 2\pi f_r$ is the demodulation frequency and i is the imaginary unit. This is the complex representation of a sine and cosine signal (phase shift 90°) forming the components of a quadrature demodulator, capable of measuring both the amplitude and the phase of the signal of interest. In principle it is possible to multiply the signal of interest with any frequency, resulting in a heterodyne operation. However the objective of the lock-in amplifier is to shift the signal as close as possible to DC, therefore the frequency of the reference and the signal is chosen similar. In literature this is called homodyne detection, synchrodyne detection, or zero-IF direct conversion.

The result of the multiplication is the signal

$$V_s(t) \cdot V_r(t) = V_s(t) \cdot \sqrt{2}e^{-i\omega_r t} = \frac{A_s}{\sqrt{2}}e^{i[(\omega_s - \omega_r)t + \theta]} + \frac{A_s}{\sqrt{2}}e^{-i[(\omega_s + \omega_r)t + \theta]} \quad (1)$$

It consists of a slow component with frequency $\omega_s - \omega_r$ and a fast component with frequency $\omega_s + \omega_r$.

The demodulated signal is then low-pass filtered with an infinite impulse response (IIR) RC filter, indicated by the symbol $\langle \cdot \rangle$. The frequency response of the filter $F(\omega)$ will let pass the low frequencies $F(\omega_s - \omega_r)$ while considerably attenuating the higher frequencies $F(\omega_s + \omega_r)$. Another way to consider the low-pass filter is an averager.

$$X + iY = \langle V_s(t) \cdot \sqrt{2}e^{-i\omega_r t} \rangle \approx F(\omega_s - \omega_r) \frac{A_s}{\sqrt{2}}e^{i[(\omega_s - \omega_r)t + \theta]} \quad (2)$$

The result after the low-pass filter is the demodulated signal $X + iY$, where X is the real and Y is the imaginary part of a signal depicted on the complex plane. These components are also called in-phase and quadrature components. The transformation of X and Y into the amplitude R and phase θ information of $V_s(t)$ can be performed with trigonometric operations.

It is interesting to note that the value of the measured signal corresponds to the RMS value of the signal, which is equivalent to $R = A_s/\sqrt{2}$.

Most lock-in amplifiers output the values (X, Y) and (R, θ) encoded in a range of -10 V to $+10$ V of the auxiliary output signals.

6.2.1. Lock-in Amplifier Applications

Lock-in amplifiers are employed in a large variety of applications. In some cases the objective is measuring a signal with good signal-to-noise ratio, and then that signal could be measured even with large filter settings. In this context the word phase sensitive detection is appropriate. In other applications, the signal is very weak and overwhelmed by noise, which forces to measure with very narrow filters. In this context the lock-in amplifier is employed for signal recovery. Also, in another context, a signal modulated on a very high frequency (GHz or THz) that cannot be measured with standard approaches, is mixed to a lower frequency that fits into the measurement band of the lock-in amplifier.

One example for measuring a small, stationary or slowly varying signal which is completely buried in the $1/f$ noise, the power line noise, and slow drifts. For this purpose a weak signal is modulated to a higher frequency, away from these sources of noise. Such signal can be efficiently mixed back and measured in the baseband using a lock-in amplifier. In Figure 6.6 this process is depicted. Many optical applications perform the up-mixing with a chopper, an electro-optical modulator, or an acousto-optical modulator. The advantage of this procedure is that the desired signal is measured in a spectral region with comparatively little noise. This is more efficient than just low-pass filtering the DC signal.

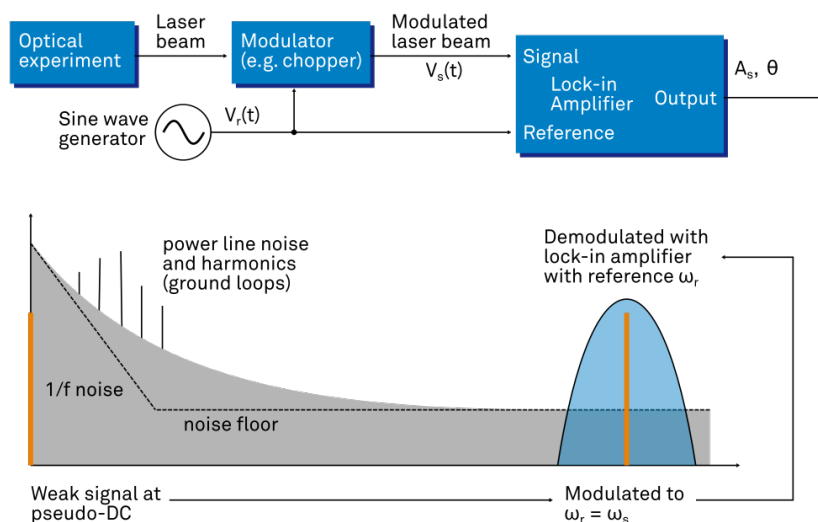


Figure 6.6: Lock-in measurement of a noisy DC signal

6.3. Signal Bandwidth

The signal bandwidth (BW) theoretically corresponds to the highest frequency components of interest in a signal. In practical signals, the bandwidth is usually quantified by the cut-off frequency. It is the frequency at which the transfer function of a system shows 3 dB attenuation relative to DC ($BW = f_{\text{cut-off}} = f_{-3\text{dB}}$); that is, the signal power at $f_{-3\text{dB}}$ is half the power at DC. The bandwidth, equivalent to cut-off frequency, is used in the context of dynamic behavior of a signals or separation of different signals. This is for instance the case for fast-changing amplitudes or phase values like in a PLL or in imaging applications, or when signals closely spaced in frequency need to be separated.

The noise equivalent power bandwidth (NEPBW) is also a useful figure, and it is distinct from the signal bandwidth. This unit is typically used for noise measurements: in this case one is interested in the total amount of power that passes through a low-pass filter, equivalent to the area under the solid curve in Figure 6.7. For practical reasons, one defines an ideal brick-wall filter that lets pass the same amount of power under the assumption that the noise has a flat (white) spectral density. This brick-wall filter has transmission 1 from DC to f_{NEPBW} . The orange and blue areas in Figure 6.7 then are exactly equal in a linear scale.

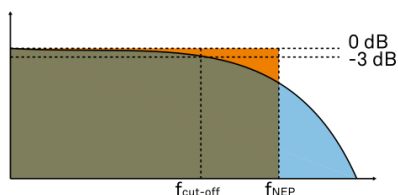


Figure 6.7: Signal bandwidth and noise equivalent power bandwidth

It is possible to establish a simple relation between the $f_{\text{cut-off}}$ and the f_{NEPBW} that only depends on the slope (or roll-off) of the filter. As the filter slope actually depends on the time constant (TC) defined for the filter, it is possible to establish the relation also to the time constant. It is intuitive to understand that for higher filter orders, the $f_{\text{cut-off}}$ is closer to the f_{NEPBW} than for smaller orders.

The time constant is a parameter used to interpret the filter response in the time domain, and relates to the time it takes to reach a defined percentage of the final value. The time constant of a low-pass filter relates to the bandwidth according to the formula

$$TC = \frac{FO}{2\pi f_{\text{cut-off}}} \quad (3)$$

where FO is said factor that depends on the filter slope. This factor, along with other useful conversion factors between different filter parameters, can be read from the following table.

Table 6.2: Summary of conversion factors for bandwidth definitions

filter order	filter roll-off	FO	$f_{\text{cut-off}}$	f_{NEPBW}	$f_{\text{NEPBW}} / f_{\text{cut-off}}$
1 st	6 dB/oct	1.0000	0.1592 / TC	0.2500 / TC	1.5708
2 nd	12 dB/oct	0.6436	0.1024 / TC	0.1250 / TC	1.2203
3 rd	18 dB/oct	0.5098	0.0811 / TC	0.0937 / TC	1.1554
4 th	24 dB/oct	0.4350	0.0692 / TC	0.0781 / TC	1.1285
5 th	30 dB/oct	0.3856	0.0614 / TC	0.0684 / TC	1.1138
6 th	36 dB/oct	0.3499	0.0557 / TC	0.0615 / TC	1.1046
7 th	42 dB/oct	0.3226	0.0513 / TC	0.0564 / TC	1.0983
8 th	48 dB/oct	0.3008	0.0479 / TC	0.0524 / TC	1.0937

6.4. Discrete-Time Filters

6.4.1. Discrete-Time RC Filter

There are many options how to implement digital low-pass filters. One common filter type is the exponential running average filter. Its characteristics are very close to those of an analog resistor-capacitor RC filter, which is why this filter is sometimes called a discrete-time RC filter. The exponential running average filter has the time constant $TC = \tau_N$ as its only adjustable parameter. It operates on an input signal $X_{\text{in}}[n,]$ defined at discrete times nT_s , $(n+1)T_s$, $(n+2)T_s$, etc., spaced at the sampling time T_s . Its output $X_{\text{out}}[n, T_s]$ can be calculated using the following recursive formula,

$$X_{\text{out}}[n, T_s] = e^{-T_s/\tau_N} X_{\text{out}}[n-1, T_s] + (1 - e^{-T_s/\tau_N}) X_{\text{in}}[n, T_s] \quad (4)$$

The response of that filter in the frequency domain is well approximated by the formula

$$H_1(\omega) = \frac{1}{1 + i \cdot \omega \cdot \tau_N} \quad (5)$$

The exponential filter is a first-order filter. Higher-order filters can easily be implemented by cascading several filters. For instance the 4th order filter is implemented by chaining 4 filters with the same time constant $TC = \tau_n$ one after the other so that the output of one filter stage is the input of the next one. The transfer function of such a cascaded filter is simply the product of the transfer functions of the individual filter stages. For an n-th order filter, we therefore have

$$H_n(\omega) = \frac{1}{(1 + i \cdot \omega \cdot \tau_n)^n} \quad (6)$$

The attenuation and phase shift of the filters can be obtained from this formula. Namely, the filter attenuation is given by the absolute value squared $|H_n(\omega)|^2$. The filter transmission phase is given by the complex argument $\arg[H_n(\omega)]$.

6.4.2. Filter Settling Time

The low-pass filters after the demodulator cause a delay to measured signals depending on the filter order and time constant $TC = \tau_n$. After a change in the signal, it will therefore take some time before the lock-in output reaches the correct measurement value. This is depicted in [Figure 6.8](#) where the response of cascaded filters to a step input signal is shown.

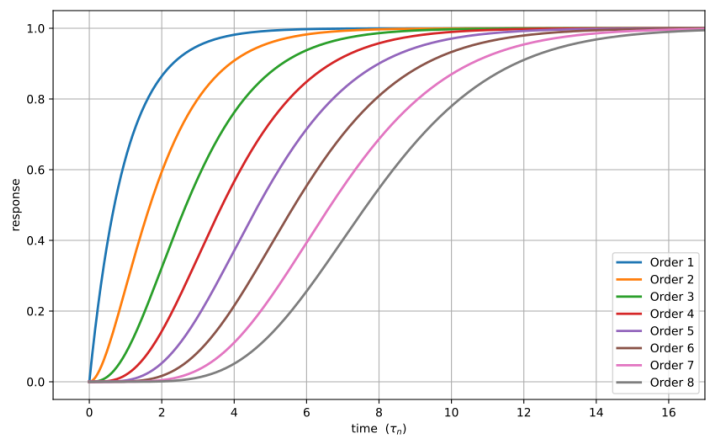


Figure 6.8: Time-domain step response of the demodulator low-pass filter for orders from 1 to 8.

More quantitative information on the settling time can be obtained from [Table 6.3](#). In this table, you find settling times in units of the 1st-order filter's time constant (**TC**) for all filter orders available with the MFIA Lock-in Amplifier. The values tell the time you need to wait for the filtered demodulator signal to reach 50%, 63%, 95% and 99% of the final value. This can help in making a quantitatively correct choice of filter parameters for example in a measurement involving a parameter sweep.

Table 6.3: Summary of Filter Settling Times

Filter order	50%	63% (1-1/e)	90%	95%	99%
1 st	0.7 · TC	1.0 · TC	2.3 · TC	3.0 · TC	4.6 · TC
2 nd	1.7 · TC	2.1 · TC	3.9 · TC	4.7 · TC	6.6 · TC
3 rd	2.7 · TC	3.3 · TC	5.3 · TC	6.3 · TC	8.4 · TC
4 th	3.7 · TC	4.4 · TC	6.7 · TC	7.8 · TC	10.0 · TC
5 th	4.7 · TC	5.4 · TC	8.0 · TC	9.2 · TC	11.6 · TC
6 th	5.7 · TC	6.5 · TC	9.3 · TC	10.5 · TC	13.1 · TC
7 th	6.7 · TC	7.6 · TC	10.5 · TC	11.8 · TC	14.6 · TC
8 th	7.7 · TC	8.6 · TC	11.8 · TC	13.1 · TC	16.0 · TC

6.5. Full Range Sensitivity

The sensitivity of the lock-in amplifier is the RMS value of an input sine that is demodulated and results in a full scale analog output. Traditionally the X, Y, or R components are mapped onto the 10 V full scale analog output. In such a case, the overall gain from input to output of the lock-in amplifier is composed of the input and output amplifier stages. Many lock-in amplifiers specify a sensitivity between 1 nV and 1 V. In other words the instrument permits an input signal between 1 nV and 1 V to be amplified to the 10 V full range output.

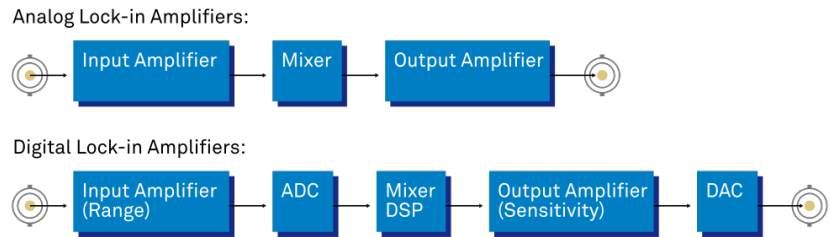


Figure 6.9: Sensitivity from signal input to signal output

In analog lock-in amplifiers the sensitivity is simple to understand. It is the sum of the analog amplification stages between in the input and the output of the instrument: in particular the input amplifier and the output amplifier.

In digital lock-in amplifiers the sensitivity is less straightforward to understand. Analog-to-digital converters (ADC) operate with a fixed input range (e.g. 1 V) and thus require a variable-gain amplifier to amplify the input signal to the range given by the ADC. This variable-gain amplifier must be in the analog domain and its capability determines the minimum input range of the instrument. A practical analog input amplifier provides a factor 1000 amplification, thus 1 V divided by 1000 is the minimum input range of the instrument.

The input range is the maximum signal amplitude that is permitted for a given range setting. The signal is internally amplified with the suited factor, e.g. (1 mV)·1000 to result in a full swing signal at the ADC. For signals larger than the range, the ADC saturates and the signal is distorted – the measurement result becomes useless. Thus the signal should never exceed the range setting.

But the input range is not the same as the sensitivity. In digital lock-in amplifiers the sensitivity is only determined by the output amplifier, which is an entirely digital signal processing unit which performs a numerical multiplication of the demodulator output with the scaling factor. The digital output of this unit is then fed to the output digital-to-analog converter (DAC) with a fixed range of 10 V. It is this scaling factor that can be retrofitted to specify a sensitivity as known from the analog lock-in amplifiers. A large scaling factor, and thus a high sensitivity, comes at a relatively small expense for digital amplification.

One interesting aspect of digital lock-in amplifiers is the connection between input resolution and sensitivity. As the ADC operates with a finite resolution, for instance 14 bits, the minimum signal that can be detected and digitized is for instance 1 mV divided by the resolution of the ADC. With 14 bits the minimum level that can be digitized would be 122 nV. How is it possible to reach 1 nV sensitivity without using a 21 bit analog-to-digital converter? In a world without noise it is not possible. Inversely, thanks to noise and current digital technology it is possible to achieve a sensitivity even below 1 nV.

Most sources of broadband noise, including the input amplifier, can be considered as Gaussian noise sources. Gaussian noise is equally distributed in a signal, and thus generates equally distributed disturbances. The noise itself can be filtered by the lock-in amplifier down to a level where it does not impact the measurement. Still, in the interplay with the signal, the noise does have an effect on the measurement. The input of the ADC is the sum of the noise and the signal amplitude. Every now and then, the signal amplitude on top of the large noise will be able to toggle the least significant bits even for very small signals, as low as 1 nV and below. The resulting digital signal has a component at the signal frequency and can be detected by the lock-in amplifier.

There is a similar example from biology. Rod cells in the human eye permit humans to see in very low light conditions. The sensitivity of rod cells in the human eye is as low as a single photon. This sensitivity is achieved in low light conditions by a sort of pre-charging of the cell to be sensitive to the single photon that triggers the cell to fire an impulse. In a condition with more surround light, rod cells are less sensitive and need more photons to fire.

To summarize, in digital lock-in amplifiers the full range sensitivity is only determined by the scaling factor capability of the digital output amplifier. As the scaling can be arbitrary big, 1 nV minimum full range sensitivity is achievable without a problem. Further, digital lock-in amplifiers exploit the input noise to heavily increase the sensitivity without impacting the accuracy of the measurement.

6.6. Sinc Filtering

As explained in [Principles of Lock-in Detection](#), the demodulated signal in an ideal lock-in amplifier has a signal component at DC and a spurious component at twice the demodulation frequency. The components at twice the demodulation frequency (called the 2ω component) is effectively removed by regular low-pass filtering. By selecting filters with small bandwidth and faster roll-offs, the 2ω component can easily be attenuated by 100 dB or more. The problem arises at low demodulation frequencies, because this forces the user to select long integration times (e.g. >60 ms for a demodulation frequency of 20 Hz) in order to achieve the same level of 2ω attenuation.

In practice, the lock-in amplifier will modulate DC offsets and non-linearities at the signal input with the demodulation frequency, resulting in a signal at the demodulation frequency (called ω component). This component is also effectively removed by the regular low-pass filters at frequencies higher than 1 kHz.

At low demodulation frequencies, and especially for applications with demodulation frequencies close to the filter bandwidth, the ω and 2ω components can affect the measurement result. Sinc filtering allows for strong attenuation of the ω and 2ω components. Technically the sinc filter is a comb filter with notches at integer multiples of the demodulation frequency (ω , 2ω , 3ω , etc.). It

removes the ω component with a suppression factor of around 80 dB. The amount of 2ω component that gets removed depends on the input signal. It can vary from entirely (e.g. 80 dB) to slightly (e.g. 5 dB). This variation is not due to the sinc filter performance but depends on the bandwidth of the input signal.

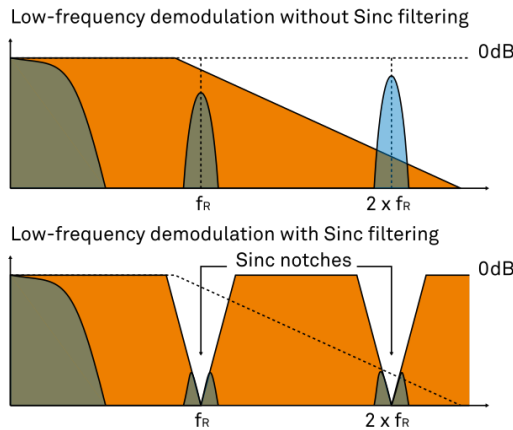


Figure 6.10: Effect of sinc filtering

Table 6.4: Artifacts in the demodulation signal

Input signal	Demodulation result before low-pass filter	Result
Signal at ω	DC component	Amplitude and phase information (wanted signal)
	2ω component	Unwanted component (can additionally be attenuated by sinc filter)
DC offset	ω component	Unwanted component (can additionally be attenuated by sinc filter)

We can observe the effect of the sinc filter by using the Spectrum Analyzer Tool of the MFIA Lock-in Amplifier. As an example, consider a 30 Hz signal with an amplitude of 0.1 V that demodulated using a filter bandwidth of 100 Hz and a filter order 8. In addition 0.1 V offset is added to the signal so that we get a significant ω component.

Figure 6.11 shows a spectrum with the sinc filter disabled, whereas for Figure 6.12 the sinc filter is enabled. The comparison of the two clearly shows how the sinc options dampens both the ω and 2ω components by about 100 dB.

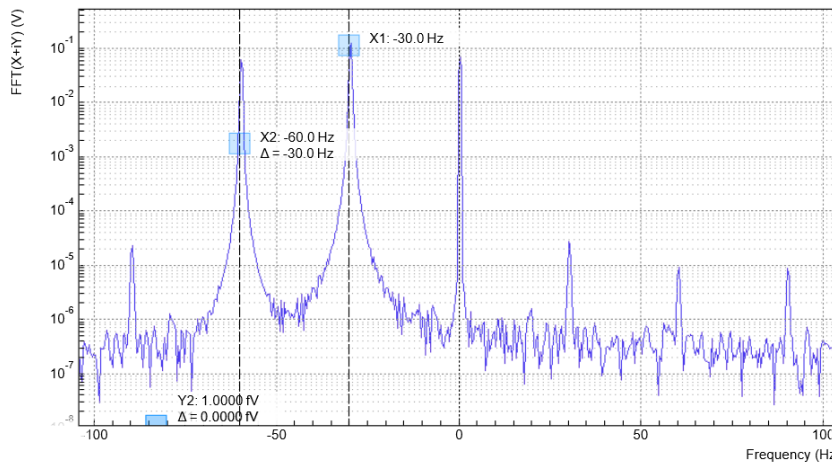


Figure 6.11: Spectrum of a demodulated 30 Hz signal without sinc filter

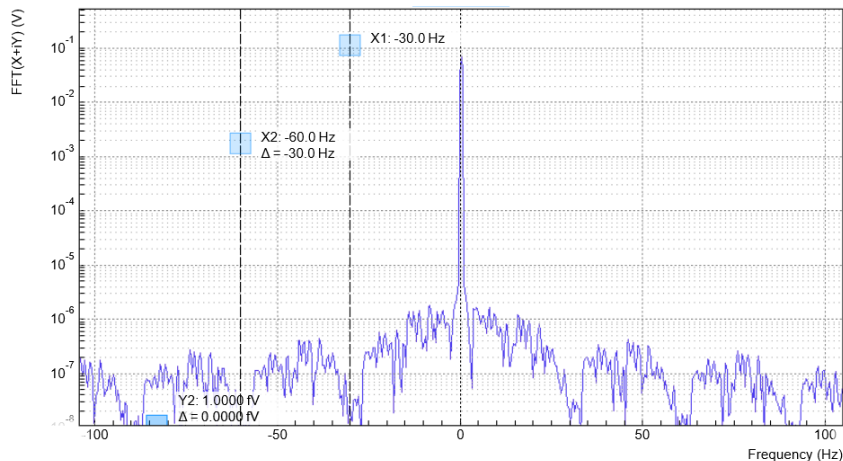


Figure 6.12: Spectrum of a demodulated 30 Hz signal with sinc filter

Note

In order to put the notches of the digital filter to ω and 2ω , the sampling rate of the filter would have to be precisely adjusted to the signal frequency. As this is technically not feasible, the generated signal frequency is adjusted instead by a very small amount.

6.7. Zoom FFT

The concept of zoom FFT allows the user to analyze the spectrum of the input signal around a particular frequency by zooming in on a narrow frequency portion of the spectrum. This is done by performing a Fourier transform of the demodulated in-phase and quadrature (X and Y) components or more precisely, on the complex quantity $X+iY$, where i is the imaginary unit. In the LabOne user interface, this functionality is available in the Spectrum tab.

In normal FFT, the sampling rate determines the frequency span and the total acquisition time determines the frequency resolution. Having a large span and a fine resolution at the same time then requires long acquisition times at high sample rates. This means that a lot of data needs to be acquired, stored, and processed, only to retain a small portion of the spectrum and discard most of it in the end. In zoom FFT, the lock-in demodulation is used to down-shift the signal frequency, thereby allowing one to use both a much lower sampling rate and sample number to achieve the same frequency resolution. Typically, to achieve a 1 Hz frequency resolution at 1 MHz, FFT would require to collect and process approximately 10^6 points, while zoom FFT only processes 10^3 points. (Of course the high rate sampling is done by the lock-in during the demodulation stage, so the zoom FFT still needs to implicitly rely on a fast ADC.)

In order to illustrate why this is so and what benefits this measurement tool brings to the user, it is useful to remind that at the end of the demodulation of the input signal $V_s(t) = A_s \cos(\omega_s t + \tau)$, the output signal is $X + iY = F(\omega_s - \omega_r)(A_s/\sqrt{2})e^{i[(\omega_s - \omega_r)t + \tau]}$ where $F(\omega)$ is the frequency response of the filters.

Since the demodulated signal has only one component at frequency $\omega_s - \omega_r$, its power spectrum (Fourier transform modulus squared) has a peak of height $(|A_s|^2/2) \cdot |F(\omega_s - \omega_r)|^2$ at $\omega_s - \omega_r$: this tells us the spectral power distribution of the input signal at frequencies close to ω_r within the demodulation bandwidth set by the filters $F(\omega)$.

Note that:

- the ability of distinguish between positive and negative frequencies works only if the Fourier transform is done on $X+iY$. Had we taken X for instance, the positive and negative frequencies of its power spectrum would be equal. The symmetry relation $G(-\omega) = G^*(\omega)$ holds for the Fourier transform $G(\omega)$ of a real function $g(t)$ and two identical peaks would appear at $\pm|\omega_s - \omega_r|$.
- one can extract the amplitude of the input signal by dividing the power spectrum by $|F(\omega)|^2$, the operation being limited by the numerical precision. This is implemented in LabOne and is activated by the Filter Compensation button: with the Filter Compensation enabled, the background noise appears white; without it, the effect of the filter roll-off becomes apparent.

The case of an input signal containing a single frequency component can be generalized to the case of multiple frequencies. In that case the power spectrum would display all the frequency components weighted by the filter transfer function, or normalized if the Filter Compensation is enabled.

When dealing with discrete-time signal processing, one has to be careful about aliasing which occurs when the signal frequencies higher than the sampling rate ω are not sufficiently suppressed. Remember that ω is the user settable readout rate, not the 2 GSa/s sampling rate of the GHFLI input. Since the discrete-time Fourier transform extends between $-\omega/2$ and $+\omega/2$, the user has to make sure that at $\pm\omega/2$ the filters provide the desired attenuation: this can be done either by increasing the sampling rate or resolving to measure a smaller frequency spectrum (i.e. with a smaller filter bandwidth).

Similarly to the continuous case, in which the acquisition time determines the maximum frequency resolution ($2\pi/T$ if T is the acquisition time), the resolution of the zoom FFT can be increased by increasing the number of recorded data points. If N data points are collected at a sampling rate ω , the discrete Fourier transform has a frequency resolution of ω/N .

7. Specifications

Important

Unless otherwise stated, all specifications apply after 30 minutes of instrument warm-up.

Important

Changes in the specification parameters are explicitly mentioned in the revision history of this document.

Important

Some specifications depend on the installed options. The options installed on a given instrument are listed in the Device tab of the LabOne user interface.

7.1. General Specifications

Table 7.1: General specifications

Parameter	Description
Storage temperature	+5°C to +65°C
Storage relative humidity	< 95%, non-condensing
Operating environment	IEC61010, indoor location, installation category II, pollution degree 2
Operating altitude	up to 2000 meters
Operating temperature	+5°C to +40°C
Operating relative humidity	< 90%, non-condensing
Specification temperature	+18°C to +28°C
Power consumption	<40 W
DC power inlet	12 V, 2 A Connector: Switchcraft 760BK, ID 2.5 mm, OD 5.5 mm
Power supply AC line	100–240 V (±10%), 50/60 Hz
Line power fuse	250 V, 2 A, fast, 5 x 20 mm, F 2 A L 250 V
Dimensions including bumper	28.3 x 23.2 x 10.2 cm 11.1 x 9.1 x 4.0 inch Rack mount on request
Weight including bumper	3.8 kg
Recommended calibration interval	2 years (see sticker on back panel)
Warranty	1 year, extensible

Table 7.2: Impedance Analyzer

Parameter	Description
Frequency range	DC to 500 kHz DC to 5 MHz, requires MF-F5M option ¹
Basic accuracy	0.05% (1 mHz to 500 kHz)
Basic temperature stability	<200 ppm/K

Parameter	Description
Test signal level	0 to 2.1 V _{rms} with monitoring
Bandwidth	276 μ Hz to 206 kHz
DC bias signal level	± 10 V (2 Terminal); ± 3 V (4 Terminal)
Compensation methods	SO, SOL, LLL, SL, L, OL
Impedance Z: range; basic accuracy	1 m Ω to 1 T Ω ; 0.05%
Admittance Y: range; basic accuracy	1 pS to 1 kS; 0.05%
Resistance Rs, Rp: range; basic accuracy	1 m Ω to 10 G Ω ; max(10 $\mu\Omega$, 0.05%) ²
Capacitance Cs, Cp: range; basic accuracy	10 fF to 1 F; max(10 fF, 0.05%) ²
Inductance Ls, Lp: range; basic accuracy	100 nH to 1 H; max(10 nH, 0.05%) ²
DC Resistance R _{DC} : range; basic accuracy	1 m Ω to 10 G Ω ; 2%
Reactance X: range; basic accuracy	1 m Ω to 10 G Ω ; 0.05%
Conductance G: range; basic accuracy	1 nS to 1 kS; max(100 nS, 0.05%)
Susceptance B: range; basic accuracy	1 nS to 1 kS; max(100 nS, 0.05%)
Loss coefficient D: range	10 ⁻⁴ to 10 ⁴
Q factor: range	10 ⁻⁴ to 10 ⁴

¹ The MFIA 5 MHz Impedance Analyzer includes the MF-F5M option.

² Accuracy valid if parameter is the dominant value of the circuit representation.

Table 7.3: Demodulators

Parameter	Description
Frequency range	DC to 500 kHz DC to 5 MHz, requires MF-F5M option
Number of demodulators	1 dual-phase (X, Y, R, Θ) 4 dual-phase, requires MF-MD option
Demodulator inputs	Signal Inputs (V/I), Auxiliary Inputs, Auxiliary Outputs, Trigger Inputs
Filter time constant	337 ns to 83 s
Filter bandwidth (-3 dB)	276 μ Hz – 206 kHz (4th order filter)
Harmonics	1 – 1023
Filter slope	6, 12, 18, 24, 30, 36, 42, 48 dB/oct
Additional filtering	Sinc filter
Phase resolution	10 μ deg
Frequency resolution	1 μ Hz
Output sample rate on Auxiliary Outputs	612 kSa/s (for each auxiliary output), 18 bit, ± 10 V
Maximum transfer rate over 1 GbE	200 kSa/s (all demodulators), 48-bit full range
Maximum rate to store on local USB drive	50 kSa/s (all demodulators), 48-bit full range
Trigger modes for data transfer	Continuous, edge, gated

Table 7.4: Reference frequencies

Parameter	Description
External reference frequency range	1 Hz to 500 kHz 1 Hz to 5 MHz, requires MF-F5M option
External reference input	Auxiliary Inputs, Trigger Inputs, Auxiliary Outputs, Current Signal Input, Voltage Signal Input
Lock time for external reference	Typically less than max(100 cycles, 1.2 ms)
Number of external references	1; 2, requires MF-MD option
Internal reference frequency range	0 to 500 kHz 0 to 5 MHz, requires MF-F5M option

Table 7.5: Scope

Parameter	Description
Input channels	Signal Inputs (V,I), Auxiliary Inputs, Auxiliary Outputs, Trigger Inputs, Trigger Outputs, Signal Output, Oscillator Phase. Demodulator R, Theta, X, Y, requires MF-DIG option
Scope modes	Time domain, frequency domain (FFT)
Number display channels	1; 2, requires MF-DIG option
Trigger channels	Signal Inputs (V,I), Auxiliary Inputs, Auxiliary Outputs, Trigger Inputs, Trigger Outputs. Demodulator R, Theta, X, Y, requires MF-DIG option
Trigger modes	Edge
Trigger hysteresis	Full input range
Pre-trigger	Full sample range
Sampling rates	1.8 kSa/s to 60 MSa/s
Vertical resolution	16 bit
Maximum number of samples per shot	16 kSa; 5 MSa, requires MF-DIG option
Minimum hold time	1 ms
Bandwidth limit mode, vertical resolution increase	Downsampling by averaging; increase vertical resolution up to 24 bit, requires MF-DIG option
Cursor math	Location, Area, Wave, Peak, Tracking, Histogram

Table 7.6: Spectrum

Parameter	Description
Center frequency range	0 to 500 kHz 0 to 5 MHz, requires MF-F5M option
Spectrum modes	FFT(X+iY), FFT@, FFT(Θ), FFT(f) and FFT((dΘ/dt)/2π)
Statistical options	Amplitude, Spectral density, Power
Averaging modes	None, Exponential moving average
Maximum number of samples per spectrum	8 kSa
Maximum span	58 kHz
Window functions	Rectangular, Hann, Hamming, Blackman Harris
Cursor math	Location, Area, Tracking, Wave, Peak, Histogram

Table 7.7: Sweeper

Parameter	Description
Sweep parameters	Oscillator frequency, Demodulator phase, Auxiliary Offset, Signal Output Offset, etc.
Parameter sweep ranges	Full range, Linear and Logarithmic
Parameter sweep resolution	Arbitrary, defined by start/stop value and number of sweep points
Display parameters	Demodulator Output (X, Y, R, θ , f), Auxiliary Input
Display options	Single Plot, Dual Plot (e.g. Bode Plot), Multi-trace
Statistical options	Amplitude, Spectral density, Power
Preset measurement modes	Parameter sweep, Noise amplitude measurement, Frequency response analyzer, 3-Omega-Sweep

Table 7.8: Voltage Signal Inputs

Parameter	Description
Connectors	2 BNC on front panel, single-ended or differential
Shield connectivity	Floating or ground
Maximum float voltage versus ground	± 1 V
Input impedance	50 Ω and 10 M Ω 27 pF for range ≥ 300 mV; 40 pF for range ≤ 100 mV
Input frequency range	DC to 500 kHz; DC to 5 MHz, requires MF-F5M option
Input A/D conversion	16 bit, 60 MSa/s
Input noise amplitude	2.5 nV/ $\sqrt{\text{Hz}}$ for frequencies > 1 kHz 7 nV/ $\sqrt{\text{Hz}}$ at 10 Hz 40 nV/ $\sqrt{\text{Hz}}$ at 1 Hz 3.3 mV input range; shorting cap on input
Input noise corner frequency	Typically 100 Hz for range ≤ 10 mV
Input bias current	Typically ± 10 pA, max ± 200 pA
Input full range sensitivity (10 V lock-in amplifier output)	1 nV to 3 V
Input AC ranges	± 1 mV, $\pm$ mV, ± 10 mV, ± 30 mV, ± 100 mV, ± 300 mV, ± 1 V, ± 3 V
AC coupling cutoff frequency	1.6 Hz
Maximum DC offset for AC coupling	± 10 V
Input DC ranges	± 1 mV, $\pm$ mV, ± 10 mV, ± 30 mV, ± 100 mV, ± 300 mV, ± 1 V, ± 3 V
Input gain inaccuracy	$< 1\%$ (< 2 MHz); for higher frequencies limited by analog input filter
Analog input filter (anti-aliasing)	1 dB suppression at 5 MHz, 3 dB at 12 MHz; 3rd order roll-off
Input amplitude stability	0.1%/°C
Input offset amplitude	$< \max(0.5 \text{ mV}, 1\% \text{ of range})$
Dynamic reserve	Up to 120 dB
Harmonic distortion (typical)	-80 dBc for frequencies ≤ 100 kHz, -65 dBc for frequencies ≤ 5 MHz; carrier amplitude -1 dBFS
Harmonic distortion (loopback, maximum)	-70 dBc for frequencies < 200 kHz, -55 dBc for frequencies < 2 MHz, -50.2 dBc for frequencies ≥ 2 MHz; input range 3 mV, output range 10 mV. -72 dBc for frequencies < 200 kHz, -55 dBc for frequencies < 2 MHz, -50.2 dBc for frequencies ≥ 2 MHz; input range 10-300 mV, output range 100-1000 mV. -72 dBc for frequencies < 200 kHz, -50.2 dBc for frequencies < 2 MHz, -46 dBc for frequencies ≥ 2 MHz; input range 3 V, output range 10 V.

Parameter	Description
Coherent pickup	< -140 dB for frequencies ≤ 5 MHz and 50 Ω input impedance; < -180 dB for frequencies ≤ 100 kHz and 50 Ω input impedance

Table 7.9: Current Signal Input

Parameter	Description
Connector	BNC on front panel, float/ground
Shield connectivity	Floating or ground
Maximum float voltage versus ground	± 1 V
Input impedance	see table_title
Input frequency range	0 to 500 kHz 0 to 5 MHz, requires MF-F5M option
Input A/D conversion	16 bit, 60 MSa/s
Input leakage current	± 10 pA
Input full range sensitivity (10 V lock-in amplifier output)	10 fA to 10 mA
Input gain inaccuracy	< 1% (for frequencies below 10% of the input bandwidth)
Input offset amplitude	1% of range
Input offset voltage	± 2.2 mV max; to shield of current input BNC connector
Dynamic reserve	up to 120 dB
Coherent pickup	< 90 G Ω for frequencies ≤ 5 MHz and 100 μ A input range < 140 T Ω for frequencies ≤ 100 kHz and 10 nA input range

Table 7.10: Current Signal Input: input ranges, transimpedance gain, bandwidth, input impedance, noise

Current input range	Transimpedance gain	Bandwidth (-3 dB)	Input impedance at DC	Input noise
10 mA	100 V/A	5 MHz	50 Ω	300 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
1 mA ¹	1 kV/A	5 MHz	50 Ω	200 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
100 μ A	10 kV/A	5 MHz	60 Ω	3.5 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
10 μ A ¹	100 kV/A	5 MHz	60 Ω	2.5 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
1 μ A	1 MV/A	450 kHz	1 k Ω	200 fA/ $\sqrt{\text{Hz}}$ at 1 kHz
100 nA ¹	10 MV/A	450 kHz	1 k Ω	150 fA/ $\sqrt{\text{Hz}}$ at 1 kHz
10 nA	100 MV/A	2 kHz	160 k Ω	20 fA/ $\sqrt{\text{Hz}}$ at 100 Hz
1 nA ¹	1 GV/A	2 kHz	160 k Ω	15 fA/ $\sqrt{\text{Hz}}$ at 100 Hz

¹ Range only available on MF Instruments with serial numbers MF-DEV3200 and higher.

Table 7.11: Signal Output

Parameter	Description
Connectors	2 BNC on front panel, single ended and differential
Output impedance	50 Ω

Parameter	Description
Output frequency range	DC to 500 kHz DC to 5 MHz (with MF-F5M option)
Output frequency resolution	1 μ Hz
Output phase range	$\pm 180^\circ$
Output phase resolution	10 μ deg
Differential outputs	Sine waves shifted by 180°
Output D/A conversion	16 bit, 60 MSa/s
Output amplitude ranges	± 10 mV, ± 100 mV, ± 1 V, ± 10 V (single ended into high impedance)
Output DC offset range	± 10 mV to ± 10 V, equal to the set output amplitude range
Output power	24 dBm (± 10 V, 250 mW), for each BNC
Output gain inaccuracy	< 1% at 100 kHz for all output ranges
Maximum output drive current	100 mA
Output offset amplitude	± 1 mV or 1% of range, whichever is bigger
Harmonic distortion (typical)	–85 dBc for frequencies <200 kHz, –60 dBc for frequencies <2 MHz, –55 dBc for frequencies <5 MHz; output range ≤ 1 V. –80 dBc for frequencies <200 kHz, –50 dBc for frequencies <2 MHz, –45 dBc for frequencies <5 MHz; output range 10 V. measured with carrier amplitude –1 dBFS, 50 Ω load, single-ended.
Analog adder	Auxiliary Input 1 can be added to the signal output, ± 10 V, DC–2 MHz

Table 7.12: Signal Output: voltage noise, ranges

Output range	Output noise density (high load impedance setting)	RMS output noise at 12 MHz bandwidth
10 mV	43 nV/ $\sqrt{\text{Hz}}$	145 μ Vrms
100 mV	43 nV/ $\sqrt{\text{Hz}}$	145 μ Vrms
1 V	48 nV/ $\sqrt{\text{Hz}}$	161 μ Vrms
10 V	104 nV/ $\sqrt{\text{Hz}}$	310 μ Vrms

Table 7.13: Auxiliary Inputs

Parameter	Description
Connectors	2 BNC on the front panel
A/D converter	16 bit, 15 MSa/s
A/D analog bandwidth	5 MHz
Input impedance	1 M Ω
Amplitude	± 10 V
Input noise amplitude	430 nV/ $\sqrt{\text{Hz}}$; frequency > 100 kHz
Resolution	0.335 mV

Table 7.14: Auxiliary Outputs

Parameter	Description
Connectors	4 BNC on the front panel
D/A converter	18 bit, 612 kSa/s

Parameter	Description
D/A analog bandwidth	200 kHz
Output impedance	50 Ω
Amplitude	± 10 V
Resolution	< 85 μ V
Drive current	20 mA
Noise density	210 nV/ $\sqrt{\text{Hz}}$ into high-impedance load; frequency > 1 kHz
RMS noise	90 μ V _{rms} into high-impedance load; measurement bandwidth 12 MHz

Table 7.15: Trigger Inputs

Parameter	Description
Connectors	2 BNC on the back panel
Trigger input impedance	1 k Ω
Frequency range external reference	1 Hz to 500 kHz; 1 Hz to 5 MHz, requires MF-F5M option
Trigger amplitude range	± 5 V
Minimum pulse width	35 ns
Trigger level	± 5 V, 3.66 mV resolution
Trigger hysteresis	< 20 mV

Table 7.16: Trigger Outputs

Parameter	Description
Connectors	2 BNC on the back panel
Trigger output impedance	50 Ω
Frequency range external reference	1 μ Hz to 500 kHz; 1 μ Hz to 5 MHz, requires MF-F5M option
Trigger amplitude	5 V

Table 7.17: 10 MHz clock synchronization

Parameter	Description
Connectors	2 BNC, 10 MHz clock input and output on the back panel
10 MHz input, impedance	50 Ω
10 MHz input, frequency range	9.98 to 10.02 MHz
10 MHz input, amplitude range	200 mV to 3 V
10 MHz output, impedance	50 Ω
10 MHz output, amplitude	Typically 1 V _{pp} into 50 Ω , sinusoidal

Table 7.18: Internal frequency reference

Parameter	Description
Type	TCXO
Initial accuracy	< ± 1.5 ppm
Long term accuracy/aging	< ± 1 ppm in the first year
Short term stability (0.1 s)	< $2 \cdot 10^{-10}$
Temperature coefficient	0.05 ppm/ $^{\circ}\text{C}$ (@23 $^{\circ}\text{C}$)
Phase noise at 1 kHz	-140 dBc/Hz

Parameter	Description
Phase noise at 10 kHz	-150 dBc/Hz

Table 7.19: Connectivity and others

Parameter	Description
Host connection	LAN, 1 GbE; USB 2.0, 480 Mbit/s
Internal drive data storage capacity	4.5 GB
USB host	2 connectors on the back panel for mass storage or WLAN modules
DIO, digital I/O	4 x 8 bit, general purpose digital input/output port, 3.3 V TTL VHDCI 68 pin female connector

Table 7.20: Maximum ratings

Parameter	Lower	Upper
Damage threshold Current Signal Input I	-5 V	+ 5 V
Damage threshold Voltage Input +V/-V Diff	-5 V	+5 V
Damage threshold Signal Output +V/-V	-12 V	+12 V
Damage threshold Aux Input 1,2	-12 V	+12 V
Damage threshold Aux Outputs 1,2,3,4	-12 V	+12 V
Damage threshold Clock 10 MHz In/Out	-5 V	+5 V
Damage threshold Trigger Out 1,2	-1 V	+6 V
Damage threshold Trigger In 1,2	-8 V	+8 V
Damage threshold DIO 32 bit	-1 V	+6 V
Damage threshold DC In	0 V	26 V

Table 7.21: LabOne UI requirements

Parameter	Description
Operating systems	Any, web browser based
Input device	Touch screen, keyboard, mouse
CPU	2+ cores, hardware accelerated rendering on browser
Browser	Edge, Firefox, Chrome, Safari, Opera
Connectivity	1 GbE, 100 MbE, USB 2.0

Table 7.22: LabOne API requirements

Parameter	Description
Operating systems	Windows 10, 11 on x86-64 macOS 10.11+ on x86-64 and ARMv8 GNU/Linux (Ubuntu 14.04+, CentOS 7+, Debian 8+) on x86-64 and ARMv8
CPU	x86-64 (Intel, AMD), ARMv8 (e.g., Raspberry Pi 4 and newer, Apple M-series)
RAM	4 GB+
Connectivity	1 GbE, 100 MbE, USB 2.0
Supported languages	LabVIEW, Python, MATLAB, .NET, C/C++

The DIO port is a VHDCI 68 pin connector as introduced by the SPI-3 document of the SCSI-3 specification. It is a female connector that requires a 32 mm wide male connector. The DIO port features 32 bits that can be configured byte-wise as inputs or outputs.

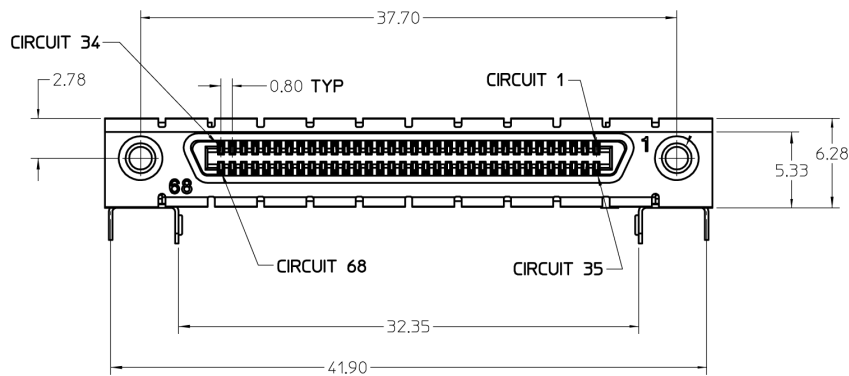


Figure 71: DIO HD 68 pin connector

Table 7.23: DIO pin assignment

Pin	Name	Description	Range specification
68	CLKI	clock input, used to latch signals at the digital input ports - can also be used to retrieve digital signals from the output port using an external sampling clock	3.3 V LVCMOS/ TTL
67	DOL	DIO output latch, 60 MHz clock signal, the digital outputs are synchronized to the falling edge of this signal	3.3 V LVCMOS/ TTL
66-59	DI[31:24]	digital input or output (set by user)	3.3 V LVCMOS/ TTL
58-51	DIO[23:16]	digital input or output (set by user)	3.3 V LVCMOS/ TTL
50-43	DIO[15:8]	digital input or output (set by user)	3.3 V LVCMOS/ TTL
42-35	DIO[7:0]	digital input or output (set by user)	3.3 V LVCMOS/ TTL
34-30	-	do not connect, for internal use only	-
29-1	GND	digital ground	-

The figure below shows the architecture of the DIO input/output. The DIO port features 32 bits that can be configured byte-wise as inputs or outputs by means of a drive signal. The digital output data is latched synchronously with the falling edge of the internal clock, which is running at 60 MHz. The internal sampling clock is available at the DOL pin of the DIO connector. Digital input data can either be sampled by the internal clock or by an external clock provided through the CLKI pin. A decimated version of the input clock is used to sample the input data. The Decimation unit counts the clocks to decimation and then latches the input data. The default decimation is 3e6, corresponding to a digital input sampling rate of 20 samples per second.

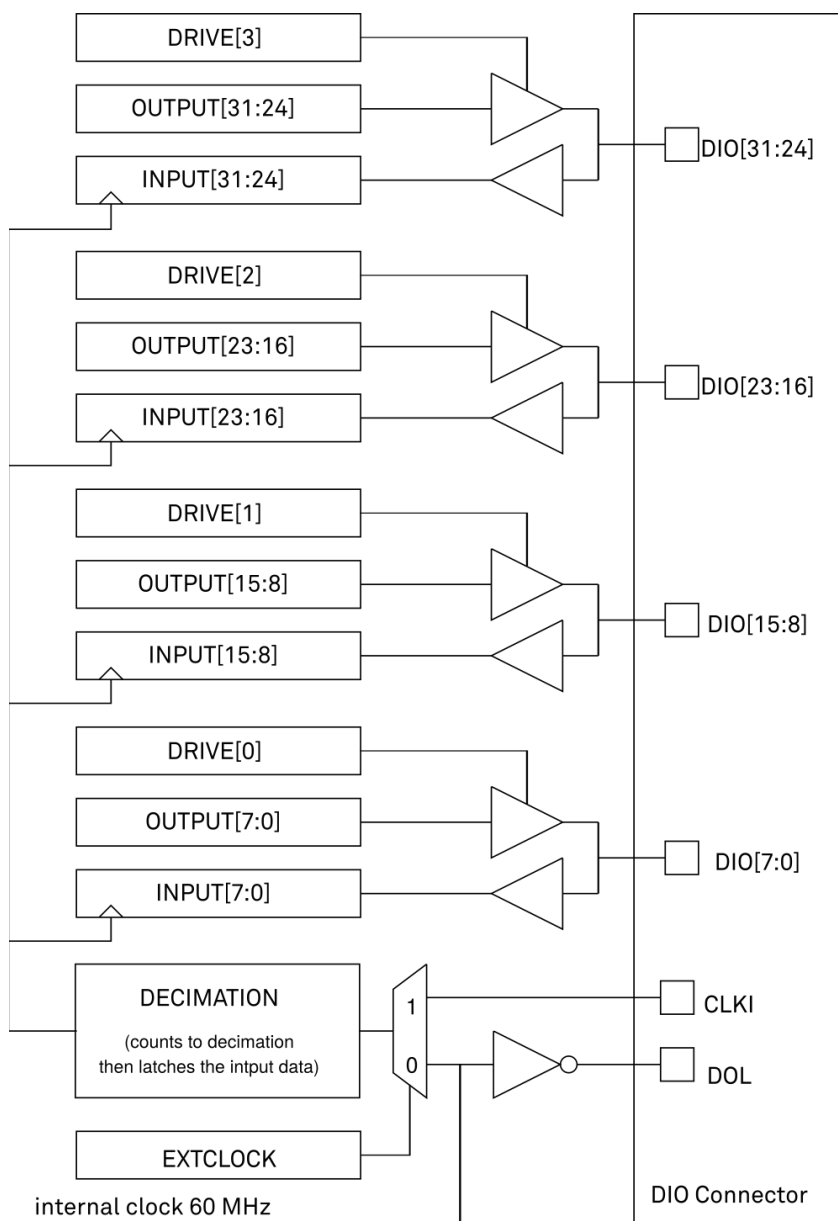


Figure 7.2: DIO input/output architecture

7.2. Performance Diagrams

Input noise amplitude depends on several parameters, and in particular on the frequency and on the input range setting. The input noise is lower for smaller input ranges, and it is recommended to use small ranges especially for noise measurements. [Figure 7.3](#) shows the noise density of the voltage input with DC coupling. The input noise with AC coupling is the same as long as the frequency is higher than the AC cutoff frequency (see). The noise is independent of the input impedance setting, 50 Ω or 10 M Ω . The corner frequency of the 1/f noise is in the range of 100 Hz and the white-noise floor is typically 2.5 nV/ $\sqrt{\text{Hz}}$ for the smallest input ranges.

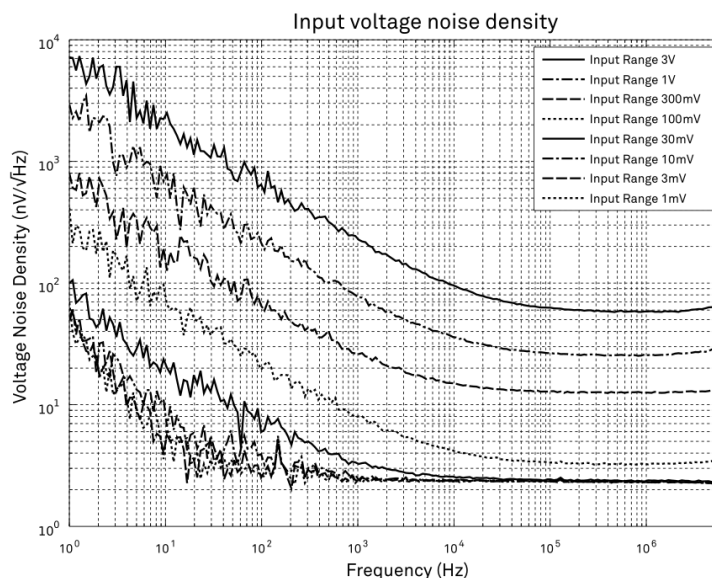


Figure 7.3: MFLI Voltage Input voltage noise density

A note on input voltage noise and AC coupling. The input voltage noise will increase at low frequencies, when using AC coupling at the signal input of the MFLI. For example, the input voltage noise of the 10 mV input range is around 10 nV/√Hz at a frequency of 10 Hz, as shown in the figure above. When switching on AC coupling, the noise increases to around 75 nV/√Hz. The reason is that the 10nF AC coupling capacitor has an impedance of 1.6 MΩ at 10 Hz. Together with the 10 MΩ input bias resistor of the MFLI input amplifier, you get the increased noise density. To work around this problem, use DC coupling for the signal input. An external large capacitor may be employed if AC coupling must be used.

Figure 7.4 shows the noise density of the current input. The frequency range of the measurements for the smaller input ranges is limited by the bandwidth of these ranges as specified in .

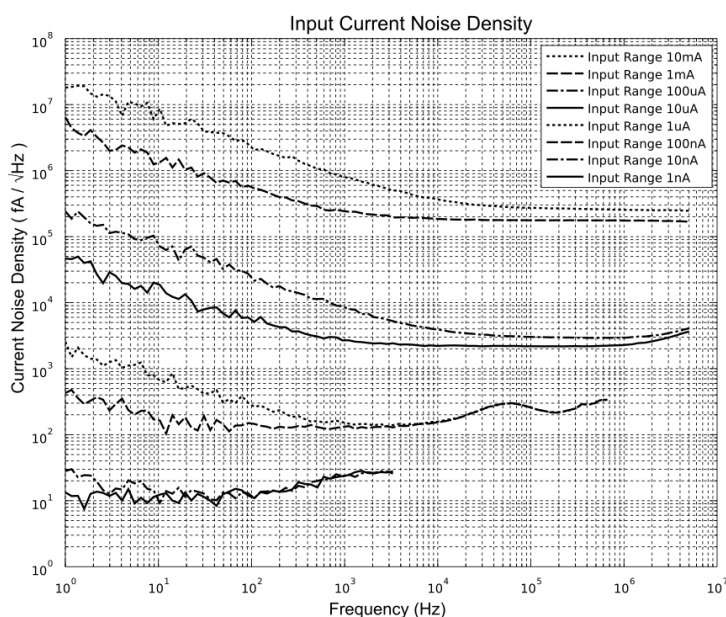


Figure 7.4: MFLI Current Input current noise density

Figure 7.5 shows the SSB phase noise measured at the signal output. For this measurement, the signal output was connected to a phase noise analyzer and the output amplitude was set to 5 V. The measured phase noise at 5 MHz and 10 kHz offset is around -153 dBc/Hz.

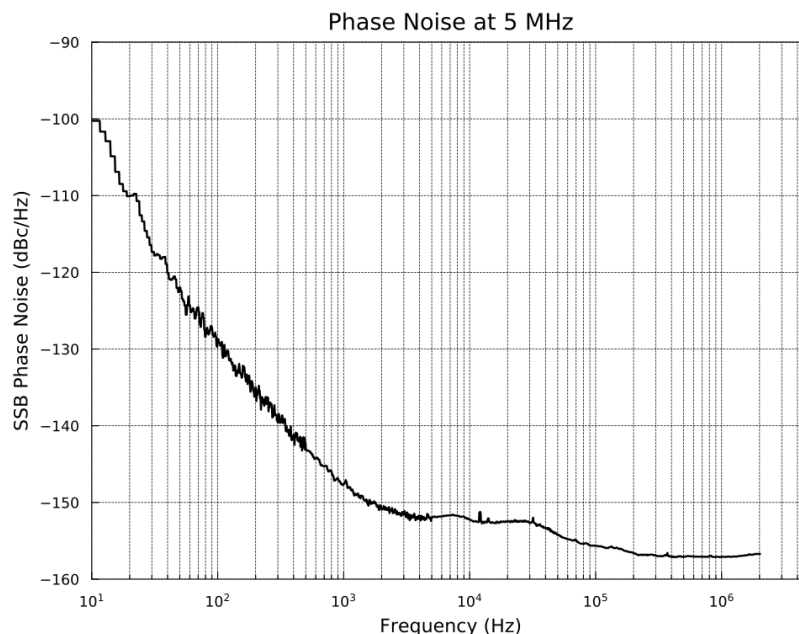


Figure 7.5: MFLI phase noise

Figure 7.6 shows the noise density of the voltage output into 50 Ω . For high-impedance loads, the voltage noise levels in the figure need to be doubled. The output noise was measured with no signal and 0 V offset. For the ranges 10 mV and 100 mV, the voltage noise is basically identical with a flat-band noise level of 14.5 nV/ $\sqrt{\text{Hz}}$ at 100 kHz into 50 Ω . For high-impedance loads, the value would be around 29 nV/ $\sqrt{\text{Hz}}$. The 10 V range shows the largest noise with 34.5 nV/ $\sqrt{\text{Hz}}$ at 100 kHz. The corner frequency of the 1/f noise is in the range of 10 to 20 kHz.

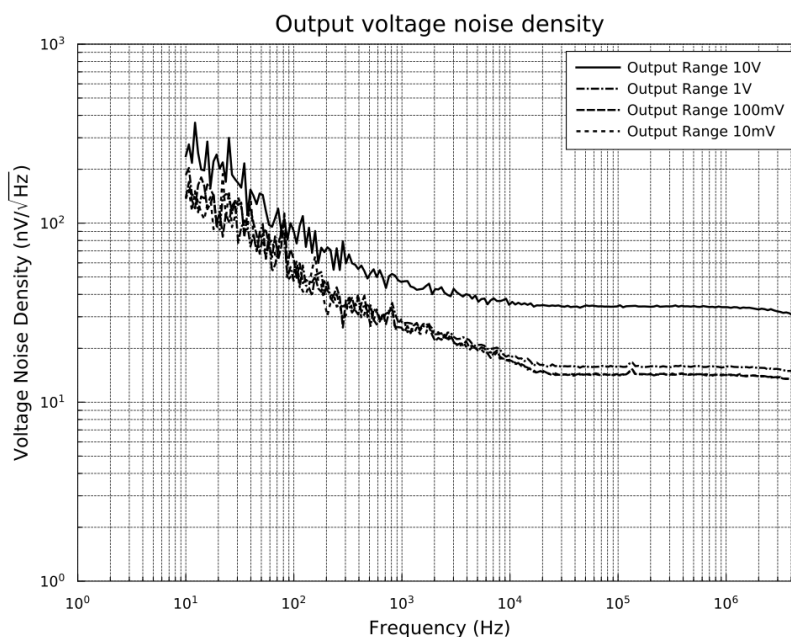


Figure 7.6: MFLI Signal Output voltage noise density

The impedance measurement accuracy and temperature stability of the MFIA Impedance Analyzer depend on the absolute value of the measured impedance, as well as the measurement frequency. The charts shown below show the minimum accuracy levels and temperature stability coefficients over the whole measurement range of the MFIA. The plots are valid in 4-terminal measurement mode with enabled automatic range control.

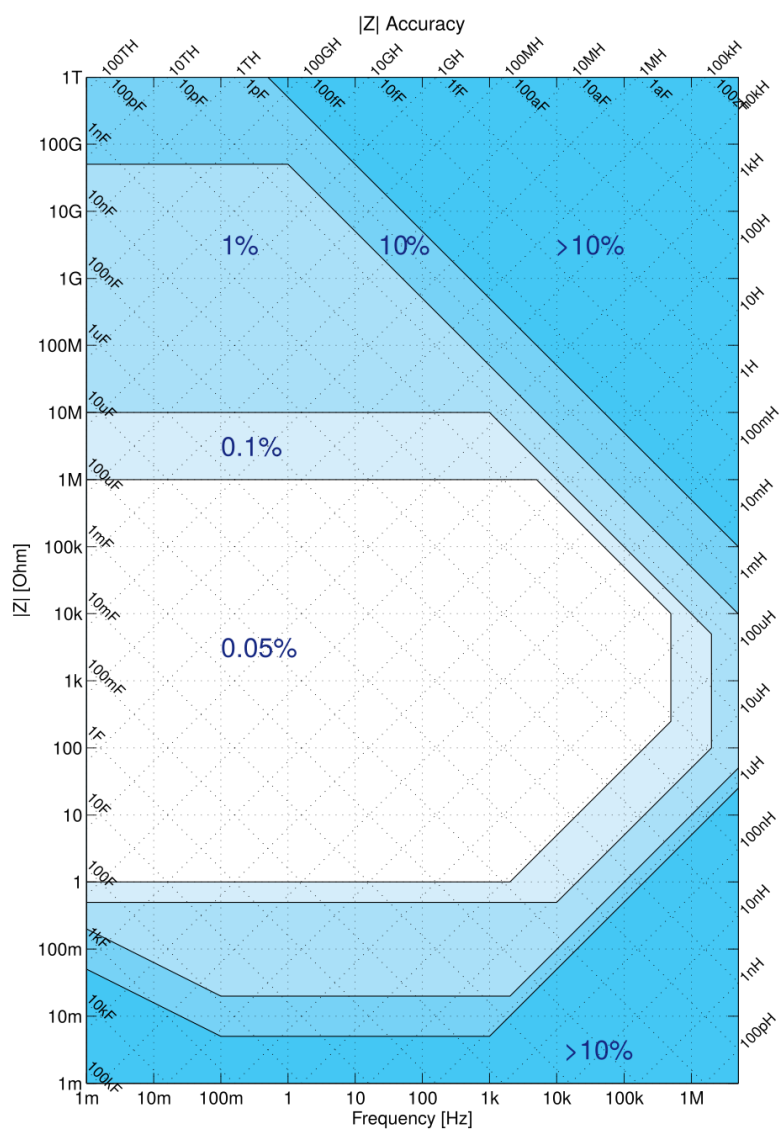


Figure 7.7: MFIA measurement accuracy as a function of impedance and frequency

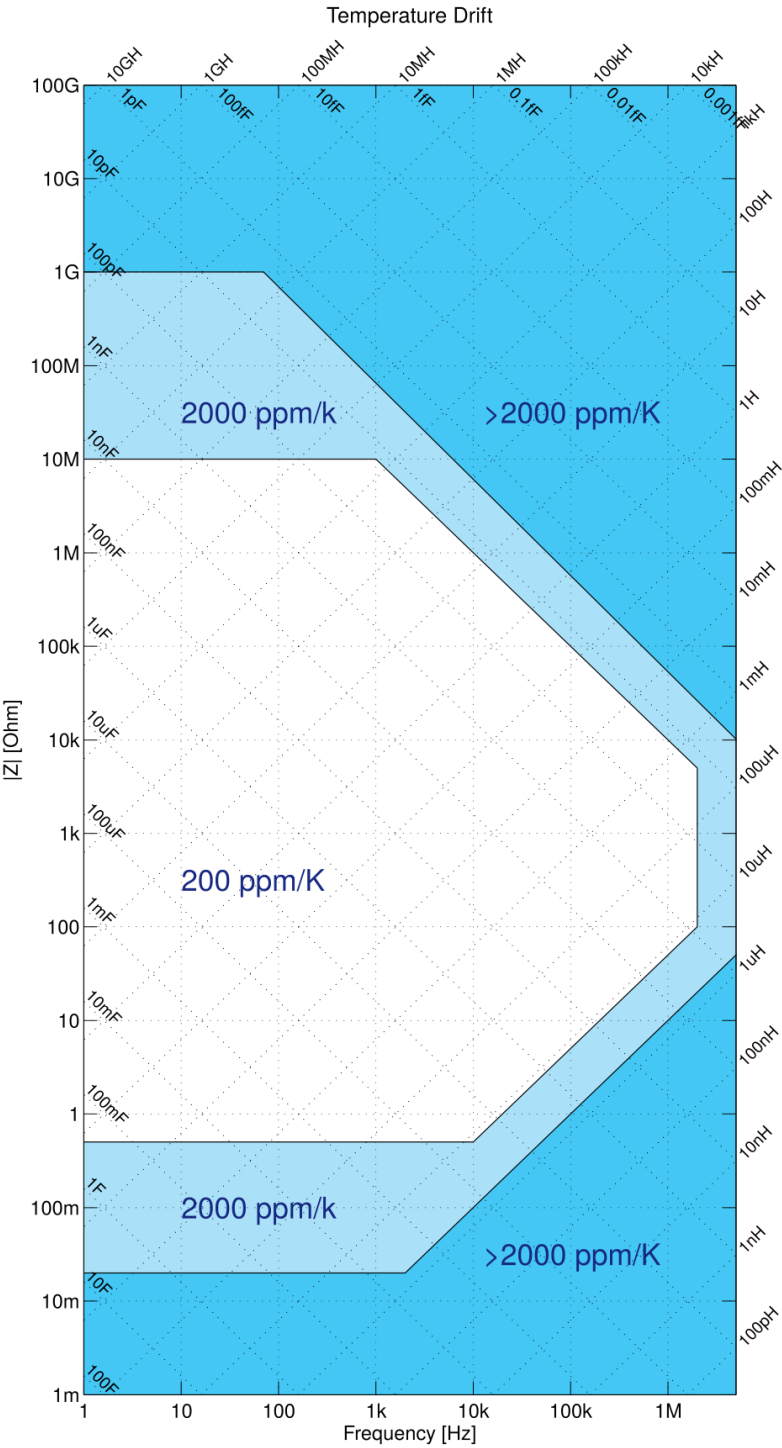


Figure 7.8: MFIA temperature stability as a function of impedance and frequency

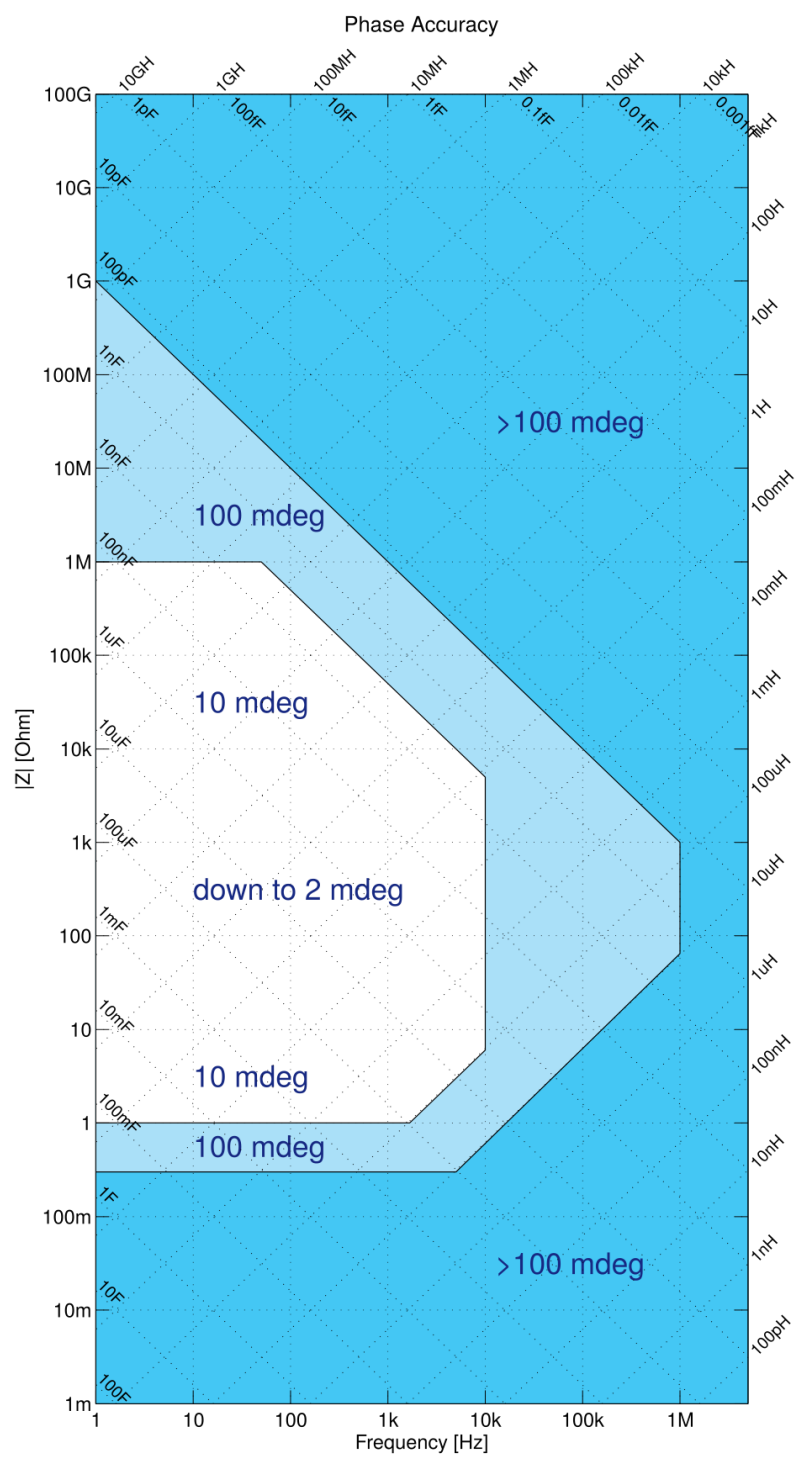


Figure 7.9: MFIA impedance phase accuracy as a function of impedance and frequency

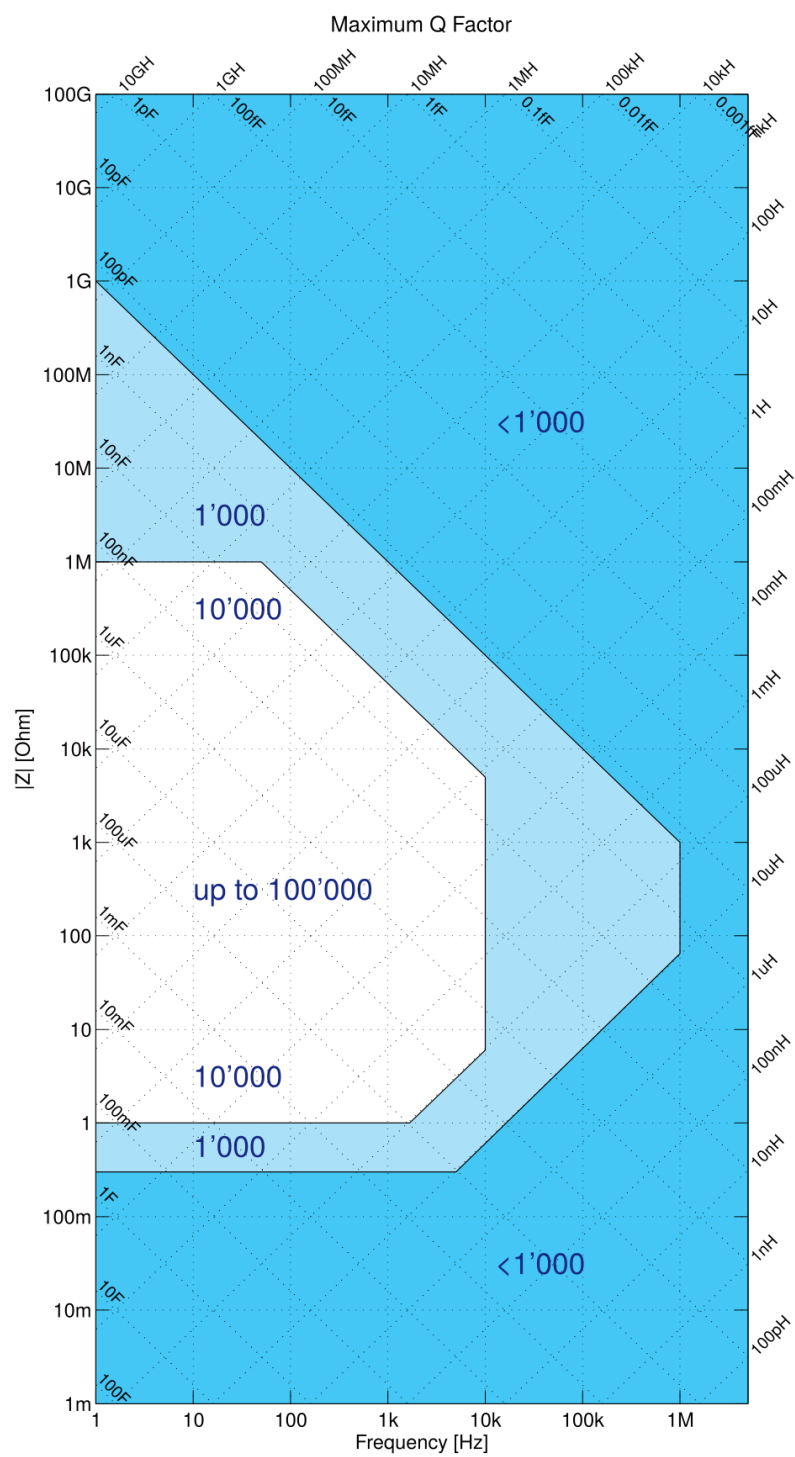


Figure 7.10: MFIA maximum measurable Q factor as a function of impedance and frequency

8. Device Node Tree

This chapter contains reference documentation for the settings and measurement data available on MFIA Instruments. Whilst [Functional Description LabOne User Interface](#) describes many of these settings in terms of the features available in the LabOne User Interface, this chapter describes them on the device level and provides a hierarchically organized and comprehensive list of device functionality.

Since these settings and data streams may be written and read using the LabOne APIs (Application Programming Interfaces) this chapter is of particular interest to users who would like to perform measurements programmatically via LabVIEW, Python, MATLAB, .NET or C.

Please see: - [Introduction](#) for an introduction of how the instrument's settings and measurement data are organized hierarchically in the Data Server's so-called "Node Tree". - [Reference Node Documentation](#) for a reference list of the settings and measurement data available on MFIA Instruments, organized by branch in the Node Tree.

8.1. Introduction

This chapter provides an overview of how an instrument's configuration and output is organized by the Data Server.

All communication with an instrument occurs via the Data Server program the instrument is connected to (see [LabOne Software Architecture](#) for an overview of LabOne's software components). Although the instrument's settings are stored locally on the device, it is the Data Server's task to ensure it maintains the values of the current settings and makes these settings (and any subscribed data) available to all its current clients. A client may be the LabOne User Interface or a user's own program implemented using one of the LabOne Application Programming Interfaces, e.g., Python.

The instrument's settings and data are organized by the Data Server in a file-system-like hierarchical structure called the node tree. When an instrument is connected to a Data Server, its device ID becomes a top-level branch in the Data Server's node tree. The features of the instrument are organized as branches underneath the top-level device branch and the individual instrument settings are leaves of these branches.

For example, the auxiliary outputs of the instrument with device ID "dev2006" are located in the tree in the branch:

```
/dev1000/auxouts/
```

In turn, each individual auxiliary output channel has its own branch underneath the "AUXOUTS" branch.

```
/dev1000/auxouts/0/  
/dev1000/auxouts/1/  
/dev1000/auxouts/2/  
/dev1000/auxouts/3/
```

Whilst the auxiliary outputs and other channels are labelled on the instrument's panels and the User Interface using 1-based indexing, the Data Server's node tree uses 0-based indexing. Individual settings (and data) of an auxiliary output are available as leaves underneath the corresponding channel's branch:

```
/dev1000/auxouts/0/demodselect  
/dev1000/auxouts/0/limitlower  
/dev1000/auxouts/0/limitupper  
/dev1000/auxouts/0/offset  
/dev1000/auxouts/0/outputselect  
/dev1000/auxouts/0/preoffset  
/dev1000/auxouts/0/scale  
/dev1000/auxouts/0/value
```

These are all individual node paths in the node tree; the lowest-level nodes which represent a single instrument setting or data stream. Whether the node is an instrument setting or data-stream and which type of data it contains or provides is well-defined and documented on a per-node basis in

the Reference Node Documentation section in the relevant instrument-specific user manual. The different properties and types are explained in [Node Properties and Data Types](#).

For instrument settings, a Data Server client modifies the node's value by specifying the appropriate path and a value to the Data Server as a (path, value) pair. When an instrument's setting is changed in the LabOne User Interface, the path and the value of the node that was changed are displayed in the Status Bar in the bottom of the Window. This is described in more detail in [Exploring the Node Tree](#).

Module Parameters

LabOne Core Modules, such as the Sweeper, also use a similar tree-like structure to organize their parameters. Please note, however, that module nodes are not visible in the Data Server's node tree; they are local to the instance of the module created in a LabOne client and are not synchronized between clients.

8.1.1. Node Properties and Data Types

A node may have one or more of the following properties:

Read	Data can be read from the node.
Write	Data can be written to the node.
Setting	The node corresponds to a writable instrument configuration. The data of these nodes are persisted in snapshots of the instrument and stored in the LabOne XML settings files.
Streaming	A node with the read attribute that provides instrument data, typically at a user-configured rate. The data is usually a more complex data type, for example demodulator data is returned as ZIDemodSample . A full list of streaming nodes is available in the Programming Manual in the Chapter Instrument Communication. Their availability depends on the device class (e.g. MF) and the option set installed on the device.

A node may contain data of the following types:

Integer	Integer data.
Double	Double precision floating point data.
String	A string array.
Integer (enumerated)	As for Integer, but the node only allows certain values.
Composite data type	For example, ZIDemodSample . These custom data types are structures whose fields contain the instrument output, a timestamp and other relevant instrument settings such as the demodulator oscillator frequency. Documentation of custom data types is available in

8.1.2. Exploring the Node Tree

In the LabOne User Interface

A convenient method to learn which node is responsible for a specific instrument setting is to check the Command Log history in the bottom of the LabOne User Interface. The command in the Status Bar gets updated every time a configuration change is made. [Figure 8.1](#) shows how the equivalent MATLAB command is displayed after modifying the value of the auxiliary output 1's offset. The format of the LabOne UI's command history can be configured in the Config Tab (MATLAB, Python and .NET are available). The entire history generated in the current UI session can be viewed by clicking the "Show Log" button.

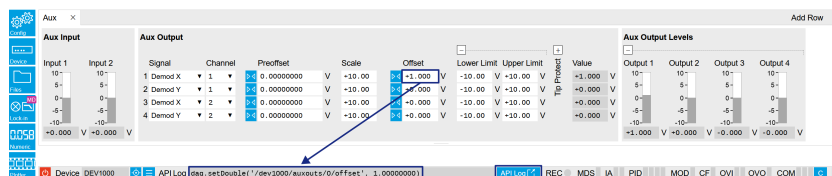


Figure 8.1: When a device's configuration is modified in the LabOne User Interface, the Status Bar displays the equivalent command to perform the same configuration via a LabOne programming interface. Here, the MATLAB code to modify auxiliary output 1's offset value is provided. When "Show Log" is clicked the entire configuration history is displayed in a new browser tab.

In a LabOne Programming Interface

A list of nodes (under a specific branch) can be requested from the Data Server in an API client using the `listNodes` command (MATLAB, Python, .NET) or `ziAPIListNodes()` function (C API). Please see each API's command reference for more help using the `listNodes` command. To obtain a list of all the nodes that provide data from an instrument at a high rate, so-called streaming nodes, the `streamingonly` flag can be provided to `listNodes`. More information on data streaming and streaming nodes is available in the LabOne Programming Manual.

The detailed descriptions of nodes that is provided in [Reference Node Documentation](#) is accessible directly in the LabOne MATLAB or Python programming interfaces using the "help" command. The `help` command is `daq.help(path)` in Python and `ziDAQ('help', path)` in MATLAB. The command returns a description of the instrument node including access properties, data type, units and available options. The "help" command also handles wildcards to return a detailed description of all nodes matching the path. An example is provided below.

```
daq = zhinst.core.ziDAQServer('localhost', 8004, 6)
daq.help('/dev2006/auxouts/0/offset')
# Out:
# /dev1000/auxouts/0/offset#
# Add the specified offset voltage to the signal after scaling. Auxiliary
Output
# Value = (Signal+Preoffset)*Scale + Offset
# Properties: Read, Write, Setting
# Type: Double
# Unit: V
```

8.1.3. Data Server Nodes

The Data Server has nodes in the node tree available under the top-level `/ZI/` branch. These nodes give information about the version and state of the Data Server the client is connected to. For example, the nodes:

- `/ZI/ABOUT/VERSION`
- `/ZI/ABOUT/REVISION`

are read-only nodes that contain information about the release version and revision of the Data Server. The nodes under the `/ZI/DEVICES/` list which devices are connected, discoverable and visible to the Data Server.

The nodes:

- /ZI/CONFIG/OPEN
- /ZI/CONFIG/PORT

are settings nodes that can be used to configure which port the Data Server listens to for incoming client connections and whether it may accept connections from clients on hosts other than the localhost.

Nodes that are of particular use to programmers are:

- /ZI/DEBUG/LOGPATH - the location of the Data Server's log in the PC's file system,
- /ZI/DEBUG/LEVEL - the current log-level of the Data Server (configurable; has the Write attribute),
- /ZI/DEBUG/LOG - the last Data Server log entries as a string array.

The Global nodes of the LabOne Data Server are listed in the Instrument Communication chapter of the LabOne Programming Manual

8.2. Reference Node Documentation

This section describes all the nodes in the data server's node tree organized by branch.

8.2.1. AUXINS

/dev..../auxins/n/averaging

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Defines the number of samples on the input to average as a power of two. Possible values are in the range [0, 16]. A value of 0 corresponds to the sampling rate of the auxiliary input's ADC.

/dev..../auxins/n/sample

Properties: Read, Stream
Type: ZIAuxInSample
Unit: V

Voltage measured at the Auxiliary Input after averaging. The data rate depends on the averaging value. Note, if the instrument has demodulator functionality, the auxiliary input values are available as fields in a demodulator sample and are aligned by timestamp with the demodulator output.

/dev..../auxins/n/values/n

Properties: Read
Type: Double
Unit: V

Voltage measured at the Auxiliary Input after averaging. The value of this node is updated at a low rate (50 Hz); the streaming node auxins/n/sample is updated at a high rate defined by the averaging.

8.2.2. AUXOUTS

/dev..../auxouts/n/demodselect

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Select the channel number of the selected signal source.

/dev..../auxouts/n/limitlower

Properties: Read, Write, Setting
Type: Double
Unit: V

Lower limit for the signal at the Auxiliary Output. A smaller value will be clipped.

/dev..../auxouts/n/limitupper

Properties: Read, Write, Setting
Type: Double
Unit: V

Upper limit for the signal at the Auxiliary Output. A larger value will be clipped.

/dev..../auxouts/n/offset

Properties: Read, Write, Setting
Type: Double
Unit: V

Add the specified offset voltage to the signal after scaling. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset

/dev..../auxouts/n/outputselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the signal source to be represented on the Auxiliary Output.

-1	"manual": Select Manual as the output option.
0	"demod_x": Select Demod X as the output option.
1	"demod_y": Select Demod Y as the output option.
2	"demod_r": Select Demod R as the output option.
3	"demod_theta": Select Demod Theta as the output option.
5	"pid": Select PID Out as the output option.
9	"pid_shift": Select PID Shift as the output option.
10	"pid_error": Select PID Error as the output option.
11	"tu_filtered": Select TU Filtered Value as the output option.
13	"tu_output": Select TU Output Value as the output option.

/dev..../auxouts/n/preoffset

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Add a pre-offset to the signal before scaling is applied. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset

/dev..../auxouts/n/scale

Properties: Read, Write, Setting
Type: Double
Unit: None

Multiplication factor to scale the signal. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset

/dev..../auxouts/n/tipprotect/enable

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Enable the Tip Protect unit.

0 "off": Switch off the Tip Protect unit.
 1 "on": Switch on the Tip Protect unit.

/dev..../auxouts/n/tipprotect/polarity

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the active level of Source signal.

0 "low": Active when the source signal is at low level.
 1 "high": Active when the source signal is at high level.

/dev..../auxouts/n/tipprotect/source

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the input source for Tip Protect.

0 "tu_output0": Select Threshold Unit 1 as the source signal.
 1 "tu_output1": Select Threshold Unit 2 as the source signal.
 2 "tu_output2": Select Threshold Unit 3 as the source signal.
 3 "tu_output3": Select Threshold Unit 4 as the source signal.

/dev..../auxouts/n/tipprotect/value

Properties: Read, Write, Setting
Type: Double
Unit: V

Signal value written to the Aux Output port when Tip Protect is active. The value overwrites Aux Output Internal value and the Lower and Upper Limits.

/dev..../auxouts/n/value

Properties: Read
Type: Double
Unit: V

Voltage present on the Auxiliary Output. Auxiliary Output Value = (Signal+Preoffset)*Scale + Offset

8.2.3. CLOCKBASE

/dev..../clockbase

Properties: Read
Type: Double
Unit: Hz

Returns the internal clock frequency of the device.

8.2.4. CURRINS

/dev..../currins/n/autorange

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Automatic adjustment of the Range to about two times the maximum current input amplitude measured over about 100 ms.

/dev..../currins/n/float

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Switches the input between floating (ON) and connected grounds (OFF). This setting applies both to the voltage and the current input. It is recommended to discharge the test device before connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.

0 "off": OFF: GND connected
 1 "on": ON: Floating

/dev..../currins/n/max

Properties: Read, Write
Type: Double
Unit: None

Gives the maximum measured input current (peak value) normalized to input range.

/dev..../currins/n/min

Properties: Read, Write
Type: Double
Unit: None

Gives the minimum measured input current (peak value) normalized to input range.

/dev..../currins/n/on

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the current input.

/dev..../currins/n/range

Properties: Read, Write, Setting
Type: Double
Unit: A

Defines the gain of the current input amplifier. The range should exceed the incoming signal by roughly a factor of two including a potential DC offset. The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used.

/dev..../currins/n/rangestep/trigger

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Switches to the next appropriate input range such that the range fits best with the measured input current.

/dev..../currins/n/scaling

Properties: Read, Write, Setting
Type: Double
Unit: None

Applies the given scale factor to the current input.

8.2.5. DEMODS

/dev..../demods/n/adselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the input signal for the demodulator.

0	"signin0", "signal_input0": Sig In 1
1	"currin0", "current_input0": Curr In 1
2	"trigin0", "trigger_input0": Trigger Input 1
3	"trigin1", "trigger_input1": Trigger Input 2
4	"auxout0", "auxiliary_output0": Aux Out 1
5	"auxout1", "auxiliary_output1": Aux Out 2
6	"auxout2", "auxiliary_output2": Aux Out 3
7	"auxout3", "auxiliary_output3": Aux Out 4
8	"auxin0", "auxiliary_input0": Aux In 1
9	"auxin1", "auxiliary_input1": Aux In 2

174 "demod_constant_input": Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).

/dev..../demods/n/enable

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Enables the data acquisition for the corresponding demodulator. Note: increasing number of active demodulators increases load on the physical connection to the host computer.

0	"off": OFF: demodulator inactive
1	"on": ON: demodulator active

/dev..../demods/n/freq

Properties: Read
Type: Double
Unit: Hz

Indicates the frequency used for demodulation and for output generation. The demodulation frequency is calculated with oscillator frequency times the harmonic factor. When the MOD option is used linear combinations of oscillator frequencies including the harmonic factors define the demodulation frequencies.

/dev..../demods/n/harmonic

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Multiplies the demodulator's reference frequency by an integer factor. If the demodulator is used as a phase detector in external reference mode (PLL), the effect is that the internal oscillator locks to the external frequency divided by the integer factor.

/dev..../demods/n/order

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the filter roll off between 6 dB/oct and 48 dB/oct.

1	1st order filter 6 dB/oct
2	2nd order filter 12 dB/oct
3	3rd order filter 18 dB/oct
4	4th order filter 24 dB/oct
5	5th order filter 30 dB/oct
6	6th order filter 36 dB/oct
7	7th order filter 42 dB/oct
8	8th order filter 48 dB/oct

/dev..../demods/n/oscselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Connects the demodulator with the supplied oscillator. Number of available oscillators depends on the installed options.

0	Oscillator 1
1	Oscillator 2
2	Oscillator 3
3	Oscillator 4

/dev..../demods/n/phaseadjust

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Adjust the demodulator phase automatically in order to read 0 degrees.

/dev..../demods/n/phaseshift

Properties: Read, Write, Setting
Type: Double
Unit: deg

Phase shift applied to the reference input of the demodulator.

/dev..../demods/n/rate

Properties: Read, Write, Setting
Type: Double
Unit: 1/s

Defines the demodulator sampling rate, the number of samples that are sent to the host computer per second. A rate of about 7-10 higher as compared to the filter bandwidth usually provides sufficient aliasing suppression. This is also the rate of data received by LabOne Data Server and saved to the computer hard disk. This setting has no impact on the sample rate on the auxiliary outputs connectors. Note: the value inserted by the user may be approximated to the nearest value supported by the instrument.

/dev..../demods/n/sample

Properties: Read, Stream
Type: ZIDemodSample
Unit: Dependent

Contains streamed demodulator samples with sample interval defined by the demodulator data rate.

/dev..../demods/n/sinc

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the sinc filter. When the filter bandwidth is comparable to or larger than the demodulation frequency, the demodulator output may contain frequency components at the frequency of demodulation and its higher harmonics. The sinc is an additional filter that attenuates these unwanted components in the demodulator output.

/dev..../demods/n/timeconstant

Properties: Read, Write, Setting
Type: Double
Unit: s

Sets the integration time constant or in other words, the cutoff frequency of the demodulator low pass filter.

/dev..../demods/n/trigger

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the acquisition mode (i.e. triggering) of the demodulator.

0	"continuous": Continuous: demodulator data is continuously streamed to the host computer.
1	"trigin0_rising", "trigger_input0_rising": Trigger Input 1: rising edge triggered.
2	"trigin0_falling", "trigger_input0_falling": Trigger Input 1: falling edge triggered.
3	"trigin0_both", "trigger_input0_both": Trigger Input 1: triggering on both rising and falling edge.
4	"trigin1_rising", "trigger_input1_rising": Trigger Input 2: rising edge triggered.
5	"trigin0or1_rising", "trigger_input0or1_rising": Trigger Input 1 or 2: rising edge triggered on either input.
8	"trigin1_falling", "trigger_input1_falling": Trigger Input 2: falling edge triggered.
10	"trigin0or1_falling", "trigger_input0or1_falling": Trigger Input 1 or 2: falling edge triggered on either input.
12	"trigin1_both", "trigger_input1_both": Trigger Input 2: triggering on both rising and falling edge.
15	"trigin0or1_both", "trigger_input0or1_both": Trigger Input 1 or 2: triggering on both rising and falling edge or either trigger input.
16	"trigin0_low", "trigger_input0_low": Trigger Input 1: demodulator data is streamed to the host computer when the level is low (TTL).
32	"trigin0_high", "trigger_input0_high": Trigger Input 1 for MF: demodulator data is streamed to the host computer when the level is high (TTL).
64	"trigin1_low", "trigger_input1_low": Trigger Input 2: demodulator data is streamed to the host computer when the level is low (TTL).
80	"trigin0or1_low", "trigger_input0or1_low": Trigger Input 1 or 2: demodulator data is streamed to the host computer when either level is low (TTL).
128	"trigin1_high", "trigger_input1_high": Trigger Input 2: demodulator data is streamed to the host computer when the level is high (TTL).
160	"trigin0or1_high", "trigger_input0or1_high": Trigger Input 1 or 2: demodulator data is streamed to the host computer when either level is high (TTL).

8.2.6. DIOS

/dev..../dios/n/auxdrive

Properties: Read
Type: Integer (64 bit)
Unit: None

Not used. Reserved for future use.

/dev..../dios/n/decimation

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Sets the decimation factor for DIO data streamed to the host computer.

/dev..../dios/n/drive

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

When on (1), the corresponding 8-bit bus is in output mode. When off (0), it is in input mode. Bit 0 corresponds to the least significant byte. For example, the value 1 drives the least significant byte, the value 8 drives the most significant byte.

/dev..../dios/n/extclk

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select DIO internal or external clocking.

0	"internal": The DIO is internally clocked with a fixed frequency of 60 MHz.
1	"external": The DIO is externally clocked with a clock signal connected to DIO Pin 68. The available range is from 1 Hz up to the internal clock frequency.

/dev..../dios/n/input

Properties: Read, Stream
Type: ZIDIOSample
Unit: None

Gives the value of the DIO input for those bytes where drive is disabled.

/dev..../dios/n/mode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select DIO mode

0	"manual": Enables manual control of the DIO output bits.
3	"threshold_unit": Enables setting of DIO output values by the threshold unit.

/dev..../dios/n/output

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Sets the value of the DIO output for those bytes where 'drive' is enabled.

8.2.7. EXTREFS

/dev..../extrefs/n/adselect

Properties: Read
Type: Integer (enumerated)
Unit: None

Indicates the input signal selection for the selected demodulator.

- | | |
|---|---|
| 0 | "sigin0", "signal_input0": Signal Input 1 is connected to the corresponding demodulator. |
| 1 | "currin0", "current_input0": Current Input 1 is connected to the corresponding demodulator. |
| 2 | "trigin0", "trigger_input0": Trigger Input 1 is connected to the corresponding demodulator. |
| 3 | "trigin1", "trigger_input1": Trigger Input 2 is connected to the corresponding demodulator. |
| 4 | "auxout0", "auxiliary_output0": Auxiliary Output 1 is connected to the corresponding demodulator. |
| 5 | "auxout1", "auxiliary_output1": Auxiliary Output 2 is connected to the corresponding demodulator. |
| 6 | "auxout2", "auxiliary_output2": Auxiliary Output 3 is connected to the corresponding demodulator. |
| 7 | "auxout3", "auxiliary_output3": Auxiliary Output 4 is connected to the corresponding demodulator. |
| 8 | "auxin0", "auxiliary_input0": Auxiliary Input 1 is connected to the corresponding demodulator. |
| 9 | "auxin1", "auxiliary_input1": Auxiliary Input 2 is connected to the corresponding demodulator. |

/dev..../extrefs/n/automode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

This defines the type of automatic adaptation of parameters in the PID used for Ext Ref.

- | | |
|---|--|
| 2 | "low_bandwidth", "pid_coefs_filter_low_bw": The PID coefficients, the filter bandwidth and the output limits are automatically set using a low bandwidth. |
| 3 | "high_bandwidth", "pid_coefs_filter_high_bw": The PID coefficients, the filter bandwidth and the output limits are automatically set using a high bandwidth. |
| 4 | "all", "pid_coefs_filter_auto_bw": The PID coefficient, the filter bandwidth and the output limits are dynamically adapted. |

/dev..../extrefs/n/demodselect

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Indicates the demodulator connected to the extref channel.

/dev..../extrefs/n/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the external reference.

/dev..../extrefs/n/locked

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates whether the external reference is locked.

/dev..../extrefs/n/oscselect

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates which oscillator is being locked to the external reference.

8.2.8. FEATURES

/dev..../features/code

Properties: Write
Type: String
Unit: None

Node providing a mechanism to write feature codes.

/dev..../features/devtype

Properties: Read
Type: String
Unit: None

Returns the device type.

/dev..../features/options

Properties: Read
Type: String
Unit: None

Returns enabled options.

/dev..../features/serial

Properties: Read
Type: String
Unit: None

Device serial number.

8.2.9. IMPS

/dev..../imps/n/ac

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Defines the input coupling for the Signal Inputs. AC coupling inserts a high-pass filter.

/dev..../imps/n/auto/bw

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enable automatic bandwidth control. If enabled the optimum bandwidth is calculated based on the frequency and measurement data.

/dev..../imps/n/auto/inputrange

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select input range control mode.

- | | |
|---|--|
| 0 | Manual: In manual mode the current and voltage input ranges are adjusted manually and separately. Use this mode with care as overload will result in inaccurate impedance results. |
| 1 | Auto: Dynamically adjust the input range according to the measured input signal strength. This optimizes the dynamic range and precision of impedance measurements. |
| 2 | Zone: This ranging option allows you to manually set the switching frequency limits for all eight current input ranges. |

/dev..../imps/n/auto/output

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

If enabled, the drive voltage amplitude is controlled by the device. If disabled it can be set manually.

/dev..../imps/n/auto/suppress

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Indicates disabled periodic auto range control. A running sweeper module takes over the range control and thus disables the periodic range checks.

/dev..../imps/n/bias/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the DC bias voltage.

/dev..../imps/n/bias/value

Properties: Read, Write, Setting
Type: Double
Unit: V

DC bias voltage applied across the device under test. Both positive and negative bias voltages are supported. In a 4-terminal measurement, the bias voltage is limited by the maximum common voltage input range of the device. In a 2-terminal measurement, the bias voltage can be larger because the voltage inputs are not connected.

/dev..../imps/n/calib/internal/active

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates if the internal calibration is applied to the measurement data.

/dev..../imps/n/calib/internal/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the internal calibration. This ensures that the input range gains match over the full frequency range. With enabled internal calibration the device fulfills the impedance accuracy specification. The internal calibration is a prerequisite to apply a user compensation.

/dev..../imps/n/calib/internal/smooth

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables smoothing of the internal calibration data. This results in lower noise in the measured data.

/dev..../imps/n/calib/user/active

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates that a valid user compensation is active. If active the impedance data streams deliver amplitude and phase corrected data based on the impedance user compensation.

/dev..../imps/n/calib/user/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the user compensation of the impedance data. The user compensation is correcting parasitics and delays caused by the external setup. The user compensation is applied on top of the internal impedance calibration.

/dev..../imps/n/calib/user/smooth

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables smoothing of the compensation data. This results in lower noise in the measured data.

/dev..../imps/n/confidence/compensation/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the indication of strong compensation in the plots. A strong compensation diminishes the measurement accuracy of the parameter.

/dev..../imps/n/confidence/compensation/ratio

Properties: Read, Write, Setting
Type: Double
Unit: None

Strength of the compensation that will trigger the strong compensation warning.

/dev..../imps/n/confidence/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables/disables all confidence indicators to check the reliability of the measured data. To enable individual indicators, open the advanced tab.

/dev..../imps/n/confidence/freqlimit/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the frequency limit detection based on the used current input range. Only relevant when Range Control is set to Manual.

/dev..../imps/n/confidence/lowdut2t/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables a warning when measuring a low impedance (100k) with a 2 point contact.

/dev..../imps/n/confidence/lowdut2t/ratio

Properties: Read, Write, Setting
Type: Double
Unit: None

When measuring with 2 point contacts, too low impedance of DUT will trigger the Low DUT 2T warning.

/dev..../imps/n/confidence/oneperiod/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the detection of unreliable data points where data sample loss leads to an invalid one-period average. Try reducing the data transfer rate.

/dev..../imps/n/confidence/opendetector/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the open terminal detection for 4-terminal measurements.

/dev..../imps/n/confidence/opendetector/ratio

Properties: Read, Write, Setting
Type: Double
Unit: None

Open terminal detection ratio. An open terminal is reported if the excitation calculated from current and voltage drop differs more than the specified factor from the driving voltage.

/dev..../imps/n/confidence/overflow/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the overload detection for current and voltage.

/dev..../imps/n/confidence/qfactor/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the detection of negative Q or D factors. Negative Q or D factors mean the measured impedance does not correspond to the chosen Representation. This can be due to an erroneous compensation, a bad choice of the Representation, or noise.

/dev..../imps/n/confidence/suppression/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

The Suppression Confidence Indicator indicates if one of the two parameters of a circuit representation cannot be calculated reliably from the measured impedance. This is the case if a small variation in one (dominant) representation parameter creates a strong variation of the other (suppressed) representation parameter. Such an error amplification indicates that the measurement of the secondary parameter is unreliable.

/dev..../imps/n/confidence/suppression/ratio

Properties: Read, Write, Setting
Type: Double
Unit: None

Error amplification limit for which a secondary parameter is marked unreliable. Larger gain values mean larger warning tolerances. A gain value between 10 and 100 is best.

/dev..../imps/n/confidence/underflow/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the underflow detection for current and voltage.

/dev..../imps/n/confidence/underflow/ratio

Properties: Read, Write, Setting
Type: Double
Unit: None

The underflow condition is met if the measured amplitude is lower than the specified ratio relative to full scale.

/dev..../imps/n/current/demodselect

Properties: Read
Type: Integer (64 bit)
Unit: None

Demodulator used for current demodulation.

/dev..../imps/n/current/inputselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the current input used for two- and four-terminal impedance measurements.

0	"currin0", "current_input0": Current Input 1
8	"auxin0", "auxiliary_input0": Aux In 1
9	"auxin1", "auxiliary_input1": Aux In 2

/dev..../imps/n/current/invert

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

If enabled, the current input signal is inverted. This is useful to switch the polarity of an input signal which can be caused by additional current amplifiers.

/dev..../imps/n/current/pid/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

If enabled, the PID value is used for the impedance calculation instead of the measured value at the current input.

/dev..../imps/n/current/range

Properties: Read, Write, Setting
Type: Double
Unit: A

Input current range used for the impedance measurement. Small current input ranges have a reduced bandwidth. In the Range Control modes 'Auto' and 'Impedance', the current range is switched automatically to a higher range if the frequency is too high.

/dev..../imps/n/current/scaling

Properties: Read, Write, Setting
Type: Double
Unit: A/V

The scaling factor will be applied to the current measurement done with an Aux In input.

/dev..../imps/n/demod/harmonic

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Multiplies the demodulator's reference frequency with the integer factor defined by this field.

/dev..../imps/n/demod/order

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the filter roll off between 6 dB/oct and 48 dB/oct of the current demodulator.

1	1st order filter 6 dB/oct
2	2nd order filter 12 dB/oct
3	3rd order filter 18 dB/oct
4	4th order filter 24 dB/oct
5	5th order filter 30 dB/oct
6	6th order filter 36 dB/oct
7	7th order filter 42 dB/oct
8	8th order filter 48 dB/oct

/dev..../imps/n/demod/oscselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Oscillator used to generate the frequency of the excitation voltage on the Hcur (+V) connector.

0	Oscillator 1
1	Oscillator 2
2	Oscillator 3
3	Oscillator 4

/dev..../imps/n/demod/rate

Properties: Read, Write, Setting
Type: Double
Unit: 1/s

Impedance data streaming rate. The same data rate is applied to the demodulators that are used for the impedance measurement.

/dev..../imps/n/demod/sinc

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the sinc filter.

/dev..../imps/n/demod/timeconstant

Properties: Read, Write, Setting
Type: Double
Unit: s

Defines low pass filter time constant.

/dev..../imps/n/discard/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Discarding impedance samples outside the indicated range.

/dev..../imps/n/discard/limitlower

Properties: Read, Write, Setting
Type: Double
Unit: None

Threshold for $\text{abs}(Z)$ below which the impedance samples are discarded.

/dev..../imps/n/discard/limitupper

Properties: Read, Write, Setting
Type: Double
Unit: None

Threshold for $\text{abs}(Z)$ above which the impedance samples are discarded.

/dev..../imps/n/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enable impedance calculation for demodulator data.

/dev..../imps/n/freq

Properties: Read, Write, Setting
Type: Double
Unit: Hz

Frequency control for the oscillator used for impedance measurement.

/dev..../imps/n/interpolation

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the interpolation method of the compensation data. The interpolation method is particularly important if the derivative changes strongly e.g at cut-off frequencies.

- | | |
|---|---|
| 0 | "linear": Linear: The linear interpolation is fastest but may create compensation errors in between the frequency points used for compensation. |
| 1 | "pchip": Piecewise Cubic Hermite (PCHIP): The piecewise cubic Hermite interpolation will result in very accurate results but requires more calculation power. |

/dev..../imps/n/maxbandwidth

Properties: Read, Write, Setting
Type: Double
Unit: Hz

Limit of the maximum bandwidth used on the demodulator filter. Values above 1 kHz can heavily diminish measurement accuracy in the high-frequency region where the amplitude is no more constant over frequency.

/dev..../imps/n/mode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select impedance measurement mode.

- | | |
|---|--|
| 0 | 4 Terminal: This method uses the current and the voltage drop across the DUT to calculate the DUT impedance. This method results in very accurate measurements as influences of series resistors on the output and current input are excluded. |
| 1 | 2 Terminal: This method uses the driving voltage and the measured current to calculate the DUT impedance. The 2 Terminal method can be beneficial when measuring very high impedance where the parasitics of the voltage measurement are limiting the frequency range. |

/dev..../imps/n/model

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Representation of the complex impedance value Z by two real values accessible as Parameter 1 and Parameter 2 on all user interface displays.

- | | |
|----|---|
| 0 | "r_c_parallel": $R_p \parallel C_p$: Impedance value Z is represented by a resistive element R_p in parallel with a capacitive element C_p . |
| 1 | "r_c_series": $R_s + C_s$: Impedance value Z is represented by a resistive element R_s in series with a capacitive element C_s . |
| 2 | "r_l_series": $R_s + L_s$: Impedance value Z is represented by a resistive element R_s in series with an inductive element L_s . |
| 3 | "g_b_parallel", "admittance": $G \parallel B$: Impedance value Z is represented by conductance $G = \text{Real}(Y)$ and Susceptance $B = \text{Imag}(Y)$ of the admittance $Y = 1/Z$. |
| 4 | "d_c_series": $D \parallel C_s$: Impedance value Z is represented by a dissipation factor $D = -\text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a series capacitive element. |
| 5 | "q_c_series": $Q \parallel C_s$: Impedance value Z is represented by a quality factor $Q = -\text{Imag}(Z)/\text{Real}(Z)$ and a series capacitive element. |
| 6 | "d_l_series": $D \parallel L_s$: Impedance value Z is represented by a dissipation factor $D = \text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a series inductive element. |
| 7 | "q_l_series": $Q \parallel L_s$: Impedance value Z is represented by a quality factor $Q = \text{Imag}(Z)/\text{Real}(Z)$ and a series inductive element. |
| 8 | "r_l_parallel": $R_p \parallel L_p$: Impedance value Z is represented by a resistive element R_p in parallel with an inductive element L_p . |
| 9 | "d_c_parallel": $D \parallel C_p$: Impedance value Z is represented by a dissipation factor $D = \text{Real}(Z)/\text{Imag}(Z)$ (loss tangent) and a parallel capacitive element C_p . |
| 10 | "dielectric": Dielectric: Impedance value Z is represented by an equivalent dielectric material with a complex permittivity. |

/dev..../imps/n/mut/area

Properties: Read, Write, Setting
Type: Double
Unit: m^2

The area of the MUT. Only relevant when using the dielectric model.

/dev..../imps/n/mut/thickness

Properties: Read, Write, Setting
Type: Double
Unit: m

The thickness of the MUT. Only relevant when using the dielectric model.

/dev..../imps/n/omegasuppression

Properties: Read, Write, Setting
Type: Double
Unit: dB

Suppression of the omega and 2-omega components. Small omega suppression can diminish measurements of very low or high impedance because the DC component can become dominant. Large omega suppression will have a significant impact on sweep time especially for low filter orders.

/dev..../imps/n/oneperiod/active

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables one-period averaging for low frequency impedance measurements. The LED is green when active.

/dev..../imps/n/oneperiod/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables one-period averaging for low frequency impedance measurements. The LED is green when active.

/dev..../imps/n/output/amplitude

Properties: Read, Write, Setting
Type: Double
Unit: V

Drive amplitude on the Signal Output.

/dev..../imps/n/output/demod

Properties: Read
Type: Integer (64 bit)
Unit: None

Demodulator unit used to generate the excitation voltage on the Signal Output.

/dev..../imps/n/output/on

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Main switch for the Signal Output corresponding to the blue LED indicator on the instrument front panel.

/dev..../imps/n/output/range

Properties: Read, Write, Setting
Type: Double
Unit: V

Selects the output voltage range.

/dev..../imps/n/output/select

Properties: Read
Type: Integer (enumerated)
Unit: None

Selects the output channel that the excitation voltage drives.

0 "sigout0", "signal_output0": Signal Output 1

/dev..../imps/n/sample

Properties: Read, Stream
Type: ZIImpedanceSample
Unit: None

Streaming node containing the impedance measurement sample data.

/dev..../imps/n/voltage/demodselect

Properties: Read
Type: Integer (64 bit)
Unit: None

Demodulator used for voltage measurement in case of a four-terminal impedance measurement.

/dev..../imps/n/voltage/inputselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the voltage input used for a four-terminal impedance measurement.

0 "voltage_input0": Voltage Input 1
 8 "auxin0", "auxiliary_input0": Aux In 1
 9 "auxin1", "auxiliary_input1": Aux In 2

/dev..../imps/n/voltage/invert

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

If enabled, the voltage input signal is inverted.

/dev..../imps/n/voltage/range

Properties: Read, Write, Setting
Type: Double
Unit: V

Input voltage range for the impedance measurement.

/dev..../imps/n/voltage/scaling

Properties: Read, Write, Setting
Type: Double
Unit: V/V

The scaling factor will be applied to the voltage measurement done with an Aux In input.

8.2.10. MODS

/dev..../mods/n/carrier/amplitude

Properties: Read, Write, Setting
Type: Double
Unit: V

Set the carrier amplitude

/dev..../mods/n/carrier/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enable the modulation

/dev..../mods/n/carrier/harmonic

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Set the harmonic of the carrier frequency. 1 = Fundamental

/dev..../mods/n/carrier/inputselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select Signal Input for the carrier demodulation

0	"signin0", "signal_input0": Sig In 1
1	"currin0", "current_input0": Curr In 1
2	"trigin0", "trigger_input0": Trigger Input 1
3	"trigin1", "trigger_input1": Trigger Input 2
4	"auxout0", "auxiliary_output0": Aux Out 1
5	"auxout1", "auxiliary_output1": Aux Out 2
6	"auxout2", "auxiliary_output2": Aux Out 3
7	"auxout3", "auxiliary_output3": Aux Out 4
8	"auxin0", "auxiliary_input0": Aux In 1
9	"auxin1", "auxiliary_input1": Aux In 2

174 "demod_constant_input": Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).

/dev..../mods/n/carrier/order

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Filter order used for carrier demodulation

1	1st order filter 6 dB/oct
2	2nd order filter 12 dB/oct
3	3rd order filter 18 dB/oct
4	4th order filter 24 dB/oct
5	5th order filter 30 dB/oct
6	6th order filter 36 dB/oct
7	7th order filter 42 dB/oct
8	8th order filter 48 dB/oct

/dev..../mods/n/carrier/oscselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the oscillator for the carrier signal.

0	Oscillator 1
1	Oscillator 2
2	Oscillator 3
3	Oscillator 4

/dev..../mods/n/carrier/phaseadjust

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Adjust the carrier demodulator phase automatically in order to read 0 degrees.

/dev..../mods/n/carrier/phaseshift

Properties: Read, Write, Setting
Type: Double
Unit: degree

Phase shift applied to the reference input of the carrier demodulator and also to the carrier signal on the Signal Outputs

/dev..../mods/n/carrier/timeconstant

Properties: Read, Write, Setting
Type: Double
Unit: s

Sets the integration time constant or in other words, the cutoff frequency of the low-pass filter for the carrier demodulation.

/dev..../mods/n/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the modulation.

/dev..../mods/n/freqdev

Properties: Read, Write, Setting
Type: Double
Unit: Hz

FM mode peak deviation value.

/dev..../mods/n/freqdevenable

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

In FM mode, choose to work with either modulation index or peak deviation. The modulation index equals peak deviation divided by modulation frequency.

0 "modulation_index": Use modulation index.
 1 "peak_deviation": Use peak deviation.

/dev..../mods/n/index

Properties: Read, Write, Setting
Type: Double
Unit: None

FM modulation index: The modulation index equals peak deviation divided by modulation frequency.

/dev..../mods/n/mode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the modulation mode.

0 "am": AM Modulation
 1 "fm": FM Modulation
 2 "manual": Manual

/dev..../mods/n/output

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select Signal Output.

0 "none": None
 1 "sigout0", "signal_output0": Signal Output 1

/dev..../mods/n/sidebands/n/amplitude

Properties: Read, Write, Setting
Type: Double
Unit: V

Set the amplitude of the sideband components.

/dev..../mods/n/sidebands/n/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enable the signal generation for the respective sideband

/dev..../mods/n/sidebands/n/harmonic

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Set harmonic of the sideband frequencies. 1 = fundamental

/dev..../mods/n/sidebands/n/inputselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select Signal Input for the sideband demodulation

0	"signin0", "signal_input0": Sig In 1
1	"currin0", "current_input0": Curr In 1
2	"trigin0", "trigger_input0": Trigger Input 1
3	"trigin1", "trigger_input1": Trigger Input 2
4	"auxout0", "auxiliary_output0": Aux Out 1
5	"auxout1", "auxiliary_output1": Aux Out 2
6	"auxout2", "auxiliary_output2": Aux Out 3
7	"auxout3", "auxiliary_output3": Aux Out 4
8	"auxin0", "auxiliary_input0": Aux In 1
9	"auxin1", "auxiliary_input1": Aux In 2
174	"demod_constant_input": Demodulate a constant input. This results in a sine wave of the frequency specified by the demodulator's oscillator with an amplitude of 1 (at lower frequencies; higher frequencies will be attenuated). The maximum possible frequency is limited by the demodulator sampling rate and bandwidth; use demodulator order 1 for the least attenuation in demodulator output. This signal may be used with the auxiliary outputs, PID and Threshold Unit for advanced measurement and control tasks. When the demodulator output is written to an auxiliary output, the resulting signal can also be used as a second output channel (for low frequencies).

/dev..../mods/n/sidebands/n/mode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Enabling of the first sideband and selection of the position of the sideband relative to the carrier frequency for manual mode.

0	"off": Off: First sideband is disabled. The sideband demodulator behaves like a normal demodulator.
1	"upper": C + M: First sideband to the right of the carrier
2	"lower": C - M: First sideband to the left of the carrier

/dev..../mods/n/sidebands/n/order

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Filter order used for sideband demodulation

1	1st order filter 6 dB/oct
2	2nd order filter 12 dB/oct
3	3rd order filter 18 dB/oct
4	4th order filter 24 dB/oct
5	5th order filter 30 dB/oct
6	6th order filter 36 dB/oct
7	7th order filter 42 dB/oct
8	8th order filter 48 dB/oct

/dev..../mods/n/sidebands/n/oscsselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the oscillator for the second sideband.

0	Oscillator 1
1	Oscillator 2
2	Oscillator 3
3	Oscillator 4

/dev..../mods/n/sidebands/n/phaseadjust

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Adjust the sideband demodulator phase automatically in order to read 0 degrees.

/dev..../mods/n/sidebands/n/phaseshift

Properties: Read, Write, Setting
Type: Double
Unit: degree

Phase shift applied to the reference input of the sideband demodulator and also to the sideband signal on the Signal Outputs

/dev..../mods/n/sidebands/n/timeconstant

Properties: Read, Write, Setting
Type: Double
Unit: s

Sets the integration time constant or in other words, the cutoff frequency of the low-pass filter for the sideband demodulation.

8.2.11. OSCS

/dev..../oscs/n/freq

Properties: Read, Write, Setting
Type: Double
Unit: Hz

Frequency control for each oscillator.

8.2.12. PIDS

/dev..../pids/n/center

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Sets the center value for the PID output. After adding the Center value, the signal is clamped to Center + Lower Limit and Center + Upper Limit.

/dev..../pids/n/d

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

PID derivative gain.

/dev..../pids/n/demod/adselect

Properties: Read
Type: Integer (enumerated)
Unit: None

Indicates the signal source which is connected to the chosen input demodulator channel.

- | | |
|---|---|
| 0 | "signin0", "signal_input0": Signal Input 1 is connected to the corresponding demodulator. |
| 1 | "currin0", "current_input0": Current Input 1 is connected to the corresponding demodulator. |
| 2 | "trigin0", "trigger_input0": Trigger Input 1 is connected to the corresponding demodulator. |
| 3 | "trigin1", "trigger_input1": Trigger Input 2 is connected to the corresponding demodulator. |
| 4 | "auxout0", "auxiliary_output0": Auxiliary Output 1 is connected to the corresponding demodulator. |
| 5 | "auxout1", "auxiliary_output1": Auxiliary Output 2 is connected to the corresponding demodulator. |
| 6 | "auxout2", "auxiliary_output2": Auxiliary Output 3 is connected to the corresponding demodulator. |
| 7 | "auxout3", "auxiliary_output3": Auxiliary Output 4 is connected to the corresponding demodulator. |
| 8 | "auxin0", "auxiliary_input0": Auxiliary Input 1 is connected to the corresponding demodulator. |
| 9 | "auxin1", "auxiliary_input1": Auxiliary Input 2 is connected to the corresponding demodulator. |

/dev..../pids/n/demod/harmonic

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Multiplier of the for the reference frequency of the current demodulator.

/dev..../pids/n/demod/order

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the filter roll off between 6 dB/oct and 48 dB/oct of the current demodulator.

- | | |
|---|----------------------------|
| 1 | 1st order filter 6 dB/oct |
| 2 | 2nd order filter 12 dB/oct |
| 3 | 3rd order filter 18 dB/oct |
| 4 | 4th order filter 24 dB/oct |
| 5 | 5th order filter 30 dB/oct |
| 6 | 6th order filter 36 dB/oct |
| 7 | 7th order filter 42 dB/oct |
| 8 | 8th order filter 48 dB/oct |

/dev..../pids/n/demod/timeconstant

Properties: Read, Write, Setting
Type: Double
Unit: s

Defines the characteristic time constant (cut off) of the demodulator filter used as an input.

/dev..../pids/n/dlimittimeconstant

Properties: Read, Write, Setting
Type: Double
Unit: s

The cutoff of the low-pass filter for the D limitation given as time constant. When set to 0, the low-pass filter is disabled.

/dev..../pids/n/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enable the PID controller.

/dev..../pids/n/error

Properties: Read
Type: Double
Unit: Dependent

Error = Set point - PID Input.

/dev..../pids/n/i

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

PID integral gain I.

/dev..../pids/n/input

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select input source of PID controller.

0	"demod_x": Demodulator X
1	"demod_y": Demodulator Y
2	"demod_r": Demodulator R
3	"demod_theta": Demodulator Theta
4	"auxin", "auxiliary_input": Aux Input
5	"auxout", "auxiliary_output": Aux Output

/dev..../pids/n/inputchannel

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Select input channel of PID controller.

/dev..../pids/n/keepint

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: Dependent

If enabled, the accumulated integral error is maintained upon restart of the PID. If is disabled, the integral error is set to zero when the PID is disabled.

/dev..../pids/n/limitlower

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Sets the lower limit for the PID output. After adding the Center value, the signal is clamped to Center + Lower Limit and Center + Upper Limit.

/dev..../pids/n/limitupper

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Sets the upper limit for the PID output. After adding the Center value, the signal is clamped to Center + Lower Limit and Center + Upper Limit.

/dev..../pids/n/mode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Sets the operation mode of the PID module.

0	"pid": PID
1	"pll": PLL (phase locked loop)
2	"extref": ExtRef (external reference)

/dev..../pids/n/output

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select output of the PID controller.

0	"sigout0_amp", "signal_output0_amplitude": Driving Signal Output 1 amplitudes
2	"oscillator_frequency": Controlling any of the internal oscillator frequencies
3	"demod_phase": Controlling any of the demodulator phase set points
5	"auxout_offset", "auxiliary_output_offset": Driving any of the 4 Auxiliary Outputs' offset
7	"sigout_offset", "signal_output_offset": Driving the main Signal Output's offset

/dev..../pids/n/outputchannel

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Select the output channel of the driven output of PID controller.

/dev..../pids/n/p

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

PID Proportional gain P.

/dev..../pids/n/phaseunwrap

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the phase unwrapping to track phase errors past the +/-180 degree boundary and increase PLL bandwidth.

/dev..../pids/n/pll/automode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

This defines the type of automatic adaptation of parameters in the PID.

- 0 "no_adaption": No automatic adaption.
- 1 "pid_coefs": The PID coefficients are automatically set based on the filter parameters.
- 2 "pid_coefs_filter_low_bw": The PID coefficients, the filter bandwidth and the output limits are automatically set using a low bandwidth.
- 3 "pid_coefs_filter_high_bw": The PID coefficients, the filter bandwidth and the output limits are automatically set using a high bandwidth.

/dev..../pids/n/pll/locked

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates when the PID, configured as PLL, is locked.

/dev..../pids/n/rate

Properties: Read, Write, Setting
Type: Double
Unit: 1/s

PID sampling rate and update rate of PID outputs. Needs to be set substantially higher than the targeted loop filter bandwidth.

/dev..../pids/n/setpoint

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

PID controller setpoint

/dev..../pids/n/setpointtoggle/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the setpoint toggle.

/dev..../pids/n/setpointtoggle/rate

Properties: Read, Write, Setting
Type: Double
Unit: Hz

Defines the rate of setpoint toggling. Note that possible values are logarithmically spaced with a factor of 4 between values.

/dev..../pids/n/setpointtoggle/setpoint

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Defines the setpoint value used for setpoint toggle.

/dev..../pids/n/shift

Properties: Read
Type: Double
Unit: Dependent

Difference between the current output value Out and the Center. $\text{Shift} = \text{PError} + \text{IInt}(\text{Error}, dt) + D * d\text{Error}/dt$

/dev..../pids/n/stream/effectiverate

Properties: Read
Type: Double
Unit: 1/s

Current rate of the PID stream data sent to PC. Defined based on Max Rate.

/dev..../pids/n/stream/error

Properties: Read, Stream
Type: Double
Unit: Dependent

PID Error = Set point - PID Input.

/dev..../pids/n/stream/overflow

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates the streaming fifo overflow state. 0 = OK, 1 = overflow.

/dev..../pids/n/stream/rate

Properties: Read, Write, Setting
Type: Double
Unit: 1/s

Target Rate for PID output data sent to PC. This value defines the applied decimation for sending data to the PC. It does not affect any other place where PID data are used.

/dev..../pids/n/stream/shift

Properties: Read, Stream
Type: Double
Unit: Dependent

Gives the difference between the current output value and the center value. $\text{Shift} = \text{PError} + \text{Int}(\text{Error}, dt) + D * d\text{Error} / dt$

/dev..../pids/n/stream/value

Properties: Read, Stream
Type: Double
Unit: Dependent

Gives the current PID output value.

/dev..../pids/n/value

Properties: Read
Type: Double
Unit: Dependent

Gives the current PID output value.

8.2.13. SCOPES

/dev..../scopes/n/channel

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Activates the scope channels.

- 1 Only channel 1 is active.
- 2 Only channel 2 is active.
- 3 "both": Both, channel 1 and 2 are active.

/dev..../scopes/n/channels/n/bwlimit

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects between sample decimation and sample averaging. Averaging avoids aliasing, but may conceal signal peaks.

0	"on": On: Selects sample averaging for sample rates lower than the maximal available sampling rate.
1	"off": OFF: Selects sample decimation for sample rates lower than the maximal available sampling rate.

/dev..../scopes/n/channels/n/fullscale

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Indicates the full scale value of the scope channel.

/dev..../scopes/n/channels/n/inputselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the scope input signal.

0	"signin0", "signal_input0": Signal Input 1
1	"currin0", "current_input0": Current Input 1
2	"trigin0", "trigger_input0": Trigger Input 1
3	"trigin1", "trigger_input1": Trigger Input 2
4	"auxout0", "auxiliary_output0": Aux Output 1. Requires an installed digitizer (DIG) option.
5	"auxout1", "auxiliary_output1": Aux Output 2. Requires an installed digitizer (DIG) option.
6	"auxout2", "auxiliary_output2": Aux Output 3. Requires an installed digitizer (DIG) option.
7	"auxout3", "auxiliary_output3": Aux Output 4. Requires an installed digitizer (DIG) option.
8	"auxin0", "auxiliary_input0": Aux Input 1
9	"auxin1", "auxiliary_input1": Aux Input 2
10	"demod1": Osc ϕ Demod 2
11	"demod3": Osc ϕ Demod 4
14	"trigout0", "trigger_output0": Trigger Output 1
15	"trigout1", "trigger_output1": Trigger Output 2
16	"demod0_x": Demod 1 X. Requires an installed digitizer (DIG) option.
17	"demod1_x": Demod 2 X. Requires an installed digitizer (DIG) option.
18	"demod2_x": Demod 3 X. Requires an installed digitizer (DIG) option.
19	"demod3_x": Demod 4 X. Requires an installed digitizer (DIG) option.
32	"demod0_y": Demod 1 Y. Requires an installed digitizer (DIG) option.
33	"demod1_y": Demod 2 Y. Requires an installed digitizer (DIG) option.
34	"demod2_y": Demod 3 Y. Requires an installed digitizer (DIG) option.
35	"demod3_y": Demod 4 Y. Requires an installed digitizer (DIG) option.
48	"demod0_r": Demod 1 R. Requires an installed digitizer (DIG) option.
49	"demod1_r": Demod 2 R. Requires an installed digitizer (DIG) option.
50	"demod2_r": Demod 3 R. Requires an installed digitizer (DIG) option.
51	"demod3_r": Demod 4 R. Requires an installed digitizer (DIG) option.
64	"demod0_theta": Demod 1 θ . Requires an installed digitizer (DIG) option.
65	"demod1_theta": Demod 2 θ . Requires an installed digitizer (DIG) option.
66	"demod2_theta": Demod 3 θ . Requires an installed digitizer (DIG) option.
67	"demod3_theta": Demod 4 θ . Requires an installed digitizer (DIG) option.
80	"pid0_value": PID 1 value. Requires an installed digitizer (DIG) option.
81	"pid1_value": PID 2 value. Requires an installed digitizer (DIG) option.
82	"pid2_value": PID 3 value. Requires an installed digitizer (DIG) option.
83	"pid3_value": PID 4 value. Requires an installed digitizer (DIG) option.
144	"pid0_shift": PID 1 Shift. Requires an installed digitizer (DIG) option.
145	"pid1_shift": PID 2 Shift. Requires an installed digitizer (DIG) option.
146	"pid2_shift": PID 3 Shift. Requires an installed digitizer (DIG) option.
147	"pid3_shift": PID 4 Shift. Requires an installed digitizer (DIG) option.
160	Reserved for future use.
161	Reserved for future use.
208	"pid0_error": PID 1 Error. Requires an installed digitizer (DIG) option.
209	"pid1_error": PID 2 Error. Requires an installed digitizer (DIG) option.
210	"pid2_error": PID 3 Error. Requires an installed digitizer (DIG) option.
211	"pid3_error": PID 4 Error. Requires an installed digitizer (DIG) option.
240	Reserved for future use.
241	Reserved for future use.
242	Reserved for future use.
243	Reserved for future use.

/dev..../scopes/n/channels/n/limitlower

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Lower limit of the scope full scale range. For demodulator, PID, Boxcar, and AU signals the limit should be adjusted so that the signal covers the specified range to achieve optimal resolution.

/dev..../scopes/n/channels/n/limitupper

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Upper limit of the scope full scale range. For demodulator, PID, Boxcar, and AU signals the limit should be adjusted so that the signal covers the specified range to achieve optimal resolution.

/dev..../scopes/n/channels/n/offset

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Indicates the offset value of the scope channel.

/dev..../scopes/n/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the acquisition of scope shots.

/dev..../scopes/n/length

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Defines the length of the recorded Scope shot in number of samples.

/dev..../scopes/n/segments/count

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Specifies the number of segments to be recorded in device memory. The maximum scope shot size is given by the available memory divided by the number of segments. This functionality requires the DIG option.

/dev..../scopes/n/segments/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enable segmented scope recording. This allows for full bandwidth recording of scope shots with a minimum dead time between individual shots. This functionality requires the DIG option.

/dev..../scopes/n/single

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Puts the Scope into single shot mode.

/dev..../scopes/n/stream/enables/n

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enable scope streaming for the specified channel. This allows for continuous recording of scope data on the plotter and streaming to disk. Note: scope streaming requires the DIG option.

/dev..../scopes/n/stream/rate

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: Hz

Streaming Rate of the scope channels. The streaming rate can be adjusted independent from the scope sampling rate. The maximum rate depends on the interface used for transfer. Note: scope streaming requires the DIG option.

4	"3.75_MHz": 3.75 MHz
5	"1.88_MHz": 1.88 MHz
6	"938_kHz": 938 kHz
7	"469_kHz": 469 kHz
8	"234_kHz": 234 kHz
9	"117_kHz": 117 kHz
10	"58.6_kHz": 58.6 kHz
11	"29.3_kHz": 29.3 kHz
12	"14.6_kHz": 14.6 kHz
13	"7.32_kHz": 7.32 kHz
14	"3.66_kHz": 3.66 kHz
15	"1.83_kHz": 1.83 kHz

/dev..../scopes/n/stream/sample

Properties: Read, Stream
Type: ZIScopeWave
Unit: None

Streaming node containing scope sample data. Note: scope streaming requires the DIG option.

/dev..../scopes/n/time

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Defines the time base of the scope from the divider exponent of the instrument's clock base. The resulting sampling time is $2^n/\text{clockbase}$.

0	60 MHz
1	30 MHz
2	15 MHz
3	7.5 MHz
4	3.75 MHz
5	1.88 MHz
6	938 kHz
7	469 kHz
8	234 kHz
9	117 kHz
10	58.6 kHz
11	29.3 kHz
12	14.6 kHz
13	7.32 kHz
14	3.66 kHz
15	1.83 kHz

/dev..../scopes/n/trigchannel

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the trigger source signal.

```

0      "signin0", "signal_input0": Signal Input 1
1      "currin0", "current_input0": Current Input 1
2      "trigin0", "trigger_input0": Trigger Input 1
3      "trigin1", "trigger_input1": Trigger Input 2
4      "auxout0", "auxiliary_output0": Aux Output 1. Requires an installed digitizer (DIG)
      option.
5      "auxout1", "auxiliary_output1": Aux Output 2. Requires an installed digitizer (DIG)
      option.
6      "auxout2", "auxiliary_output2": Aux Output 3. Requires an installed digitizer (DIG)
      option.
7      "auxout3", "auxiliary_output3": Aux Output 4. Requires an installed digitizer (DIG)
      option.
8      "auxin0", "auxiliary_input0": Aux Input 1
9      "auxin1", "auxiliary_input1": Aux Input 2
10     "demod1": Osc  $\phi$  Demod 2
11     "demod3": Osc  $\phi$  Demod 4
14     "trigout0", "trigger_output0": Trigger Output 1
15     "trigout1", "trigger_output1": Trigger Output 2
16     "demod0_x": Demod 1 X. Requires an installed digitizer (DIG) option.
17     "demod1_x": Demod 2 X. Requires an installed digitizer (DIG) option.
18     "demod2_x": Demod 3 X. Requires an installed digitizer (DIG) option.
19     "demod3_x": Demod 4 X. Requires an installed digitizer (DIG) option.
32     "demod0_y": Demod 1 Y. Requires an installed digitizer (DIG) option.
33     "demod1_y": Demod 2 Y. Requires an installed digitizer (DIG) option.
34     "demod2_y": Demod 3 Y. Requires an installed digitizer (DIG) option.
35     "demod3_y": Demod 4 Y. Requires an installed digitizer (DIG) option.
48     "demod0_r": Demod 1 R. Requires an installed digitizer (DIG) option.
49     "demod1_r": Demod 2 R. Requires an installed digitizer (DIG) option.
50     "demod2_r": Demod 3 R. Requires an installed digitizer (DIG) option.
51     "demod3_r": Demod 4 R. Requires an installed digitizer (DIG) option.
64     "demod0_theta": Demod 1  $\theta$ . Requires an installed digitizer (DIG) option.
65     "demod1_theta": Demod 2  $\theta$ . Requires an installed digitizer (DIG) option.
66     "demod2_theta": Demod 3  $\theta$ . Requires an installed digitizer (DIG) option.
67     "demod3_theta": Demod 4  $\theta$ . Requires an installed digitizer (DIG) option.
80     "pid0_value": PID 1 value. Requires an installed digitizer (DIG) option.
81     "pid1_value": PID 2 value. Requires an installed digitizer (DIG) option.
82     "pid2_value": PID 3 value. Requires an installed digitizer (DIG) option.
83     "pid3_value": PID 4 value. Requires an installed digitizer (DIG) option.
144    "pid0_shift": PID 1 Shift. Requires an installed digitizer (DIG) option.
145    "pid1_shift": PID 2 Shift. Requires an installed digitizer (DIG) option.
146    "pid2_shift": PID 3 Shift. Requires an installed digitizer (DIG) option.
147    "pid3_shift": PID 4 Shift. Requires an installed digitizer (DIG) option.
208    "pid0_error": PID 1 Error. Requires an installed digitizer (DIG) option.
209    "pid1_error": PID 2 Error. Requires an installed digitizer (DIG) option.
210    "pid2_error": PID 3 Error. Requires an installed digitizer (DIG) option.
211    "pid3_error": PID 4 Error. Requires an installed digitizer (DIG) option.
224    "tu0_filtered": TU 1 Filtered Value. Requires an installed digitizer (DIG) option.
225    "tu1_filtered": TU 2 Filtered Value. Requires an installed digitizer (DIG) option.
226    "tu2_filtered": TU 3 Filtered Value. Requires an installed digitizer (DIG) option.
227    "tu3_filtered": TU 4 Filtered Value. Requires an installed digitizer (DIG) option.

```

/dev..../scopes/n/trigdelay

Properties: Read, Write, Setting
Type: Double
Unit: s

Trigger position relative to reference. A positive delay results in less data being acquired before the trigger point, a negative delay results in more data being acquired before the trigger point.

/dev..../scopes/n/trigenable

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

When triggering is enabled scope data are acquired every time the defined trigger condition is met.

0 "off": OFF: Continuous scope shot acquisition
 1 "on": ON: Trigger based scope shot acquisition

/dev..../scopes/n/trigfalling

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

When set (1), enables falling edge triggering. This settings is synchronized with the settings done in the /TRIGSLOPE node.

/dev..../scopes/n/trigforce

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Forces a trigger event.

/dev..../scopes/n/triggate/enable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

If enabled the trigger will be gated by the trigger gating input signal. This feature requires the DIG option.

/dev..../scopes/n/triggate/inputselect

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the signal source used for trigger gating if gating is enabled. This feature requires the DIG option.

0 "trigin0_high", "trigger_input0_high": Only trigger if the Trigger Input 1 is at high level.
 1 "trigin0_low", "trigger_input0_low": Only trigger if the Trigger Input 1 is at low level.
 2 "trigin1_high", "trigger_input1_high": Only trigger if the Trigger Input 2 is at high level.
 3 "trigin1_low", "trigger_input1_low": Only trigger if the Trigger Input 2 is at low level.

/dev..../scopes/n/trigholdoff

Properties: Read, Write, Setting
Type: Double
Unit: s

Defines the time before the trigger is rearmed after a recording event.

/dev..../scopes/n/trigholdoffcount

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Defines the trigger event number that will trigger the next recording after a recording event. A value of '1' will start a recording for each trigger event.

/dev..../scopes/n/trigholdoffmode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the holdoff mode.

0 "time": Holdoff is defined as time (s).
 1 "number_of_events": Holdoff is defined as number of events.

/dev..../scopes/n/trighysteresis/absolute

Properties: Read, Write, Setting
Type: Double
Unit: V

Defines the voltage the source signal must deviate from the trigger level before the trigger is rearmed again. Set to 0 to turn it off. The sign is defined by the Edge setting.

/dev..../scopes/n/trighysteresis/mode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the mode to define the hysteresis strength. The relative mode will work best over the full input range as long as the analog input signal does not suffer from excessive noise.

0 "absolute": Selects absolute hysteresis.
 1 "relative": Selects a hysteresis relative to the adjusted full scale signal input range.

/dev..../scopes/n/trighysteresis/relative

Properties: Read, Write, Setting
Type: Double
Unit: None

Hysteresis relative to the adjusted full scale signal input range. A hysteresis value larger than 1 (100%) is allowed.

/dev..../scopes/n/triglevel

Properties: Read, Write, Setting
Type: Double
Unit: V

Defines the trigger level.

/dev..../scopes/n/trigreference

Properties: Read, Write, Setting
Type: Double
Unit: None

Trigger reference position relative to the acquired data. Default is 50% (0.5) which results in a reference point in the middle of the acquired data.

/dev..../scopes/n/trigrising

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

When set (1), enables rising edge triggering. This settings is synchronized with the settings done in the /TRIGFALLING node.

/dev..../scopes/n/trigslope

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Sets on which slope of the trigger signal the scope should trigger. This setting is synchronized with the settings done in the /TRIGFALLING and /TRIGRISING nodes.

0	"none": None
1	"rising": Rising edge triggered.
2	"falling": Falling edge triggered.
3	"both": Triggers on both the rising and falling edge.

/dev..../scopes/n/trigstate

Properties: Read
Type: Integer (64 bit)
Unit: None

When 1, indicates that the trigger signal satisfies the conditions of the trigger.

/dev..../scopes/n/wave

Properties: Read, Stream
Type: ZIScopeWave
Unit: None

Contains the scope shot data.

8.2.14. SIGINS

/dev..../sigins/n/ac

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Defines the input coupling for the Signal Inputs. AC coupling inserts a high-pass filter.

0 "dc": OFF: DC coupling
 1 "ac": ON: AC coupling

/dev..../sigins/n/autorange

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Automatic adjustment of the Range to about two times the maximum signal input amplitude measured over about 100 ms.

/dev..../sigins/n/diff

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Switches between single ended (OFF) and differential (ON) measurements.

0 "off": OFF: Single ended voltage input
 1 "on": ON: Differential voltage input

/dev..../sigins/n/float

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Switches the input between floating (ON) and connected to ground (OFF). This setting applies both to the voltage and the current input. It is recommended to discharge the test device before connecting or to enable this setting only after the signal source has been connected to the Signal Input in grounded mode.

0 "off": OFF: GND connected
 1 "on": ON: Floating

/dev..../sigins/n/imp50

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Switches between 50 Ohm (ON) and 10 M Ohm (OFF).

0 "10_MOhm": OFF: 10 M Ohm
 1 "50_Ohm": ON: 50 Ohm

/dev..../sigins/n/max

Properties: Read, Write
Type: Double
Unit: None

Indicates the maximum measured value at the input normalized to input range.

/dev..../sigins/n/min

Properties: Read, Write
Type: Double
Unit: None

Indicates the minimum measured value at the input normalized to input range.

/dev..../sigins/n/on

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the signal input.

/dev..../sigins/n/range

Properties: Read, Write, Setting
Type: Double
Unit: V

Defines the gain of the analog input amplifier. The range should exceed the incoming signal by roughly a factor two including a potential DC offset. The instrument selects the next higher available range relative to a value inserted by the user. A suitable choice of this setting optimizes the accuracy and signal-to-noise ratio by ensuring that the full dynamic range of the input ADC is used.

/dev..../sigins/n/rangestep/trigger

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Switches to the next appropriate input range such that the range fits best with the measured input signal amplitude.

/dev..../sigins/n/scaling

Properties: Read, Write, Setting
Type: Double
Unit: None

Applies the given scaling factor to the input signal.

8.2.15. SIGOUTS

/dev..../sigouts/n/add

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

The signal supplied to the Aux Input 1 is added to the signal output. For differential output the added signal is a common mode offset.

/dev..../sigouts/n/amplitudes/n

Properties: Read, Write, Setting
Type: Double
Unit: V

Sets the peak amplitude that the oscillator assigned to the given demodulation channel contributes to the signal output.

/dev..../sigouts/n/autorange

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

If enabled, selects the most suited output range automatically.

/dev..../sigouts/n/diff

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Switch between single-ended output (OFF) and differential output (ON). In differential mode the signal swing is defined between Signal Output +V / -V.

/dev..../sigouts/n/enables/n

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: V

Enables individual output signal amplitude. When the MD option is used, it is possible to generate signals being the linear combination of the available demodulator frequencies.

/dev..../sigouts/n/imp50

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the load impedance between 50 Ohm and HiZ. The impedance of the output is always 50 Ohm. For a load impedance of 50 Ohm the displayed voltage is half the output voltage to reflect the voltage seen at the load.

0	"high_impedance": HiZ
1	"50_Ohm": 50 Ohm

/dev..../sigouts/n/offset

Properties: Read, Write, Setting
Type: Double
Unit: V

Defines the DC voltage that is added to the dynamic part of the output signal.

/dev..../sigouts/n/on

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enabling/Disabling the Signal Output. Corresponds to the blue LED indicator on the instrument front panel.

/dev..../sigouts/n/over

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates that the signal output is overloaded.

/dev..../sigouts/n/range

Properties: Read, Write, Setting
Type: Double
Unit: V

Sets the output voltage range. The instrument selects the next higher available range.

8.2.16. STATS

/dev..../stats/cmdstream/bandwidth

Properties: Read
Type: Double
Unit: Mbit/s

Command streaming bandwidth usage on the physical network connection between device and data server.

/dev..../stats/cmdstream/bytesreceived

Properties: Read
Type: Integer (64 bit)
Unit: B

Number of bytes received on the command stream from the device since session start.

/dev..../stats/cmdstream/bytessent

Properties: Read
Type: Integer (64 bit)
Unit: B

Number of bytes sent on the command stream from the device since session start.

/dev..../stats/cmdstream/packetslost

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of command packets lost since device start. Command packets contain device settings that are sent to and received from the device.

/dev..../stats/cmdstream/packetsreceived

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of packets received on the command stream from the device since session start.

/dev..../stats/cmdstream/packetssent

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of packets sent on the command stream to the device since session start.

/dev..../stats/cmdstream/pending

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of buffers ready for receiving command packets from the device.

/dev..../stats/cmdstream/processing

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of buffers being processed for command packets. Small values indicate proper performance. For a TCP/IP interface, command packets are sent using the TCP protocol.

/dev..../stats/cpuload

Properties: Read
Type: Double
Unit: %

Indicates the total CPU load on the machine where the data server is running.

/dev..../stats/dataservercpuload

Properties: Read
Type: Double
Unit: %

Indicates the CPU load of the data server.

/dev..../stats/dataservermemload

Properties: Read
Type: Double
Unit: %

Indicates the memory usage of the data server.

/dev..../stats/datastream/bandwidth

Properties: Read
Type: Double
Unit: Mbit/s

Data streaming bandwidth usage on the physical network connection between device and data server.

/dev..../stats/datastream/bytesreceived

Properties: Read
Type: Integer (64 bit)
Unit: B

Number of bytes received on the data stream from the device since session start.

/dev..../stats/datastream/packetslost

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of data packets lost since device start. Data packets contain measurement data.

/dev..../stats/datastream/packetsreceived

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of packets received on the data stream from the device since session start.

/dev..../stats/datastream/pending

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of buffers ready for receiving data packets from the device.

/dev..../stats/datastream/processing

Properties: Read
Type: Integer (64 bit)
Unit: None

Number of buffers being processed for data packets. Small values indicate proper performance. For a TCP/IP interface, data packets are sent using the UDP protocol.

/dev..../stats/memload

Properties: Read
Type: Double
Unit: %

Indicates the total memory usage of the machine where the data server is running.

/dev..../stats/physical/fanspeed

Properties: Read
Type: Double
Unit: RPM

Speed of the internal cooling fan.

/dev..../stats/physical/fpga/aux

Properties: Read
Type: Double
Unit: V

Supply voltage of the FPGA.

/dev..../stats/physical/fpga/core

Properties: Read
Type: Double
Unit: V

Core voltage of the FPGA.

/dev..../stats/physical/fpga/temp

Properties: Read
Type: Double
Unit: °C

Internal temperature of the FPGA.

/dev..../stats/physical/overtemperature

Properties: Read
Type: Integer (64 bit)
Unit: None

This flag is set to 1 if the temperature of the FPGA exceeds 85°C. It will be reset to 0 after a restart of the device.

/dev..../stats/physical/temperatures/n

Properties: Read
Type: Double
Unit: °C

Internal temperature measurements.

/dev..../stats/physical/voltages/n

Properties: Read
Type: Double
Unit: V

Internal voltage measurements.

8.2.17. STATUS

/dev..../status/adc0max

Properties: Read
Type: Integer (64 bit)
Unit: None

The maximum value on Signal Input 1 (ADC0) during 100 ms.

/dev..../status/adc0min

Properties: Read
Type: Integer (64 bit)
Unit: None

The minimum value on Signal Input 1 (ADC0) during 100 ms

/dev..../status/adc1max

Properties: Read
Type: Integer (64 bit)
Unit: None

The maximum value on Signal Input 2 (ADC1) during 100 ms.

/dev..../status/adc1min

Properties: Read
Type: Integer (64 bit)
Unit: None

The minimum value on Signal Input 2 (ADC1) during 100 ms

/dev..../status/fifolevel

Properties: Read
Type: Double
Unit: None

USB FIFO level: Indicates the USB FIFO fill level inside the device. When 100%, data is lost

/dev..../status/flags/binary

Properties: Read
Type: Integer (64 bit)
Unit: None

A set of binary flags giving an indication of the state of various parts of the device. Bit 0: Signal Input 1 overflow, Bit 1: Current Input 1 overflow, Bit 2: Analog PLL fail, Bit 3: Output 1 DAC OK, Bit 4: Output 2 DAC OK, Bit 5: Signal Output 1 clipping, Bit 6: Signal Output 2 clipping, Bit 7: Ext Ref 1 Locked, Bit 8: Ext Ref 2 Locked, Bit 9: Ext Ref 3 Locked, Bit 10: Ext Ref 4 Locked, Bit 11: Sample Loss, Bits 12 - 13: Trigger In 1, Bits 14 - 15: Trigger In 2, Bits 16 - 17: Trigger In 3, Bits 18 - 19: Trigger In 4, Bit 20: PLL 1 locked, Bit 21: PLL 2 locked, Bit 22: PLL 3 locked, Bit 23: PLL 4 locked, Bit 24: Rubidium clock locked, Bit 25: AU Cartesian 1 Overflow, Bit 26: AU Cartesian 2 Overflow, Bit 27: AU Polar 1 Overflow, Bit 28: AU Polar 2 Overflow.

/dev..../status/flags/packetlosstcp

Properties: Read
Type: Integer (64 bit)
Unit: None

Flag indicating if tcp packages have been lost.

/dev..../status/flags/packetlossudp

Properties: Read
Type: Integer (64 bit)
Unit: None

Flag indicating if udp packages have been lost.

/dev..../status/time

Properties: Read
Type: Integer (64 bit)
Unit: None

The current timestamp.

8.2.18. SYSTEM**/dev..../system/activeinterface**

Properties: Read
Type: String
Unit: None

Currently active interface of the device.

/dev..../system/boardrevisions/n

Properties: Read
Type: String
Unit: None

Hardware revision of the FPGA base board

/dev..../system/compdelay/calibrate

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Perform automatic calibration of the input delay compensation.

/dev..../system/compdelay/delays/n

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Current values of the compensation delays. 0: Signal Input 0, 1: Signal Input 1, 2: Aux Inputs, 3: Trigger Inputs, 4: Loopbacks

/dev..../system/extclk

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

10 MHz reference clock source.

0 "internal": Internal 10 MHz clock is used as the frequency and time base reference.
 1 "external": An external 10 MHz clock is used as the frequency and time base reference. Provide a clean and stable 10 MHz reference to the appropriate back panel connector.

/dev..../system/fpgarevision

Properties: Read
Type: Integer (64 bit)
Unit: None

HDL firmware revision.

/dev..../system/fwlog

Properties: Read
Type: String
Unit: None

Returns log output of the firmware.

/dev..../system/fwlogenable

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Enables logging to the fwlog node.

/dev..../system/fwrevision

Properties: Read
Type: Integer (64 bit)
Unit: None

Revision of the device-internal controller software.

/dev..../system/identify

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Setting this node to 1 will cause the device to blink the power led for a few seconds.

/dev..../system/impedance/application

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the Impedance application

0	"lcr": LCR Impedance Measurement: General-purpose settings for measuring components at finite frequency.
1	"dc": DC Impedance Measurement: Settings for measuring resistance at zero frequency.

/dev..../system/impedance/calib/cablelength

Properties: Read, Write, Setting
Type: Double
Unit: m

Length of cables (One way) used in measurement set up.

/dev..../system/impedance/calib/internal/allowsmoothing

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Indicates whether smoothing can be applied to the device internal compensation data.

/dev..../system/impedance/calib/internal/data

Properties: Read, Write
Type: ZIVectorData
Unit: None

Contains the user compensation data in binary form.

/dev..../system/impedance/calib/internal/info

Properties: Read
Type: String
Unit: None

Contains general information about the device internal compensation data.

/dev..../system/impedance/calib/internal/store

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Store the compensation data persistently on the device. The compensation will be loaded automatically on every power-up.

/dev..../system/impedance/calib/internal/valid

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Indicates whether the device internal calibration data is valid (1 = valid, 0 = invalid).

/dev..../system/impedance/calib/user/allowsmoothing

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Indicates whether smoothing can be applied to the user calibration data.

/dev..../system/impedance/calib/user/data

Properties: Read, Write
Type: ZIVectorData
Unit: None

Contains the user compensation data in binary form.

/dev..../system/impedance/calib/user/info

Properties: Read
Type: String
Unit: None

Contains general information about the user compensation data.

/dev..../system/impedance/calib/user/valid

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Indicates whether the user compensation data is valid (1 = valid, 0 = invalid).

/dev..../system/impedance/filter

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Switch between application-based or manually configured impedance settings. A parameter set given by an Application mode can be fine-tuned by changing to Advanced mode. Changing back to Application mode will reset the parameters.

0 "application": Application: The impedance settings are adjusted to fit best the selected application.
 1 "advanced": Advanced: The impedance settings are manually configured.

/dev..../system/impedance/freqlimit2t/n/freq

Properties: Read
Type: Double
Unit: None

Indicates the limit of a frequency range for 2-terminal impedance measurements.

/dev..../system/impedance/freqlimit2t/n/range

Properties: Read
Type: Double
Unit: None

Indicates the minimum input current range that can be used at frequencies at or above the value in the corresponding freq node for 2-terminal impedance measurements.

/dev..../system/impedance/freqlimit4t/n/freq

Properties: Read
Type: Double
Unit: None

Indicates the limit of a frequency range for 4-terminal impedance measurements.

/dev..../system/impedance/freqlimit4t/n/range

Properties: Read
Type: Double
Unit: None

Indicates the minimum input current range that can be used at frequencies at or above the value in the corresponding freq node for 4-terminal impedance measurements.

/dev..../system/impedance/precision

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the Impedance precision and measurement speed by automatically adjusting the filter bandwidth. If the sweeper module is in impedance application mode the precision setting will also control the sweeper measurement speed.

0	"low": Low -> fast settling: Medium accuracy/precision is optimized for fast response.
1	"high": High -> medium settling: High accuracy/precision takes more settling time.
2	"very_high": Very high -> slow settling: Very high accuracy/precision uses long settling time due to very low bandwidth.

/dev..../system/interfacespeed

Properties: Read
Type: String
Unit: None

Speed of the currently active interface (USB only).

/dev..../system/jumbo

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Enables jumbo frames (4k) on the TCP/IP interface. This will reduce the load on the PC and is required to achieve maximal throughput. Make sure that jumbo frames (4k) are enabled on the network card as well. If one of the devices on the network is not able to work with jumbo frames, the connection will fail.

/dev..../system/kerneltype

Properties: Read
Type: String
Unit: None

Returns the type of the data server kernel (mdk or hpk).

/dev..../system/nics/n/defaultgateway

Properties: Read, Write
Type: String
Unit: None

Default gateway configuration for the network connection.

/dev..../system/nics/n/defaultip4

Properties: Read, Write
Type: String
Unit: None

IPv4 address of the device to use if static IP is enabled.

/dev..../system/nics/n/defaultmask

Properties: Read, Write
Type: String
Unit: None

IPv4 mask in case of static IP.

/dev..../system/nics/n/gateway

Properties: Read
Type: String
Unit: None

Current network gateway.

/dev..../system/nics/n/ip4

Properties: Read
Type: String
Unit: None

Current IPv4 of the device.

/dev..../system/nics/n/jumbo

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Stored setting for jumbo frames. Will be applied after reboot.

/dev..../system/nics/n/mac

Properties: Read
Type: String
Unit: None

Current MAC address of the device network interface.

/dev..../system/nics/n/mask

Properties: Read
Type: String
Unit: None

Current network mask.

/dev..../system/nics/n/saveip

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

If written, this action will program the defined static IP address to the device.

/dev..../system/nics/n/static

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Enable this flag if the device is used in a network with fixed IP assignment without a DHCP server.

/dev..../system/owner

Properties: Read
Type: String
Unit: None

Returns the current owner of the device (IP).

/dev..../system/porttcp

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Returns the current TCP port used for communication to the dataserver.

/dev..../system/portudp

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Returns the current UDP port used for communication to the dataserver.

/dev..../system/preampenable

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Enables the preamplifier.

/dev..../system/preset/busy

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates if presets are currently loaded.

/dev..../system/preset/default

Properties: Read, Write
Type: Integer (enumerated)
Unit: None

Indicates the preset which is used as default preset at start-up of the device.

0	"factory": Select factory preset as default preset.
1	Select the preset stored in internal flash memory at position 1 as default preset.
2	Select the preset stored in internal flash memory at position 2 as default preset.
3	Select the preset stored in internal flash memory at position 3 as default preset.
4	Select the preset stored in internal flash memory at position 4 as default preset.
5	Select the preset stored in internal flash memory at position 5 as default preset.
6	Select the preset stored in internal flash memory at position 6 as default preset.

/dev..../system/preset/erase

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Erase the selected preset.

/dev..../system/preset/error

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates if the last operation was illegal. Successful: 0, Error: 1.

/dev..../system/preset/index

Properties: Read, Write
Type: Integer (enumerated)
Unit: None

Select between factory preset or presets stored in internal flash memory.

0	"factory": Select factory preset.
1	Select the preset stored in internal flash memory at position 1.
2	Select the preset stored in internal flash memory at position 2.
3	Select the preset stored in internal flash memory at position 3.
4	Select the preset stored in internal flash memory at position 4.
5	Select the preset stored in internal flash memory at position 5.
6	Select the preset stored in internal flash memory at position 6.

/dev..../system/preset/load

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Load the selected preset.

/dev..../system/preset/records/n/features

Properties: Read
Type: Integer (64 bit)
Unit: None

Properties of the preset.

/dev..../system/preset/records/n/label

Properties: Read, Write
Type: String
Unit: None

Name of this preset.

/dev..../system/preset/records/n/timestamp

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Not used.

/dev..../system/preset/records/n/valid

Properties: Read
Type: Integer (64 bit)
Unit: None

True if a valid preset is stored.

/dev..../system/preset/save

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Save the actual setting as preset.

/dev..../system/properties/freqresolution

Properties: Read
Type: Integer (64 bit)
Unit: None

The number of bits used to represent a frequency.

/dev..../system/properties/freqscaling

Properties: Read
Type: Double
Unit: None

The scale factor to use to convert a frequency represented as a freqresolution-bit integer to a floating point value.

/dev..../system/properties/maxfreq

Properties: Read
Type: Double
Unit: None

The maximum oscillator frequency that can be set.

/dev..../system/properties/maxtimeconstant

Properties: Read
Type: Double
Unit: s

The maximum demodulator time constant that can be set. Only relevant for lock-in amplifiers.

/dev..../system/properties/minfreq

Properties: Read
Type: Double
Unit: None

The minimum oscillator frequency that can be set.

/dev..../system/properties/mintimeconstant

Properties: Read
Type: Double
Unit: s

The minimum demodulator time constant that can be set. Only relevant for lock-in amplifiers.

/dev..../system/properties/negativefreq

Properties: Read
Type: Integer (64 bit)
Unit: None

Indicates whether negative frequencies are supported.

/dev..../system/properties/timebase

Properties: Read
Type: Double
Unit: s

Minimal time difference between two timestamps. The value is equal to 1/(maximum sampling rate).

/dev..../system/saveports

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Flag indicating that the TCP and UDP ports should be saved.

/dev..../system/shutdown

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Sending a '1' to this node initiates a shutdown of the operating system on the device. It is recommended to trigger this shutdown before switching the device off with the hardware switch at the back side of the device.

/dev..../system/stall

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Indicates if the network connection is stalled.

/dev..../system/update/labone/busy

Properties: Read
Type: Integer (64 bit)
Unit: None

Device is updating.

/dev..../system/update/labone/data

Properties: Read, Write
Type: ZIVectorData
Unit: None

Node which accepts the Binary file to update the device.

/dev..../system/update/labone/done

Properties: Read
Type: Integer (64 bit)
Unit: None

Updating has finished.

/dev..../system/update/labone/error

Properties: Read
Type: Integer (64 bit)
Unit: None

An error occurred during the update.

/dev..../system/update/labone/progress

Properties: Read
Type: Double
Unit: None

A value between 0 and 1 indicating the update progress.

/dev..../system/update/labone/start

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

If set to 1 the device starts the update.

8.2.19. TRIGGERS

/dev..../triggers/in/n/autothreshold

Properties: Read, Write
Type: Integer (64 bit)
Unit: None

Automatically adjust the trigger threshold. The level is adjusted to fall in the center of the applied transitions.

/dev..../triggers/in/n/level

Properties: Read, Write, Setting
Type: Double
Unit: V

Trigger voltage level at which the trigger input toggles between low and high. Use 50% amplitude for digital input and consider the trigger hysteresis.

/dev..../triggers/out/n/pulsewidth

Properties: Read, Write, Setting
Type: Double
Unit: s

Defines the minimal pulse width for the case of Scope events written to the trigger outputs of the device.

/dev..../triggers/out/n/source

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the signal assigned to the trigger output.

0	"disabled": The output trigger is disabled.
1	"demod1_phase": Oscillator phase of demod 2 (trigger output channel 1) or demod 4 (trigger output channel 2).
2	"scope_trigger": Scope Trigger. Requires the DIG Option.
3	"scope_not_trigger": Scope /Trigger. Requires the DIG Option.
4	"scope_armed": Scope Armed. Requires the DIG Option.
5	"scope_not_armed": Scope /Armed. Requires the DIG Option.
6	"scope_active": Scope Active. Requires the DIG Option.
7	"scope_not_active": Scope /Active. Requires the DIG Option.
36	"th_logic_unit0_out": Threshold Logic Unit 1 output.
37	"th_logic_unit1_out": Threshold Logic Unit 2 output.
38	"th_logic_unit2_out": Threshold Logic Unit 3 output.
39	"th_logic_unit3_out": Threshold Logic Unit 4 output.
52	"mds_sync_out": MDS Sync Out.

8.2.20. TU

/dev..../tu/logicunits/n/inputs/n/channel

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Selects threshold channel to be used as input.

/dev..../tu/logicunits/n/inputs/n/not

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Inverts the input signal.

/dev..../tu/logicunits/n/internalvalue

Properties: Read
Type: Integer (64 bit)
Unit: None

Show the value after logical combination but before output settings. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.

/dev..../tu/logicunits/n/invert

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Changes the output signal to low-active.

/dev..../tu/logicunits/n/keep

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Hold the output state indefinitely once it has changed to the activated state.

/dev..../tu/logicunits/n/ops/n/value

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Logical operation applied to the signals to the left and right of the control.

0	"off": No logical operation selected. No additional signals will be used to generate the output.
1	"and": Use a logical AND operation.
2	"or": Use a logical OR operation.
3	"xor": Use a logical XOR operation.

/dev..../tu/logicunits/n/pulsewidth

Properties: Read, Write, Setting
Type: Double
Unit: s

Minimum pulse width for the generated output signal. The width of pulses shorter than this time will be extended.

/dev..../tu/logicunits/n/value

Properties: Read
Type: Integer (64 bit)
Unit: None

Shows the output value of the logic unit. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.

/dev..../tu/thresholds/n/activationtime

Properties: Read, Write, Setting
Type: Double
Unit: s

Minimum duration of threshold violation needed before the output is activated. This can be used as a glitch filter.

/dev..../tu/thresholds/n/deactivationtime

Properties: Read, Write, Setting
Type: Double
Unit: s

Minimum duration of threshold compliance needed before the output is deactivated.

/dev..../tu/thresholds/n/filter/abs

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Takes the absolute value of an analog input signal.

/dev..../tu/thresholds/n/filter/enable

Properties: Read
Type: Integer (64 bit)
Unit: None

Enables the low-pass filter applied to the analog input signal

/dev..../tu/thresholds/n/filter/filteredvalue

Properties: Read
Type: Double
Unit: Dependent

Shows the value after the low-pass filter.

/dev..../tu/thresholds/n/filter/mode

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Selects the analysis mode defining the output signal.

- 0 "above_upper": Enable if the Signal is above Upper Threshold.
- 1 "below_lower": Enable if the Signal is below Lower Threshold.
- 2 "outside_range": Enable if the Signal is outside the range [Lower Threshold, Upper Threshold].
- 3 "crossing_both_rising": Enable if the Signal crosses Upper Threshold from below Lower Threshold. The difference between Upper and Lower Threshold defines the threshold hysteresis.
- 4 "crossing_both_falling": Enable if the Signal crosses Lower Threshold from above Upper Threshold. The difference between Upper and Lower Threshold defines the threshold hysteresis.

/dev..../tu/thresholds/n/filter/sourcesaturated

Properties: Read
Type: Integer (64 bit)
Unit: Dependent

Flag indicating whether the source value is saturated. Values are: 0: not saturated; 1: saturation at lower bound; 2: saturation at upper bound; 3: saturation at both lower and upper bounds.

/dev..../tu/thresholds/n/filter/thresholdhigh

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Upper threshold used to generate the output. In Falling Edge mode, this parameter defines the hysteretic behavior as the output state is reset only when the signal crosses Upper Threshold from below.

/dev..../tu/thresholds/n/filter/thresholdlow

Properties: Read, Write, Setting
Type: Double
Unit: Dependent

Lower threshold used to generate the output. In Rising Edge mode, this parameter defines the hysteretic behavior as the output state is reset only when the signal crosses Lower Threshold from above.

/dev..../tu/thresholds/n/filter/timeconstant

Properties: Read, Write, Setting
Type: Double
Unit: s

Defines the characteristic time constant (cut off) of the low-pass filter applied to the analog input signal.

/dev..../tu/thresholds/n/input

Properties: Read, Write, Setting
Type: Integer (enumerated)
Unit: None

Select the signal source to be used in the threshold unit.

0	"demod_x": Demod X
1	"demod_y": Demod Y
2	"demod_r": Demod R
3	"demod_theta": Demod Theta
5	"pid_out": PID Out
6	"boxcar": Boxcar
7	"au_cartesian": AU Cartesian
8	"au_polar": AU Polar
9	"pid_shift": PID Shift
10	"pid_error": PID Error
50	"dio": DIO
51	"trigin", "trigger_input": Trigger In
52	"trigout", "trigger_output": Trigger Out
53	"input_overload_v": Input Overload (V)
54	"input_overload_i": Input Overload (I)
55	"output_overload": Output Overload
56	"auxin_overload", "auxiliary_input_overload": Aux input Overload
57	"auxout_overload", "auxiliary_output_overload": Aux Output Overload
58	"pid_output_overload": PID Output Overload
59	"tu_output": TU Output Value

/dev..../tu/thresholds/n/inputchannel

Properties: Read, Write, Setting
Type: Integer (64 bit)
Unit: None

Select the channel according to the selected signal source.

/dev..../tu/thresholds/n/internalvalue

Properties: Read
Type: Integer (64 bit)
Unit: Dependent

Shows the value after threshold but before minimum time. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.

/dev..../tu/thresholds/n/value

Properties: Read
Type: Integer (64 bit)
Unit: Dependent

Shows the threshold value at the threshold output. The value integrated over some time. Values are 1: low, 2: high, 3: was low and high in the period.